

ElonEmails

Every publicly available Elon Musk email we could source, with the published text, dates, tags and sources.

389 emails · 274 written by Musk · 250 with full text · 58,270 words · 2003–2026

Compiled October 3, 2026. An unofficial, independent compilation, not affiliated with Elon Musk or any of his companies.

How to read this archive

Each entry reproduces an email as it was published by the source linked beneath it: a court exhibit, a news outlet that obtained the email, a government release, or Musk himself. Nothing has been reconstructed or filled in.

Every entry is marked Full text, Partial text or Excerpt. Partial means the source printed most of the email or several passages; a bracketed ellipsis [...] marks a gap. Excerpt means only a line or two was ever made public. Two entries have no published text at all and carry only a summary.

Entries are grouped by company and run in date order. Replies from other people are included where they belong to a published thread, so exchanges can be read in sequence. The sender is always shown.

Most text was transcribed from web pages and court PDFs by automated retrieval. Wording was checked against a second source where one existed, but small differences in punctuation or line breaks from the originals are possible. For anything you intend to quote, follow the source link.

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Tesla

Company-wide memos, board and investor correspondence, and trial exhibits.

103 emails, July 18, 2006 to June 2024

'Incredibly insulting': Musk demands credit in Roadster press

From: Elon Musk

Date: July 18, 2006

To: Mike Harrigan (Tesla VP of marketing)

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · WIDELY REPORTED

like to talk with every major publication within reason.

[...]

The way that my role as been portrayed to date, where I am referred to merely as 'an early investor' is outrageous.

[...]

Apart from me leading the Series A & B and co-leading the Series C, my influence on the car itself runs from the headlights to the styling to the door sill to the trunk, and my strong interest in electric transport predates Tesla by a decade.

[...]

Martin should certainly be the front and center guy, but the portrayal of my role to date has been incredibly insulting.

[...]

I'm not blaming you or others at Tesla — the media is difficult to control. However, we need to make a serious effort to correct this perception.

About this email: Days before the Roadster unveiling, Musk complains to Tesla's marketing chief that press coverage calls him merely an early investor, lists his influence on the car, and demands a serious effort to correct the perception.

Notes: Business Insider published images of the emails 'pasted from court documents' (Eberhard v. Musk, San Mateo County Superior Court, 2009); the images did not render, so only the passages quoted in the article text are recorded, in article order. First passage begins mid-sentence as quoted ('as been' is as printed). Recipient's title is from general reporting, not the email header.

Sources: [Business Insider \(Drake Baer\)](#), via [Yahoo Finance](#); [Business Insider \(original\)](#)

Tags: press, founding-era, roadster, credit, eberhard

Threat to fire PR firm over New York Times article

From: Elon Musk

Date: July 20, 2006

To: Tesla's PR firm PCGC; cc Martin Eberhard and Mike Harrigan

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · WIDELY REPORTED

I was incredibly insulted and embarrassed by the NY Times article

[...]

where I am not merely unmentioned, but where Martin is actually referred to as the chairman.

[...]

If anything like this happens again, please consider the PCGC relationship with Tesla to end immediately upon publication of such a piece. Please ensure that the NYT publishes a correction as soon as possible.

About this email: After a New York Times piece on the Roadster omitted him and called Eberhard chairman, Musk tells Tesla's PR agency he was insulted and embarrassed, demands a correction, and warns the relationship will end if it happens again.

Notes: From emails filed in Eberhard v. Musk (2009) and reproduced as images by Business Insider; only article-text quotations could be retrieved. The article says the email was cc'd to Eberhard and Harrigan; the primary addressee is inferred from the reference to 'the PCGC relationship' and should be checked against the exhibit.

Sources: [Business Insider \(Drake Baer\)](#), via [Yahoo Finance](#); [Business Insider \(original\)](#)

Tags: press, founding-era, new-york-times, credit, eberhard

'This is not good': board warned about Eberhard

From: Elon Musk

Date: August 9, 2007

To: Jim Marver; Ira Ehrenpreis; Antonio J. Gracias; Kimbal Musk (Tesla board members)

Subject: Fw: meeting

FULL TEXT · BOARD, INVESTORS & ADVISERS · PUBLISHED BY MUSK · CONFIRMED

This is not good. Sounds like there are more issues that the board was not informed about. I will send out a note as soon as I talk with Marks and will ask him to email an assessment to the board as soon as he can.

Martin seems to be focused on his public image and position within Tesla, rather than solving these critical problems. If you should speak with Martin, please urge him to spend all his energy on making sure that the Roadster works and arrives on time. He doesn't seem to understand that the best way to maximize his reputation and position within the company is to help get this right.

About this email: Forwarding a message about a meeting, Musk tells fellow directors that the board was apparently not told about more problems and that Martin Eberhard is focused on his image rather than fixing the Roadster.

Notes: Header as published: 'From: Elon Musk / Sent: Thursday, August 09, 2007 08:56 AM Pacific Standard Time / To: Jim Marver; Ira Ehrenpreis; Antonio J. Gracias; kimbal / Subject: Fw: meeting'. Text is Musk's message

as reproduced by TechCrunch; the forwarded message beneath it was not reproduced. Musk published it in his 2009 Tesla blog rebuttal to the Eberhard v. Musk lawsuit; the original blog page (teslamotors.com) and its Wayback copy could not be reached in this session. The same post also printed emails TO Musk from Michael Marks (15 Nov 2007, 'completely agree with the idea that Martin shouldn't stay on the board') and Darryl Siry (8 May 2008).

Sources: [TechCrunch \(reproducing Musk's Tesla blog post of 22 June 2009\)](#); [VentureBeat](#)

Tags: eberhard, board, roadster, founding-era

Marks to Musk: Eberhard shouldn't stay on the board

From: Michael Marks (then Tesla interim CEO)

Date: November 15, 2007

To: Elon Musk

Subject: board

FULL TEXT · BOARD, INVESTORS & ADVISERS · PUBLISHED BY MUSK · CONFIRMED

I've thought about it, and completely agree with the idea that Martin shouldn't stay on the board, just fyi

About this email: Interim CEO Michael Marks tells Musk he agrees with the idea that Martin Eberhard should not remain on Tesla's board. Musk published it in 2009 to rebut Eberhard's lawsuit.

Notes: Email TO Musk, published by Musk himself; header as published: From Michael Marks, Sent Thursday, November 15, 2007 10:38 PM, To Elon Musk, Subject 'board'. Musk's side of the exchange was not published. Marks's title is from general reporting, not the header.

Sources: [TechCrunch \(reproducing Musk's Tesla blog post of 22 June 2009\)](#)

Tags: eberhard, board, founding-era, marks

Forwarding leaker Peng Zhou's apology to all staff

From: Elon Musk

Date: November 3, 2008

To: Everybody (all Tesla employees)

Subject: FW: Letter of Apology

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Below is what I believe to be a sincere apology from Peng Zhou for the leak to Valleywag.

About this email: Musk forwards to the whole company an apology letter from engineer Peng Zhou, who admitted leaking a Musk email about Tesla's finances to Valleywag and resigned.

Notes: Header as published: Monday, November 03, 2008 4:22 PM, From Elon Musk, To Everybody, Subject 'FW: Letter of Apology'. Musk's own text is the single line; the forwarded letter from Peng Zhou ('Dear Tesla Motors Employees ... I became very upset and did the very foolish thing of writing a letter to Valleywag ... I have submitted my resignation') follows it in the source. First published by Valleywag.

Sources: [The Truth About Cars](#)

Tags: leaks, valleywag, layoffs, culture

Peng Zhou's apology for the Valleywag leak

From: Peng Zhou (Tesla engineer)

Date: November 3, 2008 (Date of Musk's forward; the letter itself was sent to Musk the same day or shortly before)

To: Elon Musk (addressed to all Tesla employees; forwarded company-wide by Musk)

Subject: Re: Letter of Apology

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Dear Tesla Motors Employees,

Tesla Motors is a great company and is pioneering the green revolution. I have always been a proud Tesla employee and we all worked extremely hard towards the company mission. Throughout past two years, I also took a wild ride in the great emotional roller-coaster of WhiteStar 1.0, 2.0 and Model S. Having driven Model S mule few times, I truly believe this is the vehicle that will change the world.

However, macro economic climate changed so dramatically and Model S is getting delayed and the company is refocusing. The past month has been very difficult, sitting through planning meetings and watch employees make in or out of the layoff list. It is so sad to lose 87 employees in a week. I became very upset and did the very foolish thing of writing a letter to Valleywag. I have never thought this letter would create such an upsetting situation for Tesla Motors and I should have never sent that letter. When Elon asked the originator of the letter to step forward, I should have done so. I am deeply sorry things happened this way.

I have submitted my resignation with my manager, who had no knowledge of this. I'll try my best to make situations better for Tesla Motors.

Again, my sincere apologies.

Best Regards,

Peng

About this email: Tesla engineer Peng Zhou admits he wrote the letter to Valleywag about Tesla's cash position after the October 2008 layoffs, apologizes for not stepping forward when Musk asked, and says he has resigned. Musk forwarded it to the whole company.

Notes: Email TO Musk, included because Musk forwarded it to all employees (see the companion entry). Original wording kept as published. Musk's earlier email asking the leaker to step forward was not found in full text.

Sources: [The Truth About Cars](#)

Tags: leaks, valleywag, layoffs, apology, resignation

Explaining the Roadster option price increase to customers

From: Elon Musk

Date: January 20, 2009

To: Tesla Roadster customers

Subject: The Importance of Options

FULL TEXT · CUSTOMERS & PUBLIC · PUBLISHED BY RECIPIENT · WIDELY REPORTED

A much fuller account of the history of Tesla is worth telling at some point, but for now I will just talk about the essentials of why we needed to raise prices on options.

Fundamentally, it boils down to taking the tough steps that are difficult but necessary for Tesla to be a healthy company and not fall prey to the recession.

When the initial base price, for cars after the Signature 100 series, of \$92k was approved by the board a few years ago, it was based on an estimated vehicle cost of roughly \$65k provided by management at the time. This turned out to be wrong by a very large margin.

An audit by one of the Series D investors in the summer of 2007 found that the true cost was closer to \$140k, which was obviously an extremely alarming discovery and ultimately led to a near complete change in the makeup of the senior management team. Over the past 18 months, observers will note that Tesla has transformed from having a senior team with very little automotive experience to one with deep automotive bench strength. We now have executives with world class track records running everything from design to engineering to production to finance.

To bring the cost of the car down, we have reengineered the entire drivetrain, which is now at version 1.5 and will be at version 2 by June. The body supplier was also switched out from a little company that was charging us nutty money and had a max production of three per week to Sotira, who supplies high paint quality body panels to Lotus, Aston Martin and others. In the process, we had to pay several million dollars for a whole new set of body tooling, as the old tooling had been made incorrectly. The old HVAC system was unreliable and cost almost as much as a new compact car, so also had to be replaced. The wiring harness, seats, navigation system and instrument panel also had to be modified or replaced.

After reengineering and retooling virtually the entire Roadster and completely restructuring our supply chain, we are now finally coming to the point where the variable cost of the car (to be clear, this excludes fixed cost allocation) is between \$90k to \$100k. With a lot of additional effort by the Tesla team and the help of our suppliers, we should be at or below \$80k by this summer. There is some variability here due to exchange rate shifts. Although we gain an automatic currency hedge by selling in both Europe and the US, we are still vulnerable to the Yen, which is very strong right now.

Obviously, this still creates a serious problem for Tesla in the first half of 2009, given the \$92k to \$98k price of most cars delivered over this time period. The board and I did not want to do a retroactive increase of the base vehicle price, as that would create an unavoidable hardship for customers. Instead, apart from a \$1k destination charge increase to match our true cost of logistics, we only raised the price of the optional

elements and provided new options and a new model (Roadster Sport) to help improve the average margin per car.

The plan as currently projected, and which I believe is now realistic, shows a high likelihood of reaching profitability on the Roadster business this summer. By that time, we will be delivering cars that have a base price of \$109k plus about \$20k or so of options (having worked our way through the \$92k to \$98k early buyers) at a rate of 30 per week. We are fortunately in the position, rare among carmakers, of not having to worry too much about meeting 2009 sales targets, as we are already sold out through October and have barely touched the European market.

My paramount duty is to ensure that we get from here to there without needing to raise more money in this capital scarce environment, even if things don't go as well as expected. I firmly believe that the plan above will achieve that goal and that it strikes a reasonable compromise between being fair to early customers and ensuring the viability of Tesla, which is obviously in the best interests of all customers. It's also important to note that the price increases will affect 400 customers, all of whom will take delivery after Jan. 1 and receive a \$7,500 federal tax credit. We made the pricing changes to ensure the viability of Tesla in the long term, regardless of government incentives, but we hope the credit will offset the increase for most customers.

There is one additional point that relates to the government loans that Tesla is seeking for the Model S program, a much more affordable sedan that we are trying to bring to market as soon as possible. A key requirement is that any company applying be able to show that it is viable without the loans. If we allow ourselves to lose money on the cars we are shipping today, we place those loans at risk. Mass market electric cars have been my goal from the beginning of Tesla. I don't want and I don't think the vast majority of Tesla customers want us to do anything to jeopardize that objective.

Elon Musk

CEO & Product Architect

About this email: Musk explains to Roadster reservation holders why Tesla raised option prices: the car's true cost was far above the original estimate, the Roadster had to be re-engineered, and Tesla must show viability to obtain government loans for the Model S.

Notes: Mass email to Roadster reservation holders; header as reproduced by recipient: From 'Elon Musk, Tesla Motors', Tue, 20 Jan 2009 12:45 pm PST.

Sources: [Tom Saxton's blog \(Roadster customer\)](#); [The Truth About Cars](#); [Jalopnik](#)

Tags: roadster, pricing, customers, cost-cutting, doe-loan

Leak crackdown: sign a declaration or confess for amnesty

From: Elon Musk

Date: March 2009 (Published by The Truth About Cars on 2009-03-06 as 'recently sent'; exact send date not given)

To: All Tesla employees and contractors (Bay Area)

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

I'm a big believer in trusting employees and sharing information widely within the company, rather than confining it to a narrow set of senior execs and giving everyone else the mushroom treatment. Providing people with an understanding of what problems need to be overcome helps them align and prioritize their actions in pursuit of the greater good. It also ensures that all employees feel included and part of the same team.

This is why I'm so concerned about the continuing leaks to media. It really hurts free communication when even minor issues are leaked and blown way out of proportion. It is nutty that a company like Tesla, which is doing really well right now (how many companies can say that they're sold out through October?) should suffer from misleading articles on blog sites that would have no credibility, but for a purported inside leak. The leaks often aren't even accurate!

This kills trust and creates a negative atmosphere within Tesla. It has to stop.

Today, the legal department will circulate a declaration form to all employees and contractors within the Bay Area. People will be asked to provide their word of honor and signature that they haven't knowingly leaked any Tesla confidential information to the media. They'll be reminded in clearly written language of the substantial liability they would incur for disclosure of confidential information in willful violation of the confidentiality agreement they signed with Tesla. If someone does not tell the full truth here, please take my word that you will be prosecuted to the fullest extent of the law.

Alternatively, people will be given the option of listing every leak they have made, whether published or not. If you fully disclose any leak you have done, the consequences will be precisely nothing. You will be completely forgiven and, unlike Peng, won't be asked to publicly apologize to the company.

The actions of any one person can't be allowed to hurt the vast majority of people at Tesla who are working incredibly hard to make a difference in the world.

Elon

About this email: Musk tells staff that continuing leaks to the media are destroying trust, announces that legal will circulate a declaration every Bay Area employee and contractor must sign, and offers full amnesty to anyone who discloses past leaks.

Notes: TTAC inserted its own editorial aside '(ed: as opposed to convoluted plumb-and-purge operations?)' after 'negative atmosphere within Tesla.'; that aside is not Musk's and has been removed here. TTAC reported each copy carried subtle textual variations to trace leakers, so wording of other copies may differ slightly. Originally surfaced via Valleywag/Gawker.

Sources: [The Truth About Cars](#)

Tags: leaks, confidentiality, legal, culture

Disputed: alleged Musk email to employee who missed event for child's birth

From: Elon Musk

Date: 2015 (send date unknown; 2015 is the year the book quoting it was published)

To: Unnamed Tesla employee

EXCERPT · EXECUTIVE & INTERNAL · BOOK/BIOGRAPHY · SINGLE-SOURCE

That is no excuse. I am extremely disappointed. You need to figure out where your priorities are. We're changing the world and changing history, and you either commit or you don't.

About this email: Ashlee Vance's 2015 biography quotes an anonymous Tesla employee's recollection of an email from Musk rebuking him for missing a company event to attend his child's birth. Musk publicly denied ever writing or saying it.

Notes: AUTHENTICITY DISPUTED. The wording is an anonymous employee's recollection as printed in Vance's book and first reported by the Washington Post; no copy of the email has been published. Musk's denial on Twitter (May 2015): "I have never written or said this. It is total BS & hurtful to claim that I told a guy to miss his child's birth just to attend a company meeting." Tesla-related, so it may overlap with the Tesla slice.

Sources: [CNNMoney \(quoting Ashlee Vance, 'Elon Musk: Tesla, SpaceX, and the Quest for a Fantastic Future'\)](#); [CBS San Francisco](#); [Jezebel](#)

Tags: ashlee-vance, biography, work-ethic, disputed

Launching the owner referral program

From: Elon Musk

Date: July 29, 2015

To: Tesla Model S owners

Subject: Trying something new (plus party at the Gigafactory and a Founder Series Model X)

FULL TEXT · CUSTOMERS & PUBLIC · PUBLISHED BY RECIPIENT · WIDELY REPORTED

Word of mouth has always been a major part of how Tesla sales have grown. When I meet Tesla owners, one of the first things they often tell me is how they have convinced many others to buy the car. As you may already know, Tesla does not advertise or pay for endorsements or product placement. Maybe by doing so we could sell more cars, but I don't like the idea of trying to trick people into buying a product by false association. If you see somebody famous driving a Model S, it is because they genuinely like the car.

If you see it in a movie or TV show, it is because the people associated with that production genuinely like the car. Besides word of mouth, another way that our cars are sold is through stores. These will always be important to allow people to check out new models and ask our product specialists detailed questions. However, stores are quite expensive to set up and operate. In reviewing the Tesla cost of sales, we found that it is approximately \$2,000 to sell a car through our stores, higher in some regions and lower in others.

Both ways of reaching potential customers are important, but, if we can amplify word of mouth, then we don't need to open as many new stores in the future. So, we are going to try an experiment. This is similar to the customer growth program that I worked on at PayPal/X.com back in '99. What worked for PayPal may not work for Tesla, but it is

worth trying, as the net result would be lowering our costs by \$2,000, allowing us to give that money to our customers.

From now through October 31st, if someone buys a new Model S through your link, they will get \$1,000 off the purchase price and you will get a \$1,000 credit in your Tesla account, which can be applied to a future car purchase, service charge or accessories. To put some limits on the experiment, each Tesla owner can grant a maximum of ten \$1,000 discounts.

Just for fun, there will also be some things that money can't buy. If five of your friends order a Model S, you and a guest will receive an invitation to tour the Gigafactory in Nevada – the world's biggest factory by footprint – and attend the grand opening party. This will be awesome. At ten orders, you get the right to purchase a Founder Series Model X, which is not available to the public, with all options free (value of about \$25,000). The first person to reach ten will get the entire car for free.

Elon

About this email: Musk emails Model S owners to announce an experimental referral program: \$1,000 off for a referred buyer and \$1,000 credit for the owner, with a Gigafactory opening invitation at five referrals and a Founder Series Model X at ten.

Notes: Mass email to owners; header as reproduced: From "Elon Musk" <elon@teslamotors.com>, July 29, 2015 at 2:05:31 PM PDT. Paragraph breaks follow Teslarati's reproduction and may differ from the original.

Sources: [Teslarati](#)

Tags: referral-program, customers, sales, gigafactory, model-x

'Save costs or we are doomed'

From: Elon Musk

Date: February 14, 2016

To: Tesla executives (recipients not named in reporting)

EXCERPT • EXECUTIVE & INTERNAL • COURT RECORD • WIDELY REPORTED

I just gave a company talk making it clear that we need to save costs or we are fucking doomed. This [sic] words need to be taken to heart. There is zero appetite for investors to give Tesla more money.

About this email: In a February 2016 email later used in the SolarCity shareholder litigation, Musk says he has just told the company it must cut costs and that investors have no appetite to give Tesla more money.

Notes: Quoted to Musk during his 2019 deposition in In re Tesla Motors, Inc. Stockholder Litigation (Del. Ch. C.A. No. 12711-VCS); '[sic]' is in the source. Recipients were not identified in the reporting retrieved.

Sources: [ProMarket \(Stigler Center\)](#)

Tags: cost-cutting, cash, solarcity, fundraising

To Fortune: Autopilot death 'not material', 'do the bloody math'

From: Elon Musk

Date: July 4, 2016 (Sent while Fortune sought comment on July 4, 2016; article published July 5)

To: Carol J. Loomis (Fortune)

EXCERPT · PRESS & JOURNALISTS · PUBLISHED BY RECIPIENT · CONFIRMED

is not material to the value of Tesla.

[...]

Indeed, if anyone bothered to do the math (obviously, you did not) they would realize that of the over 1M auto deaths per year worldwide, approximately half a million people would have been saved if the Tesla autopilot was universally available. Please, take 5 mins and do the bloody math before you write an article that misleads the public.

About this email: Challenged by Fortune over Tesla selling stock without disclosing the fatal May 2016 Autopilot crash, Musk replies that the death is not material to Tesla's value and berates the writer for not doing the math on lives Autopilot could save.

Notes: Only the passages Fortune quoted from the exchange are public; the first fragment begins mid-sentence.

Sources: [Fortune](#)

Tags: autopilot, press, fortune, stock-sale, safety

Q3 2016 push: 'throw a pie in the face of all naysayers'

From: Elon Musk

Date: August 29, 2016

To: All Tesla employees

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

I thought it was important to write you a note directly to let you know how critical this quarter is. The third quarter will be our last chance to show investors that Tesla can be at least slightly positive cash flow and profitable before the Model 3 reaches full production.

[...]

on the razor's edge of achieving a good Q3, but it requires building and delivering every car we possibly can, while simultaneously trimming any cost that isn't critical, at least for the next 4.5 weeks.

[...]

We will need to raise additional cash in Q4 to complete the Model 3 vehicle and the Gigafactory.

[...]

The simple reality of it is that we will be in a far better position to convince potential investors to bet on us if the headline is not 'Tesla Loses Money Again,' but rather 'Tesla

Defies All Expectations and Achieves Profitability.' That would be amazing!

[...]

It would be awesome to throw a pie in the face of all naysayers on Wall Street who keep insisting that Tesla will always be a money loser!

About this email: Musk tells employees the third quarter of 2016 is Tesla's last chance to show positive cash flow and profit before Model 3, urging them to deliver every possible car and cut non-critical costs ahead of a Q4 capital raise.

Notes: Email obtained by Bloomberg, which printed only passages. Passages assembled from Electrek and the Bloomberg reprint; their original order within the email is not certain. The 'razor's edge' passage begins mid-sentence as quoted.

Sources: [Bloomberg](#); [Electrek](#); [Fortune](#); [David Stockman's Contra Corner \(Bloomberg reprint\)](#); [CNN Money](#)

Tags: end-of-quarter, profitability, cost-cutting, fundraising, model-3

No discounts on new cars, ever

From: Elon Musk

Date: September 28, 2016 (Date Musk tweeted an image of the email; it was sent to employees that week)

To: All Tesla employees

PARTIAL TEXT · COMPANY-WIDE MEMO · PUBLISHED BY MUSK · CONFIRMED

That said, it is absolutely vital that we adhere to the no negotiation and no discount policy that has been true since we first started taking orders ten years ago. This is fundamental to our integrity and we maintained this policy even through the terrible depths of the great recession of 2008/2009.

It is fine to have a discount on cars that have been floor models , were used in test drives or were damaged before delivery. All we are doing there is assigning an accurate price to the vehicle. However, there can never — and I mean never — be a discount on a new car coming out of the factory in pristine condition, where there is no underlying rationale. This is why I always pay full price when I buy a car and the same applies to my family, friends and celebrities, no matter how famous or influential.

[...]

The acid test is that if you can't explain to a customer who paid full price why another customer didn't without being embarrassed, then it is not right. We either win in a way that is fair and right or we lose with our honor intact and accept the consequences.

[...]

I'm sorry for the Draconian language, as I am super grateful for your hard work, but there is nothing that matters more than our integrity as a company... Customers need to know with absolute certainty that they can always trust Tesla to do the right thing.

About this email: After a Reddit/analyst report of end-of-quarter discounting, Musk tells employees the no-negotiation, no-discount policy is fundamental to Tesla's integrity: floor models and damaged cars may be marked down, but never a pristine new car.

Notes: Musk posted the email as an image on Twitter; the image text could not be read in this session, so the text is assembled from passages quoted by Electrek (first two paragraphs), Autoblog ('acid test') and Jalopnik

('Draconian' passage, whose ellipsis is Jalopnik's). Fortune also quotes Musk saying the quarter 'is likely to be the best ever in Tesla history.' The opening of the email is missing.

Sources: [Electrek](#); [Autoblog](#); [Jalopnik](#); [Fortune](#); [CNN Money](#)

Tags: discounts, pricing, sales, integrity, end-of-quarter

Saudi PIF asks Musk for a meeting with Yasir Al-Rumayyan

From: Saad Al-Jarboa (Saudi Public Investment Fund)

Date: November 7, 2016

To: Elon Musk (via EM Desk - SpaceX); cc: Turqi Alnowaiser and a redacted recipient

Subject: Request to Meet with H.E. Yasir Al

FULL TEXT · BOARD, INVESTORS & ADVISERS · COURT RECORD · CONFIRMED

Dear Mr. Musk,

My name is Saad Al-Jarboa and I work with H.E. Mr. Yasir Al-Rumayyan at the Public Investment Fund (PIF).

H.E. has asked me to reach out to you in order to inquire about your availability to meet with him next week. HE would be willing to meet with you in either London or New York.

I would appreciate it if you could get back to us with some times that might work for you next week.

Best,

Saad Al-Jarboa

PIF

About this email: An aide at the Saudi Public Investment Fund writes to Musk's office asking when he could meet PIF governor Yasir Al-Rumayyan in London or New York the following week.

Notes: Trial Exhibit 105. Email TO Musk (addressed 'Dear Mr. Musk', sent to emdesk@spacex.com) that he answered internally on Nov 10, 2016. Standard PIF disclaimer footer omitted by the transcription.

Sources: [U.S. District Court, N.D. Cal. - admitted trial exhibits, In re Tesla Inc. Securities Litigation \(18-cv-04865-EMC\)](#); [N.D. Cal. admitted-evidence directory for In re Tesla Inc. Securities Litigation](#)

Tags: saudi-pif, yasir-al-rumayyan, scheduling, funding-secured

Musk declines to travel to meet Saudi PIF's Yasir Al-Rumayyan

From: Elon Musk

Date: November 10, 2016

To: Sam Teller (cc: Emma Gallagher)

Subject: Re: NEXT WEEK - FW: Request to Meet with H.E. Yasir Al

FULL TEXT · BOARD, INVESTORS & ADVISERS · COURT RECORD · CONFIRMED

I appreciate the offer, but my obligations prevent me from traveling to those locations on those dates. Let's just schedule something when he is next in California. This is not an urgent matter.

About this email: Replying to a forwarded request from the Saudi Public Investment Fund for a meeting with Yasir Al-Rumayyan in London or New York, Musk tells his chief of staff he cannot travel and that the matter is not urgent.

Notes: Trial Exhibit 105. Sent from erm@spacex.com, Thursday Nov 10, 2016, 3:35 PM. The thread begins with a Nov 7, 2016 email from PIF's Saad Al-Jarboa to EM Desk asking for a meeting with H.E. Yasir Al-Rumayyan; Sam Teller then told Emma Gallagher: "Don't say not urgent / Just say not possible unfortunately / Hope to soon".

Sources: [U.S. District Court, N.D. Cal. - admitted trial exhibits, In re Tesla Inc. Securities Litigation \(18-cv-04865-EMC\)](#); [N.D. Cal. admitted-evidence directory for In re Tesla Inc. Securities Litigation](#)

Tags: saudi-pif, yasir-al-rumayyan, scheduling, funding-secured

Musk sets time for San Francisco meeting with Yasir

From: Elon Musk

Date: January 31, 2017

To: Emma Gallagher (cc: Sam Teller)

Subject: Re: Ya sir wants to meet in SF

FULL TEXT · BOARD, INVESTORS & ADVISERS · COURT RECORD · CONFIRMED

Let's aim for 11:30

About this email: A one-line scheduling reply about a San Francisco meeting with PIF's Yasir Al-Rumayyan; his assistant then confirmed 11:30am at OpenAI.

Notes: Trial Exhibit 106. Sent from erm@spacex.com on Jan 31, 2017, 7:56 AM, with an attached image (image1.PNG, apparently a screenshot). Emma Gallagher replied at 4:47 PM: "We are confirmed for 11:30am at OpenAI." Subject line spelled 'Ya sir' in the original.

Sources: [U.S. District Court, N.D. Cal. - admitted trial exhibits, In re Tesla Inc. Securities Litigation \(18-cv-04865-EMC\)](#); [N.D. Cal. admitted-evidence directory for In re Tesla Inc. Securities Litigation](#)

Tags: saudi-pif, yasir-al-rumayyan, scheduling

Response to UAW drive: safety, pay, frozen yogurt, roller coaster

From: Elon Musk

Date: February 23, 2017 (Sent Thursday night, 23 Feb 2017, per BuzzFeed News; published 24 Feb)

To: All Tesla employees

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

For Tesla to become and remain one of the great companies of the 21st century, we must have an environment that is as safe, fair and fun as possible. It is incredibly important to me that you look forward to coming to work every day. For that, we must be a fair and just company – the only kind worth creating.

This is vital to succeed in our mission to accelerate the advent of a clean, sustainable energy future. The forces arrayed against us are many and incredibly powerful. This is David vs Goliath if David were six inches tall! Only by being smarter, faster and working well as a tightly integrated team do we have any chance of success. We should never forget the history of car startups originating in the United States: dozens have gone bankrupt and only two, Tesla and Ford, have not. Despite the odds being strongly against us, my faith in you is why I am confident that we will succeed.

That is why I was so distraught when I read the recent blog post promoting the UAW, which does not share our mission and whose true allegiance is to the giant car

companies, where the money they take from employees in dues is vastly more than they could ever make from Tesla.

The tactics they have resorted to are disingenuous or outright false. I will address their underhanded attacks below. While this discussion focuses on Fremont, these same principles apply to every Tesla facility worldwide.

Safety First

The workplace issue that comes before any other is safety. If you do not have your health, then nothing else matters. Simply due to size and bad luck, there will always be some injuries in a company with over 30,000 employees, but our goal is simple: to have as close to zero injuries as possible and be the safest factory in the auto industry by far. The Tesla executive team and I are absolutely committed to this goal.

That is why I was particularly troubled by the safety claim in last week's blog post, which said: "A few months ago, six out of eight people in my work team were out on medical leave at the same time due to various work-related injuries. I hear the ergonomics are even more severe in other areas of the factory."

Obviously, this cannot be true: if three quarters of his team suddenly went on medical leave, we would not be able to operate that part of the factory. Furthermore, if things were really even worse in other departments, that would mean something like 80% or more of the factory would be out on injury, production would drop to virtually nothing and the parking lot would be almost empty. As you know firsthand, we have the *opposite* problem – there is never enough room to park! In fact, we are working at top speed to build more parking. Also, hopefully our darn BART train station will open before all hell freezes over!

After looking into this claim, not only was it untrue for this individual's team, it was untrue for any of the hundreds of teams in the factory.

That said, reducing excess overtime and improving safety are extremely important. This is why we hired thousands of additional team members to create a third shift, which has reduced the burden on everyone. Moreover, since the beginning of Tesla production at Fremont five years ago, there have been dedicated health and safety experts covering the factory and we hold regular safety meetings with operations leaders. Since the majority of the injuries in the factory are ergonomic in nature, we have an ergonomics department focused exclusively on this issue.

The net result is that since January 1st, our total recordable incident rate (TRIR) is under 3.3, which is less than half the industry average of 6.7.

Of course, the goal is to have as close to zero injuries as humanly possible, so we need to keep improving. If you have a safety concern or an idea on how to make things better, please let your manager, safety representative or HR partner know. You can also send an anonymous note through the Integrity Hotline (it applies broadly to any problems you notice at our company) or you can email safety@tesla.com.

Compensation

At Tesla, we believe it is important for everyone to be an owner of the company. This is your company. That is why, unlike other car companies, everyone is awarded shares and you get to buy stock at a discount compared to the public through the employee stock purchase program. Last year, stock equity grants were increased significantly and it will happen again later this year once Model 3 achieves high volume.

The chart below contrasts the total comp received by a Tesla production team member who started on January 1, 2013 against the total comp received over the same period at GM, Ford, and Fiat Chrysler (FCA). A four year period is used because that's the vesting length of a new hire equity grant. I believe the equity gain over the next four years will be similar. As shown below, a Tesla team member earned between \$70,000 and \$100,000 more in total compensation than the employees at other US auto companies!

Work Hours

Another issue raised in the UAW blog was hours worked. First, I want to recognize how hard you worked to make our company successful. Those hours mattered to you, to your family and to our company, and I can't tell you how much I appreciate them.

However, the pace needs to be sustainable. This is why the third shift was established and why we created alternate work schedules based on feedback from various teams in the factory.

These changes have had a big impact. The average amount of hours worked by production team members this year is about 43 hours per week. The percentage of overtime hours has declined by almost 50% since the super tough time we had last year achieving rate on the Model X, which is probably the hardest car to build in history. What an amazing accomplishment! It is also a lesson learned, which is why Model 3 is designed to be dramatically easier to manufacture.

Fun

As we get closer to being a profitable company, we will be able to afford more and more fun things. For example, as I mentioned at the last company talk, we are going to hold a really amazing party once Model 3 reaches volume production later this year. There will also be little things that come along like free frozen yogurt stands scattered around the factory and my personal favorite: a Tesla electric pod car roller coaster (with an optional loop the loop route, of course!) that will allow fast and fun travel throughout our Fremont campus, dipping in and out of the factory and connecting all the parking lots. It's going to get crazy good :)

Thanks again for all your effort and I look forward to working alongside you to create an amazing future!

Elon

About this email: Responding to worker Jose Moran's pro-UAW blog post, Musk disputes its injury and overtime claims, defends Tesla's safety record and equity-based pay, and promises perks including free frozen yogurt stands and an electric pod-car roller coaster at Fremont.

Notes: First reported by BuzzFeed News; Electrek printed the full text. The original email included a compensation comparison chart that is not reproduced. Section headings (Safety First, Compensation, Work

Hours, Fun) were bold in the source.

Sources: [Electrek](#); [BuzzFeed News](#); [The Truth About Cars](#); [Engadget](#)

Tags: union, uaw, safety, compensation, fremont

Musk tells Straubel he plans 'something really crazy' on pay

From: Elon Musk

Date: May 12, 2017

To: JB Straubel

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

planning on something really crazy, but also high risk

About this email: Replying to an email from Tesla CTO JB Straubel about equity incentives, Musk describes the compensation plan he has in mind for himself.

Notes: Quoted in paragraph 62 of the amended derivative complaint (March 9, 2022): Straubel had emailed Musk under the subject 'Equity Incentives' asking him to 'consider giving [Straubel] the same vesting milestones that [E. Musk] and the board set up for [E. Musk]'. Only this fragment of Musk's reply is public.

Sources: [Tornetta v. Musk, C.A. No. 2018-0408-KSJM \(Del. Ch. Jan. 30, 2024\), post-trial opinion](#); [Harvard Law School Forum on Corporate Governance \(reproduces the opinion text\)](#); [Amended derivative complaint \(Bernstein Litowitz\)](#)

Tags: pay-package, compensation, jb-straubel

'Doing the right thing': don't be a jerk, accept apologies

From: Elon Musk

Date: May 31, 2017

To: All Tesla employees

Subject: Doing the right thing

PARTIAL TEXT · COMPANY-WIDE MEMO · COURT RECORD · CONFIRMED

Part of not being a huge jerk is considering how someone might feel who is part of an historically less represented group. This doesn't mean there is a different standard of performance or that you can't give critical feedback. You should -- doing anything else would be an insult to the hard work it took to get there -- but don't ever intentionally allow someone to feel excluded, uncomfortable or unfairly treated. Sometimes these things happen unintentionally, in which case you should apologize.

In fairness, if someone is a jerk to you, but sincerely apologizes, it is important to be thick-skinned and accept that apology. If you are part of a less represented group, you don't get a free pass on being a jerk yourself. We have had a few cases at Tesla where someone in a less represented group was actually given a job or promoted over more qualified highly represented candidates and then decided to sue Tesla for millions of dollars because they felt they weren't promoted enough. That obviously is not cool.

About this email: In a company-wide email on workplace conduct, Musk says employees must consider how members of historically less represented groups feel, should apologize for unintended offence and accept sincere apologies, and that minority employees get no free pass; he complains some have sued Tesla after being promoted. A racial-harassment suit quoted it and Tesla then released it in full.

Notes: First quoted in the [Vaughn v. Tesla class-action complaint](#) (Alameda County, Nov 2017); Tesla then published the whole email in its 14 Nov 2017 blog post 'Hotbed of Misinformation', which is no longer on

tesla.com. Fortune and Teslarati reproduce the full email only as an image that could not be read in this session, so the text here is the contiguous passage printed by Bloomberg (two consecutive quotations; paragraph break is Bloomberg's). 'apologies' is as printed by Bloomberg; Jalopnik prints 'apologizes' and '[a] historically'. The opening and close of the email are missing. The safety email Musk sent the same day ('No words can express how much I care about your safety...') may be the first part of this same message; that could not be confirmed.

Sources: [Jalopnik](#); [Bloomberg](#), via [IndustryWeek](#); [The Root](#); [Fortune](#) (image of full email); [Engadget](#); [Teslarati](#)

Tags: harassment, culture, discrimination, lawsuit, conduct

Report every injury to me; I will do the same task

From: Elon Musk

Date: May 31, 2017 (Teslarati reports it was sent on 31 May 2017; Electrek (2 June) said 'this week')

To: All Tesla employees

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

No words can express how much I care about your safety and wellbeing. It breaks my heart when someone is injured building cars and trying their best to make Tesla successful.

Going forward, I've asked that every injury be reported directly to me, without exception. I'm meeting with the safety team every week and would like to meet every injured person as soon as they are well, so that I can understand from them exactly what we need to do to make it better. I will then go down to the production line and perform the same task that they perform.

This is what all managers at Tesla should do as a matter of course. At Tesla, we lead from the front line, not from some safe and comfortable ivory tower. Managers must always put their team's safety above their own.

About this email: After reports about injury rates at Fremont, Musk tells employees that every injury must be reported directly to him, that he will meet each injured worker and then perform the same task on the line himself.

Notes: Electrek and Inc. printed these three paragraphs; no greeting, subject or sign-off was published. Teslarati says the email first surfaced as an image on the Model 3 Owners Club Facebook page. Tesla's company-wide email 'Doing the right thing' was sent the same day (31 May 2017), so these paragraphs may be the opening of that email rather than a separate message; this could not be confirmed because the full email is only available as an image.

Sources: [Electrek](#); [CNBC](#); [The Truth About Cars](#); [The Drive](#); [Teslarati](#); [Inc.](#)

Tags: safety, injuries, fremont, leadership

Musk on hiring Karpathy: 'The OpenAI guys are gonna want to kill me'

From: Elon Musk

Date: June 2017 (MIT Technology Review gives only the year 2017; a summary of the trial exhibit list (Ex. 773) dates it June 2017.)

To: A Tesla vice president (unnamed in coverage)

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · WIDELY REPORTED

The OpenAI guys are gonna want to kill me. But it had to be done

About this email: After hiring OpenAI founding member Andrej Karpathy to lead Tesla's vision team, Musk told a Tesla vice president that OpenAI's people would be furious but that it had to be done. The email was shown to the jury in Musk v. Altman.

Notes: Trial exhibit 773 in Musk v. Altman (2026). Only this line was quoted in coverage I could retrieve; the full exhibit (published by The Verge in its trial-evidence roundup) could not be accessed.

Sources: [MIT Technology Review: Musk v. Altman week 1 \(May 1, 2026\)](#); [Neura News summary of trial exhibits \(Ex. 773\)](#)

Tags: karpathy, hiring, tesla-ai, openai, poaching, musk-v-openai

Musk asks to put his pay discussions on hold

From: Elon Musk

Date: July 30, 2017 (July 30, 2017 per the amended complaint)

To: Todd Maron

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

on hold for a few weeks[.]

About this email: Two days before a planned Compensation Committee meeting, Musk asks Tesla's general counsel to pause work on his new compensation plan.

Notes: Known only through the passages quoted in Chancellor McCormick's post-trial opinion (factual background); the full email is not public. The amended complaint (para. 114) quotes it as "Let's put [my Grant] on hold for a few weeks[.]" and dates it July 30, 2017; the retrieved copy of the opinion text reads 'June 30', but the surrounding passage concerns late July 2017.

Sources: [Tornetta v. Musk, C.A. No. 2018-0408-KSJM \(Del. Ch. Jan. 30, 2024\), post-trial opinion](#); [Harvard Law School Forum on Corporate Governance \(reproduces the opinion text\)](#); [Amended derivative complaint \(Bernstein Litowitz\)](#)

Tags: pay-package, compensation, todd-maron

Maron to Musk: would 'rather keep cranking' on pay plan

From: Todd Maron

Date: July 30, 2017 (reply to Musk's email of about July 30, 2017; exact date not given)

To: Elon Musk

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

rather keep cranking on it . . . because there's a fair amount to it that we've been working on with the board and there's lead time involved.

About this email: Tesla's general counsel pushes back on Musk's request to pause, saying the board work has lead time.

Notes: Known only through the passages quoted in Chancellor McCormick's post-trial opinion (factual background); the full email is not public. Maron's reply to Musk's request to put the compensation discussion on hold.

Sources: [Tornetta v. Musk, C.A. No. 2018-0408-KSJM \(Del. Ch. Jan. 30, 2024\)](#), post-trial opinion; [Harvard Law School Forum on Corporate Governance \(reproduces the opinion text\)](#); [Amended derivative complaint \(Bernstein Litowitz\)](#)

Tags: pay-package, compensation, todd-maron

Musk: added pay is for 'putting the money towards Mars'

From: Elon Musk

Date: July 30, 2017 (reply sent about July 30-31, 2017; exact date not given in the retrieved text)

To: Todd Maron

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

[J]ust want[ed] to make sure Tesla interests come first.

[...]

[T]he added comp is just so that I can put as much as possible towards minimizing existential risk by putting the money towards Mars if I am successful in leading Tesla to be one of the world's most valuable companies. This is kinda crazy, but it is true.

About this email: Agreeing to let Maron continue, Musk says he wants Tesla's interests to come first and that the extra compensation is meant to fund Mars and reduce existential risk.

Notes: Known only through the passages quoted in Chancellor McCormick's post-trial opinion (factual background); the full email is not public. Bracketed alterations are the court's. The final sentence ('This is kinda crazy, but it is true.') appears at the end of the same passage of the opinion and is read here as the continuation of the block quote.

Sources: [Tornetta v. Musk, C.A. No. 2018-0408-KSJM \(Del. Ch. Jan. 30, 2024\)](#), post-trial opinion; [Harvard Law School Forum on Corporate Governance \(reproduces the opinion text\)](#); [Amended derivative complaint \(Bernstein Litowitz\)](#)

Tags: pay-package, compensation, todd-maron

Communication Within Tesla: talk to anyone, no chain of command

From: Elon Musk

Date: August 30, 2017 (Date of first publication by Inc.; send date unknown. Inc. said Musk sent it 'a few years ago'; it is often dated to around 2013 but that could not be verified)

To: All Tesla employees

Subject: Communication Within Tesla

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · CONFIRMED

There are two schools of thought about how information should flow. By far the most common way is chain of command, which means that you always flow communication through your manager. The problem with this approach is that, while it enhances the power of the manager, it fails to serve the company.

To solve a problem quickly, two people in different depts should simply talk and make the right thing happen. Instead, people are forced to talk to their manager, who talks to their manager, who talks to the manager in the other dept, who talks to someone on his

team. Then the info has to flow back the other way again. This is incredibly dumb. Any manager who allows this to happen, let alone encourages it, will soon find themselves working at another company. No kidding.

Anyone at Tesla can and should email/talk to anyone else according to what they think is the fastest way to solve a problem for the benefit of the whole company.

You can talk to your manager's manager without his permission, you can talk directly to a VP in another dept, you can talk to me, you can talk to anyone without anyone else's permission. Moreover, you should consider yourself obligated to do so until the right thing happens. The point here is to ensure that we execute ultra-fast and well. We obviously cannot compete with the big car companies in size, so we must do so with intelligence and agility.

One final point is that managers should work hard to ensure that they are not creating silos within the company that create an "us vs. them" mentality, or impede communication in any way. This is unfortunately a natural tendency and needs to be actively fought. How can it possibly help Tesla for depts to erect barriers between themselves, or see their success as relative within the company instead of collective?

We are all in the same boat. Always view yourself as working for the good of the company and never your dept.

Thanks,
Elon

About this email: Musk tells all staff that information should flow by the shortest path, not the chain of command: anyone may email or talk to anyone else, including him, and managers who enforce chain-of-command communication or build silos will be out.

Notes: Inc. described it as 'a previously unpublished email Musk sent to Tesla employees a few years ago' and said Tesla verified it was sent to all employees. Text taken from a reprint; sign-off rendered there as 'Thanks, Elon'.

Sources: [Inc. \(Justin Bariso\)](#); [Alexander Jarvis \(full text reprint\)](#); [InsideEVs](#); [GIGAZINE](#)

Tags: communication, management, culture, silos

Musk restarts pay talks 'in a reduced manner'

From: Elon Musk

Date: November 9, 2017

To: Todd Maron

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

I'd like to move forward with that now, but in a reduced manner from before.

About this email: In the early hours of November 9, 2017, Musk tells Maron he wants to move ahead with his compensation plan but at a reduced size.

Notes: Known only through the passages quoted in Chancellor McCormick's post-trial opinion (factual background); the full email is not public. The opinion quotes it as wanting to 'move forward with [his compensation plan] now, but in a reduced manner from before'; the sentence as given here is from para. 115 of the amended complaint.

Sources: [Tornetta v. Musk, C.A. No. 2018-0408-KSJM \(Del. Ch. Jan. 30, 2024\), post-trial opinion](#); [Harvard Law School Forum on Corporate Governance \(reproduces the opinion text\)](#); [Amended derivative complaint](#)

(Bernstein Litowitz)

Tags: pay-package, compensation, todd-maron

Musk proposes 10% stake increment at \$550B valuation

From: Elon Musk

Date: November 9, 2017

To: Todd Maron

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

[I'd] like to take board action as soon as possible if they feel comfortable and then it would go to shareholders.

[...]

I think the amount should be reduced to a 10% increment in my Tesla ownership if I can get us to a \$550B valuation, but that should be a fully diluted 10%, factoring in that it dilutes me too.

About this email: Later the same morning Musk asks for prompt board action and proposes cutting the award to a fully diluted 10% increase in his ownership if Tesla reaches a \$550 billion valuation.

Notes: Known only through the passages quoted in Chancellor McCormick's post-trial opinion (factual background); the full email is not public. The two passages are quoted separately in the opinion; their order within the email is not certain.

Sources: [Tornetta v. Musk, C.A. No. 2018-0408-KSJM \(Del. Ch. Jan. 30, 2024\)](#), post-trial opinion; [Harvard Law School Forum on Corporate Governance \(reproduces the opinion text\)](#); [Amended derivative complaint \(Bernstein Litowitz\)](#)

Tags: pay-package, compensation, todd-maron

Musk: shareholders 'will be super happy' with pay plan

From: Elon Musk

Date: November 9, 2017

To: Todd Maron

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Given that this will all go to causes that at least aspirationally maximize the probability of a good future for humanity, plus all Tesla shareholders will be super happy, I think this will be received well.

About this email: Moments after his revised proposal, Musk tells Maron the plan will be well received because the money will go to causes that improve humanity's future and shareholders will be happy.

Notes: Known only through the passages quoted in Chancellor McCormick's post-trial opinion (factual background); the full email is not public.

Sources: [Tornetta v. Musk, C.A. No. 2018-0408-KSJM \(Del. Ch. Jan. 30, 2024\)](#), post-trial opinion; [Harvard Law School Forum on Corporate Governance \(reproduces the opinion text\)](#); [Amended derivative complaint \(Bernstein Litowitz\)](#)

Tags: pay-package, compensation, todd-maron

Musk again asks to 'pause' pay plan

From: Elon Musk

Date: November 14, 2017

To: Todd Maron

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

pause for a week or two

[...]

terrible timing

About this email: Musk asks Maron to pause work on his compensation plan for a week or two because of bad timing.

Notes: Known only through the passages quoted in Chancellor McCormick's post-trial opinion (factual background); the full email is not public. Only these two fragments are quoted.

Sources: [Tornetta v. Musk, C.A. No. 2018-0408-KSJM \(Del. Ch. Jan. 30, 2024\), post-trial opinion](#); [Harvard Law School Forum on Corporate Governance \(reproduces the opinion text\)](#); [Amended derivative complaint \(Bernstein Litowitz\)](#)

Tags: pay-package, compensation, todd-maron

Musk: 'That is more than intended' - cuts award to 20.915M shares

From: Elon Musk

Date: December 1, 2017

To: Todd Maron

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

That is more than intended. Let's go with 10% of the current FDS number, so 20.915M.

About this email: After Maron sent share calculations on November 29, Musk replies that the figure is more than he intended and sets the award at 10% of the then-current fully diluted share count. At trial he called this 'me negotiating against myself'.

Notes: Known only through the passages quoted in Chancellor McCormick's post-trial opinion (factual background); the full email is not public. FDS = fully diluted shares. Also quoted at para. 120 of the amended complaint.

Sources: [Tornetta v. Musk, C.A. No. 2018-0408-KSJM \(Del. Ch. Jan. 30, 2024\), post-trial opinion](#); [Harvard Law School Forum on Corporate Governance \(reproduces the opinion text\)](#); [Amended derivative complaint \(Bernstein Litowitz\)](#)

Tags: pay-package, compensation, todd-maron

Musk on M&A adjustment: 'no chance I will game the economics'

From: Elon Musk

Date: January 16, 2018 (on or shortly after January 16, 2018)

To: Todd Maron and Deepak Ahuja

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

I don't think we will be making big acquisitions[.]

[...]

[T]here is no chance I will game the economics here, so I'm fine with limits that prevent that.

About this email: In negotiating how acquisitions would be excluded from the pay plan's milestones, Musk proposes a stricter threshold and says he does not expect big acquisitions.

Notes: Known only through the passages quoted in Chancellor McCormick's post-trial opinion (factual background); the full email is not public. The opinion says Musk 'told Maron and Ahuja' this in countering Maron's January 16, 2018 proposal; it does not expressly say whether the message was an email, so treat the medium as probable rather than certain.

Sources: [Tornetta v. Musk, C.A. No. 2018-0408-KSJM \(Del. Ch. Jan. 30, 2024\), post-trial opinion](#); [Harvard Law School Forum on Corporate Governance \(reproduces the opinion text\)](#); [Amended derivative complaint \(Bernstein Litowitz\)](#)

Tags: pay-package, compensation, acquisitions

Musk 'very offended' by stockholders voting against pay plan

From: Elon Musk

Date: March 20, 2018 (on or about March 20, 2018 (the day before the stockholder vote))

To: Todd Maron

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

very offended by their action if they choose to vote that way, but but [sic] by all means do so.

[...]

convince them to divest from Tesla and any of [his] companies ever. They are not welcome.

About this email: Told that two large stockholders would vote against his 2018 pay package, Musk asks Maron to tell one he is very offended and says he will try to convince them to divest from all his companies.

Notes: MEDIUM NOT CONFIRMED: the opinion says Musk 'asked Maron' this in response to Maron's update and quotes his written words (with a '[sic]'), but the retrieved text does not say whether it was an email or a text message. Bracketed alteration is the court's. Fragments only.

Sources: [Tornetta v. Musk, C.A. No. 2018-0408-KSJM \(Del. Ch. Jan. 30, 2024\), post-trial opinion](#); [Harvard Law School Forum on Corporate Governance \(reproduces the opinion text\)](#)

Tags: pay-package, stockholder-vote, institutional-investors

Model 3 passes 2,000 cars per week

From: Elon Musk

Date: April 2, 2018 (Sent about 3:01 a.m. PDT on Monday April 2, 2018)

To: All Tesla employees

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

It has been extremely difficult to pass the 2000 cars per week rate for Model 3, but we are finally there. If things go as planned today, we will comfortably exceed that number over a seven-day period!

[...]

This is already double the pace of 2017!

[...]

By the end of the year, I believe we will be producing vehicle sat least four times faster than last year.

[...]

It took five years to reach the 2000/week production rate for S and X combined, but only nine months to achieve that output with Model 3. Mind-blowing progress!

About this email: Musk tells staff that Model 3 production has finally passed 2,000 cars a week, double the 2017 pace, and predicts output four times faster by year-end.

Notes: Only excerpts were published. Passages combined from Electrek (first and last) and Jalopnik (middle two); order within the email is not certain. 'vehicle sat least' typo is as quoted by Jalopnik.

Sources: [Electrek](#); [Jalopnik](#); [TechCrunch](#)

Tags: model-3, production, milestone

Model 3 ramp to 6,000/week, 24/7 shifts, productivity rules

From: Elon Musk

Date: April 17, 2018

To: Everybody (all Tesla employees)

Subject: Progress, Precision and Profit

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Progress

First, congratulations are in order! We have now completed our third full week of producing over 2000 Model 3 vehicles. The first week was 2020, the second was 2070 and we just completed 2250 last week, along with 2000 Model S/X vehicles.

This is more than double Tesla's weekly production rate last year and an amazing feat in the face of many challenges! It is extremely rare for an automotive company to grow the production rate by over 100% from one year to the next. Moreover, there has simultaneously been a significant improvement in quality and build accuracy, which is reflected in positive owner feedback.

Starting today at Giga and tomorrow at Fremont, we will be stopping for three to five days to do a comprehensive set of upgrades. This should set us up for Model 3 production of 3000 to 4000 per week next month.

Another set of upgrades starting in late May should be enough to unlock production capacity of 6000 Model 3 vehicles per week by the end of June. Please note that all areas of Tesla and our suppliers will be required to demonstrate a Model 3 capacity of ~6000/week by building 850 sets of car parts in 24 hours no later than June 30th.

Any Tesla department or supplier that is unable to do this will need to have a very good explanation why not, along with a plan for fixing the problem and present that to me directly. If anyone needs help achieving this, please let me know as soon as possible. We are going to find a way or make a way to get there.

The reason that the burst-build target rate is 6000 and not 5000 per week in June is that we cannot have a number with no margin for error across thousands of internally and externally produced parts and processes, amplified by a complex global logistics chain. Actual production will move as fast as the least lucky and least well-executed part of the entire Tesla production/supply chain system.

By having a Model 3 subsystem burst-build requirement of 6k by the end of June, we will lay the groundwork for achieving a steady 6k/week across the whole Model 3 system a few months later.

As part of the drive towards 6k, all Model 3 production at Fremont will move to 24/7 operations. This means that we will be adding another shift to general assembly, body and paint. Please refer anyone you know who you think meets the Tesla bar for talent, drive and trust. Between Fremont and Giga, Tesla will be adding about 400 people per week for several weeks.

Precision

Most of the design tolerances of the Model 3 are already better than any other car in the world. Soon, they will all be better. This is not enough. We will keep going until the Model 3 build precision is a factor of ten better than any other car in the world. I am not kidding.

Our car needs to be designed and built with such accuracy and precision that, if an owner measures dimensions, panel gaps and flushness, and their measurements don't match the Model 3 specs, it just means that their measuring tape is wrong.

Some parts suppliers will be unwilling or unable to achieve this level of precision. I understand that this will be considered an unreasonable request by some. That's ok, there are lots of other car companies with much lower standards. They just can't work with Tesla.

Profit

A fair criticism leveled at Tesla by outside critics is that you're not a real company unless you generate a profit, meaning simply that revenue exceeds costs. It didn't make sense to do that until reaching economies of scale, but now we are there.

Going forward, we will be far more rigorous about expenditures. I have asked the Tesla finance team to comb through every expense worldwide, no matter how small, and cut everything that doesn't have a strong value justification.

All capital or other expenditures above a million dollars, or where a set of related expenses may accumulate to a million dollars over the next 12 months, should be considered on hold until explicitly approved by me. If you are the manager responsible, please make sure you have a detailed, first principles understanding of the supplier quote, including every line item of parts & labor, before we meet.

I have been disappointed to discover how many contractor companies are interwoven throughout Tesla. Often, it is like a Russian nesting doll of contractor, subcontractor, sub-subcontractor, etc. before you finally find someone doing actual work. This means a

lot of middle-managers adding cost but not doing anything obviously useful. Also, many contracts are essentially open time & materials, not fixed price and duration, which creates an incentive to turn molehills into mountains, as they never want to end the money train.

There is a very wide range of contractor performance, from excellent to worse than a drunken sloth. All contracting companies should consider the coming week to be a final opportunity to demonstrate excellence. Any that fail to meet the Tesla standard of excellence will have their contracts ended on Monday.

Btw, here are a few productivity recommendations:

- Excessive meetings are the blight of big companies and almost always get worse over time. Please get of all large meetings, unless you're certain they are providing value to the whole audience, in which case keep them very short.
- Also get rid of frequent meetings, unless you are dealing with an extremely urgent matter. Meeting frequency should drop rapidly once the urgent matter is resolved.
- Walk out of a meeting or drop off a call as soon as it is obvious you aren't adding value. It is not rude to leave, it is rude to make someone stay and waste their time.
- Don't use acronyms or nonsense words for objects, software or processes at Tesla. In general, anything that requires an explanation inhibits communication. We don't want people to have to memorize a glossary just to function at Tesla.
- Communication should travel via the shortest path necessary to get the job done, not through the "chain of command". Any manager who attempts to enforce chain of command communication will soon find themselves working elsewhere.
- A major source of issues is poor communication between depts. The way to solve this is allow free flow of information between all levels. If, in order to get something done between depts, an individual contributor has to talk to their manager, who talks to a director, who talks to a VP, who talks to another VP, who talks to a director, who talks to a manager, who talks to someone doing the actual work, then super dumb things will happen. It must be ok for people to talk directly and just make the right thing happen.
- In general, always pick common sense as your guide. If following a "company rule" is obviously ridiculous in a particular situation, such that it would make for a great Dilbert cartoon, then the rule should change.

If there is something you think should be done to make Tesla execute better or allow you to look forward to coming to work more (same thing in the long term), please send a note to [redacted]

Thanks for being such a kickass team and accomplishing miracles every day. It matters. We are burning the midnight oil to burn the midnight oil.

Elon

About this email: Musk congratulates staff on three weeks above 2,000 Model 3s, sets a 6,000/week burst-build target for end of June with 24/7 operations at Fremont, puts all spending above \$1M on hold pending his

approval, threatens underperforming contractors, and lists his now-famous productivity recommendations (leave useless meetings, no acronyms, ignore chain of command).

Notes: Electrek obtained and printed the full email; feedback email address redacted by the outlet. 'Please get of all large meetings' typo is in the published text.

Sources: [Electrek](#); [CNBC](#); [TechCrunch](#)

Tags: model-3, production, productivity, meetings, contractors, cost-cutting

Contractor purge: 'Time to scrub off the barnacles'

From: Elon Musk

Date: May 2018 (Reported by Electrek on Sunday May 6, 2018; sent in the preceding days (it sets a deadline of 'Monday'))

To: All Tesla employees

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Please note my comment below about contractor companies and consultants. I extended the performance evaluation deadline to provide more opportunity to demonstrate excellence, but now time is up.

Please send a note to HR before Monday justifying the excellence, necessity and trustworthiness by individual (not just the contractor company as a whole) of every non-Tesla person who has badge access to our buildings or network access to our systems.

By default, anyone who does not have a Tesla employee putting their reputation on the line for them will be denied access to our facilities and networks on Monday morning. This applies worldwide.

Time to scrub off the barnacles.

Thanks, Elon

About this email: Following up on his April 17 warning to contractors, Musk says time is up: every non-Tesla person with badge or network access must be personally vouched for by a Tesla employee or lose access worldwide on Monday morning.

Notes: 'My comment below' refers to the contractor paragraphs of the April 17, 2018 'Progress, Precision and Profit' email, which this message forwarded.

Sources: [Electrek](#); [Electrek \(reorganization follow-up\)](#)

Tags: contractors, cost-cutting, security, access

Reorganization: flattening the management structure

From: Elon Musk

Date: May 14, 2018

To: All Tesla employees

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

To ensure that Tesla is well prepared for the future, we have been undertaking a thorough reorganization of our company. As part of the reorg, we are flattening the management structure to improve communication, combining functions where sensible and trimming activities that are not vital to the success of our mission.

To be clear, we will continue to hire rapidly in critical hourly and salaried positions to support the Model 3 production ramp and future product development.

About this email: Short memo telling staff Tesla is undergoing a thorough reorganization, flattening management, combining functions and trimming non-vital activities, while continuing to hire for the Model 3 ramp.

Notes: First reported by the Wall Street Journal. CNBC labels this 'the full memo'; it is two paragraphs with no greeting or sign-off as published.

Sources: [CNBC](#); [Electrek](#); [CleanTechnica](#)

Tags: reorganization, management, hiring

Model 3 likely to exceed 500 cars per day this week

From: Elon Musk

Date: May 15, 2018 (Date Electrek and CNBC obtained/reported it)

To: Tesla employees

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

It is looking quite likely that we will exceed 500 vehicles per day across all Model 3 production zones this week.

About this email: Musk tells employees Model 3 output will probably exceed 500 vehicles a day across all production zones that week and asks to be told of any specific bottlenecks.

Notes: Only one sentence was published verbatim; Electrek paraphrases the request to flag 'any specific bottlenecks'.

Sources: [Electrek](#); [CNBC](#)

Tags: model-3, production, burst-build, bottlenecks

Layoff of about 9% of Tesla's workforce

From: Elon Musk

Date: June 12, 2018

To: All Tesla employees

FULL TEXT · COMPANY-WIDE MEMO · PUBLISHED BY MUSK · CONFIRMED

As described previously, we are conducting a comprehensive organizational restructuring across our whole company. Tesla has grown and evolved rapidly over the past several years, which has resulted in some duplication of roles and some job functions that, while they made sense in the past, are difficult to justify today.

As part of this effort, and the need to reduce costs and become profitable, we have made the difficult decision to let go of approximately 9% of our colleagues across the company. These cuts were almost entirely made from our salaried population and no production associates were included, so this will not affect our ability to reach Model 3 production targets in the coming months.

Given that Tesla has never made an annual profit in the almost 15 years since we have existed, profit is obviously not what motivates us. What drives us is our mission to accelerate the world's transition to sustainable, clean energy, but we will never achieve that mission unless we eventually demonstrate that we can be sustainably profitable.

That is a valid and fair criticism of Tesla's history to date.

This week, we are informing those whose roles are impacted by this action. We made these decisions by evaluating the criticality of each position, whether certain jobs could be done more efficiently and productively, and by assessing the specific skills and abilities of each individual in the company. As you know, we are also continuing to flatten our management structure to help us communicate better, eliminate bureaucracy and move faster.

In addition to this company-wide restructuring, we've decided not to renew our residential sales agreement with Home Depot in order to focus our efforts on selling solar power in Tesla stores and online. The majority of Tesla employees working at Home Depot will be offered the opportunity to move over to Tesla retail locations.

I would like to thank everyone who is departing Tesla for their hard work over the years. I'm deeply grateful for your many contributions to our mission. It is very difficult to say goodbye. In order to minimize the impact, Tesla is providing significant salary and stock vesting (proportionate to length of service) to those we are letting go.

To be clear, Tesla will still continue to hire outstanding talent in critical roles as we move forward and there is still a significant need for additional production personnel. I also want to emphasize that we are making this hard decision now so that we never have to do this again.

To those who are departing, thank you for everything you've done for Tesla and we wish you well in your future opportunities. To those remaining, I would like to thank you in advance for the difficult job that remains ahead. We are a small company in one of the toughest and most competitive industries on Earth, where just staying alive, let alone growing, is a form of victory (Tesla and Ford remain the only American car companies who haven't gone bankrupt). Yet, despite our tiny size, Tesla has already played a major role in moving the auto industry towards sustainable electric transport and moving the energy industry towards sustainable power generation and storage. We must continue to drive that forward for the good of the world.

Thanks,
Elon

About this email: Musk announces Tesla is cutting roughly 9% of staff, almost entirely salaried, as part of a company-wide restructuring to reduce costs and reach profitability; also ends the Home Depot solar sales agreement.

Notes: Musk posted the email himself on Twitter after it began leaking ('Difficult, but necessary Tesla reorg underway. My email to the company has already leaked to media. Here it is unfiltered').

Sources: [TechCrunch](#); [CNBC](#); [Axios](#)

Tags: layoffs, restructuring, profitability, home-depot

'Radical improvements' still needed for 5k/week

From: Elon Musk

Date: June 15, 2018 (Sent June 15, 2018 at 8:27 pm)

To: Everybody (all Tesla employees)

Subject: Only 8 days left to reach 700 cars/day or 5k/week

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

It's getting very exciting! All parts of the Model 3 production system are now above 500 and some are almost at 700 cars already. Congratulations to all on making so much progress!

That said, radical improvements are still needed in paint shop output, GA3, bringing up the new GA4, End of Line and Module Zone 4 at Giga. We also need to achieve sustained, 700+ per week on the body line.

Wherever you are in the company, if you feel you can help out in any of those areas, please check in with Jat Dhillon on GA3, Jerome Guillen on GA4 and Omead on EoL and JB Straubel or Christ Lister on Module Zone 4.

I will be at our Fremont factory almost 24/7 for the next several days checking in with those groups to make sure they have as many resources as they can handle.

Thanks,

Elon

About this email: Eight days before the end of Q2 2018, Musk says all parts of the Model 3 system are above 500 cars a day but radical improvements are needed in paint, general assembly and Gigafactory modules, and says he will be at Fremont almost 24/7.

Notes: 'Christ Lister' and '700+ per week on the body line' are as published.

Sources: [Business Insider \(via Yahoo News\)](#); [FenderBender](#)

Tags: model-3, production, fremont, gigafactory

Employee 'sabotage' warning before Model 3 ramp

From: Elon Musk

Date: June 17, 2018 (Sent Sunday June 17, 2018 at 11:57 p.m.)

To: Everybody (all Tesla employees)

Subject: Some concerning news

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

I was dismayed to learn this weekend about a Tesla employee who had conducted quite extensive and damaging sabotage to our operations. This included making direct code changes to the Tesla Manufacturing Operating System under false usernames and exporting large amounts of highly sensitive Tesla data to unknown third parties.

The full extent of his actions are not yet clear, but what he has admitted to so far is pretty bad. His stated motivation is that he wanted a promotion that he did not receive. In light of these actions, not promoting him was definitely the right move.

However, there may be considerably more to this situation than meets the eye, so the investigation will continue in depth this week. We need to figure out if he was acting alone or with others at Tesla and if he was working with any outside organizations.

As you know, there are a long list of organizations that want Tesla to die. These include Wall Street short-sellers, who have already lost billions of dollars and stand to lose a lot more. Then there are the oil & gas companies, the wealthiest industry in the world — they don't love the idea of Tesla advancing the progress of solar power & electric cars. Don't want to blow your mind, but rumor has it that those companies are sometimes not super nice. Then there are the multitude of big gas/diesel car company competitors. If they're willing to cheat so much about emissions, maybe they're willing to cheat in other ways?

Most of the time, when there is theft of goods, leaking of confidential information, dereliction of duty or outright sabotage, the reason really is something simple like wanting to get back at someone within the company or at the company as a whole. Occasionally, it is much more serious.

Please be extremely vigilant, particularly over the next few weeks as we ramp up the production rate to 5k/week. This is when outside forces have the strongest motivation to stop us.

If you know of, see or suspect anything suspicious, please send a note to [email address removed for privacy] with as much info as possible. This can be done in your name, which will be kept confidential, or completely anonymously.

Looking forward to having a great week with you as we charge up the super exciting ramp to 5000 Model 3 cars per week!

Will follow this up with emails every few days describing the progress and challenges of the Model 3 ramp.

Thanks for working so hard to make Tesla successful,

Elon

About this email: Musk tells staff an employee made unauthorized code changes to the Manufacturing Operating System and exported sensitive data, warns that short-sellers, oil companies and rival automakers want Tesla to die, and asks for vigilance during the 5k/week ramp.

Notes: CNBC obtained the email; the employee was later identified as Martin Tripp. Reporting email address removed by the outlets. Text retrieved from Armenpress and Teslarati reprints, which match.

Sources: [CNBC](#); [Armenpress \(reprint of CNBC text\)](#); [Teslarati](#); [Jalopnik](#)

Tags: sabotage, security, martin-tripp, short-sellers, model-3

Factory fire: 'Only the paranoid survive'

From: Elon Musk

Date: June 18, 2018 (Sent Monday June 18, 2018 at 9:38 a.m.)

To: Everybody (all Tesla employees)

Subject: Some concerning news

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Late last night we had another strange incident that was hard to explain. Small fire on the body-in-white production line. No one was in the area and there were no injuries or significant equipment damage, but it was enough to stop the body production line for several hours.

Could just be a random event, but as Andy Grove said, "Only the paranoid survive." Please be on the alert for anything that's not in the best interests of our company. If you aren't getting a response from the emdesk email, please send me a note directly.

Thanks,

Elon

About this email: Follow-up to the sabotage email reporting a small overnight fire on the body-in-white line that stopped production for several hours, and asking staff to stay alert.

Notes: Text retrieved from Jalopnik's reproduction of the email first obtained by CNBC.

Sources: [CNBC](#); [Jalopnik](#); [Inverse](#)

Tags: sabotage, fire, fremont, security

Tripp to Musk: 'you have what's coming to you'

From: Martin Tripp

Date: June 20, 2018

To: Elon Musk

Subject: Termination/Lawsuit

FULL TEXT · OUTSIDE & PERSONAL · LEAKED TO PRESS · CONFIRMED

Don't worry, you have what's coming to you for the lies you have told to the public and investors.

About this email: The former Tesla technician Musk had accused of sabotage emails Musk directly on the day Tesla sued him.

Notes: Exchange took place the day Tesla sued former Gigafactory technician Martin Tripp (Tesla v. Tripp, D. Nev. 3:18-cv-00296); it was first reported by the Washington Post and later featured in the litigation. Sent June 20, 2018 at 8:57:29 AM PDT.

Sources: [TechCrunch](#) (email exchange obtained by TechCrunch and confirmed by Tesla); [Jalopnik](#); [BGR](#); [Inverse](#)

Tags: martin-tripp, whistleblower, sabotage, gigafactory, lawsuit

Musk to Tripp: 'Threatening me only makes it worse for you'

From: Elon Musk

Date: June 20, 2018

To: Martin Tripp

Subject: Re: Termination/Lawsuit

FULL TEXT · OUTSIDE & PERSONAL · LEAKED TO PRESS · CONFIRMED

Threatening me only makes it worse for you

About this email: Musk's first reply to Martin Tripp treats Tripp's email as a threat.

Notes: Exchange took place the day Tesla sued former Gigafactory technician Martin Tripp (Tesla v. Tripp, D. Nev. 3:18-cv-00296); it was first reported by the Washington Post and later featured in the litigation. Sent at 9:42 AM.

Sources: [TechCrunch \(email exchange obtained by TechCrunch and confirmed by Tesla\)](#); [Jalopnik](#); [BGR](#); [Inverse](#)

Tags: martin-tripp, whistleblower, sabotage, gigafactory, lawsuit

Tripp: 'I never made a threat'

From: Martin Tripp

Date: June 20, 2018

To: Elon Musk

Subject: Re: Termination/Lawsuit

FULL TEXT · OUTSIDE & PERSONAL · LEAKED TO PRESS · CONFIRMED

I never made a threat. I simply told you that you have what's coming. Thank you for this gift!!!!

About this email: Tripp denies threatening Musk.

Notes: Exchange took place the day Tesla sued former Gigafactory technician Martin Tripp (Tesla v. Tripp, D. Nev. 3:18-cv-00296); it was first reported by the Washington Post and later featured in the litigation. Sent at 9:59 AM.

Sources: [TechCrunch \(email exchange obtained by TechCrunch and confirmed by Tesla\)](#); [Jalopnik](#); [BGR](#); [Inverse](#)

Tags: martin-tripp, whistleblower, sabotage, gigafactory, lawsuit

Musk to Tripp: 'You're a horrible human being'

From: Elon Musk

Date: June 20, 2018

To: Martin Tripp

Subject: Re: Termination/Lawsuit

FULL TEXT · OUTSIDE & PERSONAL · LEAKED TO PRESS · CONFIRMED

You should be ashamed of yourself for framing other people. You're a horrible human being.

About this email: Musk accuses Tripp of framing colleagues.

Notes: Exchange took place the day Tesla sued former Gigafactory technician Martin Tripp (Tesla v. Tripp, D. Nev. 3:18-cv-00296); it was first reported by the Washington Post and later featured in the litigation. Sent at 10:00 AM. 'should be ashamed' is as printed by TechCrunch.

Sources: [TechCrunch \(email exchange obtained by TechCrunch and confirmed by Tesla\)](#); [Jalopnik](#); [BGR](#); [Inverse](#)

Tags: martin-tripp, whistleblower, sabotage, gigafactory, lawsuit

Tripp: 'I NEVER framed anyone'

From: Martin Tripp

Date: June 20, 2018

To: Elon Musk

Subject: Re: Termination/Lawsuit

FULL TEXT · OUTSIDE & PERSONAL · LEAKED TO PRESS · CONFIRMED

I NEVER 'framed' anyone else or even insinuated anyone else as being involved in my production of documents of your MILLIONS OF DOLLARS OF WASTE, Safety concerns, lying to investors/the WORLD. Putting cars on the road with safety issues is being a horrible human being!

About this email: Tripp denies framing anyone and accuses Tesla of waste, safety problems and lying to investors.

Notes: Exchange took place the day Tesla sued former Gigafactory technician Martin Tripp (Tesla v. Tripp, D. Nev. 3:18-cv-00296); it was first reported by the Washington Post and later featured in the litigation. Sent at 10:03 AM.

Sources: [TechCrunch \(email exchange obtained by TechCrunch and confirmed by Tesla\)](#); [Jalopnik](#); [BGR](#); [Inverse](#)

Tags: martin-tripp, whistleblower, sabotage, gigafactory, lawsuit

Musk to Tripp: 'So it goes. Be well.'

From: Elon Musk

Date: June 20, 2018

To: Martin Tripp

Subject: Re: Termination/Lawsuit

FULL TEXT · OUTSIDE & PERSONAL · LEAKED TO PRESS · CONFIRMED

There are literally injuries with Model 3. It is by far the safest car in the world for any midsize vehicle. And of course a company with billions of dollars in product is going to have millions of dollars in scrap. This is not news. However, betraying your word of honor, breaking the deal you had when Tesla gave you a job and framing your colleagues are wrong and some come with legal penalties. So it goes. Be well.

About this email: Musk defends the Model 3's safety and scrap levels and tells Tripp that betraying his word and framing colleagues carries legal penalties.

Notes: Exchange took place the day Tesla sued former Gigafactory technician Martin Tripp (Tesla v. Tripp, D. Nev. 3:18-cv-00296); it was first reported by the Washington Post and later featured in the litigation. Sent at about 10:28 AM. Musk meant 'literally no injuries' and corrected himself in a follow-up.

Sources: [TechCrunch \(email exchange obtained by TechCrunch and confirmed by Tesla\)](#); [Jalopnik](#); [BGR](#); [Inverse](#)

Tags: martin-tripp, whistleblower, sabotage, gigafactory, lawsuit

Musk corrects himself: 'Meant to say no injuries'

From: Elon Musk

Date: June 20, 2018

To: Martin Tripp

Subject: Re: Termination/Lawsuit

FULL TEXT · OUTSIDE & PERSONAL · LEAKED TO PRESS · CONFIRMED

Meant to say 'no injuries'

About this email: A one-line correction to his previous email about Model 3 injuries.

Notes: Exchange took place the day Tesla sued former Gigafactory technician Martin Tripp (Tesla v. Tripp, D. Nev. 3:18-cv-00296); it was first reported by the Washington Post and later featured in the litigation. Sent later on June 20, 2018 (about 5:16-5:17 PM as shown by TechCrunch).

Sources: [TechCrunch \(email exchange obtained by TechCrunch and confirmed by Tesla\)](#); [Jalopnik](#); [BGR](#); [Inverse](#)

Tags: martin-tripp, whistleblower, sabotage, gigafactory, lawsuit

'We did it!!' - 5,000 Model 3s in a week

From: Elon Musk

Date: July 1, 2018 (Sent July 1, 2018 at 12:37 PM)

To: Everybody (all Tesla employees)

Subject: 7000

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

We did it!! What an incredible job by an amazing team. Couldn't be more proud to work with you. It is an honor.

The level of dedication and creativity was mind-blowing. We either found a way or, by will and inventiveness, created entirely new solutions that were thought impossible.

Intense in tents. Transporting entire production lines across the world in massive cargo planes. Whatever. It worked.

Not only did we factory gate over 5000 Model 3's, but we also achieved the S & X production target for a combined 7000 vehicle week!

What's more, with the widespread productivity gains throughout Tesla and the new production lines spooling up, we are on track to reach 6k/week for Model 3 next month.

I think we just became a real car company ...

Thank you for your hard work and dedication,

Elon

About this email: Musk celebrates Tesla building over 5,000 Model 3s and 7,000 total vehicles in the last week of Q2 2018 and says the company is on track for 6k/week.

Sources: [EVANNEX \(reprinting email obtained by Electrek/Bloomberg\)](#); [Bloomberg](#); [CNBC](#); [Fortune](#)

Tags: model-3, production, milestone, celebration

Musk's \$420 take-private offer to Tesla's board

From: Elon Musk

Date: August 2, 2018 (Exhibit header: sent 8/2/2018 3:26:59 PM)

To: Tesla Board of Directors; Todd Maron; Deepak Ahuja

Subject: Offer to Take Tesla Private at \$420

FULL TEXT · BOARD, INVESTORS & ADVISERS · COURT RECORD · CONFIRMED

It is my firm belief that Tesla can operate more effectively as a private company for the next several years. In the future, once Tesla enters a phase of slower, more predictable growth, it will likely make sense to return to the public markets.

However, being public during this period of rapid and often unpredictable growth:

1. Subjects Tesla to constant defamatory attacks by the short-selling community, resulting in great harm to our valuable brand.
2. Puts great pressure on Tesla to make decisions for a given quarter that are suboptimal for long-term shareholder value creation.
3. Causes wild swings in our share price, which are a constant distraction to Tesla employees, all of whom are shareholders, making it harder to focus on useful work.

The difference in operational efficiency between Tesla and SpaceX, the latter being privately held, is dramatic.

Unless another bidder comes forward with a better offer, I would ask that this matter be put to a shareholder vote at the earliest opportunity. This offer expires in 30 days.

Please note that I fully support any shareholders who wish to remain shareholders of Tesla as a private company retaining their shares. This offer is just for those who do not wish to remain shareholders of Tesla as a private company.

Regards,

Elon Musk
Chairman & CEO
Tesla, Inc.

About this email: Five days before the 'funding secured' tweet, Musk formally tells the board he believes Tesla should be private, cites short-seller attacks, quarterly pressure and stock swings, and asks for a shareholder vote on a \$420 offer that expires in 30 days.

Notes: Sent from erm@tesla.com. Transcribed from the court's PDF exhibit; two exhibits (81 and 802) carry identical text.

Sources: [U.S. District Court N.D. Cal., In re Tesla Inc. Securities Litigation, Trial Exhibit 81](#); [Trial Exhibit 802 \(same email with Gracias reply\)](#); [Court exhibit index](#)

Tags: take-private, funding-secured, board, short-sellers, 420

Gracias acknowledges Musk's take-private offer

From: Antonio J. Gracias

Date: August 2, 2018 (Exhibit header: 8/2/2018 4:00:21 PM)

To: Elon Musk (cc: Board of Directors; Todd Maron; Deepak Ahuja)

Subject: Re: Offer to Take Tesla Private at \$420

FULL TEXT · BOARD, INVESTORS & ADVISERS · COURT RECORD · CONFIRMED

Thank you. We are in receipt of your offer.

We will schedule a board meeting in the next 24 hours to discuss this matter and get back to you with the appropriate next steps.

Antonio

About this email: Lead independent director Antonio Gracias confirms receipt of Musk's offer and says the board will meet within 24 hours.

Notes: Reply to Musk's email of the same day; included as part of the thread.

Sources: [U.S. District Court N.D. Cal., In re Tesla Inc. Securities Litigation, Trial Exhibit 802](#)

Tags: take-private, board, 420

Ron Baron asks to stay invested in a private Tesla

From: Ron Baron (Baron Funds)

Date: August 7, 2018

To: Elon Musk

Subject: Tesla +\$11

FULL TEXT · BOARD, INVESTORS & ADVISERS · COURT RECORD · CONFIRMED

If you decide to purchase Tesla in a "going private" transaction, I hope you will allow us to continue to be an investor in your newly formed business. I want to make sure you know that we remain optimistic about the prospects for Tesla and want to continue to invest with you. Thanks in advance for considering us as one of your partners.

About this email: Minutes after the 'funding secured' tweet, fund manager Ron Baron asks Musk to let Baron Funds remain an investor if Tesla goes private.

Notes: Trial Exhibit 1001. Baron's message is quoted beneath Musk's reply with the line 'On Aug 7, 2018, at 10:55 AM, Ron Baron <rbaron@baronfunds.com> wrote:'. The exhibit also contains internal Baron Funds emails; Baron wrote to colleagues that a take-private 'would be a terrible result for us ... unless we were included in the buying group, of course.'

Sources: [U.S. District Court, N.D. Cal. - admitted trial exhibits, In re Tesla Inc. Securities Litigation \(18-cv-04865-EMC\)](#); [N.D. Cal. admitted-evidence directory for In re Tesla Inc. Securities Litigation](#)

Tags: take-private, investors, ron-baron, funding-secured

'Taking Tesla Private' email to employees

From: Elon Musk

Date: August 7, 2018

To: All Tesla employees

Subject: Taking Tesla Private

FULL TEXT • COMPANY-WIDE MEMO • PUBLISHED BY MUSK • CONFIRMED

Earlier today, I announced that I'm considering taking Tesla private at a price of \$420/share. I wanted to let you know my rationale for this, and why I think this is the best path forward.

First, a final decision has not yet been made, but the reason for doing this is all about creating the environment for Tesla to operate best. As a public company, we are subject to wild swings in our stock price that can be a major distraction for everyone working at Tesla, all of whom are shareholders. Being public also subjects us to the quarterly earnings cycle that puts enormous pressure on Tesla to make decisions that may be right for a given quarter, but not necessarily right for the long-term. Finally, as the most shorted stock in the history of the stock market, being public means that there are large numbers of people who have the incentive to attack the company.

I fundamentally believe that we are at our best when everyone is focused on executing, when we can remain focused on our long-term mission, and when there are not perverse incentives for people to try to harm what we're all trying to achieve.

This is especially true for a company like Tesla that has a long-term, forward-looking mission. SpaceX is a perfect example: it is far more operationally efficient, and that is largely due to the fact that it is privately held. This is not to say that it will make sense for Tesla to be private over the long-term. In the future, once Tesla enters a phase of slower, more predictable growth, it will likely make sense to return to the public markets.

Here's what I envision being private would mean for all shareholders, including all of our employees.

First, I would like to structure this so that all shareholders have a choice. Either they can stay investors in a private Tesla or they can be bought out at \$420 per share, which is a 20% premium over the stock price following our Q2 earnings call (which had already increased by 16%). My hope is for all shareholders to remain, but if they prefer to be bought out, then this would enable that to happen at a nice premium.

Second, my intention is for all Tesla employees to remain shareholders of the company, just as is the case at SpaceX. If we were to go private, employees would still be able to periodically sell their shares and exercise their options. This would enable you to still share in the growing value of the company that you have all worked so hard to build over time.

Third, the intention is not to merge SpaceX and Tesla. They would continue to have separate ownership and governance structures. However, the structure envisioned for Tesla is similar in many ways to the SpaceX structure: external shareholders and

employee shareholders have an opportunity to sell or buy approximately every six months.

Finally, this has nothing to do with accumulating control for myself. I own about 20% of the company now, and I don't envision that being substantially different after any deal is completed.

Basically, I'm trying to accomplish an outcome where Tesla can operate at its best, free from as much distraction and short-term thinking as possible, and where there is as little change for all of our investors, including all of our employees, as possible.

This proposal to go private would ultimately be finalized through a vote of our shareholders. If the process ends the way I expect it will, a private Tesla would ultimately be an enormous opportunity for all of us. Either way, the future is very bright and we'll keep fighting to achieve our mission.

Thanks,
Elon

About this email: Hours after tweeting 'funding secured', Musk explains to employees why he is considering taking Tesla private at \$420 a share, promising that shareholders and employees could stay invested and that no final decision has been made.

Notes: Tesla published the email on its blog under the note 'The following email was sent to Tesla employees today'. Court exhibit 503 shows the text was drafted by communications chief Sarah O'Brien and approved by Musk. The tesla.com blog page itself could not be opened in this session (403); text taken from TechCrunch's reproduction.

Sources: [TechCrunch \(reproducing the Tesla blog post\)](#); [CleanTechnica](#); [Court exhibit 12 \(Tesla blog post\)](#); [CNBC](#)

Tags: take-private, funding-secured, 420, short-sellers, shareholders

Comms chief warns Musk of FT story on Saudi stake

From: Dave Arnold

Date: August 7, 2018 (Exhibit header: August 7, 2018, 8:16 AM)

To: Elon Musk

Subject: FT article re: Saudi investment in Tesla

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Elon,

FYI - The Financial Times is preparing to run a story today saying that sources have confirmed that Saudi Arabia (via the Saudi Public Investment Fund) has built a 3-5% stake in Tesla after buying shares on the open market earlier this year. This comes after a visit to the US by the crown prince in March. The FT will also report that the Saudis previously approached Tesla about buying new shares but that we declined because we weren't looking to raise capital. The reporter says the Saudis "first approached Tesla after investment in Uber in 2016."

Their deadline is 9am PT, and we're planning to decline to comment to avoid confirming their reporting.

Thanks,

Dave

About this email: Tesla communications head Dave Arnold alerts Musk that the Financial Times is about to report the Saudi PIF has built a 3-5% Tesla stake and that Tesla plans to decline comment. Musk tweeted 'funding secured' later that day.

Notes: Email to Musk, included because he replied in the same exhibit.

Sources: [U.S. District Court N.D. Cal., In re Tesla Inc. Securities Litigation, Trial Exhibit 87](#)

Tags: saudi-pif, take-private, press, funding-secured

Musk's one-word reply on FT Saudi-stake story

From: Elon Musk

Date: August 7, 2018 (Exhibit header: sent 8/7/2018 4:20:20 PM (header time zone not stated))

To: Dave Arnold (cc: Deepak Ahuja; Todd Maron; Sarah O'Brien; Martin Viecha; EMDesk)

Subject: Re: FT article re: Saudi investment in Tesla

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Ok

About this email: Musk replies 'Ok' to the plan to decline comment on the Financial Times report about the Saudi PIF's Tesla stake.

Notes: Sent from erm@tesla.com.

Sources: [U.S. District Court N.D. Cal., In re Tesla Inc. Securities Litigation, Trial Exhibit 87](#)

Tags: saudi-pif, take-private, press, funding-secured

CFO tells Musk investors and SEC are asking about 'funding secured'

From: Deepak Ahuja

Date: August 7, 2018 (Exhibit header: sent 8/7/2018 3:11:18 PM)

To: Elon Musk (cc: Todd Maron)

Subject: FW: hey

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Elon,

FYI. We are getting a lot of enquiries from investors, SEC and the media to better understand the comment "funding secured".

Deepak

[Forwarded message] From: Owuraka Koney (Jennison Associates) / Date: Tuesday, August 7, 2018 at 2:15 PM / To: deepak@tesla.com / Subject: hey

Hey Deepak,

I hope all is well. Today (and the last few months!) has been a bit of a whirlwind for us as I imagine for you and the firm as well. We are obviously quite surprised by the intent to take Tesla private but we need more information. Specifically, we want to know if there is some kind of commitment letter not just a verbal agreement to provide the funding to go private? This is a very serious matter and saying it on a tweet is just not sufficient.

As a side note, I just want you to know that Martin is doing an absolutely fantastic job running IR. His level of knowledge and seriousness with which he approaches the job has been very impressive. Multiples over the last few months I've done calls with him and my PM team and we can't praise him enough.

Owuraka

About this email: CFO Deepak Ahuja forwards Musk an email from Jennison Associates' Owuraka Koney asking whether there is a commitment letter for the take-private funding, and tells Musk that investors, the SEC and media are asking what 'funding secured' means.

Notes: Email TO Musk. The court labels the exhibit '8.7.18 from Musk to Maron', but the transcription retrieved in this session showed only Ahuja's note to Musk and the forwarded investor email (reproduced below it); no Musk-authored text was recovered. The forwarded message's legal disclaimer is omitted.

Sources: [U.S. District Court N.D. Cal., In re Tesla Inc. Securities Litigation, Trial Exhibit 337](#)

Tags: funding-secured, take-private, sec, investors

Musk to investor Ron Baron: 'Absolutely'

From: Elon Musk

Date: August 7, 2018 (Exhibit header: sent 8/7/2018 6:20:45 PM)

To: Ron Baron (cc: David Baron; Michael Baron)

Subject: Re: Tesla +\$11

FULL TEXT · BOARD, INVESTORS & ADVISERS · COURT RECORD · CONFIRMED

Absolutely

About this email: On the day of the 'funding secured' tweet, Musk sends a one-word reply to longtime investor Ron Baron on a chain in which Baron had told colleagues a buyout would be 'a terrible result for us ... unless we were included in the buying group'.

Notes: The exhibit shows Musk's reply atop an internal Baron Funds chain (Jodi Montgomery: 'SAUDI SOVEREIGN WEALTH FUND IS SAID TO BUILD \$2B TESLA STAKE:FT'; Jamie Stone relaying the tweet; Ron Baron: 'That would be a terrible result for us ... unless we were included in the buying group, of course.'). Baron's direct message to Musk, if any, was not visible in the transcription retrieved.

Sources: [U.S. District Court N.D. Cal., In re Tesla Inc. Securities Litigation, Trial Exhibit 1001](#)

Tags: take-private, ron-baron, investors, funding-secured

Musk approves the drafted 'Taking Tesla Private' email

From: Elon Musk

Date: August 7, 2018 (Exhibit shows 11:49 AM for 'Great' and a header time of 8/7/2018 7:11:15 PM for 'Will send')

To: Sarah O'Brien (cc: Todd Maron; Deepak Ahuja; Dave Arnold; Sam Teller)

Subject: Re: Draft Email 420

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Great

[...]

Will send

About this email: Communications chief Sarah O'Brien sends Musk a draft of the employee email explaining the \$420 take-private idea; Musk replies 'Great', and after she suggests he send it to employees before it goes on

the blog, he answers 'Will send'.

Notes: Two very short Musk replies in one chain, combined here. Successive machine reads of the PDF disagreed on which header carries which reply, so the pairing of 'Great' (reply to the draft) and 'Will send' (final reply) should be checked against the PDF. O'Brien's messages: 'Below is a draft email for your consideration. Immediately after you send something to employees, we'll get it up on the blog.'; 'Apologies Elon, use this version:'; 'Shall we leave you send this to employees first and then we'll post it on the blog as a message from to our employees? To me, that feels like the best approach.'

Sources: [U.S. District Court N.D. Cal., In re Tesla Inc. Securities Litigation, Trial Exhibit 503](#)

Tags: take-private, 420, communications, funding-secured

Goldman's Dan Dees offers help on take-private

From: Dan Dees (Goldman Sachs)

Date: August 7, 2018 (Quoted header: Aug 7, 2018, 9:07 PM)

To: Elon Musk

Subject: really exciting day for tesla!

FULL TEXT · BOARD, INVESTORS & ADVISERS · COURT RECORD · CONFIRMED

I thought your letter today was excellent. Very clear.

At GS, we are standing by and ready to help.

Call me any time-24/7.

This whole topic feels very reminiscent of a conversation that we started together, many many months ago at Larry's house!

We are ready to fly in and share some views and dig in to help whenever you're ready. We know we could be very helpful throughout this process.

Great stuff!!

About this email: Goldman Sachs banker Dan Dees praises Musk's letter to employees and says Goldman is standing by to help with a Tesla take-private.

Notes: Email to Musk, included because he replied. Dees's later reply of Aug 8 (6:45 AM) is partly cut off in the exhibit as transcribed and is not included as a separate entry.

Sources: [U.S. District Court N.D. Cal., In re Tesla Inc. Securities Litigation, Trial Exhibit 243](#)

Tags: take-private, goldman-sachs, bankers, funding-secured

Musk to Goldman's Dees: 'Will call'

From: Elon Musk

Date: August 8, 2018 (Exhibit header: Wednesday, August 08, 2018 12:32 AM)

To: Dan Dees (Goldman Sachs)

Subject: Re: really exciting day for tesla!

FULL TEXT · BOARD, INVESTORS & ADVISERS · COURT RECORD · CONFIRMED

Sounds good. Will call as soon as I have a spare moment.

About this email: Musk briefly replies to Goldman Sachs banker Dan Dees's offer of help on the take-private.

Notes: Sent from Musk's SpaceX address (erm@spacex.com).

Sources: [U.S. District Court N.D. Cal., In re Tesla Inc. Securities Litigation, Trial Exhibit 243](#)

Tags: take-private, goldman-sachs, bankers

Musk to Goldman: Saudi Arabia 'would do the entire thing'

From: Elon Musk

Date: August 11, 2018 (Exhibit header: Saturday, August 11, 2018 3:39:59 PM)

To: Dan Dees (Goldman Sachs)

Subject: Re: Good to see u yesterday

FULL TEXT · BOARD, INVESTORS & ADVISERS · COURT RECORD · CONFIRMED

Sounds good. Will call this evening. Just a short call between us that maybe leads to a larger group call. Key question to solve is: can we put together a diverse enough investor/lender group, such that no single investor can demand special terms? Saudi Arabia would do the entire thing of course, as this is strategically critical to their future, but they would require that we build a Gigafactory there sooner than we would prefer. Hopefully, we can avoid that requirement.

About this email: Musk tells Goldman's Dan Dees the key question is whether a diverse investor group can be assembled so no single investor can demand special terms, saying Saudi Arabia would fund the whole take-private but would want a Gigafactory there.

Notes: In the machine transcription of the PDF, Musk's reply runs directly into the quoted earlier message from Dees ('Anytime. On mobile. Or if u have a set time that works best for u, I can line up a dial in and get the broader team. Either way works'); that trailing text is treated here as Dees's quoted message and excluded. Boundary should be checked against the PDF.

Sources: [U.S. District Court N.D. Cal., In re Tesla Inc. Securities Litigation, Trial Exhibit 256](#)

Tags: take-private, saudi-pif, goldman-sachs, gigafactory, funding-secured

Musk to general counsel: won't take Yasir's calls

From: Elon Musk

Date: August 12, 2018 (Exhibit header: sent 8/12/2018 7:12:53 PM)

To: Todd Maron (cc: Sam Teller)

Subject: Privileged

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Yasir has tried to call me several times today, but I will not take the call or agree to a meeting unless he takes action to correct the public record.

There is no question that PIF said they were was all-in for a Tesla take-private when we met two weeks ago and I clarified whether Yasir was the decision-maker and he said yes unequivocally and, moreover, that this was fully backed by the crown Prince.

About this email: Forwarding screenshots of his texts with PIF head Yasir Al-Rumayyan, Musk tells Tesla's general counsel he will not take Yasir's calls until he corrects the public record, insisting PIF said it was all-in on a take-private.

Notes: Attachments were six PNG screenshots of text messages (not transcribed here; texts are out of scope). 'they were was all-in' is as transcribed from the exhibit.

Sources: [U.S. District Court N.D. Cal., In re Tesla Inc. Securities Litigation, Trial Exhibit 825](#)

Tags: saudi-pif, take-private, funding-secured, yasir-al-rumayyan

Musk agrees to Goldman's take-private engagement terms

From: Elon Musk

Date: August 14, 2018 (Exhibit header: Tuesday, August 14, 2018 1:03:58 PM)

To: Dan Dees (Goldman Sachs) (cc: Sam Teller; Sam Britton)

Subject: Re:

FULL TEXT · BOARD, INVESTORS & ADVISERS · COURT RECORD · CONFIRMED

Agreed

About this email: Musk replies 'Agreed' to a Goldman Sachs email setting conditions for advising on the take-private, including that all his future tweets and public communications be pre-cleared with legal and financial advisers.

Notes: The Goldman email he was answering is a separate entry in this thread.

Sources: [U.S. District Court N.D. Cal., In re Tesla Inc. Securities Litigation, Trial Exhibit 261](#)

Tags: take-private, goldman-sachs, tweets, pre-clearance

Goldman's conditions for advising Musk on take-private

From: Dan Dees (Goldman Sachs)

Date: August 14, 2018 (Date of Musk's reply; send time of the Goldman email itself not shown in the transcription)

To: Elon Musk

FULL TEXT · BOARD, INVESTORS & ADVISERS · COURT RECORD · CONFIRMED

To ensure there is no misunderstanding, we wanted to follow up on our conversation this morning with this email. We are pleased to accept the mandate to work with you on the basis of the following mutual understanding.

We want to remind you that GS, alongside other banks, is a counterparty to the Company and its subsidiaries on multiple equity derivative transactions, including multiple call spread transactions and a capped call transaction. A change of control of the Company would trigger a termination/unwind of the call spreads that could result in a payment to or from GS and others and could result in varying p+I consequences to GS depending on our hedging strategy. You should also know that your recent announcements may trigger potential adjustments under those transactions. We will review with you and your team potential implications of a take-private on the equity derivatives. As is typical under these transactions, you understand that GS will continue to act as an arms'-length counterparty to you and the Company and as such would not be in a position to advise you on the equity derivative transactions.

In addition, you know that GS has an ABL outstanding with the company which has a change of control provision. We will need to discuss with you plans for the ABL in connection with the new financing to be put in place in connection with the take private.

We would expect to sign an engagement letter with you as soon as practical, which would include agreement on fees, our standard indemnification, reimbursement of legal expenses and a waiver of conflicts relating to the equity derivatives and would expect that you require any buy side consortium members to also sign on the obligations under the engagement letter. In addition, we would look to be indemnified and have our legal

expenses paid to the extent GS becomes involved in connection with investigations into the Company and its CEO or associated shareholder lawsuits.

We understand you have received the necessary permission and will keep your financial advisors and deal counsel fully informed on the status of all litigation and governmental investigations involving the recent tweets and production issues as well as any shareholder or other material litigation.

As we have previously discussed, you understand that we will neither advise nor participate in any capital raises, rollovers or restructuring, debt or equity, unless we believe appropriate disclosure and timely disclosure has been made.

We agree to a working arrangement whereby

- You will keep the special committee fully apprised of your work and to abide by their requests
- All future tweets and public communications will be pre-cleared with your legal and financial advisors.
- Our name will not be used without our consent.

You understand that our ongoing involvement with the you and the consortium is based on this understanding.

About this email: Goldman Sachs sets out the understanding on which it will advise Musk on taking Tesla private: disclosure of its derivative positions, indemnification, and a working arrangement under which all his future tweets and public communications are pre-cleared with advisers.

Notes: Email to Musk, included because he replied 'Agreed'. Wording such as 'p+I consequences' and 'with the you' is as transcribed from the exhibit.

Sources: [U.S. District Court N.D. Cal., In re Tesla Inc. Securities Litigation, Trial Exhibit 261](#)

Tags: take-private, goldman-sachs, tweets, pre-clearance, derivatives

Management changes; 'most amazing quarter in our history'

From: Elon Musk

Date: September 7, 2018

To: All Tesla employees

Subject: Company Update

FULL TEXT • COMPANY-WIDE MEMO • PUBLISHED BY MUSK • CONFIRMED

First, I'd like to thank you for your incredible efforts and contribution as we enter the final few weeks of this quarter. This is a very exciting time for Tesla, to say the least, and you are the reason for the tremendous progress we've made in a relatively short period of time. It is easy to forget that our company was almost unknown 10 years ago and didn't even exist until after the start of the 21st century!

We are about to have the most amazing quarter in our history, building and delivering more than twice as many cars as we did last quarter. For a while, there will be a lot of fuss and noise in the media. Just ignore them. Results are what matter and we are creating the most mind-blowing growth in the history of the automotive industry. Even the Ford Model T, which held the world record for the fastest growing car in history,

didn't grow as fast in sales or production as the Model 3.

Moreover, we also have the most exciting new product lineup of any company in the world. There is the Model Y, the Tesla (pickup) Truck, the Semi and the new Roadster. Then there is the Solar Roof, which is spooling up in production, and continued advancements in Powerwall and Powerpack. And that's just what people know about ...

To help achieve our goals, after discussing this in-depth with the Tesla board of directors and executive staff over the past several weeks, I am excited to announce a number of management changes:

Jerome Guillen has been promoted to President, Automotive, reporting directly to me. In his new role, Jerome will oversee all automotive operations and program management, as well as coordinate our extensive automotive supply chain. Jerome has made major contributions and acquired deep knowledge of Tesla's operations over the past eight years at our company, from being the first Model S Program Manager to managing all vehicle programs, then all vehicle engineering and worldwide sales & service. Recently, Jerome played a critical role in ramping Model 3 production, leading what almost all thought was impossible: creation of an entire high-volume General Assembly line for Model 3 in a matter of weeks. Before coming to Tesla, Jerome was responsible for creating and running the most successful semi truck program in history at Daimler's Freightliner division.

Kevin Kassekert has been promoted to VP of People and Places, with responsibility for Human Resources, Facilities, Construction, and Infrastructure Development. Kevin has been with Tesla for 6 years and was previously VP, Infrastructure Development. Kevin led the construction and development of our Gigafactory in Nevada, turning what was a pile of rocks in the Sierra Nevada mountains into a factory employing 12 thousand people with greater output than the entire rest of the world's battery factories combined, in roughly three years. That is insanely badass. Prior to joining Tesla, Kevin engineered and built multi-billion dollar production plants in the semiconductor industry. As many of you know, Tesla's Chief People Officer Gaby Toledano has been on leave for a few months to spend more time with her family and has decided to continue doing so for personal reasons. She's been amazing and I'm very grateful for everything she's done for Tesla.

Chris Lister joined Tesla last year and was key to solving our production problems at the Gigafactory and ramping Model 3 production. He is being promoted to VP, Gigafactory Operations. In this role, he leads our production and manufacturing engineering at Giga. Before Tesla, Chris ran several ultra high volume factories at PepsiCo.

Felicia Mayo, Sr. HR Director and head of our Diversity and Inclusion program, has been promoted to VP, reporting to both Kevin and me. Felicia has done great work promoting the importance of having one of the most diverse workforces in the world and ensuring fairness across hundreds of roles and Tesla operations in over 50 countries. She previously worked in senior HR roles at PwC and Oracle, and, most recently was Vice President, Global Talent Acquisition and Diversity for Juniper Networks.

Laurie Shelby, Tesla's VP, Environmental, Health and Safety (EHS) will also report directly to me. Your safety and just generally making sure that you love coming to work is extremely important, which is why EHS will report directly to me. We are working hardcore on having the safest (and most fun) work environment in the automotive industry by far. Laurie came to Tesla last year after a 25+ year career at Alcoa, where she made tremendous progress in workplace safety. Since joining, she has already overseen a reduction in our employee injury rate, as well as major improvements in our EHS program.

Cindy Nicola, VP of Global Recruiting, will report to both Kevin and me. Cindy has overseen the company's global recruiting efforts since 2015, bringing on tens of thousands of highly talented people. Prior to Tesla, Cindy led worldwide Corporate Recruiting at Apple, and before that was VP, Global Talent Acquisition at Electronic Arts.

Finally, Dave Arnold has been promoted to Sr. Director, Global Communications, overseeing Tesla's communications team. Dave was previously Director of Corporate Communications at Virgin America.

Thanks again for your incredible work in making Tesla successful. What you are doing is vital to achieving an amazing and sustainable energy future for all of humanity and life on Earth.

Elon

About this email: Hours after his Joe Rogan appearance and the resignation of two executives, Musk tells staff to ignore media 'fuss and noise', predicts a record quarter, and announces promotions including Jerome Guillen to President, Automotive.

Notes: Tesla posted the email on its own blog on Sept 7, 2018. The subject line 'Company Update' is the blog post title; the actual email subject was not published.

Sources: [Tesla blog \('Company Update'\)](#), reproduced by [CNBC](#); [Electrek](#); [Quartz](#)

Tags: management, promotions, jerome-guillen, media, q3-2018

'Ignore all distractions' after SEC lawsuit

From: Elon Musk

Date: September 28, 2018

To: All Tesla employees

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Ignore all distractions. One more hardcore weekend and we will all be victorious.
Thanks for being amazing.

Elon

About this email: Sent the day after the SEC sued Musk over the 'funding secured' tweet, a one-line email urging staff to ignore distractions and push through the last weekend of Q3 2018.

Sources: [CNBC](#); [Fortune](#)

Tags: sec, end-of-quarter, deliveries, morale

'Very close to achieving profitability' - go all out Sunday

From: Elon Musk

Date: September 29, 2018 (Sent Saturday Sept 29 (it refers to 'tomorrow (Sunday)'); Tesla filed it with the SEC as an 8-K exhibit dated September 30, 2018)

To: All employees of Tesla, Inc.

FULL TEXT · COMPANY-WIDE MEMO · PUBLISHED BY MUSK · CONFIRMED

We are very close to achieving profitability and proving the naysayers wrong, but, to be certain, we must execute really well tomorrow (Sunday).

If we go all out tomorrow, we will achieve an epic victory beyond all expectations.

Go Tesla!!!

Thanks for all your hard work,
Elon

About this email: Musk tells employees Tesla is very close to profitability and asks for an all-out final day of Q3 2018; Tesla filed the email with the SEC.

Notes: Tesla itself filed the email as Exhibit 99.1 to a Form 8-K ('email to all employees of Tesla, Inc.'). Exact send date inferred from the text; CNBC describes it as sent over the weekend.

Sources: [SEC EDGAR \(Tesla 8-K, Exhibit 99.1\)](#); [Bloomberg](#); [CNBC](#)

Tags: profitability, end-of-quarter, deliveries, naysayers

'Now we need to prove financial sustainability'

From: Elon Musk

Date: November 2018 (Described by Bloomberg on Nov 16, 2018 as a 'separate email to staff'; exact date not given)

To: Tesla employees

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · SINGLE-SOURCE

Last quarter was great, but now we need to prove financial sustainability.

About this email: In a separate November 2018 email to staff, Musk says that after a great third quarter Tesla must now prove it is financially sustainable.

Notes: Only this sentence was published.

Sources: [Bloomberg \(via BNN Bloomberg\)](#); [IndustryWeek \(Bloomberg\)](#)

Tags: profitability, q4-2018, finance

Push for 1,000 Model 3s a day by Nov 28

From: Elon Musk

Date: November 15, 2018 (Bloomberg describes it as sent Thursday (Nov 15, 2018), the day Electrek reported it)

To: All Tesla employees

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

By the 28th, all Model 3 production subsystems need to be at 50+ UPH steady, which is what's needed for a true 1000 vehicles per day rate (taking equipment uptime into account). To be clear, this includes all lines for a particular subsystem, so, for example, general assembly UPH would include both GA3 lines plus GA4.

[...]

If you can't see a way to get there, please let me know as soon as possible, so that I can help solve for this outcome. This is extremely important for our prosperity.

About this email: Musk sets a deadline of November 28, 2018 for every Model 3 production subsystem to run steadily at 50+ units per hour, the rate needed for a true 1,000 cars a day (7,000 a week).

Notes: Only excerpts were published: the first passage via Electrek/BGR, the second via Bloomberg. 'subsystems' typo as quoted.

Sources: [Electrek](#); [BGR](#); [The Driven](#); [Bloomberg \(via BNN\)](#)

Tags: model-3, production, uph, targets

1,000 Model 3s a day and the 'game of pennies'

From: Elon Musk

Date: November 30, 2018 (Electrek describes it as sent to employees on November 30, 2018)

To: All Tesla employees

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · SINGLE-SOURCE

If you are able to help in any way with getting Model 3 production to a steady 1000 per day at excellent quality, everyone at the company should please consider this their top priority. Body production currently appears to be our limiting factor, so it needs the most support right now. Please focus on simplification and reducing cycle time first and then uptime.

[...]

It's important to bear in mind that the cost of the car is made up of about 10,000 unique parts and processes. depending on how you count it, the current cost of a standard range Model 3 would be around \$38,000, so each part or process step only costs around \$3.80. That means finding cost efficiencies is a game of pennies, even though it might not seem so.

About this email: Musk asks everyone to make a steady 1,000 Model 3s per day at excellent quality their top priority, names body production as the bottleneck, and explains that with about 10,000 parts and processes in a roughly \$38,000 car, cost reduction is 'a game of pennies'.

Notes: Two passages as quoted by Electrek; the rest of the email was not published. The lower-case 'depending' after a full stop is as published.

Sources: [Electrek](#); [Electrek \(earlier Nov 2018 email\)](#)

Tags: model-3, production, cost-cutting, body-line

7% layoff: 'the road ahead is very difficult'

From: Elon Musk

Date: January 18, 2019

To: All Tesla employees

Subject: Company Update

FULL TEXT · COMPANY-WIDE MEMO · PUBLISHED BY MUSK · CONFIRMED

As we all experienced first-hand, last year was the most challenging in Tesla's history. However, thanks to your efforts, 2018 was also the most successful year in Tesla's history: we delivered almost as many cars as we did in all of 2017 in the last quarter alone and nearly as many cars last year as we did in all the prior years of Tesla's existence combined! Model 3 also became the best-selling premium vehicle of 2018 in the US. This is truly remarkable and something that few thought possible just a short time ago.

Looking ahead at our mission of accelerating the advent of sustainable transport and energy, which is important for all life on Earth, we face an extremely difficult challenge: making our cars, batteries and solar products cost-competitive with fossil fuels. While we have made great progress, our products are still too expensive for most people. Tesla has only been producing cars for about a decade and we're up against massive, entrenched competitors. The net effect is that Tesla must work much harder than other manufacturers to survive while building affordable, sustainable products.

In Q3 last year, we were able to make a 4% profit. While small by most standards, I would still consider this our first meaningful profit in the 15 years since we created Tesla. However, that was in part the result of preferentially selling higher priced Model 3 variants in North America. In Q4, preliminary, unaudited results indicate that we again made a GAAP profit, but less than Q3. This quarter, as with Q3, shipment of higher priced Model 3 variants (this time to Europe and Asia) will hopefully allow us, with great difficulty, effort and some luck, to target a tiny profit.

However, starting around May, we will need to deliver at least the mid-range Model 3 variant in all markets, as we need to reach more customers who can afford our vehicles. Moreover, we need to continue making progress towards lower priced variants of Model 3. Right now, our most affordable offering is the mid-range (264 mile) Model 3 with premium sound and interior at \$44k. The need for a lower priced variants of Model 3 becomes even greater on July 1, when the US tax credit again drops in half, making our car \$1,875 more expensive, and again at the end of the year when it goes away entirely.

Sorry for all these numbers, but I want to make sure that you know all the facts and figures and understand that the road ahead is very difficult. This is not new for us - we have always faced significant challenges - but it is the reality we face. There are many companies that can offer a better work-life balance, because they are larger and more mature or in industries that are not so voraciously competitive. Attempting to build affordable clean energy products at scale necessarily requires extreme effort and

relentless creativity, but succeeding in our mission is essential to ensure that the future is good, so we must do everything we can to advance the cause.

As a result of the above, we unfortunately have no choice but to reduce full-time employee headcount by approximately 7% (we grew by 30% last year, which is more than we can support) and retain only the most critical temps and contractors. Tesla will need to make these cuts while increasing the Model 3 production rate and making many manufacturing engineering improvements in the coming months. Higher volume and manufacturing design improvements are crucial for Tesla to achieve the economies of scale required to manufacture the standard range (220 mile), standard interior Model 3 at \$35k and still be a viable company. There isn't any other way.

To those departing, thank you for everything you have done to advance our mission. I am deeply grateful for your contributions to Tesla. We would not be where we are today without you.

For those remaining, although there are many challenges ahead, I believe we have the most exciting product roadmap of any consumer product company in the world. Full self-driving, Model Y, Semi, Truck and Roadster on the vehicle side and Powerwall/pack and Solar Roof on the energy side are only the start.

I am honored to work alongside you.

Thanks for everything,

Elon

About this email: Musk announces a roughly 7% cut in full-time headcount, explaining that Tesla must make cheaper Model 3 variants while tax credits fall, and that Q4 2018 profit was lower than Q3.

Notes: Tesla published the email on its blog ('This morning, the following email was sent to all Tesla employees'). The tesla.com page returned 403 in this session; text taken from Electrek's reproduction. Subject shown is the blog post title.

Sources: [Electrek \(reproducing the Tesla blog post 'Company Update'\)](#); [Tesla blog \(Company Update\)](#); [Al Jazeera](#); [KTVU](#)

Tags: layoffs, profitability, model-3, affordability

\$35,000 Model 3 and shift to online-only sales

From: Elon Musk

Date: February 28, 2019

To: All Tesla employees

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Last month, I noted in my email that the fundamental issue Tesla must overcome is that our products remain too expensive for most people. We know there are many people who want to buy Model 3, but simply can't afford to do so.

That is why we're excited to announce today that we are now offering the standard Model 3 at \$35,000. This is a significant milestone for Tesla, the culmination of years of hard work by employees across the company, and something of which you should all be very proud. You can read the details of the announcement on our blog: [LINK HERE](#)

In addition, we are also making the decision to shift all sales worldwide to online only.

Last year, 78% of all Model 3 orders were placed online, rather than in a store, and 82% of customers bought their Model 3 without ever having taken a test drive. Customers can now buy a Tesla in North America via their phone in about 1 minute, and that capability will soon be extended worldwide. We are also making it much easier to try out and return a Tesla without a test drive. You can now return a car within 7 days or 1,000 miles for a full refund. Customers are becoming increasingly comfortable making purchases online, and that is especially true for Tesla — which is a testament to the products we make.

As a result, over the next few months, we will be winding down many of our stores and significantly reducing our spend on sales and marketing, which will help make the price changes we've announced today possible. Shifting all sales online combined with other ongoing cost efficiency will enable us to lower all vehicle prices by about 6% on average, allowing us to achieve the \$35,000 Model 3 price point.

A small number of stores in high-traffic locations will remain as galleries, showcases and Tesla information centers. At the same time, we will be increasing our investment in the Tesla service system and manufacturing, and I expect that headcount to grow next year.

Unfortunately, this means that some jobs will be impacted or transitioned to other areas of the business. This is a hard decision, but it necessary to make our cars more affordable. Our sales team has fought on the front lines of advancing our mission and has been our connection to hundreds of thousands of customers along the way. I want to express my sincere gratitude for all that you've done.

Kn the coming weeks, we will be evaluating all of our sales and marketing organization to understand where there are operation efficiencies, and how best to support the transition to online sales while also continuing to deliver a truly awesome and education Tesla buying experience.

We'll be sharing more information on this transition soon.

Thank you,
Elon

About this email: Musk tells employees Tesla is launching the \$35,000 Model 3, moving all sales online worldwide, winding down many stores and cutting sales and marketing spend, which will affect some jobs.

Notes: Typos ('Kn the coming weeks', 'it necessary', 'education Tesla buying experience') and the 'LINK HERE' placeholder are as published by CNBC.

Sources: [CNBC](#); [TechSpot](#); [TechCrunch](#)

Tags: stores, online-sales, model-3, layoffs, pricing

All hands on deliveries for last ten days of Q1 2019

From: Elon Musk

Date: March 21, 2019

To: Everybody (all Tesla employees)

Subject: Vehicle Delivery Help Needed!

FULL TEXT • COMPANY-WIDE MEMO • LEAKED TO PRESS • WIDELY REPORTED

For the last ten days of this quarter, please consider your primary priority to be helping with vehicle deliveries. This applies to everyone. As challenges go, this is a good one to have, as we've built the cars and people have bought the cars, so we just need to get the cars to their new owners!

What has made this particularly difficult is that Europe and China are simultaneously experiencing the same massive increase in delivery volume that North America experienced last year. In some locations, the delivery rate is over 600% higher than its previous peak! This was further exacerbated by supplier shortages of EU spec components and a sticker printing error on our part in China that were only resolved in the past few weeks.

North America is also stressed, as the final month of this quarter is almost all North America builds. Moreover, for the first two weeks of March, most cars were sent from our factory in California to the East Coast to ensure arrival before the end of quarter.

The net result is a massive wave of deliveries needed throughout Europe, China and North America. This is the biggest wave in Tesla's history, but it is primarily a function of our first delivery of mass manufactured cars on two continents simultaneously, and will not be repeated in subsequent quarters.

To help, please contact Sanjay Shah in North America, Robin Ren in China and Ashley Harris in Europe.

About this email: Musk asks every employee to treat helping with vehicle deliveries as their primary priority for the final ten days of Q1 2019, citing the first simultaneous delivery wave in Europe, China and North America.

Notes: First reported by Business Insider. No sign-off appears in the text as published by CNBC.

Sources: [CNBC](#); [Carscoops](#); [IBTimes](#)

Tags: deliveries, end-of-quarter, europe, china, logistics

Clarifying store closures: 'seeds on barren ground'

From: Elon Musk

Date: March 27, 2019 (Date of Electrek's publication; sent that week)

To: Tesla employees

FULL TEXT • COMPANY-WIDE MEMO • LEAKED TO PRESS • WIDELY REPORTED

There still some uncertainty around Tesla stores and the sales team. Hopefully, this note clears things up. Please let me know if there's anything I've forgotten to address.

– Stores with a high visitation rate and that lead to significant sales will absolutely not be closed down. It would not make any sense to do so, except in rare cases where the

rent is absurdly high. Moreover, Tesla will continue to open stores throughout the world that meet the above criteria.

- Stores that are in a location with low visitation rates (ie empty most of their opening hours) and lead to low sales will gradually be closed down. This is analogous to seeds on barren ground. There is no reasonable way to justify keeping such stores open.
- Stores that are somewhere in the middle will be evaluated over time to see there is some way to allow them to cover their costs. If there is, they will remain open, otherwise not. However, these stores will be given a fair opportunity to prove their case.

The above principles also apply to the sales team. No one who is a major contributor to demand generation will be let go. That would make no sense. However, sometimes, in a company with 45,000 people, things happen that make no sense. I will do my best to remedy issues when brought to my attention directly or through [redacted email address].

What is meant by "all sales will be online" is just that the act of purchasing a Tesla will always be done via the potential new owner's phone or computer. This is true whether they are at home or in a store. Unlike buying from other carmakers, ordering a Tesla doesn't require any physical paperwork.

This is very different from normal expectations for buying from other carmakers and is simply meant to emphasize that ordering a Tesla is super easy and can be done in 2 minutes from your phone or laptop at Tesla.com. Ordering a Tesla is not much harder than ordering an Uber, but hardly anyone knows this!

However, many potential Tesla owners will still want to talk to a Tesla representative in person or want a test drive from a Tesla representative. Stores also have a small number of Tesla vehicles available to drive away immediately for customers that want a car right then and there.

This is why stores and Tesla product specialists and owner advisors will always be of critical importance to our long term success.

Thanks,

Elon

About this email: Musk clarifies the store-closure plan: high-traffic stores stay open, empty ones close gradually, and 'all sales online' only means the purchase itself is made on a phone or computer.

Notes: Internal email address redacted by Electrek. 'There still some uncertainty' is as published.

Sources: [Electrek](#); [Engadget](#)

Tags: stores, retail, sales-team, online-sales

'Hardcore' review of every Tesla expense

From: Elon Musk

Date: May 16, 2019

To: All Tesla employees

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

examine every expenditure at Tesla, no matter how small

[...]

This is a lot of money, but actually only gives us about 10 months at the Q1 burn rate to achieve breakeven!

[...]

This is hardcore, but it is the only way for Tesla to become financially sustainable and succeed in our goal of helping make the world environmentally sustainable

About this email: After raising \$2.7 billion, Musk tells staff the money gives Tesla only about 10 months at the Q1 burn rate, and that he and CFO Zach Kirkhorn will personally review and sign off on every expenditure, including parts, salary, travel and rent.

Notes: The full email was reported by Reuters, CNBC and Electrek; only the quoted passages below could be retrieved verbatim in this session. A fuller text exists and should be added.

Sources: [Electrek](#); [Fossbytes](#); [CleanTechnica](#)

Tags: cost-cutting, cash, zach-kirkhorn

50,000 net new orders; record quarter within reach

From: Elon Musk

Date: May 22, 2019 (Header as published: Wed 5/22/2019 10:45 PM)

To: Everybody (all Tesla employees)

Subject: Exciting Goal!

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

As of yesterday, we had over 50,000 net new orders for this quarter. Based on current trends, we have a good chance of exceeding the record 90,700 deliveries of Q4 last year and making this the highest deliveries/sales quarter in Tesla history!

In order to achieve this, we need sustained output of 1,000 Model 3's per day. Almost all parts of the Model 3 production system have exceeded 1000 units on multiple days (congratulations!!) and we've averaged about 900/day this week, so we're only about 10% away from 7000/week.

If we rally hard, we can do it!

Thanks for your great work,
Elon

About this email: Musk tells employees Tesla has over 50,000 net new orders in Q2 2019 and could beat its 90,700 delivery record if Model 3 output is sustained at 1,000 a day.

Notes: First obtained by Electrek.

Sources: [Business Insider](#); [CleanTechnica](#)

Tags: orders, deliveries, model-3, production, q2-2019

'A shot at' first 100,000-delivery quarter

From: Elon Musk

Date: September 26, 2019 (Date of Electrek's report; sent that week)

To: All Tesla employees

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

We have a shot at achieving our first 100,000 vehicle delivery quarter, which is an incredibly exciting milestone for our company!

[...]

The challenge is making sure that we have the right car variants in the right locations and rallying as much as our company resources as possible to help with the end of the quarter deliveries.

About this email: Musk tells employees Tesla could reach 100,000 deliveries in a quarter for the first time and asks them to rally resources for end-of-quarter deliveries.

Notes: Only these two passages were published. 'as much as our company resources' is as quoted.

Sources: [Electrek](#)

Tags: deliveries, end-of-quarter, q3-2019, milestone

December 2019 priorities: deliveries and solar

From: Elon Musk

Date: December 4, 2019 (Date of Electrek's report; sent that week)

To: All Tesla employees

Subject: Tesla's two most critical priorities for December

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · SINGLE-SOURCE

The first [priority] is making sure all cars are delivered to their customers before end of year.

[...]

The second, just as important, is that we immediately increase the rate of solar deployments by a significant degree.

About this email: Musk tells employees Tesla's two most critical priorities for December 2019 are delivering all cars to customers before year-end and sharply increasing the rate of solar deployments.

Notes: Only two passages were published. The bracketed '[priority]' is Electrek's insertion.

Sources: [Electrek](#)

Tags: deliveries, solar, end-of-year, priorities

Coronavirus: 'panic far exceeds' harm of the virus

From: Elon Musk

Date: March 16, 2020 (Reported March 17, 2020; sent the previous evening)

To: All Tesla employees

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

First, I'd like to be super clear that if you feel the slightest bit ill or even uncomfortable, please don't feel obligated to come to work. I will personally be at work, but that is just me. Totally OK if you want to stay home for any reason.

A lot of rumors are flying around, but, to the best of our knowledge, no one at Tesla (over 56,000 people) has tested positive for COVID-19. I will inform you immediately if anything changes.

My frank opinion remains that the harm from the coronavirus panic far exceeds that of the virus itself. If there is a massive redirection of medical resources out of proportion to the danger, it will result in less available care to those with critical medical needs, which does not serve the greater good.

What the COVID-19 coverage does show is that the danger of colds and flu in general can be high for the elderly, especially if there is prior lung damage. Much is made of public gatherings, but please be cautious of family gatherings, too. What is relatively harmless to a child can be dangerous to grandparents.

My best guess, for what it is worth, based on the latest Center for Disease Control data, is that confirmed COVID-19 (this specific form of the common cold) cases will not exceed 0.1% of the US population. Moreover, I do not think, when we look back on 2020, that the causes of death or serious injury will have changed much from 2017, for example...

[...]

not feel obligated to come to work if they have any concern

About this email: As Bay Area shelter-in-place orders took effect, Musk tells staff they need not come to work if ill or uncomfortable, says no Tesla employee has tested positive, and argues the harm from coronavirus panic exceeds that of the virus.

Notes: REPLACES: 2020-03-16 Coronavirus: 'panic far exceeds' harm of the virus. Adds the closing fragment Electrek quotes from the end of the email. Electrek printed the bulk of the email in block quotes; the fifth passage is cut off with an ellipsis as published, and Electrek says Musk then reiterated that employees should 'not feel obligated to come to work if they have any concern' (last fragment below). The sign-off was not reproduced.

Sources: [Electrek](#)

Tags: covid-19, fremont, health

Plan to restart Fremont production

From: Elon Musk

Date: May 7, 2020 (Sent the night of May 7, 2020 (reported May 8))

To: Tesla employees (Fremont)

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

In light of Governor Newsom's statement earlier today approving manufacturing in California, we will aim to restart production in Fremont tomorrow afternoon.

[...]

I will be on the line personally helping wherever I can. However, if you feel uncomfortable coming back to work at this time, please do not feel obligated to do so. These are difficult times, so thanks very much for working hard to make Tesla successful!

About this email: Citing Governor Newsom's statement allowing manufacturing, Musk tells employees Tesla will aim to restart Fremont production the next afternoon and that those uncomfortable returning need not come in.

Notes: Two passages as quoted by Electrek; Alameda County blocked the restart the same day.

Sources: [Electrek](#); [Electrek \(county blocks restart\)](#)

Tags: covid-19, fremont, reopening, production

Model Y ramp is top priority; 'minimize rectification'

From: Elon Musk

Date: June 2020 (Reported by Electrek on June 8, 2020; sent shortly before)

To: Tesla production employees

Subject: Model Y production

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

It is extremely important for us to ramp up Model Y production and minimize rectification needs. I want you to know that it really makes a difference to Tesla right now.

Model Y, especially GA (stands for General Assembly), is the top priority for both production and manufacturing engineering. GA4 (stands for General Assembly Line 4) is also top priority for facility improvements. For those working in GA4, thank you for bearing with tough conditions. Will get better fast. I will be walking the line personally every week.

We are doing reasonably well with S, X, and 3, but there are production and supply chain ramp challenges with Model Y, as is always the case with new products.

Please let me know if there's anything I can do to help.

Thanks,

Elon

About this email: Musk tells production staff that ramping Model Y and minimizing rework is the top priority for production and manufacturing engineering, acknowledges tough conditions on the GA4 line, and says he will walk the line weekly.

Notes: The parentheticals '(stands for General Assembly)' and '(stands for General Assembly Line 4)' appear in Electrek's published text and are probably the outlet's explanatory insertions rather than Musk's words.

Sources: [Electrek](#)

Tags: model-y, production, quality, fremont, ga4

'Time to go all out' on Tesla Semi volume production

From: Elon Musk

Date: June 9, 2020 (Sent the night before Electrek's June 10, 2020 report)

To: All Tesla employees

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

It's time to go all out and bring the Tesla Semi to volume production. It's been in limited production so far, which has allowed us to improve many aspects of the design.

[...]

Production of the battery and powertrain would take place at Giga Nevada, with most of the other work probably occurring in other states.

[...]

Jerome and I are very excited to work with you to bring this amazing product to market!

About this email: Musk tells employees it is time to bring the Tesla Semi to volume production, with battery and powertrain built at Giga Nevada.

Notes: Also reported by Reuters and CNBC. Three passages as quoted by Electrek; the email was short and these appear to be most of it.

Sources: [Electrek](#)

Tags: semi, production, giga-nevada, jerome-guillen

Q2 2020: 'Breaking even is looking super tight'

From: Elon Musk

Date: June 29, 2020

To: All Tesla employees

Subject: Down to the last few days

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Breaking even is looking super tight. Really makes a difference for every car you build and deliver. Please go all out to ensure victory!

About this email: With two days left in Q2 2020, Musk tells staff that breaking even will be very close and every car built and delivered matters.

Notes: Also reported by CNBC. The body as published is three sentences; any sign-off was not reproduced. Electrek also mentions an earlier, 'more cryptic' email that week urging staff to go 'all out', whose text was not published.

Sources: [Electrek](#)

Tags: profitability, end-of-quarter, deliveries, covid-19

Stock could be 'crushed like a soufflé under a sledgehammer'

From: Elon Musk

Date: December 1, 2020

To: All Tesla employees

FULL TEXT • COMPANY-WIDE MEMO • LEAKED TO PRESS • WIDELY REPORTED

At a time like this, when our stock is reaching new highs, it may seem as though spending carefully is not as important. This is definitely not true.

When looking at our actual profitability, it is very low at around 1% for the past year. Investors are giving us a lot of credit for future profits, but if, at any point, they conclude that's not going to happen, our stock will immediately get crushed like a soufflé under a sledgehammer!

Much more important, in order to make our cars affordable, we have to get smarter about how we spend money. This a tough Game of Pennies - requiring thousands of good ideas to improve part cost, a factory process or simply the design, while increasing quality and capabilities. A great idea would be on that saves \$5, but the vast majority are 50 cents here or 20 cents there.

In order to make the electric revolution happen we much make electric cars, stationary batteries and solar affordable to all.

Thanks and great working with you as always,
Elon

About this email: With Tesla's stock at record highs, Musk warns employees that profitability is only about 1% and that investors will crush the stock if they lose faith in future profits, urging a 'Game of Pennies' on costs.

Notes: Also reported by CNBC and Reuters. Typos ('This a tough', 'would be on that saves', 'we much make') are as published by Electrek.

Sources: [Electrek](#)

Tags: cost-cutting, stock-price, profitability, affordability

'High-class problem': demand exceeds production

From: Elon Musk

Date: December 11, 2020 (Date of Electrek's report; sent that week)

To: All Tesla employees

FULL TEXT • COMPANY-WIDE MEMO • LEAKED TO PRESS • SINGLE-SOURCE

We are fortunate to have a high-class problem of demand being quite a bit higher than production this quarter.

To ensure that we have the best possible outcome and earn the trust of the customers and investors who have placed their faith and hard-earned money with us, we need to increase production for the remainder of this quarter as much as possible.

I would only send this note if it really mattered.

Super Appreciated,,
Elon

Btw, please send me a note directly if you see ways to improve output, but feel that your voice is not being heard.

About this email: Musk tells employees demand is well above production in Q4 2020 and asks them to increase output as much as possible for the rest of the quarter.

Notes: Double comma in 'Super Appreciated,,,' is as published.

Sources: [Electrek](#)

Tags: production, demand, end-of-quarter, q4-2020

Q2 2021: 'the next 12 days are super important'

From: Elon Musk

Date: June 18, 2021 (Date of Electrek's report; the email speaks of 'the next 12 days' of the quarter)

To: Tesla employees

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · SINGLE-SOURCE

Overall, we are executing well, but the next 12 days are super important for production and delivery this quarter. Please go all out!

Thanks,
Elon

About this email: A two-line end-of-quarter rallying email asking Tesla staff to go all out on production and delivery for the last 12 days of Q2 2021.

Notes: Text as reproduced by Electrek; subject line and exact send time not published.

Sources: [Electrek](#); [Electrek \(follow-up on end-of-quarter practices\)](#)

Tags: deliveries, production, end-of-quarter, q2-2021

Q3 2021: 'biggest wave in Tesla history'

From: Elon Musk

Date: September 8, 2021

To: All Tesla employees

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

The end of the quarter delivery wave is unusually high this time as we suffered (like the rest of the industry) from extremely severe parts shortages earlier this quarter. This meant building a lot of cars with missing parts that needed to be added later. I'd like to thank all Tesla techs and contractors that helped add back the missing parts, often in very difficult conditions. You rock!

And we are for sure dying to reduce the size of the end-of-the-quarter delivery wave! The goal is to do so in Q4 (allowing some Q4 production to spill over to delivery in Q1). However, early Q3 production was so challenging that we need to go super hardcore to make up for it over the next ~22 days to ensure a decent Q3 delivery number. This is the biggest wave in Tesla history, but we got to get it done.

Much appreciated,
Elon

About this email: Musk explains that severe parts shortages early in Q3 2021 left many cars needing parts added later, creating an unusually large end-of-quarter delivery wave, and asks staff to go 'super hardcore' for the remaining 22 days.

Notes: Also reported by CNBC.

Sources: [Electrek](#)

Tags: deliveries, end-of-quarter, parts-shortage, supply-chain

Three actions allowed when Musk sends explicit directions

From: Elon Musk

Date: October 4, 2021 (Date from the email header)

To: Everybody (all Tesla employees)

Subject: Please Note

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

If an email is sent from me with explicit directions, there are only three actions allowed by managers.

1. Email me back to explain why what I said was incorrect. Sometimes, I'm just plain wrong!
2. Request further clarification if what I said was ambiguous.
3. Execute the directions.

If none of the above are done, that manager will be asked to resign immediately.

Thank you,
Elon

About this email: Musk tells Tesla managers that an email from him with explicit directions leaves three options: explain why he is wrong, ask for clarification, or execute. Managers who do none of these will be asked to resign.

Notes: Text transcribed from the image of the email in the X post that prompted this archive. The email was first reported by CNBC in November 2021; that article could not be retrieved for this archive, so the wording rests on the posted image.

Sources: [Image of the email posted on X by @StephNass](#)

Tags: managers, directions, ultimatum, management

Minimize delivery cost, not the end-of-quarter rush

From: Elon Musk

Date: November 26, 2021 (Header as published: Nov. 26, 2021 [time stamp redacted])

To: Everybody (all Tesla employees)

Subject: Q4 deliveries vs. cost efficiency

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Per my email several weeks ago, our focus this quarter should be on minimizing cost of deliveries rather than spending heavily on expedite fees, overtime and temporary contractors just so that cars arrive in Q4.

What has happened historically is that we sprint like crazy at end of quarter to maximize deliveries, but then deliveries drop massively in the first few weeks of the next quarter. In effect, looked at over a six month period, we won't have delivered any extra cars but we will have spent a lot of money and burned ourselves out to accelerate deliveries in the last two weeks of each quarter.

We will still have quite a big wave of deliveries in the last few weeks of December, as we don't yet have high volume production either in Europe or Texas, which means a lot of cars on boats from China to Europe and on trucks [and/or] rail from California to the East Coast arriving late in the quarter, but this is nonetheless the right time to start reducing the size of the wave in favor of a steadier and more efficient pace of deliveries.

The right principle is take the most efficient action, as though we were not publicly-traded and the notion of "end of quarter" didn't exist.

Thanks,
Elon

About this email: Musk tells employees to stop spending heavily on expedite fees, overtime and temporary contractors just to land cars in Q4, arguing the end-of-quarter sprint adds no deliveries over six months, and to act as if Tesla were not publicly traded.

Notes: First published by CNBC; body text retrieved from Electrek's reproduction, and the header (From/To/Subject/Date) confirmed on the CNBC page. The word 'cost' in the first sentence is emphasized in the published version. '[and/or]' and the redacted time stamp are as published. The email refers to an earlier 'email several weeks ago' on the same subject, whose text was not found.

Sources: [CNBC](#); [Electrek](#)

Tags: deliveries, end-of-quarter, cost-cutting, logistics, q4-2021

'Remote work is no longer acceptable'

From: Elon Musk

Date: May 31, 2022

To: Tesla executive staff (per press reports)

Subject: Remote work is no longer acceptable

FULL TEXT · EXECUTIVE & INTERNAL · LEAKED TO PRESS · CONFIRMED

Anyone who wishes to do remote work must be in the office for a minimum (and I mean *minimum*) of 40 hours per week or depart Tesla. This is less than we ask of factory workers.

If there are particularly exceptional contributors for whom this is impossible, I will review and approve those exceptions directly.

Moreover, the "office" must be a main Tesla office, not a remote branch office unrelated to the job duties, for example being responsible for Fremont factory human relations, but having your office be in another state.

Thanks,
Elon

About this email: Musk tells Tesla staff that anyone wishing to work remotely must first be in a main Tesla office at least 40 hours a week or leave the company, with exceptions approved only by him.

Notes: The subject line's typo ('acceptble') is in the original. The email leaked on Twitter and Musk effectively confirmed it there ('They should pretend to work somewhere else'). Bloomberg and others reported it was addressed to executive staff ('ExecStaff'); the recipient line is not shown in the text retrieved.

Sources: [Electrek](#)

Tags: remote-work, return-to-office, policy

'To be super clear': 40 hours in the office or resign

From: Elon Musk

Date: May 31, 2022

To: All Tesla employees

Subject: To be super clear

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · CONFIRMED

Everyone at Tesla is required to spend a minimum of 40 hours in the office per week. Moreover, the office must be where your actual colleagues are located, not some remote pseudo office. If you don't show up, we will assume you have resigned.

The more senior you are, the more visible must be your presence. That is why I lived in the factory so much – so that those on the line could see me working alongside them. If I had not done that, Tesla would long ago have gone bankrupt.

There are of course companies that don't require this, but when was the last time they shipped a great new product? It's been a while.

Tesla has and will create and actually manufacture the most exciting and meaningful products of any company on Earth. This will not happen by phoning it in.

Thanks,
Elon

About this email: Follow-up to all staff: everyone must spend at least 40 hours a week in the office where their colleagues are, or be assumed to have resigned; Musk says he lived in the factory so workers could see him.

Notes: Second of two emails sent the same day; widely reported (Bloomberg, CNBC, The Verge).

Sources: [Electrek](#)

Tags: remote-work, return-to-office, policy, work-ethic

'Super bad feeling' about economy; pause all hiring

From: Elon Musk

Date: June 2, 2022

To: Tesla executives

Subject: pause all hiring worldwide

EXCERPT · EXECUTIVE & INTERNAL · LEAKED TO PRESS · WIDELY REPORTED

pause all hiring worldwide

[...]

super bad feeling

About this email: In an email to executives seen by Reuters, Musk says he has a 'super bad feeling' about the economy, orders a worldwide hiring pause and says Tesla needs to cut about 10% of staff.

Notes: Reuters, which saw the email, quoted only the subject line and the phrase 'super bad feeling'; the full text was never published. Reuters' page was blocked in this session, so the words were taken from Electrek's relay of the Reuters report. Reuters dated the email Thursday, June 2, 2022.

Sources: [Electrek \(relaying the Reuters exclusive\)](#)

Tags: hiring, layoffs, economy, headcount

10% salaried headcount reduction

From: Elon Musk

Date: June 3, 2022

To: All Tesla employees

Subject: Headcount Reduction

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Tesla will be reducing salaried headcount by 10% as we have become overstaffed in many areas.

Note, this does not apply to anyone actually building cars, battery packs or installing solar. Hourly headcount will increase.

Elon

About this email: Musk tells all employees Tesla will cut salaried headcount by 10% because it has become overstaffed, while hourly headcount will increase.

Notes: Also reported by Reuters and CNBC.

Sources: [Electrek](#)

Tags: layoffs, headcount, salaried, hourly

'More respect for you than the richest person on Wall St'

From: Elon Musk

Date: June 11, 2022 (Sent the weekend of June 11-12, 2022 (reported June 12))

To: All Tesla employees

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · SINGLE-SOURCE

We *actually* build great, real products that people love and make their lives and the world better. That is so profound. It is an honest day's work that you can feel in your heart. Whatever else is going on in this messed-up world, know that at least what you are doing is pure goodness and that I have infinitely more respect and admiration for you than the richest person on Wall St.

About this email: A morale email telling Tesla workers that building real products people love is 'pure goodness' and that he respects them more than the richest person on Wall Street.

Notes: Electrek quoted this passage from the first of two emails sent the same weekend; it may not be the complete email.

Sources: [Electrek](#)

Tags: morale, work-ethic, wall-street, products

'Very tough quarter' - rally hard to recover

From: Elon Musk

Date: June 11, 2022 (Sent the weekend of June 11-12, 2022 (reported June 12))

To: All Tesla employees

Subject: Production and Delivery

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

This has been a very tough quarter, primarily due to supply chain and production challenges in China, so we need to rally hard to recover!

I'd like to congratulate the Fremont team for achieving a new all-time record production day last week and Berlin for making almost 1000 cars last week! This is great stuff.

Shanghai is returning to full strength and Austin is spooling up. Onward to victory!

Elon

About this email: Musk says Q2 2022 has been very tough because of supply-chain and production problems in China, congratulates Fremont and Berlin on production records, and urges staff to rally.

Sources: [Electrek](#)

Tags: production, shanghai, supply-chain, q2-2022

Don't be bothered by 'stock market craziness'

From: Elon Musk

Date: December 28, 2022

To: All Tesla employees

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Please go all out for the next few days and volunteer to help deliver if at all possible. It will make a real difference!

[...]

Btw, don't be too bothered by stock market craziness. As we demonstrate continued excellent performance, the market will recognize that.

About this email: With Tesla shares down nearly 70% for the year, Musk tells employees not to be bothered by the stock market and asks them to volunteer to help deliver cars in the last days of 2022.

Notes: Also reported by CNBC and Reuters, which quoted additional lines (e.g. his belief that Tesla will long-term be the most valuable company on Earth); only the two passages below were retrieved verbatim in this session, and their order in the email is not certain.

Sources: [Electrek](#)

Tags: stock-price, deliveries, end-of-quarter, morale

Musk must personally approve every new hire

From: Elon Musk

Date: May 15, 2023

To: Tesla employees

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

No one can join Tesla, even as a contractor, until you receive my email approval.

About this email: Musk tells Tesla staff that, to better understand hiring, no one may join the company, even as a contractor, without his emailed approval.

Notes: Also reported by CNBC, which published more of the text (including his request for weekly hiring-request lists from executives); only this sentence was retrieved verbatim in this session.

Sources: [Electrek](#)

Tags: hiring, approvals, headcount, contractors

Cybertruck parts to 'sub 10 micron accuracy'

From: Elon Musk

Date: August 23, 2023 (Date of Electrek's report; sent that week)

To: Tesla employees

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Due to the nature of Cybertruck, which is made of bright metal with mostly straight edges, any dimensional variation shows up like a sore thumb. All parts for this vehicle, whether internal or from suppliers, need to be designed and built to sub 10 micron accuracy. That means all part dimensions need to be to the third decimal place in millimeters and tolerances need be specified in single digit microns. If LEGO and soda cans, which are very low cost, can do this, so can we. Precision predicates perfectionism.

About this email: Musk tells employees every Cybertruck part must be designed and built to sub-10-micron accuracy because the truck's flat metal panels expose any variation: 'If LEGO and soda cans ... can do this, so can we.'

Notes: The email first surfaced on the Cybertruck Owners Club forum. Text as published by Electrek, with no greeting or sign-off shown; 'tolerances need be specified' is as published.

Sources: [Electrek](#)

Tags: cybertruck, quality, precision, manufacturing

Mandatory FSD demo ride before every delivery

From: Elon Musk

Date: March 25, 2024

To: Tesla employees (sales and delivery, North America)

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Going forward, it is mandatory in North America to install and activate FSD V12.3.1 and take customers on a short test ride before handing over the car.

[...]

Almost no one actually realizes how well (supervised) FSD actually works.

[...]

I know this will slow down the delivery process, but it is nonetheless a hard requirement.

About this email: Musk makes it mandatory in North America to install and activate FSD V12.3.1 and give every customer a short test ride before handing over a new car, acknowledging it will slow deliveries.

Notes: Also reported by Bloomberg and Teslarati. Three passages as quoted by Electrek; order within the email is not certain and the sign-off was not reproduced.

Sources: [Electrek](#)

Tags: fsd, deliveries, autopilot, sales

'Absolutely hard core' about headcount; Supercharger team cut

From: Elon Musk

Date: April 2024 (Reported by The Information on April 29-30, 2024; exact send date not given in the source retrieved)

To: Tesla executives

EXCERPT · EXECUTIVE & INTERNAL · LEAKED TO PRESS · WIDELY REPORTED

absolutely hard core

[...]

don't obviously pass the excellent, necessary and trustworthy test

About this email: In an email to executives first reported by The Information, Musk finalizes a further round of layoffs, including the entire Supercharger team, saying Tesla must be 'absolutely hard core' about headcount reduction and targeting staff who 'don't obviously pass the excellent, necessary and trustworthy test'.

Notes: The Information is paywalled; only the phrases below were retrieved verbatim, via Electrek's relay. Longer passages were quoted by The Information, Reuters and CNBC and should be added from those sources.

Sources: [The Information \(as relayed by Electrek\)](#); [Electrek](#); [Electrek \(follow-up\)](#)

Tags: layoffs, headcount, supercharger, executives

Layoff of more than 10% of global workforce

From: Elon Musk

Date: April 14, 2024 (Sent late Sunday April 14, 2024 (reported April 15))

To: All Tesla employees

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Over the years, we have grown rapidly with multiple factories scaling around the globe. With this rapid growth there has been duplication of roles and job functions in certain areas. As we prepare the company for our next phase of growth, it is extremely important to look at every aspect of the company for cost reductions and increasing productivity.

As part of this effort, we have done a thorough review of the organization and made the difficult decision to reduce our headcount by more than 10% globally. There is nothing I hate more, but it must be done. This will enable us to be lean, innovative and hungry for

the next growth phase cycle.

I would like to thank everyone who is departing Tesla for their hard work over the years. I'm deeply grateful for your many contributions to our mission and we wish you well in your future opportunities. It is very difficult to say goodbye.

For those remaining, I would like to thank you in advance for the difficult job that remains ahead. We are developing some of the most revolutionary technologies in auto, energy and artificial intelligence. As we prepare the company for the next phase of growth, your resolve will make a huge difference in getting us there.

Thanks,
Elon

About this email: Musk announces Tesla is cutting more than 10% of its global headcount after a review found duplicated roles, saying 'There is nothing I hate more, but it must be done.'

Notes: Also reported by Reuters, CNBC and Bloomberg.

Sources: [Electrek](#)

Tags: layoffs, cost-cutting, restructuring, headcount

Stock option grants for 'exceptional performance'

From: Elon Musk

Date: June 2024 (Electrek (June 18, 2024) says Musk made the announcement 'last week')

To: Tesla employees

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Over the next few weeks, Tesla will be doing a comprehensive review to provide stock option grants for exceptional performance. There will also be an ongoing program to award spot option grants for anyone who does something outstanding for the company. Thanks for everything you're doing to make Tesla successful.

About this email: After the 2024 layoffs and suspension of stock awards, Musk tells employees Tesla will run a comprehensive review to award stock options for exceptional performance and start an ongoing spot-grant program.

Notes: FLAG: the Electrek page retrieved does not state the medium; other outlets (CNBC, Bloomberg) reported this as a company-wide email from Musk, but that could not be re-verified in this session. Treat the 'email' classification as unconfirmed.

Sources: [Electrek](#)

Tags: stock-options, compensation, performance, retention

SpaceX

Memos to SpaceX staff and correspondence tied to the company.

65 emails, May 9, 2003 to February 2, 2026

Merlin firings, design review, first customers, human spaceflight goal

From: Elon Musk

Date: May 9, 2003 (SpaceRef reprint date (9 May 2003). HobbySpace RLV News (12 May 2003) says the status report was emailed 'yesterday' and was not yet on spacex.com; HobbySpace later calls it SpaceX's 'April update'.)

To: SpaceX update mailing list

Subject: SpaceX Update

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · CONFIRMED

Much has happened in the past few months, so there is a lot of good information in the update below. We have had several successful firings of the Falcon main engine (named Merlin), the first stage is almost complete, the second stage is in fabrication, our prototype fairing structure is done and a lot of the small, but important pieces arrived.

In March, we ran a three day Falcon design review for potential customers and the regulatory authorities, which seemed to be very well received. Representatives from NASA, the Dept of Defense and several commercial satellite builders were present on the customer side. On the regulatory front, we had Air Force Range & Safety and the FAA in attendance, and I would like to thank both of those organizations for their continuing support in helping us move Falcon through the regulatory approval process.

At this point, we have a verbal launch commitment from a Dept of Defense customer and a written launch commitment from an international government customer. While still preliminary, this seems to confirm that there is at least modest demand for launching small satellites.

We are starting to have more definite numbers for the payload capability of Falcon and indications are that it will do at least 1,250lbs (570kg) to 200km LEO from Cape Canaveral. This is 25% above our initial target of a 1,000lb payload capability. Pricing for the Falcon remains constant at \$6 million per launch.

Also, starting in Q4 2004, SpaceX will offer Falcon with two liquid strap-on boosters in a configuration similar to Boeing's Delta IV Heavy. Falcon's capability to LEO would increase substantially to above 4000lbs (1820kg) and enable GTO payloads of approximately 1300lbs (590kg). Pricing and final payload capabilities are not yet determined and a more formal announcement will follow in coming months.

I have been asked several times whether we intend to pursue human spaceflight. The answer is that SpaceX will definitely pursue that market in the long term. In fact, part of the reason why we have spent substantial capital on reliability — some would argue

more than is necessary for satellites — is that we would like Falcon to be safe enough for human transportation. However, we believe it makes sense to prove reliability with satellites before putting people on the rocket. 99% reliability is considered exceptionally good in the satellite delivery world, but is not very satisfactory for people.

Over the next several months, everyone at SpaceX will be heads down working hard towards our first satellite launch. Before too long, we should be able to release details of those launch plans and the partnership that has come together to make it happen.

Elon

Merlin at full thrust with a copper chamber

Merlin with a flight weight ablative chamber

After several weeks of testing, we are at about 94.5% combustion efficiency on a copper heatsink chamber. We will be firing with an ablative chamber in about six weeks, once we correct an issue with the metal insert that joins the injector head to the chamber. That should bump that up efficiency by a few percentage points and put us above our target of 96%. Including gas generator losses, vacuum specific impulse would then be about 310s with a modest expansion ratio.

Merlin Turbo-pump

Also in May, we expect to run the Merlin turbo-pump and verify pressures & flow rates. As mentioned before, our approach is a gas generator cycle, single shaft, single turbine wheel combined LOX/kerosene pump. Like the rest of our development, all components except those used for calibration are at or near flight weight. At 150lbs, the turbo-pump may be the best ever pump in its class for thrust (max 85,000lbf vacuum) to weight.

First Stage Primary Structure Arrives

Despite delays due to the transport company truck breaking down on the way to pick up the first stage in Wisconsin, it arrived in time for the design review in good shape. Our shipping fixture with additional bracing from Spincraft did a great job of supporting the stage against lateral loads from the road. We could measure no meaningful deviation from circularity when it arrived, despite the fact that it was shipped unpressurized. This was actually an important practical test of structural stiffness as compared to an Atlas II/III, which requires pressure stabilization in transit to maintain its shape.

Falcon 1st Stage Primary Structure

Avionics Ring, Satellite Adapter Cone and Separation System

The vehicle avionics and flight termination system (apart from the linear shaped charges) are mounted on the cone, which is in turn mounted to the top of the second stage. On top of the cone is the satellite release system. We have decided to use a low shock, non-pyrotechnic device called Lightband, with essentially zero debris that could contaminate a satellite or spacecraft's systems.

Payload Adapter and Satellite Release System on Mounting Tool

Test Stand Instrumentation and Setup in Texas

The Merlin horizontal pressure fed test stand is now fully set up at our propulsion testing facility in Texas. You can see it on the left under construction in this picture taken in February during a cold snap (that is ice on the ground). The large white pressurant ball, formerly used for wind tunnel tests, is made of one foot thick steel and weighs about 35 tons. In the middle of the picture is the vertical test stand for Merlin and to the right is the Kestrel vertical test stand.

This is intended to be a world class stand on par or better than those in use by Boeing or Northrop, so it is quite expensive and a lot of work to build and check out. In the end, however, we should be able to characterize very accurately the flows, pressures and temperatures throughout the thrust chamber assembly and fine tune the injector for maximum efficiency.

Test Stand Instrumentation and Setup in Texas

Payload Fairing

The tooling for joining the fairing skin panels, ring frames and stringers has been built and both fairing halves assembled. Our small construction crew, most of whom are former Boeing veterans from Huntington Beach, are doing a great job. They've built dozens of fairings in the past for everything from Delta II & IV to the Titan IV.

The first unit is an engineering prototype, which we will place in a test harness and verify all the expected load cases. Depending on how well the physical testing compares to our finite element model predictions, we will consider reducing the aluminum skin thickness in follow on units.

About this email: Musk reports several successful Merlin firings, a three-day Falcon design review, launch commitments from a Defense Department customer and an international government, a 1,250 lb payload figure at a constant \$6 million price, and a planned strap-on-booster version. He says SpaceX will pursue human spaceflight in the long term, then gives hardware notes on the turbo-pump, first stage, avionics, test stand and fairing.

Notes: Signed 'Elon' after the opening letter; the hardware sections that follow are unsigned and may have been written with SpaceX staff. Short lines such as 'Merlin at full thrust with a copper chamber' are photo captions kept as published by SpaceRef (images not reproduced). SpaceRef's old URL was spaceref.com/news/viewsr.html?pid=9113, which no longer resolves to this report.

Sources: [SpaceRef](#); [HobbySpace RLV News, 12 May 2003 \(highlights with matching quotes\)](#)

Tags: [spacex-update](#), [falcon-1](#), [merlin](#), [customers](#), [human-spaceflight](#), [turbopump](#)

June 2003 update: range safety system quote

From: Elon Musk

Date: June 2003 (Update 'covering events through June' 2003; HobbySpace reported its arrival on 21 July 2003.)

To: SpaceX update mailing list

EXCERPT · NEWSLETTER · PUBLISHED BY MUSK · SINGLE-SOURCE

it is a bit sad to spend so much money and effort on a system that we hope will never be used.

About this email: Only one sentence of this newsletter survives in the source found: on the flight termination system, Musk remarks that it is a little sad to spend so much on something SpaceX hopes never to use. HobbySpace's paraphrased summary says the update also covered Falcon Heavy pricing, a slip of the first launch to January, ablative chamber tests and turbopump progress.

Notes: Excerpt only: these are the passages that HobbySpace's RLV News quoted verbatim from the newsletter when it came out; HobbySpace's own paraphrased highlights are not included. The complete newsletter was posted on spacex.com (Updates section), which is no longer online, and no full-text reprint was found. The rest of the HobbySpace item (21 July 2003) is a paraphrased summary by Jonathan Goff. The quote is a sentence fragment as HobbySpace printed it.

Sources: [HobbySpace RLV News \(Clark S. Lindsey\)](#)

Tags: spacex-update, falcon-1, range-safety, turbopump

July 2003 update: Kestrel upper stage versus Delta II

From: Elon Musk

Date: July 2003 ('Update for July' 2003; HobbySpace noted on 29 August 2003 that it had been posted on spacex.com.)

To: SpaceX update mailing list

EXCERPT · NEWSLETTER · PUBLISHED BY MUSK · SINGLE-SOURCE

Delta II rocket upper stage has a specific impulse of 319s and a mass efficiency of 86%, compared to the expected Falcon specific impulse of 325s and mass efficiency of 91%. Our inert mass being one third better than Boeing's is very significant as that translates to a pound for pound improvement in actual payload to orbit

[...]

biggest single advantage we have is our use of advanced aluminum-lithium for the propellant tanks rather than steel in the Boeing case. We also employ a blended system of both heated helium and gaseous oxygen to pressurize the tanks, significantly reducing our pressurant weight.

About this email: In the passages quoted by HobbySpace, Musk compares the Falcon upper stage with the Delta II's: 325s specific impulse and 91% mass efficiency against 319s and 86%, crediting aluminum-lithium tanks and a blended heated-helium and gaseous-oxygen pressurization system.

Notes: Excerpt only: these are the passages that HobbySpace's RLV News quoted verbatim from the newsletter when it came out; HobbySpace's own paraphrased highlights are not included. The complete newsletter was posted on spacex.com (Updates section), which is no longer online, and no full-text reprint was found. HobbySpace (29 August 2003) also reports that this update covered the first-stage parachute recovery system, the thrust vector control test stand, a Vandenberg launch animation, the vertical test stand in Texas and an addendum to Musk's congressional testimony.

Sources: [HobbySpace RLV News \(Clark S. Lindsey\)](#)

Tags: spacex-update, falcon-1, kestrel, upper-stage, aluminum-lithium

August 2003 update: turbo-pump at full power, TVC complete

From: Elon Musk

Date: August 2003 ('Update for August' 2003; HobbySpace says it arrived by email on Friday 12 September 2003, before it was posted on spacex.com.)

To: SpaceX update mailing list

EXCERPT · NEWSLETTER · PUBLISHED BY MUSK · SINGLE-SOURCE

flying colors

[...]

Merlin to be the best performing engine of its class (LOX/Kerosene, GG cycle turbo-pump)

[...]

Development and testing of the thrust vector control (TVC) system that steers the main engine is now complete.

[...]

integrating those three subsystems together and attaching Merlin to the Falcon stage 1 primary flight structure

[...]

first stage hold-down flight simulation firing in late October or early November

[...]

as close to an actual launch as the vehicle will get before the real thing.

About this email: Quoted fragments from the August 2003 newsletter: the Merlin turbo-pump passed a full 180-second duty cycle, thrust vector control development is complete, and the next milestone is integrating the engine with the first stage for a hold-down firing in late October or early November.

Notes: Excerpt only: these are the passages that HobbySpace's RLV News quoted verbatim from the newsletter when it came out; HobbySpace's own paraphrased highlights are not included. The complete newsletter was posted on spacex.com (Updates section), which is no longer online, and no full-text reprint was found. The quotes are short fragments embedded in HobbySpace's 13 September 2003 summary, given here in the order HobbySpace printed them.

Sources: [HobbySpace RLV News \(Clark S. Lindsey\)](#)

Tags: spacex-update, falcon-1, merlin, turbopump, thrust-vector-control

Autumn 2003 update: TacSat-1 funding, FPGA avionics board

From: Elon Musk

Date: September 2003 (HobbySpace reported this update as having 'just come out' on 17 October 2003 without naming the month; it falls between the August update (emailed 12 September) and the October update (emailed mid-November), so it is taken to be the September 2003 update.)

To: SpaceX update mailing list

EXCERPT · NEWSLETTER · PUBLISHED BY MUSK · SINGLE-SOURCE

Falcon was one of only a few technologies that the Defense Department felt was sufficiently important to receive funding outside of the normal federal budgetary cycle, which is quite an honor. It is decisions like this that really help move the space program forward.

[...]

erosion rates of [the first stage engine] Merlin's composite ablative chamber & nozzle under various film cooling scenarios. Erosion rates so far are within the predicted range.

[...]

..with the addition of ram and a boot rom, this one little board will perform all avionics and vehicle control functions, apart from the ring laser gyroscopes, GPS and radios, which is really amazing. The Saturn V Moon rocket had substantially less capability and the avionics were the size of a house

[...]

double-walled, liquid oxygen feed tunnel that goes from the common bulkhead to the turbo-pump inlet through the center of the kerosene tank.

[...]

multiple sets of aluminum bellows to take out the differential expansion caused by liquid oxygen at -300F on the one hand and room temperature kerosene at 70F on the other.

About this email: Quoted passages: Musk calls it an honor that the Defense Department funded the TacSat-1 Falcon launch outside the normal budget cycle; Merlin ablative erosion rates are within predictions; a single FPGA board will run nearly all avionics; and the first stage's double-walled LOX feed tunnel uses aluminum bellows.

Notes: Excerpt only: these are the passages that HobbySpace's RLV News quoted verbatim from the newsletter when it came out; HobbySpace's own paraphrased highlights are not included. The complete newsletter was posted on spacex.com (Updates section), which is no longer online, and no full-text reprint was found. '[the first stage engine]' is HobbySpace's bracketed insertion. HobbySpace attributes the first quote to 'Elon'. Its summary also mentions the planned December unveiling in Washington DC and the choice of a first-stage recovery contractor.

Sources: [HobbySpace RLV News \(Clark S. Lindsey\)](#)

Tags: spacex-update, falcon-1, tacsat-1, avionics, merlin, defense-department

October 2003 update: first engine lost in testing

From: Elon Musk

Date: October 2003 ('SpaceX update for October' 2003; HobbySpace says it arrived by email on 14 November 2003 and was online by 16 November 2003.)

To: SpaceX update mailing list

EXCERPT · NEWSLETTER · PUBLISHED BY MUSK · SINGLE-SOURCE

A fuel valve actuator failed to open properly, allowing liquid oxygen to flow back up the fuel injector and ignite inside the engine plumbing.

[...]

[B]y the end of the month, the first Falcon rocket will roll off the assembly line, be mounted on the mobile launcher and sent to Washington DC.

[...]

finished coding of the Falcon vehicle flight simulator.

About this email: Quoted passages: a fuel valve actuator failure let liquid oxygen ignite inside the engine plumbing, the first engine SpaceX lost in testing; the first Falcon is to roll off the line by month's end and go to Washington DC; and the vehicle flight simulator code is finished.

Notes: Excerpt only: these are the passages that HobbySpace's RLV News quoted verbatim from the newsletter when it came out; HobbySpace's own paraphrased highlights are not included. The complete newsletter was posted on spacex.com (Updates section), which is no longer online, and no full-text reprint was found. '[B]y' is HobbySpace's bracketed capitalisation. HobbySpace's 14 November 2003 summary also mentions the 4 December unveiling at the Smithsonian Air & Space Museum, a DARPA/Air Force FALCON phase-one contract and in-house machining.

Sources: [HobbySpace RLV News \(Clark S. Lindsey\)](#)

Tags: spacex-update, falcon-1, merlin, engine-failure, washington-dc-unveiling

Jan/Feb 2004 update: thrust-cutoff approval, final stretch

From: Elon Musk

Date: February 2004 ('Update for Jan/Feb' 2004; HobbySpace says it 'just came out' on 5 March 2004, a day or two before it was posted on spacex.com.)

To: SpaceX update mailing list

EXCERPT · NEWSLETTER · PUBLISHED BY MUSK · SINGLE-SOURCE

final stretch

[...]

Should happen in the next four to six weeks.

[...]

approval to fly the rocket with only thrust cutoff, rather than explosive termination.

[...]

[This] improves hazardous procedures in transportation, on the launch pad and particularly on recovery of the first stage.

[...]

due to the all liquid fuel configuration and six-fold valve redundancy.

[...]

significant cavitation margin

[...]

on track for a maiden launch around the middle of this year.

About this email: Quoted fragments: SpaceX is in the 'final stretch'; regulators approved flying with thrust cutoff instead of explosive termination; the first-stage hold-down test should happen in four to six weeks; and the company is on track for a maiden launch around mid-2004.

Notes: Excerpt only: these are the passages that HobbySpace's RLV News quoted verbatim from the newsletter when it came out; HobbySpace's own paraphrased highlights are not included. The complete newsletter was posted on spacex.com (Updates section), which is no longer online, and no full-text reprint was found. The quotes are short fragments embedded in HobbySpace's 5 March 2004 summary, in the order printed; '[This]' is HobbySpace's insertion. The summary also reports Kestrel reaching 96.4% efficiency and 327s Isp, first-stage pressurization tests in Texas, and a Falcon User's Conference on 25-26 March 2004.

Sources: [HobbySpace RLV News](#) (Clark S. Lindsey)

Tags: spacex-update, falcon-1, flight-termination, kestrel, merlin

Falcon V sold to Bigelow, turbopump fixed, Kistler protest

From: Elon Musk

Date: April 2004 (Covers March-April 2004. SpaceRef's page carries a nominal date of 1 April 2004; HobbySpace says the update 'just came out' by email on 5 May 2004, and the text refers to 5 May as upcoming.)

To: SpaceX update mailing list

Subject: March/April Update

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · CONFIRMED

March/April Update

First Flight of Falcon V Sold!

SpaceX is pleased to announce that we have a signed contract and a deposit for the maiden launch of Falcon V. The customer is a US commercial company (Bigelow Aerospace) and launch date is set for November 2005 from our launch complex at Vandenberg Air Force Base in California. Further details will be provided by the customer at their discretion in coming months.

Falcon V with Standard Fairing

Slight Price Reduction for Falcon I

We also anticipate announcing additional customers for Falcon I in the next few months. As mentioned in the Aviation Week cover story, the Falcon I price has been reduced to \$5.9 million plus range costs that vary by launch location.

The cost reduction is driven by the fact that Air Force range safety is allowing us to use thrust termination rather than explosive termination. In fact, Falcon will be the only US

launch vehicle approved for thrust termination. Our continuing aspiration is to find other ways to reduce price without affecting reliability. This is a case where there is actually a very slight improvement in reliability by removing the explosives.

Per our standard policy, any price reductions are made public and are applied retroactively to customers, so a customer can always be sure of having the best price.

Two Major Problems Solved

After a series of hiccups debugging our main vertical test stand, the propulsion team was able to run the Merlin turbopump with the new impeller. This was the third impeller design and third time lucky. There is more detail below in the propulsion section, but suffice to say that we have overcome the final major issue with our Merlin engine required for flight!

This has been a pretty tough development and there is good reason for it. Merlin will be only the second new American (orbital) rocket booster engine to see flight in over twenty-five years. The other engine is the RS-68 from Boeing and the one before that is the Space Shuttle Main Engine.

The other problem solved is the flight termination receiver, where we have worked out a compromise that appears to satisfy all parties, both for first and future flights. It involves some modest additional expenditures and time on our part, but much less than earlier estimates. We'd like to thank Air Force JAG, AF Range Safety and Boeing for working through this with us. Also, Cincinnati Electronics is definitely high on our preferred vendor list.

Falcon User's Conference

Our first user's conference took place in late March and seemed to go quite well. Attendees were present from a number of commercial companies as well as the Air Force and NASA. We received a lot of good feedback from customers on satellite processing, as well as areas for improvement in flight environment and mechanical/electrical interfaces.

In particular, we took back the need for a good size clean room with an overhead crane in close proximity to the launch pad. Fortunately, Lockheed, our nearby pad mate at Vandenberg AFB, just gave up such a building and we will be taking it over. They are welcome to borrow a cup of liquid oxygen from us any time ? With a little refurbishment, it will be a first rate satellite processing facility.

More Testimony

Last month, I was asked to testify before the House of Representatives Subcommittee on Space regarding NASA/DoD cooperation and before the President's Commission on Moon, Mars and Beyond. On May 5, I'm testifying before the Senate Subcommittee on Space regarding the future of space launch vehicles, which will be webcast live.

Protesting the Kistler Deal

As some may have read, SpaceX is protesting the contract offered to Kistler Aerospace by NASA. Although we have a high regard for NASA as a whole, we are against this particular contract. We believe it does not support a healthy future for American space exploration. However, this should be viewed as an isolated difference of opinion between NASA and SpaceX, and we look forward to working with NASA in other areas.

As background for those unfamiliar with the Kistler deal, the approximately quarter billion dollars involved in the contract would be awarded for flight demonstrations and technology showing the potential to resupply the Space Station and possibly for transportation of astronauts down the road.

That all sounds well and good. The reason SpaceX is opposing the contract and asking the General Accounting Office to intervene is that it was awarded on a sole source, uncompleted basis to Kistler instead of undergoing a full, fair and open competition. SpaceX and other companies (Lockheed and Spacehab also raised objections) should have been offered, but were denied the opportunity to compete on a level playing field. We are only asking for a fair shot at meeting the objectives, not demanding that we win the contract.

Giving a quarter billion dollars to a company like Kistler, which declared bankruptcy last year and has not managed to build a complete prototype in eleven years of operation, without even holding a competition sends a very negative signal to the marketplace.

As it is, we already have a means of space station access via the Russian Soyuz and the European Automated Transfer Vehicle. If what is sought is American access, since US tax dollars are paying for this, the deal makes even less sense, since their intent is to operate from Australia and use exclusively Russian engines.

Our suggested solution to this problem, given that the Kistler contract is already performance based, is to open it up and turn the objectives into a prize that can be won by any company. That company might be Kistler, SpaceX, Lockheed or an entity yet to be formed (perhaps by a reader of this newsletter). Given how powerful the \$10 million X Prize has been in generating entrepreneurial vigor around suborbital flight, I'm confident that a quarter billion dollar prize would have an incredibly beneficial effect on orbital flight.

Elon

TECHNICAL DEVELOPMENT UPDATES

Propulsion

A Brief History of the Turbopump

It is almost a canon of rocket development that turbopumps are the toughest technical challenge. In fact, a friend of mine recently characterized pump fed launch vehicles as a turbopump development with a rocket attached. It's not quite that bad, but there is some truth to the comment.

Let me provide a brief history of our pump development and tell you where we went wrong and how we made mistakes:

Right from the beginning, Tom and I expected the turbopump to bite us in the rear (as it turns out, we weren't disappointed). So the first thing we did, even before we had an office, was initiate the pump development in mid 2002.

In May of 2003, about 12 months after starting development, we had a completely new turbopump designed, built and operating in the test stand. When we were able to bring the pump up to almost full power only three months later, it seemed like we had miraculously sailed through the development with barely a hitch. Soon, we would be able to join the turbopump with the thrust chamber and perform an integrated engine test.

However, in September the pump started showing significant problems, including severe vibration and a few cases where the shaft itself seized up completely. What had actually happened was that we had misdiagnosed the nature of the problem, confusing cause with effect.

Between May and August last year, we had noticed that the seals were rubbing on the shaft and had concluded that this was overloading the bearings, resulting in vibration and occasional seizure. So the fix we put in was to increase clearance between the shaft and the seals and make up for it by increasing the amount of helium purge.

However, after making the changes, problems continued with only modest improvement. That was the point at which we realized that we had the problem backwards. What was actually happening was that the bearings in the liquid oxygen were overheating (despite being -300F, liquid oxygen is a poor coolant). This caused them to melt their housings and become very loose against the shaft. Now that the shaft was not held rigidly in place, it vibrated against the seals, causing them to cut into and eventually seize the shaft.

We redesigned the turbopump to dramatically increase cooling to the LOX bearings and the problems went away. The seal design was also changed to float instead of being fixed, so even with severe vibration the seals would not cut into the shaft metal.

That solved the operating life issues and in December we were able to run the pump at will with no indications of excess vibration or shaft seizure. However, that uncovered one last problem, which took a few iterations to fix.

Now that we were able to run the pump for extended periods, we found that the cavitation margin on the liquid oxygen portion of the turbopump was lower than required. Cavitation is what occurs when the pump starts creating bubbles in the liquid as it spins, resulting in little "cavities" of vapor. Beyond a certain point, the small bubbles become a big bubble and the blades pump a combination of vapor and liquid, which causes the shaft to overspeed.

It took two more iterations on the LOX impeller/inducer design and in April we solved the final problem with the turbopump. In fact, the LOX pump is now significantly overperforming, showing a 43% margin over the minimum design target for suction specific speed ($N_{ss} = \text{shaft rpm} * \text{Flow}^{.5} / \text{Net Positive Suction Head}^{.75}$). This bodes well for future thrust upgrades of the Merlin engine.

In addition, the last few months of testing were with flight tanks in the vertical configuration, proving out not just the turbopump, but also several structural components and flight valves.

Turbopump in Vertical Stand 1 attached to the thrust frame

Turbopump test video in Vertical Stand 1
(the cloud coming towards you is LOX)

Structures

Stage Separation

We were able to do a stage separation unit test in April, which performed very well. Our current high def video and pictures show too much proprietary detail, so we can't put that on the web :) The next update will have a video clip for public viewing with the secret sauce obscured.

Our approach is to use pneumatic pushers that are pressure equalized before separation to ensure each pusher is exerting exactly the same force. This is both lighter and more reliably balanced than a spring system.

The two stages are held together by redundant explosive bolts, which are independently wired from both the first and second stage batteries. The first and second stage batteries simultaneously initiate the explosive bolts, ensuring that, even if there is a wiring or power problem from one side, the stages will still separate.

As a further precaution, the upper stage engine nozzle is made of niobium metal sheet, rather than a more brittle carbon-carbon (such as the Space Shuttle leading edge or the RL-10B2 nozzle). If the nozzle hits the inter-stage at separation, it will only dent, which has no meaningful effect on performance, rather than fracture.

Upper Stage

Pictured below is an upper stage on our large five axis mill for some final machining. Since inert mass is a pound for pound trade against payload on the upper stage, it makes sense to machine away all excess material, which we don't do on our large lower stage. There the trade is about seven pounds of stage mass for one pound of payload.

Note, this is actually two tanks in one, because we use a common bulkhead to separate the liquid oxygen and RP-1 kerosene.

Another Fairing in Production

This picture shows a set of fairing skins in production on the skin tooling fit check tool. The golden sheen comes from the alodine corrosion protection coating.

Avionics, Guidance and Control

Engine Computer

The engine computer, which includes our own relay board driven by an FPGA, was subjected to a thermal test for over a week. Once we fixed problems with heat

dissipation on the simulated output loads, we cycled the unit between -40 deg. C and +80 deg. C. These temperatures are well beyond the specified range for some of the components, and we expect far more moderate temperatures during the flight.

Engine computer, relay board and power supply
(with a little frost)

Hardware-in-the-Loop

Pictured below is our Hardware-in-the-Loop test simulator. The test setup simulates all the signals that the vehicle needs to believe that it is flying, such as tank and engine pressures, GPS and gyro data. It also records the valve and pyrotechnics that are switched on and off by the flight computer and allows us to verify the software.

Part of the Hardware-in-the-Loop Simulator

Command Trailer

This is our mobile command trailer, which contains personnel critical to the launch. There will be enough workstations inside for SpaceX, range safety and customer needs. It will also carry all the communication equipment for interfacing with the rocket and range infrastructure. The trailer has its own power generators and a backup UPS to ensure continuous power during the countdown sequence.

IMU (Inertial Measurement Unit) Test

A common method to align the vehicle's IMU to "the world" is by measuring the local gravity vector with the IMU's accelerometers and comparing it to the theoretical local gravity vector. The angle between the 2 vectors is the vehicle's tilt angle. Here, the IMU was placed on a sine table, and tilted in 20 arc-second increments – the local gravity vector was computed/recorded each time. This experiment verified the fact that our IMU is accurate enough to "self-align," as it showed respectabilities of a few arc-seconds and absolute accuracies well under one arc minute.

The IMU is the little gold soup can object
sitting on the sine table

About this email: Musk announces a signed contract with Bigelow Aerospace for the first Falcon V, a Falcon I price cut to \$5.9 million, and fixes for the Merlin turbopump and flight termination receiver. He explains why SpaceX is protesting NASA's sole-source Kistler contract and proposes turning it into a prize, then gives a candid history of the turbopump development and other technical updates.

Notes: Signed 'Elon' before the technical sections; the turbopump history is also in the first person ('Tom and I'). Other technical sections may have been written with SpaceX staff. Photo captions are kept as published (images not reproduced). The '?' after 'any time' is an encoding artifact in SpaceRef's copy, probably a smiley in the original; the phrase 'a turbopump development with a rocket attached' may have been in quotation marks in the original.

Sources: [SpaceRef](#); [HobbySpace RLV News, 5 May 2004 \(summary by Jonathan Goff\)](#)

Tags: spacex-update, falcon-v, merlin, turbopump, kistler-protest, bigelow

June/July 2004 update: first integrated Merlin firing

From: Elon Musk

Date: July 2004 ('June/July update' 2004; HobbySpace noted on 16 July 2004 that it was available on spacex.com.)

To: SpaceX update mailing list

EXCERPT · NEWSLETTER · PUBLISHED BY MUSK · SINGLE-SOURCE

not launch until all engineers are two thumbs up, so that date may get pushed back.

[...]

do not use a permanent tower on the pad, employing instead our mobile erector/launcher platform.

[...]

fired Merlin (our main engine) in the fully integrated configuration, with both the thrust chamber assembly, turbo-pump and gas generator attached to a set of flight tanks. This is one of the biggest milestones before launch and the culmination of a tremendous amount of work by the propulsion team.

[...]

rock solid reliable

[...]

[W]e are spending more than planned on this piece of equipment, we expect to be able to reuse it essentially forever (i.e. thousands of flights), so long as the stage itself is recovered.

[...]

we can't yet obtain the non-explosive separation nuts with sufficient strength to hold the stages together under maximum load.

[...]

probably the most powerful rocket flight computer in the world, since it is the most recently designed and the only one to use current 21st century technology.

[...]

The rocket is truly a flying radio station, with two C-band, four S-band, two UHF and two GPS antennas, as well as a beacon antenna on the first stage for location by the recovery ship

About this email: Quoted passages: Merlin was fired in the fully integrated configuration on flight tanks; the first launch will not happen until every engineer is 'two thumbs up'; the pad uses a mobile erector instead of a permanent tower; the titanium thrust frame should be reusable for thousands of flights; and Musk describes the flight computer and the rocket's ten antennas.

Notes: Excerpt only: these are the passages that HobbySpace's RLV News quoted verbatim from the newsletter when it came out; HobbySpace's own paraphrased highlights are not included. The complete newsletter was posted on spacex.com (Updates section), which is no longer online, and no full-text reprint was found. '[W]e' is HobbySpace's bracketed capitalisation. HobbySpace's 16 July 2004 summary also mentions a September

launch target, a public viewing for the launch and a launch manifest of three Falcon I and two Falcon V flights.

Sources: [HobbySpace RLV News \(Clark S. Lindsey\)](#)

Tags: spacex-update, falcon-1, merlin, vanderberg, avionics, stage-separation

Falcon I on the Vandenberg pad; Falcon V redesign

From: Elon Musk

Date: September 2004 (Covers August-September 2004. Posted on spacex.com and reprinted by SpaceRef on 13 October 2004; HobbySpace previewed it on 6 October 2004.)

To: SpaceX update mailing list

Subject: SpaceX August-September 2004 Update

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · CONFIRMED

Falcon I on the Launch Pad!

The Falcon I actual flight vehicle is now on the SpaceX launch pad at Vandenberg Air Force Base (near Santa Barbara, California). This is a huge milestone for our company and is the result of a monumental effort by everyone here.

Below are a series of pictures showing Falcon being erected by the mobile launcher on to the quadropod hold down fixture. Time to rotate vertical is about 15 minutes, although we should be able to reduce this to only a few minutes once we're comfortable with the process. The white building on the launch deck (shown in pictures four and five) is the retractable hangar, which provides shelter to technicians for final checkout and satellite integration. It moves back on rails shortly before launch. Note the new carbon fiber inter-stage and the ablative covering the forward portion of the payload fairing.

Falcon on mobile launcher ready to rotate vertical

About halfway to vertical (the telescoping hydraulic ram is now visible)

Vertical and fixed to the hydraulic launch hold down clamps

Perspective looking straight down the flame trench

Oblique view (white structure on the deck is the retractable hangar)

Side view (large tank in the foreground contains the liquid oxygen propellant supply)

The remaining milestones are (not necessarily in this order):

1. Complete the formal flight qualification program for the engines.
2. Receive range safety approval to launch.
3. Do a full vehicle hold down firing on the launch pad as a final verification.
4. Integrate the TacSat-1 satellite.
5. Launch to a 500km orbit

At this point, it looks like the first available launch window that works for both SpaceX and Vandenberg is mid to late January.

Mobile Control Center

We built our control center in a trailer for maximum flexibility, which means we don't need to spend capital building up this capability at each range. It also minimizes the ongoing overhead costs of having to pay rent or maintenance for a control center when we only use it for a few weeks at a time during launch.

The MCC contains up to twelve personnel and is fully self-supporting with its own generator, UPS and air-conditioning. It contains everything necessary to interface with the Air Force Base, all voice and data com, vehicle telemetry displays, etc.

Inside of Mobile Control Center

Outside of Mobile Control Center

Launch Manifest

Every update going forward will have the expected launch schedule, but showing only firm contracts for which we have received a deposit. There are a lot of discussions underway with potential customers and I expect that in 2006 we will manifest at least four flights again, with volume picking up significantly above that in 2007. It will take a few years for the satellite market to adjust to the availability of a low cost launch vehicle.

Customer | Launch Date | Vehicle | Departure Point

US Defense Dept (OSD/NRL) | Q1 2005 | Falcon I | Vandenberg

US Defense Dept (DARPA) | Q2 2005 | Falcon I | Marshall Islands

Malaysia (ATSB) | Q4 2005 | Falcon I | Marshall Islands

Bigelow Aerospace | Q4 2005 | Falcon V | Vandenberg

Falcon V Progress

The information on Falcon V that has been on the website for several months is now significantly out of date. Following discussions with customers, we decided earlier this year to make some important changes.

We are switching from a dual Kestrel upper stage to a single Merlin upper stage. This has a major effect on mass to orbit due to improved mass fraction, higher specific impulse and better staging efficiency. The improved mass fraction results from having a pump-fed upper stage with thin tank walls and low helium requirements vs. pressure-fed with thick tank walls and high helium requirements. The higher specific impulse comes from the much higher chamber pressure of Merlin and much higher expansion ratio. The Merlin vacuum specific impulse is expected to be 340s vs. 325s for Kestrel.

Other changes not included in prior numbers will also boost payload to orbit. During the development process, the Merlin engine has show itself capable of at least an 80,000 lb sea level thrust vs. the nominal 72,000 lb. Also, both the inter-stage and fairing will be made from a carbon fiber sandwich composite (vs. just the inter-stage in Falcon I), improving mass fraction. Following completion of the Merlin 1 qualification, we will begin work on a Merlin 2 with a sea level thrust target in excess of 100,000 lb as well as a slight increase in Isp.

The estimated payload to orbit with the Merlin 2 engine, including both an Isp and mass fraction sandbag, is as follows:

Orbit | Payload

200 km, 28.5 deg | 6020 kg / 13,260 lb

400 km, 51 deg | 5450 kg / 12,010 lb

700 km, sun-synch | 4780 kg / 10,530 lb

GTO, 9 deg | 1920 kg / 4,240 lb

Escape velocity (0 C3, no kick stage) | 1200 kg / 2,650 lb

The above performance to GTO & escape can be improved significantly and the burnout g load reduced by using a kick stage, such as a Star motor from ATK. Note, the initial version of Falcon V, which is expected to launch in late 2005, will have approximately 20% less capability than the above numbers (official update to the website coming shortly). These performance numbers are applicable to launches occurring in late 2006 when the Merlin 2 engine upgrade is flight ready.

I expect the Falcon V development to be considerably faster and easier than the Falcon I development, because it uses most of the same components. Unlike Falcon I, where we had to develop two complete engines from scratch, Falcon V in its initial version requires only minor adjustments to the existing Merlin engine. The avionics are the same as Falcon I and the airframe architecture on Falcon V is just a wide body version of Falcon I. Also, most of our launch site infrastructure and the environmental permits already accommodate both vehicles.

Noteworthy items on the Falcon V development:

1. The upper stage Merlin engine is in the fabrication cycle and should be complete in a few months, except for the vacuum skirt extension.
2. LOX and RP-1 manifold parts for the five first stage engines are done.
3. Our custom-made friction stir welder, which will be used for the Falcon V tanks, has arrived and is producing excellent welds with literally zero defects.
4. Spin form tooling for the tank domes is on order and expected in January.

Elon

TECHNICAL UPDATES

Propulsion

The past few months have mostly consisted of testing and retesting the engines. As reported earlier, we noticed cracks developing in the cast aluminum fuel manifold. As a result, we decided to switch to inconel, a nickel-chromium-iron alloy that has several times the strength of cast aluminum and even greater toughness, at the cost of a few extra pounds of mass. The area under the stress-strain curve (definition of toughness) for our inconel manifold is an order of magnitude greater than the aluminum casting.

Testing at this point is focused mostly on ablative life and making sure that we are not at risk of burning through the chamber wall. Depending on how this goes, we may need to increase film cooling to the engine walls and sacrifice a few seconds of specific impulse.

Maximum thrust achieved so far during testing of Merlin is 81,000 lb and in fact the engine has to be detuned to bring thrust down to within specifications for Falcon I. This is quite good news, because we have been able to build that extra thrust into the Falcon V design and boost payload well above numbers currently advertised on our website.

Structures

Launch Pad

All major launch pad construction is complete, including concrete, steel work, propellant supply tanks, launch mount, etc. We also built a retractable hangar on the upper deck to provide shelter to people working on the rocket during final checkout and integration of the satellite. This rolls back on rails just prior to launch.

As shown above in launch pad pictures, LOX and RP-1 tanks sit on either side of the launch deck, along with helium pressurant and nitrogen purge supply tanks. Pad electronics are in a small room under the launch quadropod. The work remaining consists mainly of finishing the electrical and communication wiring, as well as the plumbing from the supply tanks to the vehicle.

Our pad infrastructure is probably more than an order of magnitude less complicated than most other US facilities and bears a closer resemblance to the current Russian/Ukrainian approach or the early US Thor architecture. In fact, we drew some of our ideas from an old Thor rocket and its mobile launcher that are sitting in a museum at Vandenberg. It is not clear to me why those ideas were abandoned.

Mass Improvements

We have made significant mass savings in first stage, particularly in the inter-stage structure. By moving from an aluminum design to a carbon fiber composite design, we cut the weight of the inter-stage in half. Now, we started with a poorly optimized aluminum design and then moved to a really optimized carbon fiber/honeycomb design, so ordinarily you would not see that much of an improvement. The icing on the cake here is that it actually costs us less to build the carbon fiber inter-stage than the aluminum one! Long term, we will probably switch our aluminum fairing design to composite as well.

Other areas of first stage mass improvement:

- Parachute recovery system mounts
- Helium bottle mounts
- Pressurant plumbing changed from steel to titanium
- Separation rings
- Thrust skirt
- Removal of an unnecessary ring at the base of the vehicle
- New (and better in every way) pre-valve design
- Thinner slosh baffles
- Thinner tunnel covers
- Thrust frame changed from steel to titanium
- Gimbal joint changed from steel to titanium

Some parts of the main engine have grown in mass, but the net effect is that we are likely to meet our propellant mass fraction target of 94% for the first stage, including the inter-stage and recovery system. This is among the best mass fraction numbers for any boost stage and is particularly good when considering that our first stage weight incorporates a recovery system and increased design margins for reusability.

Avionics

We integrated our full suite of avionics with the aluminum honeycomb support structure, which located just below the fairing. This was then mounted to the flight vehicle for transfer to Vandenberg. After ironing out a few nits, communication through the umbilical cable from the Mobile Command Center to the vehicle flight computer worked flawlessly. We were also able to obtain a very precise alignment of the vehicle using a theodolite.

Below is a picture of the avionics tray nearing final assembly. The upper half contains vehicle avionics necessary for guidance & control and transmission of telemetry & video. The lower half contains a completely independent flight termination system that will be used by range safety to terminate thrust in an emergency.

About this email: Musk reports that the Falcon I flight vehicle is on the SpaceX pad at Vandenberg, lists the remaining milestones before a launch window in mid to late January, and publishes the first launch manifest. He explains the Falcon V switch to a single Merlin upper stage with new payload numbers, followed by propulsion, structures and avionics updates.

Notes: Signed 'Elon' before the technical sections, which may have been written with SpaceX staff. Photo captions are kept as published (images not reproduced); the two tables are rendered with ' | ' between columns. 'we-re' and 'don-t' are encoding artifacts in SpaceRef's copy (apostrophes in the original). The subject is SpaceRef's headline; the newsletter's own first heading is 'Falcon I on the Launch Pad!'. SpaceRef's old URL was spaceref.com/news/viewsr.html?pid=14216, which no longer resolves to this report.

Sources: [SpaceRef](#); [HobbySpace RLV News](#), 6 and 13 October 2004 (preview with matching quotes)

Tags: spacex-update, falcon-1, falcon-v, vandenberg, merlin, launch-manifest

January-May 2005 update: Falcon I development essentially complete

From: Elon Musk

Date: June 2005 (Covers January through May 2005. Space.com called it 'a just-issued SpaceX update' on June 8, 2005; HobbySpace noted on June 8, 2005 that 'the latest SpaceX update is now available'.)

To: SpaceX update mailing list

EXCERPT · NEWSLETTER · PUBLISHED BY MUSK · SINGLE-SOURCE

Falcon I development is essentially complete

[...]

There are also a number of other customer discussions that are quite mature. It appears quite likely that we will receive one or two more launch contracts over the next few months

[...]

Not yet mentioned on our manifest or in any press release is our classified launch business, for obvious reasons

[...]

JSC is the primary NASA center for human spaceflight and, since human transportation is the primary long term goal of SpaceX, we are likely to be doing a lot of work in partnership

[...]

Now that development of Falcon I is almost over, we are turning our focus to Falcon 5

[...]

We have tripled in size from a year ago and are now at about 130 employees

About this email: Musk's first long newsletter of 2005, issued after the May 27 Falcon I hold-down firing at Vandenberg. Only the sentences Space.com quoted from it could be retrieved: Falcon I development is essentially complete, more launch contracts are expected, attention is turning to Falcon 5, and the company has about 130 employees.

Notes: Full newsletter not retrieved; no surviving reprint was found. These are only the passages Space.com quoted, in the order the article gives them, each as quoted (some are sentence fragments and end where the article's quotation ends). The article attributes all of them to the written update. The newsletter's own title was not seen on any retrieved page, so subject is left blank. The update also described the May 27 pad firing as 'picture perfect'. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [Space.com](#) (Leonard David, 'SpaceX Rocket Readied For Maiden Flight', June 8, 2005); [HobbySpace RLV News, June 2005 archive](#)

Tags: spacex-update, falcon-1, falcon-5, launch-manifest, nasa, hiring

Titan IV delay moves first Falcon I launch to Kwajalein

From: Elon Musk

Date: June 24, 2005 (Dated 'June 24' on the SpaceX updates page, as quoted by HobbySpace on June 24, 2005.)

To: SpaceX update mailing list

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

June 24: We were just informed that the Titan IV flight will launch no earlier than September and may very well be delayed until October or November, depending upon what issues arise (due to overflight concerns, Falcon I is required to launch after Titan IV). As a result, you can expect that our first launch will now be from our island launch complex in the Kwajalein Atoll. Nominal launch date is late September.

About this email: A short note on the SpaceX updates page: because the last Titan IV at Vandenberg has slipped to September or later and Falcon I must fly after it, the maiden launch moves to the company's island pad at Kwajalein Atoll, nominally in late September.

Notes: Unsigned note posted on [spacex.com/updates.php](#); HobbySpace introduces it as 'the following note posted at SpaceX' and quotes it whole. SpaceRef ran it as 'Inaugural SpaceX Falcon 1 Flight Moved to Kwajalein', but that page now returns HTTP 410. Authorship by Musk is assumed from the updates page, not stated. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [HobbySpace RLV News \(June 24, 2005\)](#), quoting the [SpaceX updates page](#); [NASA Watch, June 2005 archive](#) ('Inaugural SpaceX Falcon 1 Flight Moved to Kwajalein'); [Spaceflight Now, 'First flight of Falcon 1 rocket moved to tropical paradise' \(June 29, 2005\)](#)

Tags: spacex-update, falcon-1, kwajalein, vandenberg, launch-delay

Falcon 1 maiden launch bumped a day by Army range

From: Elon Musk

Date: November 24, 2005 (Update line is stamped '24 Nov 2005 9:10 pm EST'; SpaceRef posted it November 24, 2005.)

To: SpaceX update mailing list

Subject: SpaceX Falcon Launch Delayed by 24 Hours

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

Update 24 Nov 2005 9:10 pm EST: All prelaunch preparation is complete. The final task of test firing the torch igniters worked on the first attempt, making us green for launch. Since our launch was bumped by a day to Saturday (California time), we will make tomorrow a shorter day and do some of the optional checkouts of systems that might cause a halt in the countdown.

SpaceX was just informed that "in order to facilitate preparations for a missile defense launch", the Army Space and Missile Defense Command (SMDC) has bumped the SpaceX Falcon 1 maiden flight from its officially scheduled launch date of Friday, November 25 at 1 p.m. PST (9 p.m. GMT).

The new launch time is Saturday, November 26 1 p.m. PST (9 p.m. GMT).

About this email: On the eve of the first Falcon 1 countdown, SpaceX reports all prelaunch work is done and the torch igniters fired on the first try, but the Army's missile defense command has bumped the launch from Friday November 25 to Saturday November 26.

Notes: Unsigned; SpaceRef filed it under press releases. The first paragraph is a first-person launch-site update, the rest is SpaceX's statement about the range bump, which Spaceflight Now also quoted. Attribution to Musk personally is not stated by the source. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [SpaceRef](#); [Spaceflight Now Falcon 1 mission status center \(Nov. 25, 2005, 0125 GMT entry\)](#)

Tags: spacex-update, falcon-1, kwajalein, launch-delay, maiden-flight

First Falcon 1 launch attempt scrubbed over LOX and helium loss

From: Elon Musk

Date: November 26, 2005 (NASA Watch gives the time of the statement as November 26, 2005 at 5:11 p.m. PST.)

To: SpaceX update mailing list

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

The launch is scrubbed for today. As I warned, the likelihood of an all new rocket launching from an all new launch pad on its first attempt is low.

What happened was that an auxiliary liquid oxygen (LOX) fill tank had a manual vent valve incorrectly set to vent. The time it took to correct the problem resulted in

significant LOX boiloff and loss of helium, and it was the latter that caused the launch abort. LOX is used to chill the helium bottles, so we lose helium if there is no LOX to cool the bottles.

Although we were eventually able to refill the vehicle LOX tanks, the rate at which we could add helium was slower than the rate at which LOX was boiling away. There was no way to close the gap, so the launch had to be called off. In addition, we experienced an anomaly with the main engine computer that requires further investigation and was arguably reason in and of itself to postpone launch.

We are anticipating rescheduling the launch within a week at the earliest but probably longer as we need to bring in LOX and helium from Hawaii. Our LOX plant on Omelek will not replenish in time.

About this email: Musk's note after the first Falcon 1 countdown was called off. A manual vent valve on a ground liquid oxygen tank was left open; fixing it cost so much LOX and helium that the launch had to be abandoned, and an engine computer anomaly also needs investigating.

Notes: NASA Watch introduces it as 'According to Elon Musk at SpaceX: on November 26, 2005 at 5:11 p.m. PST'. Kimbal Musk's blog quotes the two middle paragraphs as 'the summary of the cause of the abort from Elon', with a different closing sentence ('The next launch attempt is expected to occur in approximately mid-December, depending on the time required to resupply LOX and helium.'), which suggests the posted text was revised. A missing space in the reader output ('Omelekwil') was restored. SpaceRef's copy ('SpaceX Falcon 1 Launch Scrubbed') now returns HTTP 410. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [NASA Watch \('SpaceX Launch Scrubbed', November 26, 2005\)](#); [Kwajalein Atoll and Rockets \(Kimbal Musk's blog\)](#), 'Post-Launch Summary'; [Spaceflight Now Falcon 1 mission status center](#)

Tags: spacex-update, falcon-1, kwajalein, launch-scrub, maiden-flight

Why Falcon 1 ran out of liquid oxygen; new launch date

From: Elon Musk

Date: December 2005 (Undated in the source. Issued between the November 26 scrub and December 11, 2005, when Spaceflight Now reported its contents; the launch was formally reset to December 19 on December 15.)

To: SpaceX update mailing list

PARTIAL TEXT · NEWSLETTER · PUBLISHED BY MUSK · SINGLE-SOURCE

The new launch date is approximately December 20, depending on when the Missile Defense Agency testing is complete. As soon as we have a firm time, it will be posted here.

[...]

Liquid Oxygen

Regarding liquid oxygen (LOX) supplies, we expect to have enough on hand this time to fill the rocket four or five times over. This should account for almost any issue with a particular storage tank as well as an extended hold on the pad. We chartered a C-17 to fly two of our empty high quality LOX containers to Hawaii, sourced another high quality LOX container on Hawaii and put all three on the barge to Kwajalein. In addition, our LOX plant on Kwajalein has been repaired and is producing LOX on island again.

Many have asked how we could run out of LOX on a remote tropical island on the last launch attempt. The team tried hard to avoid it, but several issues conspired to create the problem:

- The additional month of Merlin testing resulted in additional LOX boil-off on island. Even though it is stored in vacuum jacketed containers, LOX at -300F degrees does not like being on a tropical island at 85F.
- The SpaceX LOX plant on island broke down a few weeks prior to launch, which meant we could not top up.
- We ordered replacement LOX from Hawaii, but the container quality was poor, so only 20% of what we ordered actually arrived.
- Ground winds were unusually high on launch day, which amplifies the boil-off rate significantly, since the Falcon's first stage LOX tank is uninsulated.
- All of the above would not have mattered if our final storage tank did not have a small, manual vent valve incorrectly in the open position. Somewhat agonizingly, we were only a few percent away from being full. We just needed a little sip from the last tank.
- After a while, we were able to close the vent and fill the vehicle's LOX tanks. However, we use LOX to chill our onboard helium and the absence of ground LOX to do so resulted in the helium heating up and venting back to storage. In the end, we did not have enough LOX to stay filled on the rocket and chill & pressurize the helium.

Engine Computer

The engine computer reboot anomaly was definitively traced to a ground power problem. Importantly, this would have had no effect on flight, since we switch to vehicle power before the autosequence begins. The reason it cropped up at Kwajalein was that the higher load on the longer umbilical (three times longer than in prior tests) coupled with high temperatures in Kwajalein resulted in increased resistance in the ground umbilical. This was just enough to lower the voltage below minimums and cause an engine computer reset when drawing maximum power. The same max power test was repeated on internal vehicle batteries with no problem at all.

This problem has been solved by slightly increasing voltage on the ground umbilical.

About this email: Before the second launch attempt Musk explains how SpaceX ran short of liquid oxygen on a tropical island, what has been shipped in so the rocket can be filled four or five times over, and that the engine computer reboot was a ground power problem that would not have affected flight.

Notes: The source page is a SpaceX media release that embeds this update; it shows no headline or date. Kept here: the opening launch-date paragraph and the first-person 'Liquid Oxygen' and 'Engine Computer' sections. Left out at the [...] and after the end: a third-person recap of the November 26 scrub, a 'SpaceX Launch Overview' section and media viewing logistics, which are press-release boilerplate. Unsigned; Spaceflight Now attributed the same explanations to Musk. The page was read in three passes and the bullet layout is the reader's rendering. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [SpaceNews archive copy of a SpaceX release \(formerly on SpaceRef\)](#); [Spaceflight Now](#), 'SpaceX targeting Falcon 1 launch for around Dec. 19' (December 11, 2005)

Tags: spacex-update, falcon-1, kwajalein, liquid-oxygen, launch-delay

Second Falcon 1 attempt scrubbed: fuel tank sucked inward

From: Elon Musk

Date: December 19, 2005 (Statement issued December 19, 2005 (Spaceflight Now logged it at 2045 GMT); a follow-up on the cause came later that day.)

To: SpaceX update mailing list

PARTIAL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

Launch is scrubbed until early next year, as there is a structural issue with the 1st stage fuel tank that will require repair.

I will provide further comment as soon as this has been carefully analyzed. Consistent with our policy, we must be 100% green for launch with no outstanding concerns whatsoever. It is not just a matter of repairing the damage, but also understanding at a fundamental level how to ensure it never happens again.

We will also do another full review of all the vehicle systems, including propulsion, structures, avionics, software and ground support systems. Therefore, I expect that the earliest that launch would occur is late January.

Third time's the charm.

[...]

Due to high winds, we placed the countdown on hold and began draining the fuel tank. As we drained fuel from the 1st stage tank, a faulty pressurization valve caused a vacuum condition in the tank.

This caused a fuel tank barrel section to deform and suck slightly inward. It is important to note that the root cause is an electrical fault with a valve, not structural design.

At this point, it appears that no other damage was sustained to the vehicle or the satellite. The rocket will be lowered down this afternoon and placed in its hangar for further inspection.

About this email: Musk calls off the second Falcon 1 launch attempt until early 2006 after the first stage fuel tank was damaged. A follow-up explains that a faulty pressurization valve created a vacuum while fuel was being drained during a wind hold, and the tank barrel deformed inward.

Notes: Two statements from the same day joined at [...]. The first is the statement Spaceflight Now printed as issued by Musk (wording identical in NASASpaceFlight's quotation, whose paragraph breaks are used). The second is the follow-up explanation as quoted by NASASpaceFlight; Spaceflight Now says it came from 'a company status report'. How the two were laid out on spacex.com was not verified. NBC and Space.com quote an earlier wording of the first line: 'Launch is scrubbed for the year. We noticed a structural issue with the first stage fuel tank that will require repair.' SpaceRef's copy ('Falcon 1 Launch Scrubbed') now returns HTTP 410. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [Spaceflight Now Falcon 1 mission status center \(Dec. 19, 2005, 2045 GMT entry\)](#); [NASASpaceFlight.com, 'Falcon scrub: Rocket deformed'](#); [Spaceflight Now, 'Damage puts first SpaceX rocket launch on hold'](#); [NBC News, 'Rocket damage puts a dent in SpaceX plans'](#)

Tags: spacex-update, falcon-1, kwajalein, launch-scrub, maiden-flight

New Falcon 1 launch date and what went wrong on Dec. 19

From: Elon Musk

Date: January 10, 2006 (SpaceRef republication date; NASASpaceFlight reported the update the same day.)

To: SpaceX update mailing list

Subject: January 2006 SpaceX Update: New Falcon 1 Launch Date

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · CONFIRMED

New Launch Time

The new launch time is February 8 at 4:30 p.m. California time with Feb. 9 as a backup day. We will actually be ready to launch earlier, but are planning to spend extra time reviewing and double-checking all vehicle systems.

Following the problem on Dec. 19, we flew a whole new first stage to Hawaii via C-5 just in time to catch the barge from there to Kwaj a few days before New Year's Eve. The new stage should arrive at Kwaj in about a week, whereupon we will switch it out with the damaged unit, which will be sent back to California for repair. The repair is not particularly difficult or expensive, but can only be done properly in a factory setting.

What Was the Problem?

As previously reported, we traced the problem to the failure of an electronic component in one of the first stage fuel tank pressurization valves. Although we have triple redundant pressure sensors and dual redundant pressurization valves, when this component shorted, it caused the valve controller board to reboot, effectively eliminating the redundancy.

This is the first time in 3.5 years of hard testing that we have ever seen this happen. Moreover, the component in question has a cycle life and power rating far in excess of the theoretical load that it should see. To address this specific problem, we are replacing the component with one that has a quasi-infinite lifespan and taking a few other steps that will isolate any issue with this component if it goes wrong in the future.

However, as I mentioned in an earlier update, we are not simply going to address this particular point problem and then merrily jump back into a countdown sequence. Throughout January, the SpaceX team will be doing another full review of vehicle systems, including propulsion, structures, avionics, software and ground support systems. We will be conducting additional engine tests, stage separation tests and avionics tests to once again attempt to flush out any issues. Even if we find nothing, the exercise is worthwhile.

Wind Delays Suck (Literally)

It is worth noting that we would have caught the problem without any damage to the vehicle if we had entered the final countdown sequence as planned. The sucked in tank damage only occurred because we partly drained the fuel tank due to the hold for high winds.

High winds are not a limitation of the rocket, which is designed to be essentially "all weather" and handle ground winds in excess of 50 mph (watch out for flying coconuts!). The ground winds limitation is actually due to the need to avoid a collision with the launch stand hold down arms, which grab the rocket at the base of the fuel tank, as the rocket lifts off.

To alleviate this problem, we have redesigned the launch stand so that the hold down arms retract out of the way on liftoff, activated by a breakwire. This gives us something very close to 100% winds availability from Kwaj. The retraction force is low, so even if there were an early activation of the actuator, it would not damage the rocket.

Another bothersome problem is the high rate of liquid oxygen (LOX) boiloff. This is not surprising when LOX is at -300F and there is a stiff wind impinging on the vehicle at 85F. To minimize boiloff, we will wrap the LOX tank in low cost cryo insulation attached with velcro straps that tear away on liftoff.

Lessons Learned on F1 Apply to F9

The challenges to date I think vindicate the strategy of building a small launch vehicle before a large one. If we had started out with an F9 class vehicle, the cost of every mistake would be multiplied by as much as an order of magnitude. As it is, we are able to overcome problems comparatively quickly and cheaply.

With the benefit of lessons learned on F1, it is taking far less time, effort and money to create F9. Despite the distraction of the F1 launch countdowns, I still anticipate a flight F9 first stage firing later this year and a maiden launch in late 2007.

Great Expectations

Those familiar with the launch business will know that countdown scrubs are a way of life. It's often said that the safest time to schedule your vacation is around launch day and that's true more often than not. Even rockets that have launched hundreds of times from launch pads that are in heavy use have multiple scrubs. Not too long ago, there was a Titan launch that had eleven scrubs and Delta launch that had six.

Reasons range from hard to avoid technical glitches, like the Shuttle fuel sensor malfunction on its last launch attempt, to silly false alarms. A Titan countdown was once aborted when someone spotted a "bag of suspicious liquid" on the mobile service tower. It turned out that the latrine had simply been a bridge too far for one of the technicians.

Given that Falcon 1 is an all new rocket and is launching from an all new launch pad on a remote tropical island, countdown scrubs in the first few attempts were very likely. As it is, we have had one abort due to a launch pad issue and one due to the rocket. If this next attempt succeeds in getting to t-zero, SpaceX will be reasonably fortunate in the scheme of things.

About this email: Musk sets February 8 for the next Falcon 1 attempt and explains the December 19 damage: a shorted electronic component rebooted the valve controller and removed its redundancy. He describes a new first stage shipped to Kwajalein, launch stand and insulation changes, and argues the setbacks vindicate building a small rocket before Falcon 9.

Notes: First-person throughout ('as I mentioned in an earlier update'); no signature appears in the SpaceRef copy. Five passages quoted by NASASpaceFlight match this text word for word. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [SpaceRef](#); [NASASpaceFlight.com](#), 'Third time lucky for SpaceX' (January 10, 2006)

Tags: spacex-update, falcon-1, falcon-9, kwajalein, launch-delay, maiden-flight

Static fire set for Feb. 9, launch for Feb. 10

From: Elon Musk

Date: February 6, 2006 (Dated February 6, 2006 in its title; SpaceRef posted it February 7, 2006.)

To: SpaceX update mailing list

Subject: Falcon 1 Maiden Flight Update: February 6, 2006

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

As an additional reliability measure, SpaceX will conduct a full test of vehicle systems, including initiating the flight countdown autosequence and briefly firing the main engine on the launch stand.

This test will take place Thursday, February 9 (California time).

If no flight critical anomalies are detected, launch will take place on Friday, February 10. The launch window on Friday is 1- 7 p.m. (California time).

About this email: A three-sentence schedule note: SpaceX will run a full countdown and briefly fire the main engine on the stand on February 9, and launch on February 10 if nothing critical turns up.

Notes: Unsigned note from the 'Falcon 1 Maiden Flight Update' page on spacex.com; authorship by Musk is assumed from that page, not stated by the source. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [SpaceRef](#); [NASA Watch](#), February 2006 archive ('SpaceX Launch Status')

Tags: spacex-update, falcon-1, kwajalein, static-fire, maiden-flight

Launch postponed after static fire countdown

From: Elon Musk

Date: February 9, 2006 (NASA Watch labels it 'Posted February 9, 2006 (via SpaceX)'. Kimbal Musk's blog and NASASpaceFlight place the completed hold-down firing and the postponement on February 10 (California time).)

To: SpaceX update mailing list

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

After analyzing data from the static fire countdown, we decided to postpone the launch. The vehicle is being lowered for further investigation. Once we have thoroughly checked out all systems, I will post an update on what was found and when the next countdown attempt will occur. Based on range availability and logistics constraints, a rough guess would be two to four weeks.

About this email: After reviewing data from the static fire countdown Musk postpones the launch, has the rocket lowered for inspection, and guesses the next countdown is two to four weeks away.

Notes: First person ('I will post an update'); unsigned in the NASA Watch copy. NASASpaceFlight paraphrases the same note and attributes it to Musk. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [NASA Watch \('SpaceX Launch Status', February 2006\)](#); [NASASpaceFlight.com, 'SpaceX postpones launch - heads to Court' \(February 10, 2006\)](#)

Tags: spacex-update, falcon-1, kwajalein, static-fire, launch-delay

Musk on the Falcon 1 hold-down firing

From: Elon Musk

Date: February 10, 2006 (Date of Kimbal Musk's blog post quoting it (Friday, February 10, 2006, 8:09 PM).)

To: SpaceX update mailing list

EXCERPT · NEWSLETTER · PUBLISHED BY MUSK · SINGLE-SOURCE

We were very happy to be able to execute a flight countdown all the way to lighting the engine. Although there wasn't a launch this time, we made a lot of progress refining the rocket and launch pad -- all work that needed to be done anyway.

About this email: Musk's statement after the Falcon 1 hold-down firing at Kwajalein: the team ran a flight countdown all the way to engine ignition and used the exercise to refine the rocket and pad.

Notes: Kimbal Musk introduces it as 'Elon's official statement on the launch'. It may come from a press statement rather than the update page, and it may be only part of a longer note; treat as an excerpt. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [Kwajalein Atoll and Rockets \(Kimbal Musk's blog\)](#), ['Hold Down Fire Video and Photos'](#)

Tags: spacex-update, falcon-1, kwajalein, static-fire

'We had a great static fire today'

From: Elon Musk

Date: March 21, 2006 (Says 'today'; Kimbal Musk's blog reported the same static fire on Tuesday, March 21, 2006 (California time). [NASASpaceFlight](#) quoted the update on March 22.)

To: SpaceX update mailing list

EXCERPT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

We had a great static fire today. Falcon was held down for almost three seconds of thrust (T+0.5s), part of which was under autonomous thrust vector control. All systems were green and no aborts were triggered.

About this email: Musk reports a clean static fire of the Falcon 1 at Kwajalein: almost three seconds of thrust, partly under autonomous thrust vector control, with no aborts.

Notes: [NASASpaceFlight](#) quotes this as Musk's online update. Only the quoted sentences were retrieved, so the note may have been longer. Kimbal Musk's blog paraphrases the same wording. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [NASASpaceFlight.com, 'SpaceX launch moved to Friday' \(March 22, 2006\)](#); [Kwajalein Atoll and Rockets \(Kimbal Musk's blog\)](#), ['Static Fire Analysis'](#)

Tags: spacex-update, falcon-1, kwajalein, static-fire, maiden-flight

Falcon 1 launch slipped a day to review static fire data

From: Elon Musk

Date: March 22, 2006 (NASASpaceFlight reported the statement on March 22, 2006; NASA Watch and Spaceflight Now carried it on March 23.)

To: SpaceX update mailing list

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

No major issues were discovered following the static fire, but, as a cautionary measure, we are going to take one more day to review data and verify system functionality. Launch is now scheduled for Friday at 1 p.m. California time.

About this email: Nothing serious turned up after the static fire, but Musk takes one more day to review data and moves the launch to Friday, March 24.

Notes: Spaceflight Now and NASASpaceFlight attribute the statement to Elon Musk; they quote only the first sentence. A missing space in the reader output ('andverify') was restored. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [NASA Watch \('SpaceX Update', March 2006\)](#); [NASASpaceFlight.com, 'SpaceX launch moved to Friday'](#); [Spaceflight Now Falcon 1 mission status center \(March 23, 2006 entry\)](#)

Tags: spacex-update, falcon-1, kwajalein, launch-delay, maiden-flight

Falcon 1 lost during first stage burn

From: Elon Musk

Date: March 24, 2006 (Headed 'Posted March 24, 2006 3:00 pm' on the SpaceX website, per NASA Watch.)

To: SpaceX update mailing list

Subject: Falcon 1 Maiden Flight Update

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · CONFIRMED

We had a successful liftoff and Falcon made it well clear of the launch pad, but unfortunately the vehicle was lost later in the first stage burn. More information will be posted once we have had time to analyze the problem.

About this email: Musk's two-sentence notice minutes after the first Falcon 1 flight: the rocket cleared the pad but was lost during the first stage burn, with analysis to follow.

Notes: NASA Watch: 'The following Statement was posted on the SpaceX website by Elon Musk', under the heading 'Falcon 1 Maiden Flight Update: Posted March 24, 2006 3:00 pm'. Spaceflight Now and NASASpaceFlight print the same two sentences. A missing space in the reader output ('thefirst') was restored. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [NASA Watch \('Falcon I Fails', March 24, 2006\)](#); [Spaceflight Now Falcon 1 mission status center \(2326 GMT entry\)](#); [NASASpaceFlight.com, 'Falcon 1 suffers launch failure'](#)

Tags: spacex-update, falcon-1, kwajalein, launch-failure, maiden-flight

Falcon 1 failure: fuel leak, fire, engine shutdown at T+29s

From: Elon Musk

Date: March 25, 2006 (Posted the morning after the March 24, 2006 launch; SpaceRef dated it March 25, 2006.)

To: SpaceX update mailing list

Subject: Preliminary Fault Analysis

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · CONFIRMED

(Note: There will be a formal US Government led anomaly investigation in partnership with SpaceX. The report issued by that team will reflect the fully considered official opinion. The website updates reflect only the preliminary analysis of SpaceX and are provided pursuant to our policy of early and full disclosure.)

The good news is that all vehicle systems, including the main engine, thrust vector control, structures, avionics, software, guidance algorithm, etc. were picture perfect. Falcon's trajectory was within 0.2 degrees of nominal during powered flight.

However, at T+25s, a fuel leak of currently unknown origin caused a fire around the top of the main engine that cut into the first stage helium pneumatic system. On high resolution imagery, the fire is clearly visible within seconds after liftoff. Once the pneumatic pressure decayed below a critical value, the spring return safety function of the pre-valves forced them closed, shutting down the main engine at T+29s.

It does not appear as though the first stage insulation played a negative role, nor are any other vehicle anomalies apparent from either the telemetry or imaging. Falcon was executing perfectly on all fronts until fire impaired the first stage pneumatic system.

Our plan at this point is to analyze data and debris to be certain that the above preliminary analysis is correct and then isolate and address all possible causes for the fuel leak. In addition, we will do another ground up systems review of the entire vehicle to flush out any other potential issues.

I cannot predict exactly when the next flight will take place, as that depends on the findings of this investigation and ensuring that our next customer is comfortable that all reasonable steps have been taken to ensure reliability. However, I would hope that the next launch occurs in less than six months.

It is perhaps worth noting that those launch companies that succeeded also took their lumps along the way. A friend of mine wrote to remind me that only 5 of the first 9 Pegasus launches succeeded; 3 of 5 for Ariane; 9 of 20 for Atlas; 9 of 21 for Soyuz; and 9 of 18 for Proton. Having experienced firsthand how hard it is to reach orbit, I have a lot of respect for those that persevered to produce the vehicles that are mainstays of space launch today.

I am very encouraged and grateful that our launch customers took the time to call and express their support of SpaceX when their reaction could easily have been the opposite. We will stand by them as they have stood by us. SpaceX is in this for the long haul and, come hell or high water, we are going to make this work.

As SpaceX is a company that believes in maximum disclosure (within the boundaries of proprietary data and ITAR restrictions), I will try to post as much as possible about this launch attempt over the coming weeks.

— Elon

About this email: Musk's first analysis of the Falcon 1 loss: every vehicle system worked, but a fuel leak started a fire above the main engine that cut the helium pneumatic system and shut the engine down at T+29 seconds. He hopes to fly again within six months, notes that other launchers also failed early, and promises full disclosure.

Notes: Signed 'Elon'. Kimbal Musk's blog introduces the same text with 'From Elon this morning:' under the heading 'Preliminary Fault Analysis' and signs it '--- Elon'; his copy lacks SpaceRef's opening parenthetical note, which was presumably added on the website. NASASpaceFlight's quotation reads 'forced them to close, shutting down the the main engine' where SpaceRef has 'forced them closed, shutting down the main engine'. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [SpaceRef \('Preliminary SpaceX Internal Analysis: Falcon I Flight'\)](#); [Kwajalein Atoll and Rockets \(Kimbal Musk's blog\)](#), ['Post-Launch Analysis'](#); [NASASpaceFlight.com](#), ['SpaceX: Come Hell or High Water'](#); [NASA Watch, March 2006 archive](#)

Tags: spacex-update, falcon-1, launch-failure, failure-analysis, maiden-flight

Return to flight findings coming; Falcon 9 tooling 90% done

From: Elon Musk

Date: July 6, 2006 (Headed 'July 6, 2006 Update'; reposted on the Space.com forum the same day.)

To: SpaceX update mailing list

Subject: July 6, 2006 Update

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

July 6, 2006 Update

My apologies for the long delay in providing an update. Between the Falcon 1 return to flight, Falcon 9 development and the NASA COTS program, this has been an inordinately busy period for SpaceX.

I will post the findings of the DARPA/SpaceX Return to Flight Board on the SpaceX website in about two weeks. The final meeting of the board was last week and the results will be posted after the DARPA senior leadership is briefed.

No major surprises were uncovered in the course of the investigation, so perhaps more interesting is the broader set of design & process improvements that constitute "Falcon 1.1". I will post whatever I can, excluding only proprietary or ITAR restricted information.

At the end of July, I will post a detailed update on Falcon 9. We've made a tremendous amount of progress on that front. Except for the fairing (nosecone), we are 90% done with all the manufacturing tooling and should have serial number 1 of the first stage built within three to four months. We are targeting a stage hold down, multi-engine firing in about six months.

On the business front, SpaceX now has ten launches on manifest and is on track to be cash flow positive in 2006, our fourth full year of operation.

---Elon

About this email: A short catch-up note. Musk says the DARPA/SpaceX Return to Flight Board has finished and its findings will be posted in about two weeks, previews the 'Falcon 1.1' improvements and a Falcon 9 update, and reports ten launches on the manifest with positive cash flow expected in 2006.

Notes: Signed '---Elon'. The forum post reproduces the whole update; NASASpaceFlight quotes several of the same sentences the next day. Text was retrieved through a page reader, so punctuation and paragraph breaks may differ slightly from the original page.

Sources: [Space.com forums \(post by 'mikejz', July 6, 2006\)](#); [NASASpaceFlight.com, 'Space X - investigation heading for closure' \(July 7, 2006\)](#)

Tags: spacex-update, falcon-1, falcon-9, return-to-flight, cots, darpa

Static fire slips to Friday, launch to January 22

From: Elon Musk

Date: January 2007 (Posted between January 16 and January 19, 2007 (it moves the static fire planned for Thursday January 18 to Friday January 19); exact day not shown in the source.)

To: SpaceX update mailing list

Subject: DemoFlight 2 launch update (static fire moved)

EXCERPT · NEWSLETTER · PUBLISHED BY MUSK · SINGLE-SOURCE

The static fire has moved to Friday (California time) and launch to Monday, January 22. We have not encountered any new issues – the shift in timing is primarily to provide for additional risk reduction activities on site, as we continue to operate with a healthy paranoia.

About this email: A short note on the SpaceX site moves the Demo Flight 2 static fire to Friday and the launch to Monday January 22, saying no new problems were found and the extra time is for risk reduction.

Notes: Two sentences quoted by a forum member 'From <http://www.spacex.com>'. Probably the whole of a short posting, but that cannot be confirmed, so it is marked as an excerpt. Unsigned; authorship by Musk is assumed from the other updates in this series.

Sources: [CosmoQuest \(BAUT\) forum, quoting spacex.com](#)

Tags: spacex-update, falcon-1, demo-flight-2, static-fire, kwajalein

Demo Flight 2: readiness review and wet dress done

From: Elon Musk

Date: January 16, 2007 (Dated 'Posted January 16, 2007' in the text.)

To: SpaceX update mailing list

Subject: DemoFlight 2 Launch Update

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

Posted January 16, 2007

DemoFlight 2 Launch Update

We recently completed our first major pre-launch milestone, the Flight Readiness Review (FRR). This confirmed that all Vehicle, Payload, Ground Support Equipment, and Personnel are ready for launch. Following the FRR, the Falcon 1 was rolled out to the

pad and we completed our wet dress rehearsal, which is a full countdown that holds approximately one minute before T-0.

The upgrades to the vehicle, Kwajalein launch facilities and our launch procedures have proven quite effective, and we have had a relatively smooth launch campaign up to this point. A number of non-critical issues arose during the wet-dress and are being resolved, but we have seen nothing to date that should delay our launch schedule.

Going forward, a static fire is planned Thursday, January 18 (California time). We have a launch window on January 21 and 22 (California time) and are working with the range to secure a couple of additional days as contingency. Should we go beyond that, which is still a good possibility as we work with the upgraded vehicle, pad, and procedures, the next available launch window is mid February.

There will be a live webcast and a media call in line for the launch. Details will be provided shortly.

About this email: SpaceX reports that Falcon 1's second demo flight has passed its Flight Readiness Review and wet dress rehearsal on Omelek, with a static fire planned January 18 and a launch window January 21-22, or mid-February if that slips.

Notes: Unsigned in the forum copy; Space News/NBC attribute the wording to Musk ('Musk wrote'). Forum poster's own framing lines omitted.

Sources: [CosmoQuest \(BAUT\) forum](#), quoting the [spacex.com updates page](#); [NBC News / Space News](#) ('SpaceX gears up for launch within days', attributes the update to Musk)

Tags: spacex-update, falcon-1, demo-flight-2, kwajalein, launch-campaign

Kwajalein range staffing pushes Demo Flight 2 to March

From: Elon Musk

Date: February 2007 (Space.com calls it 'the Thursday update' in an early-February 2007 article (probably February 8, possibly February 1); exact day not confirmed.)

To: SpaceX update mailing list

Subject: DemoFlight 2 launch update (range delay to March)

EXCERPT · NEWSLETTER · PUBLISHED BY MUSK · SINGLE-SOURCE

We have recently been informed by the Kwajalein Army Range that they do not have sufficient resources to support our launch in mid to late [February].

[...]

Several range personnel critical to the launch safety process will be unavailable in that timeframe.

About this email: Musk tells subscribers that the Kwajalein Army range cannot support a mid-to-late February launch because key safety personnel will be away, pushing Demo Flight 2 to no earlier than March 9.

Notes: Only the two sentences Space.com quoted from Musk's written status update; the bracketed '[February]' is Space.com's. The original Space.com page was not retrieved directly; the text comes from a forum repost of the article. The rest of the update was not found.

Sources: [Space.com article](#) ('SpaceX Pushes Second Falcon 1 Rocket Test to Early March') as reposted on the [CosmoQuest forum](#)

Tags: spacex-update, falcon-1, demo-flight-2, kwajalein, launch-delay

Static fire clean; Demo Flight 2 launch next week

From: Elon Musk

Date: March 16, 2007 (SpaceRef titles it 'Update 16 March 2007'; the same page carries an '18 March Update' paragraph added on top after the flight readiness review.)

To: SpaceX update mailing list

Subject: Falcon 1 Demoflight 2 Update 16 March 2007

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · CONFIRMED

18 March Update: The flight readiness review conducted tonight shows all systems are go for a launch attempt at 4pm California time (11pm GMT) tomorrow (Monday). The webcast can be seen at spacex.com/webcast.php and will start at T-60 minutes. Please check back for updates, as the launch will be postponed if we have even the tiniest concern.

At present, it appears that we are on track for a launch next week, possibly as soon as Monday. We had a very successful static fire yesterday that proceeded smoothly with no aborts.

Our initial review of the data showed that the rocket functioned almost perfectly. The only remaining concern is that the GPS portion of the guidance system showed an anomaly about 15 minutes *after* the static fire. Falcon 1 is designed to achieve its target orbit purely on inertial navigation, so the GPS, while helpful for improving orbit insertion accuracy, is not flight critical.

We are carefully analyzing the GPS and, more importantly, are making sure that the GPS problem does not hint at some larger issue. I will send out an update on Sunday confirming what day we will have our first countdown attempt. In the meantime, here are two videos of the static fire, one from a medium distance camera and one from a high speed close up camera. There will be a total of 12 cameras looking at the rocket on launch day, including two thermal imaging and two vehicle cams, so no shortage visual data.

I know it has been a year since our last launch and some people are wondering if launch 3 will also be a year away if something goes wrong this time. The answer is definitely no. The reason it took us a year is that the vehicle on the pad and the ground support equipment have hundreds of robustness upgrades — this is really Falcon 1 version 2.

There is nothing significant that we can think of to improve the vehicles under construction for the Dept of Defense and Malaysian satellite launches later this year. Therefore, no matter what happens, I do not expect there to be a significant delay in their approximate end of summer and mid fall launch dates.

On a separate note, we have made tremendous progress with the Falcon 9 development and I'm way overdue in posting a big update on progress. Should have that out sometime next month. Between ramping up for NASA COTS and the F1 launch, time has been a little tight around here n

-Elon-

About this email: Musk reports a smooth static fire with only a non-critical GPS anomaly, says a failure would not mean another year's delay because the vehicle is effectively 'Falcon 1 version 2', and promises a Falcon 9 progress update. An 18 March addendum confirms all systems go for a Monday launch attempt.

Notes: Signed '-Elon-'. The 16 March update and the 18 March addendum are kept together as SpaceRef published them. The 'π' before the signature and the typo 'inseration' are as rendered in the SpaceRef copy (the former is probably a mis-encoded smiley).

Sources: [SpaceRef](#); [NASA Watch \(excerpt\)](#); [CosmoQuest forum \(18 March paragraph\)](#); [Space.com \(quotes the 18 March paragraph as Musk's mission update\)](#)

Tags: spacex-update, falcon-1, demo-flight-2, static-fire, kwajalein

Why Monday's countdown aborted; go for Tuesday

From: Elon Musk

Date: March 20, 2007 (Posted Tuesday March 20, 2007 ('T-0 at 4pm California time today (Tues)'), the day after the March 19 terminal-count abort.)

To: SpaceX update mailing list

Subject: DemoFlight 2 launch update (abort explanation)

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

All systems are now go for launch with T-0 at 4pm California time today (Tues). The abort that occurred a few minutes before T-0 was triggered by our ground control software. It commanded a switchover of range telemetry from landline to radio, which took place correctly, however, because of the hardware involved, this transition takes a few hundred milliseconds. Before it had time to complete, our system verification software examined state and aborted.

Our simulations done beforehand all passed, because the simulator did not account for a hardware driven delay in the transition. We considered putting the vehicle into a safe state yesterday and updating the ground control software to make the very minor fix needed, but the safer course of action was to stand down.

Yesterday afternoon and evening (Kwaj time), our launch team updated the software to address the timing issue and verified that there were no similar problems elsewhere. We ran the software through several simulated countdowns and then once again with the rocket and range in the loop.

About this email: Musk explains that Monday's abort came from ground control software checking state before a telemetry switchover from landline to radio had finished, a delay the simulator had not modelled. The software was fixed and re-run overnight and all systems are go for a Tuesday launch.

Notes: Unsigned in the reprints; NASA Watch attributes it to SpaceX, Samizdata to Elon Musk. Three paragraphs as quoted by NASA Watch; it is possible the spacex.com posting had further lines. No title survives; subject is descriptive.

Sources: [NASA Watch](#); [CosmoQuest forum \(first two paragraphs, 'From SpaceX updates page'\)](#); [Samizdata \(first paragraph, attributed 'According to Elon Musk'\)](#)

Tags: spacex-update, falcon-1, demo-flight-2, launch-abort, software

Demo Flight 2 reaches space: 'not perfect, but pretty good'

From: Elon Musk

Date: March 20, 2007 (Launch-day statement ('took place today at 6:10 pm California time'), March 20, 2007 PDT; NASA Watch posted it the same day.)

To: SpaceX update mailing list

Subject: Post-flight Comments Regarding Second Falcon 1 Launch

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · CONFIRMED

The second test launch of Falcon 1 took place today at 6:10 pm California time. The launch was not perfect, but certainly pretty good. Given that the primary objectives were demonstrating responsive launch and gathering test data in advance of our first operational satellite launch later this year, the outcome was great. Operationally responsive (ie fast) launch has become an increasingly important national security objective, so demonstrating rapid loading of propellents and launch in less than an hour, as well as a rapid recycle following the first engine ignition are major accomplishments.

We retired almost all of the significant development risk items, in particular:

- 1st stage ascent past max dynamic pressure
- avionics operation in vacuum and under radiation
- stage separation
- 2nd stage ignition
- fairing separation
- 2nd stage nozzle/chamber at steady state temp in vacuum

Falcon flew far beyond the "edge" of space, typically thought of as around 60 miles. Our altitude was approximately 200 miles, which is just 50 miles below the International Space Station. The second stage didn't achieve full orbital velocity, due to a roll excitation late in the burn, but that should be a comparatively easy fix once we examine the flight data. Since it is impossible to ground test the second stage under the same conditions it would see in spaceflight, this anomaly was also something that would have been very hard to determine without a test launch.

All in all, this test has flight proven 95+ percent of the Falcon 1 systems, which bodes really well for our upcoming flights of Falcon 1 and Falcon 9, which uses similar hardware. We do not expect any significant delay in the upcoming flights at this point. The Dept of Defense satellite launch is currently scheduled for late summer and the Malaysian satellite for the Fall.

I'd like to thank DARPA and the Air Force for buying the two test flights and helping us work through a number of challenges over the past year. I'd also like to express my appreciation for the efforts of the Kwajalein Army Range (Reagan Test Site) and we look forward to many more launches in the future.

Finally, thank you to everyone at SpaceX for working so hard to make this a great test. This is a big leap forward for commercial spaceflight!

About this email: Hours after Falcon 1's second test launch, Musk says the flight reached about 200 miles altitude and retired nearly all major development risks, though a late roll excitation kept the second stage short of orbital velocity. He thanks DARPA, the Air Force, the Kwajalein range and the SpaceX team.

Notes: First-person statement ('I'd like to thank DARPA and the Air Force...'); no signature in the SpaceRef copy. Title is SpaceRef's.

Sources: [SpaceRef](#); [NASA Watch](#) (first paragraph)

Tags: spacex-update, falcon-1, demo-flight-2, launch-day, kwajalein

Demo Flight 2 post-flight analysis: LOX slosh ended the burn

From: Elon Musk

Date: March 27, 2007 (Space.com describes it as an update Musk posted Tuesday (March 27, 2007); Spaceflight Now reprinted it March 28 and SpaceDaily March 29.)

To: SpaceX update mailing list

Subject: DemoFlight 2 post launch update

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · CONFIRMED

Having had several days to examine the data, the second test launch of Falcon 1 is looking increasingly positive. Post flight review of telemetry has verified that oscillation of the second stage late in the mission is the only thing that stopped Falcon 1 from reaching full orbital velocity.

The second stage was otherwise functioning well and even deployed the satellite mass simulator ring at the end of flight! Actual final velocity was 5.1 km/s or 11,000 mph, whereas 7.5 km/s or 17,000 mph is needed for orbit. Altitude was confirmed to be 289 km or 180 miles, which is certainly enough for orbit and is about where the Space Shuttle enters its initial parking orbit.

This confirms the end of the test phase for Falcon 1 and the beginning of the operational phase. The next Falcon 1 flight will carry the TacSat 1 satellite for the US Navy, with a launch window that begins in September, followed by Razaksat for the Malaysian Space Agency in November. Beyond that, we have another nine missions on manifest for F1 and F9. Note, the first F9 mission will also be a test flight and the three NASA F9/Dragon missions are all test flights for Dragon.

Telemetry shows that engine shutdown occurred only about a minute and a half before schedule (roughly T + 7.5 mins), due to the oscillations causing propellant to slosh away from the sump. When the liquid level in the tank was low, this effectively starved the engine of propellant. A disproportionate amount of the velocity gain occurs in the final few minutes of flight, when the stage is very light, which is why the velocity difference is greater than just linearly subtracting 1.5 mins from the burn time.

Except for a few blips here and there, we have now cleaned up the raw data feed and recovered video and telemetry for the entire mission well past 2nd stage shutdown. Including all the launch pad video and ground support equipment data, we have somewhere close to a terabyte of information to review.

This was far too much to send over the T1 satellite link from Kwaj and had to be brought over in person after the flight. Given that a number of our engineers have only just returned from Kwaj, please consider this still a preliminary analysis:

In a nutshell, the data shows that the increasing oscillation of the second stage was likely due to the slosh frequency in the liquid oxygen (LOX) tank coupling with the thrust

vector control (engine steering) system. This started out as a pitch-yaw movement and then transitioned into a corkscrewing motion.

For those that aren't engineers, imagine holding a bowl of soup and moving it from side to side with small movements, until the entire soup mass is shifting dramatically. Our simulations prior to flight had led us to believe that the control system would be able to damp out slosh, however we had not accounted for the perturbations of a contact on the stage during separation, followed by a hard slew to get back on track.

The nozzle impact during stage separation occurred due to a much higher than expected vehicle rotation rate of about 2.5 deg/sec vs. max expected of 0.5 deg/sec. As the 2nd stage nozzle exited the interstage, the first stage was rotating so fast that it contacted the niobium nozzle. There was no apparent damage to the nozzle, which is not a big surprise given that niobium is tough stuff.

The unexpectedly high rotation rate was due to not knowing the shutdown transient of the 1st stage engine (Merlin) under flight conditions. The actual shutdown transient had a very high pitch over force, causing five times the max expected rotation rate.

We definitely intend to have both the diagnosis and cure vetted by third party experts, however we believe that the slosh issue can be dealt with in short order by adding baffles to our 2nd stage LOX tank and adjusting the control logic. Either approach separately would do the trick (eg. the Atlas-Centaur tank has no baffles), but we want to ensure that this problem never shows up again.

The Merlin shutdown transient can be addressed by initiating shutdown at a much lower thrust level, albeit at some risk to engine reusability. Provided we have a good set of slosh baffles, even another nozzle impact at stage separation would not pose a significant flight risk, although obviously we will work hard to avoid that.

I will be posting another DemoFlight 2 post launch update within a week, which will include a list of all subsystems color coded for status: green = good, yellow = cause for concern, red = flight failure if unchanged, black = untested. Of the hundreds of subsystems on the rocket, only the 2nd stage LOX tank slosh baffles are clearly red right now, but that could change with further analysis. As much as is reasonably possible (subject to ITAR and proprietary info), SpaceX will provide full disclosure with respect to the findings of the mission review team.

The Difference Between a Test Flight and an Operational Satellite Mission

There seems to be a lot of confusion in the media about what constitutes a success. The critical distinction is that a test flight has many gradations of success, whereas an operational satellite mission does not.

Although we did our best at SpaceX to be clear about last week's launch, including naming it DemoFlight 2 and explicitly not carrying a satellite, a surprising number of people still evaluated the test launch as though it were an operational mission.

This is neither fair nor reasonable. Test flights are used to gather data before flying a "real" satellite and the degree of success is a function of how much data is gathered. The

problem with our first launch is that, although it taught us a lot about the first stage, ground support equipment and launch pad, we learned very little about the second stage, apart from the avionics bay.

However, that first launch was still a partial success, because of what we learned and, as shown by flight two, that knowledge was put to good use: there were no flight critical issues with the first stage on flight two.

The reason that flight two can legitimately be called a near complete success as a test flight is that we have excellent data throughout the whole orbit insertion profile, including well past second stage shutdown, and met all of the primary objectives established beforehand by our customer (DARPA/AF).

This allows us to wrap up the test phase of the Falcon 1 program and transition to the operational phase, beginning with the TacSat mission at the end of summer. Let me be clear here and now that anything less than orbit for that flight or any Falcon 1 mission with an operational satellite will unequivocally be considered a failure.

This is not "spin" or some clever marketing trick, nor is this distinction an invention of SpaceX -- it has existed for decades. The US Air Force made the same distinction a few years ago with the demonstration flight of the Delta IV Heavy, which also carried no primary satellite.

Although the Delta IV Heavy fell materially short of its target velocity and released its secondary satellites into an abnormally low altitude, causing reentry in less than one orbit, it was still correctly regarded by Boeing and the Air Force as a successful test launch, because sufficient data was obtained to transition to an operational phase.

It is perhaps worth drawing an analogy with more commonplace consumer products. Before software is released, it is beta tested in non-critical applications, where bugs are worked out, before being released for critical applications, although some companies have been a little loose with this rule. :) Cars go through a safety and durability testing phase before being released for production. Rockets may involve rocket science, but are no different in this regard.

--Elon--

About this email: A week after Falcon 1's second test launch, Musk reports that LOX slosh coupling with the second-stage steering, set off by a nozzle bump at stage separation, starved the engine about 90 seconds early. He declares the test phase over and argues a test flight should be judged differently from an operational mission.

Notes: Signed '--Elon--'. Wording and signature follow the Spaceflight Now reprint ('statement from SpaceX founder Elon Musk'); paragraph breaks follow the SpaceDaily reprint, whose wording is identical apart from omitting the ':' and signature. Space.com's same-day quotation reads 'smacked the niobium nozzle' where both reprints read 'contacted the niobium nozzle', suggesting SpaceX revised the wording after first posting. No newsletter title survives in the reprints; subject is descriptive, taken from Musk's own phrase.

Sources: [Spaceflight Now](#); [SpaceDaily](#); [Space.com](#) (Brian Berger, long quotations)

Tags: spacex-update, falcon-1, demo-flight-2, post-flight-analysis, slosh

Plan Going Forward: 'I will never give up' after Falcon 1 Flight 3

From: Elon Musk

Date: August 2, 2008 (Header as published by NASA Watch: Sent Saturday, August 02, 2008 9:45 PM)

To: Space Exploration Technologies (all-company list)

Subject: Plan Going Forward

FULL TEXT • COMPANY-WIDE MEMO • LEAKED TO PRESS • CONFIRMED

From: Elon Musk

Sent: Saturday, August 02, 2008 9:45 PM

To: Space Exploration Technologies

Subject: Plan Going Forward

It was obviously a big disappointment not to reach orbit on this flight. On the plus side, the flight of our first stage, with the new Merlin 1C engine that will be used in Falcon 9, was picture perfect. Unfortunately, a problem occurred with stage separation, causing the stages to be held together. This is under investigation and I will send out a note as soon as we understand exactly what happened.

The most important message I'd like to send right now is that SpaceX will not skip a beat in execution going forward. We have flight four of Falcon 1 almost ready for flight and flight five right behind that. I have also given the go ahead to begin fabrication of flight six. Falcon 9 development will also continue unabated, taking into account the lessons learned with Falcon 1. We have made great progress this past week with the successful nine engine firing.

As a precautionary measure to guard against the possibility of flight 3 not reaching orbit, SpaceX recently accepted a significant investment. Combined with our existing cash reserves, that ensures we will have more than sufficient funding on hand to continue launching Falcon 1 and develop Falcon 9 and Dragon. There should be absolutely zero question that SpaceX will prevail in reaching orbit and demonstrating reliable space transport. For my part, I will never give up and I mean never.

Thanks for your hard work and now on to flight four.

Elon

About this email: Hours after the third Falcon 1 failed to reach orbit because of a stage-separation problem, Musk emailed the whole company that SpaceX would not skip a beat, that flights four and five were nearly ready and flight six approved, and that a recent investment guaranteed funding. He vowed never to give up.

Notes: REPLACES: 2008-08-02 Falcon 1 Flight 3 failure: 'I will never give up'. NASA Watch reproduces the message with its email header (From/Sent/To/Subject), which settles that it was an all-company email with the subject 'Plan Going Forward'; the first-pass entry lacked the header and flagged the medium as uncertain. Sign-off is 'Elon' on NASA Watch ('--Elon--' in Universe Today's copy). The same text was also released to media by SpaceX.

Sources: [NASA Watch](#); [Universe Today](#); [Spaceflight Now Falcon 1 Flight 3 status](#)

Tags: falcon-1, launch-failure, morale, funding

Flight 3 failure traced to Merlin 1C thrust transient

From: Elon Musk

Date: August 6, 2008 (Released by SpaceX on August 6, 2008 (date of the reprints); covers the August 2, 2008 launch.)

To: SpaceX update mailing list

Subject: Message from Elon Musk: Falcon 1, Flight 3 Mission Summary

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · CONFIRMED

On August 2nd, Falcon 1 executed a picture perfect first stage flight, ultimately reaching an altitude of 217 km, but encountered a problem just after stage separation that prevented the second stage from reaching orbit. At this point, we are certain as to the origin of the problem. Four methods of analysis – vehicle inertial measurement, chamber pressure, onboard video and a simple physics free body calculation – all give the same answer.

The problem arose due to the longer thrust decay transient of our new Merlin 1C regeneratively cooled engine, as compared to the prior flight that used our old Merlin 1A ablatively cooled engine. Unlike the ablative engine, the regen engine had unburned fuel in the cooling channels and manifold that combined with a small amount of residual oxygen to produce a small thrust that was just enough to overcome the stage separation pusher impulse.

We were aware of and had allowed for a thrust transient, but did not expect it to last that long. As it turned out, a very small increase in the time between commanding main engine shutdown and stage separation would have been enough to save the mission.

The question then is why didn't we catch this issue? Unfortunately, the engine chamber pressure is so low for this transient thrust -- only about 10 psi – that it barely registered on our ground test stand in Texas where ambient pressure is 14.5 psi. However, in vacuum that 10 psi chamber pressure produced enough thrust to cause the first stage to recontact the second stage.

It looks like we may have flight four on the launch pad as soon as next month. The long gap between flight two and three was mainly due to the Merlin 1C regen engine development, but there are no technology upgrades between flight three and four.

Good Things About This Flight

- Merlin 1C and overall first stage performance was excellent
- The stage separation system worked properly, in that all bolts fired and the pneumatic pushers delivered the correct impulse
- Second stage ignited and achieved nominal chamber pressure
- Fairing separated correctly
- We discovered this transient problem on Falcon 1 rather than Falcon 9
- Rocket stages were integrated, rolled out and launched in seven days
- Neither the near miss potential failures of flight two nor any new ones were present
- The only untested portion of flight is whether or not we have solved the main problem of flight two, where the control system coupled with the slosh modes of the liquid oxygen

tank. Given the addition of slosh baffles and significant improvements to the control logic, I feel confident that this will not be an issue for the upcoming flight four.

About this email: Musk explains that Falcon 1 Flight 3 was lost because residual thrust from the new regeneratively cooled Merlin 1C lasted longer than expected and pushed the first stage back into the second after separation. He lists what went right and says Flight 4 could be on the pad the following month.

Notes: Text taken from Kimbal Musk's blog, which introduces it as 'Official report from SpaceX on the Falcon 1 Vehicle 3 launch'; the title is from the Space.com forum repost. Samizdata carries the identical text. Written in the first person; none of the three copies has a signature line. Wikipedia cites it as 'Flight 3 mission summary', August 6, 2008, on spacex.com. The closing 'The only untested portion...' sentence is the last bullet in the blog and Samizdata copies (a separate paragraph in the forum copy).

Sources: [Kwajalein Atoll and Rockets](#) (Kimbal Musk's blog); [Samizdata](#) (Dale Amon, Aug 8, 2008; full text, 'the official word from SpaceX'); [Space.com forums](#) (post by docm, titled 'MESSAGE FROM ELON MUSK: FALCON 1, FLIGHT 3 MISSION SUMMARY'); [Universe Today](#) (quotes two paragraphs)

Tags: spacex-update, falcon-1, flight-3, failure-analysis, merlin-1c

Flight 4 countdown expected in late September

From: Elon Musk

Date: September 2008 (Posted on spacex.com a few days before the Saturday September 20, 2008 static fire (Space News covered it about September 18-19; Samizdata reprinted it September 20). Exact day not shown in the sources.)

To: SpaceX update mailing list

Subject: Flight 4 of Falcon 1

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

Flight 4 of Falcon 1

As mentioned in my update last month, we do expect to conduct a launch countdown in late September — as scheduled.

Having said that, it is still possible that we encounter an issue that needs to be investigated, which would delay launch until the next available window in late October. If preparations go smoothly, we will conduct a static fire on Saturday and launch sometime between Tuesday and Thursday (California time).

The SpaceX team worked hard to make this launch window, but we also took the time to review data from Flight 3 in detail. In addition to us reviewing the data, we had several outside experts check the data and conclusions. No flight critical problems were found apart from the thrust transient issue.

Flight 5 production is well underway with an expected January completion date, Flight 6 parts are on order and Flight 7 production will begin early next year. We are now in steady state production of Falcon 1 at a rate of one vehicle every four months, which we will probably step up to one vehicle every two to three months in 2010.

-Elon Musk

About this email: Musk says Falcon 1 Flight 4 is on track for a late-September countdown, with a static fire on Saturday and launch between Tuesday and Thursday. Outside experts reviewed the Flight 3 data and found no flight-critical problem beyond the thrust transient; Flights 5-7 are in production.

Notes: Signed '-Elon Musk' in the forum copy. SpaceRef's page of this title (spaceref.com/status-report/spacex-update-flight-4-of-falcon-1/) now serves machine-rewritten text, not the

original, and was not used.

Sources: [Space.com forums](#) (post by tanstaaf176 quoting [spacex.com/updates.php](#)); [Samizdata](#) (Dale Amon, Sept 20, 2008; full text, signed '- Elon Musk'); [spacetoday.net](#) summary (Sept 21, 2008)

Tags: spacex-update, falcon-1, flight-4, kwajalein, production

Flight 4 static fire done; launch slips a few days

From: Elon Musk

Date: September 23, 2008 (Reprinted by Kimbal Musk on September 23, 2008; Wikipedia's citation of the [spacex.com](#) updates archive dates the 'Flight 4 Launch Update' September 23, 2008. Reports the Saturday September 20 static fire.)

To: SpaceX update mailing list

Subject: Flight 4 Launch Update

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

The static fire took place on Saturday [20 Sep 2008, CA time], as expected, and no major issues came up. However, after a detailed analysis of data, we decided to replace a component in the 2nd stage engine LOX supply line. There is a good chance we would be ok flying as is, but we are being extremely cautious.

This adds a few extra days to the schedule, so the updated launch window estimate is now Sept 28th through Oct 1st [CA time].

—Elon—

About this email: Musk reports the Flight 4 static fire went as expected, but SpaceX will replace a component in the second-stage LOX supply line as a precaution, moving the launch window to September 28 through October 1.

Notes: Signed '-Elon-'. Bracketed date clarifications are as they appear in the source copy.

Sources: [Kwajalein Atoll and Rockets](#) (Kimbal Musk's blog); [Astroengine](#) (paraphrase with quote)

Tags: spacex-update, falcon-1, flight-4, static-fire, kwajalein

Flight 4 cleared for liftoff Sunday

From: Elon Musk

Date: September 27, 2008 (Posted Saturday September 27, 2008 ('tomorrow, Sunday September 28th'); Wikipedia cites it as 'Falcon 1 Flight 4', Elon Musk, September 27, 2008, at [spacex.com/F1-004.php](#).)

To: SpaceX update mailing list

Subject: Falcon 1 Flight 4

EXCERPT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

Falcon 1 is currently cleared for liftoff sometime between 4 p.m. and 9 p.m. (California time) tomorrow, Sunday September 28th. Of course, if we see anything that requires investigation, the launch will be postponed, but we'll let you know as soon as we know. As with prior flights, you can access the webcast from the SpaceX site:

About this email: On the eve of Flight 4, Musk says Falcon 1 is cleared to lift off Sunday September 28 between 4 and 9 p.m. California time, with a postponement if anything needs investigating, and points readers to the webcast.

Notes: Parabolic Arc introduces it as an update Elon Musk 'posted on his blog'; the quotation stops at the webcast link, so the original may have continued. NBC's Cosmic Log also carried it but is closed to the fetch tool.

Sources: [Parabolic Arc \('SpaceX Could Launch Falcon 1 Today'\)](#); [Spaceflight Now preview \(reports the update, no quotation\)](#)

Tags: spacex-update, falcon-1, flight-4, launch-day, kwajalein

Falcon 9 second stage fired; 2009 in review

From: Elon Musk

Date: January 4, 2010 (Dated 'Monday, January 4, 2010' in the update itself (the Daily Kos copy reproduces the date line with the typo '2009').)

To: SpaceX update mailing list

Subject: Inaugural Falcon 9 / Dragon Flight Hardware Update

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

Inaugural Falcon 9 / Dragon Flight Hardware Update

Monday, January 4, 2010

The SpaceX team kicked off 2010 with the successful full duration orbit insertion firing of the Falcon 9 second stage at our Texas test site (details below). This was the final stage firing required for launch, so the second stage will soon be packaged for shipment and should arrive at Cape Canaveral by end of month. Depending on how well full vehicle integration goes, launch should occur one to three months later.

2009 was an exciting year for SpaceX. In July, with the successful launch of RazakSAT, Falcon 1 became the first privately developed liquid fuel rocket to put a commercial satellite in orbit. That same month, DragonEye — SpaceX's Laser Imaging Detection and Ranging (LIDAR) sensor — launched on NASA's STS-127 shuttle mission and successfully completed flight system trials in preparation for guiding the Dragon spacecraft as it approaches the International Space Station. We also hosted the first astronaut training day at our Hawthorne headquarters in preparation for flights to the Space Station.

Last year also saw the successful arc jet testing of PICA-X, SpaceX's high performance heat shield material developed in collaboration with NASA, which will be used to protect our Dragon spacecraft on reentry. And our Merlin Vacuum engine demonstrated the highest efficiency ever for an American hydrocarbon rocket engine. SpaceX also signed deals with several key customers, including CONAE (Argentina's National Commission on Space Activity), Astrium and Orbcomm.

The ongoing evolution of the commercial space industry was recently featured as the cover story ("The New Space Rush") in Popular Science magazine. The article provided a great perspective on the industry as a whole, but I disagree with the subheading, "Who Needs NASA?". If you read the article, it's clear their intent was just to convey excitement for the developments in commercial space, but obviously NASA is and always will be critical to the future of space exploration, particularly at the outer edge where there is no commercial market. Without NASA, SpaceX certainly would not be where it is today.

As we get closer to our first Falcon 9 launch, SpaceX would like to thank NASA, the Air Force, the FAA, and our commercial customers for their continued support. And, of course, I would like to thank the whole SpaceX team for their unwavering commitment to our company and our mission, especially over these last few months. Through their hard work and dedication, 2010 promises to be another great year.

-Elon-

Falcon 9 First Stage

Prior to arrival at the Cape, the Falcon 9 first stage arrived at our Texas Test Site. There, we did a full checkout, raised it up to the top of the 72 meter (235 foot) tall test stand, and conducted two successful nine engine test firings — the first 10 seconds long, followed by a 30 second long firing three days later.

Test firing of the full flight first stage of Falcon 9, conducted Oct 16, 2009 at the SpaceX Texas Test Facility in McGregor.

[Click to play video](#), and note the engines gimbaling (steering) in the upper left camera.

Everything performed as planned; we then shipped the first stage to Florida and have commenced final processing in the hangar at the SpaceX launch site. Once all propulsion and avionics checkout processes are complete, we will move forward with stage mate, to be followed closely by vehicle transfer to the transporter erector, and a static fire shortly thereafter.

Falcon 9 first stage arriving in the hangar at Space Launch Complex 40, Cape Canaveral, Florida.

Photo credit: SpaceX.

Falcon 9 Second Stage

Flight hardware for the Falcon 9 second stage also shipped to Texas, where it completed static load testing, and then was integrated with the previously tested Merlin Vacuum second stage engine. After performing system checkouts, we raised the stage up on to the newly completed Upper Stage test stand.

Installing the Falcon 9 second stage into the newest test stand at our Texas test site.

Photo credit: Chris Thompson, SpaceX.

In November we conducted the initial second stage test firing lasting forty seconds. This test involved a new test stand, a new flight stage, and it occurred as planned, on the first attempt without aborts or recycles.

First test firing of the full flight second stage of Falcon 9, conducted at the SpaceX Texas Test Facility in McGregor.

On January 2, 2010, the team completed a full duration orbit insertion firing (329 seconds) of the integrated Falcon 9 second stage. At full power, the Merlin Vacuum engine generates 411,000 N (92,500 lbs force) of thrust, and operates with the highest performance ever for an American-made hydrocarbon rocket engine.

Full duration orbit insertion firing of the Falcon 9 second stage, conducted on January 2, 2010. [Click to play video.](#)

(Click to play video)

Having multiple stands for testing individual engines, first and second stages, and Draco thrusters allows us great freedom in processing hardware for flight. Our manifest currently lists more than twenty-five Falcon 1e and Falcon 9 missions, seventeen of those with Dragon spacecraft, so all of our stands will be kept very active.

Merlin Vacuum Engine Expansion Nozzle

We recently fabricated and formed the first flight expansion nozzle for the Merlin Vacuum second stage engine. Made of a thin, high temperature alloy, the large expansion nozzle extends from the regeneratively cooled portion of the engine, and improves its performance in the vacuum of space. Standing 2.7 meters (9 feet) tall and 2.4 m (8 ft) in diameter, it resembles the nozzle used on our Falcon 1's second stage engine, only larger.

The Merlin Vacuum engine expansion nozzle measures 2.7 meters (9 feet) tall, and most of it has a wall thickness of about 1/3 of a millimeter (1/64 of an inch). Photo credit: SpaceX.

Interstage

The interstage physically joins the first and second stages, and houses the Merlin Vacuum engine during first stage ascent. The carbon composite cylinder measures 3.6 meters (12 feet) in diameter and nearly 8 m (26 ft) tall.

The top edge of the interstage contains a set of clamping collets that join the first and second stages during liftoff and ascent. After the first stage shuts down, the collets release, and three pneumatic pushers smoothly and forcefully separate the stages, clearing the second stage engine for ignition.

We recently conducted a series of full-scale tests verifying the performance of the separation system under a variety of load conditions. We placed the fully configured interstage in the Falcon 9 structural test stand in Texas, and mounted a large mass on top to simulate the second stage. During testing, the collets release the stage and the pushers force the simulated second stage high into the air. See video below.

The Falcon 9 Interstage (black cylinder at lower center) pushes away the simulated second stage (blue cylinder above).

A series of restraining cables and counterweights capture the load and prevent it from falling downwards.

(Click to play video)

This stage separation system resembles a larger version of the one successfully used on our Falcon 1 vehicle. Note that this system uses no explosives, making it safer to assemble and deploy, and increasing its overall reliability, as we can conduct multiple tests of every flight component, whereas an individual explosive device carries the risk of being fully testable only once — in actual use.

In addition to the stage separation system, the interstage also houses the parachute system that will aid in first stage recovery. Our Cape team has mated the interstage to the first stage and continues to finalize vehicle wiring in preparation for complete vehicle integration.

The Falcon 9 flight interstage in the Cape Canaveral launch site hangar prior to mating with the first stage.

Photo credit: SpaceX.

Dragon Qualification Spacecraft

As mentioned above, the inaugural Falcon 9 flight will loft our Dragon qualification spacecraft into orbit. After completing testing in Texas, the Dragon spacecraft shipped to the Cape in preparation for first flight.

First flight Dragon nosecone (tan, at left), spacecraft (middle) and trunk (right) in process at the SLC-40

launch pad hangar in Florida. Photo credit: SpaceX.

In preparation for flight, the Dragon spacecraft was mated to the trunk (see below), which in future flights will house both unpressurized payloads and the vehicle's solar panels. By flying the Dragon spacecraft configuration, we will obtain valuable data about its performance during the climb to orbit, which will support the following Falcon 9 flight — the first launch under the NASA Commercial Orbital Transportation Services (COTS) program. On that flight, an operational Dragon spacecraft will make several orbit of the Earth, followed by reentry and splashdown in the Pacific Ocean off the coast of California.

Pressurized portion of the Dragon spacecraft, top, mated to the cylindrical unpressurized trunk

section below, with nose cap in foreground. As Dragon has been designed from the start for human

transport, even the cargo and demonstration versions include windows

(circle at top, covered for protection during painting). Photo credit: SpaceX.

Launch Operations – Cape Canaveral SLC-40

As the flight hardware converges on Florida, many significant activities continue around our launch site in preparation for first flight.

Launch Mount

As with our Falcon 1 rocket, the Falcon 9 uses a "hold before launch" system where the launch mount firmly restrains the rocket as it develops full thrust. Once engine performance is verified, the rocket commands the launch mount to set it free.

The Falcon 9's four-part launch mount assembly performs several significant tasks. At rest, it supports the fully fueled Falcon 9, with a mass of over 330,000 kilograms (nearly three-quarters of a million pounds). Next, as the first stage's nine Merlin engines fire and reach full power of nearly 5 MN (over 1 million pounds force), the mounts must hold the vehicle down against the upward thrust.

Finally, upon command, the mounts release the rocket and then move out of the way, giving the nine engines maximum clearance as they lift the vehicle away from Earth.

Months of construction and testing converged into a series of final tests of the launch mount system. The four mount towers were attached to the base of the Transporter / Erector, and their hydraulically powered actuators checked to verify performance.

We then conducted a set of live load tests that simulated the significant downward and upward forces present during the launch sequence. We placed an actual Falcon 9 truss (the structure that joins the nine Merlin engines to the vehicle) into the launch mount, and used a crane and pneumatic cylinders to simulate the forces at liftoff. On command, the launch restraints let the truss fly free. See the video below.

Launch mount system test, with a crane pulling up on a Falcon 9 engine mount truss to simulate the

forces it will experience at liftoff. After releasing the rocket, the mount towers move back to give

maximum clearance to the departing vehicle.

(Click to play video)

Recovery Preparations

Both the Falcon 9 first stage and Dragon spacecraft are designed to be recovered. For this first demonstration flight, the Dragon spacecraft will remain in orbit but our team will attempt recovery of the Falcon 9 first stage and has commenced with recovery testing operations (see photo below).

Flotation testing of a portion of the recovery raft that will aid in returning the Falcon 9 first stage to land after flight.

Photo credit: SpaceX.

Other progress at SLC-40 includes:

- Nearing completion of a new hydraulic system to provide pressurized RP-1 propellant in support of hangar and pad checkout of vehicle Thrust Vector Control (TVC) systems.
- Nearing completion of new gaseous nitrogen system (used for pressurization, line purges, etc.), and a new helium system (used for vehicle pressurization, cooling and engine startup).
- Completion of the liquid nitrogen delivery system and final fill of 4,900 gallons to the site's storage tank.
- Installing new Payload Environmental Control System on the pad to keep future cargo loads comfortable during processing and preparation for launch.
- Functional testing of the new helium fill system. During loading, we chill the Falcon 9's helium storage tanks down to minus 184 degrees C (minus 300 degrees F).
- Multiple test deployments of the Transporter / Erector system (shown above), and the addition of vehicle fill and drain plumbing and umbilical support systems.
- Completed installation of a new dual-redundant, fault tolerant digital information network in support of mission operations and launch pad systems.
- Flow tests verifying the systems that will apply large amounts of water to the launch

pad to provide noise and fire suppression during liftoff.

Mission Operations

Radio Tests

Back at our Hawthorne CA headquarters in mid-October we conducted a complete end-to-end test of our Dragon radio communications system with the NASA geosynchronous Tracking and Data Relay Satellite System (TDRSS).

From SpaceX's Hawthorne headquarters, Dragon's 20 watt transmitter and separate receiver antenna (rectangles at left) communicate with NASA's TDRS 5 satellite on orbit 35,800 km (22,240 mi) above the Earth.

Photo credit: SpaceX.

The SpaceX communications flight hardware, developed with subcontractors Delta Microwave (Low Noise Amplifier), Quasonix (transmitter and receiver), and Haigh-Farr (antennas), emulated a complete Dragon spacecraft comm link, and successfully sent and received data through the TDRSS network. Commands were dispatched from our Hawthorne headquarters command station, to NASA JSC in Houston, across Texas to the TDRSS White Sands Ground Terminal, up to the TDRS 5 Spacecraft in geosynchronous orbit, and back down to the Dragon receiver on the ground in Hawthorne.

The test series demonstrated telemetry and command transmission at a variety of data rates up to 2.1 Mbps, and paves the way for using TDRSS on all fifteen of our Dragon missions for the COTS and Commercial Resupply Services (CRS) programs.

COTS Flight 2 Rehearsals

Also in Hawthorne, we recently completed a very successful joint mission simulation with NASA's Mission Operations Directorate where the team rehearsed the operations that will be conducted during the second COTS flight (the third Falcon 9 launch).

During that mission, dubbed "C2", a Dragon Spacecraft will approach within 10 kilometers (6 miles) of the International Space Station, and check out navigation, communication and control systems in preparation for actual approach and berthing with the ISS.

Computer illustration showing a Dragon spacecraft approaching the ISS.

Image credit: SpaceX.

These tests help us progress towards the day when SpaceX will begin a series of twelve CRS cargo delivery missions for NASA to support the continued operation of the ISS.

Stay tuned for more Falcon 9 updates in the coming weeks as we head for launch in early 2010.

About this email: Musk opens 2010 by reporting the full-duration firing of the Falcon 9 second stage, reviews SpaceX's 2009 milestones, and pushes back on a Popular Science subheading by saying NASA is critical to SpaceX. Detailed hardware sections on the first Falcon 9, the Dragon qualification unit and the Cape Canaveral pad follow.

Notes: Attribution: the opening letter is written in the first person ('I disagree with the subheading', 'I would like to thank the whole SpaceX team') and the copy reproduced on Daily Kos on 8 Jan 2010 ends with the signature '-Elon-'; Daily Kos calls it 'a personal message from SpaceX founder and owner Elon Musk'. The erps.org mirror carries the complete update including the technical sections but drops the signature line; the signature is placed here where the Daily Kos copy has it, directly after the opening letter. The technical sections after the signature were probably written by SpaceX staff. 'Inaugural Falcon 9 / Dragon Flight Hardware Update' was the running page heading SpaceX used for this series (the unsigned 14 March 2010 update carries the same heading). Photo and video captions are kept as they appear in the mirror; the text was retrieved in four consecutive parts and stitched at section headings.

Sources: [Experimental Rocket Propulsion Society \(mirror of \[spacex.com/updates.php\]\(http://spacex.com/updates.php\)\)](#); [Daily Kos, 8 Jan 2010 \(reproduces the opening letter with the '-Elon-' signature\)](#); [Space.com forums, 'SpaceX updates' thread \(quotes the update from the SpaceX website\)](#)

Tags: spacex-update, falcon-9, dragon, cape-canaveral, nasa, year-in-review

Musk backs Obama's cancellation of Ares I/Orion

From: Elon Musk

Date: April 15, 2010 (Issued by SpaceX on 15 April 2010, the day of President Obama's space policy speech at Kennedy Space Center ('Today, the President will articulate...'); NASA Watch posted it that day and the SpaceDaily reprint is datelined Apr 16, 2010.)

To: SpaceX update mailing list

Subject: At Long Last, an Inspiring Future for Space Exploration

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

At Long Last, An Inspiring Future For Space Exploration

The Apollo Moon landing was one of humanity's greatest achievements. Millennia from now, when the vast majority of the 20th century is reduced to a few footnotes known only to erudite scholars of history, they will still remember that was when we first set foot upon a heavenly body. It was a mere 66 years after the first powered airplane flight by the Wright brothers.

In the 41 years that have passed since 1969, we have yet to surpass that achievement in human spaceflight. Since then, our capability has actually declined considerably and to a degree that would yield shocked disbelief from anyone in that era.

By now, we were supposed to have a base on the Moon, perhaps even on Mars, and have sent humans traveling on great odysseys to the outer planets. Instead, we have been confined to low Earth orbit and even that ends this year with the retirement of the Space Shuttle.

In 2003, following the Columbia accident, President Bush began development of a system to replace the Shuttle, called the Ares I rocket and Orion spacecraft. It is important to note that this too would only have been able to reach low Earth orbit. Many in the media mistakenly assumed it was capable of reaching the Moon. As is not unusual with large government programs, the schedule slipped by several years and costs ballooned by tens of billions.

By the time President Obama cancelled Ares I/Orion earlier this year, the schedule had already slipped five years to 2017 and completing development would have required another \$50 billion. Moreover, the cost per flight, inclusive of overhead, was estimated

to be at least \$1.5 billion compared to the \$1 billion of Shuttle, despite carrying only four people to Shuttle's seven and almost no cargo.

The President quite reasonably concluded that spending \$50 billion to develop a vehicle that would cost 50% more to operate, but carry 50% less payload was perhaps not the best possible use of funds.

To quote a member of the Augustine Commission, which was convened by the President to analyze Ares/Orion, "If Santa Claus brought us the system tomorrow, fully developed, and the budget didn't change, our next action would have to be to cancel it," because we can't afford the annual operating costs.

Cancellation was therefore simply a matter of time and thankfully we have a President with the political courage to do the right thing sooner rather than later. We can ill afford the expense of an "Apollo on steroids", as a former NASA Administrator referred to the Ares/Orion program.

A lesser President might have waited until after the upcoming election cycle, not caring that billions more dollars would be wasted. It was disappointing to see how many in Congress did not possess this courage. One senator in particular was determined to achieve a new altitude record in hypocrisy, claiming that the public option was bad in healthcare, but good in space!

Thankfully, as a result of funds freed up by this cancellation, there is now hope for a bright future in space exploration. The new plan is to harness our nation's unparalleled system of free enterprise (as we have done in all other modes of transport), to create far more reliable and affordable rockets.

Handing over Earth orbit transport to American commercial companies, overseen of course by NASA and the FAA, will free up the NASA resources necessary to develop interplanetary transport technologies. This is critically important if we are to reach Mars, the next giant leap in human exploration of the Universe.

Today, the President will articulate an ambitious and exciting new plan that will alter our destiny as a species. I believe this address could be as important as President Kennedy's 1962 speech at Rice University. For the first time since Apollo, our country will have a plan for space exploration that inspires and excites all who look to the stars. Even more important, it will work.

About this email: On the day of President Obama's Kennedy Space Center speech, Musk argues that cancelling Ares I/Orion was the right call and that handing Earth-orbit transport to commercial companies frees NASA for interplanetary exploration.

Notes: Attribution: SpaceDaily runs it under the byline 'by Elon Musk', datelined 'Hawthorne CA (SPX) Apr 16, 2010', with the footer 'This editorial is a public domain media release from SpaceX'; SpaceRef and NASA Watch carried it as 'Elon Musk: At Long Last, an Inspiring Future for Space Exploration'. This is an essay-style statement that SpaceX distributed as a media release; no retrieved source states that it also went to the update mailing list, and no signature line appears in the SpaceDaily copy. The SpaceDaily page opens with a pull-quote that repeats the fifth paragraph; that layout duplicate is not repeated here. The title line uses SpaceDaily's capitalisation.

Sources: [SpaceDaily](#); [NASA Watch](#), 15 Apr 2010, 'SpaceX Sees Something Big Ahead' (quotes the closing paragraph, links SpaceRef); [SpaceNews / SpaceRef press-release archive](#) (page exists but is disallowed by robots.txt for our fetcher); [steve.blog](#) (Steve Jurvetson), 16 Apr 2010, quoting the closing paragraph

Tags: spacex-update, space-policy, obama, constellation, commercial-crew, nasa

Acronyms Seriously Suck

From: Elon Musk

Date: May 2010 (Ashlee Vance dates the email to May 2010; exact day not given)

To: All SpaceX employees

Subject: Acronyms Seriously Suck

FULL TEXT · COMPANY-WIDE MEMO · BOOK/BIOGRAPHY · CONFIRMED

There is a creeping tendency to use made up acronyms at SpaceX. Excessive use of made up acronyms is a significant impediment to communication and keeping communication good as we grow is incredibly important. Individually, a few acronyms here and there may not seem so bad, but if a thousand people are making these up, over time the result will be a huge glossary that we have to issue to new employees. No one can actually remember all these acronyms and people don't want to seem dumb in a meeting, so they just sit there in ignorance. This is particularly tough on new employees. That needs to stop immediately or I will take drastic action—I have given enough warnings over the years. Unless an acronym is approved by me, it should not enter the SpaceX glossary. If there is an existing acronym that cannot reasonably be justified, it should be eliminated, as I have requested in the past. For example, there should be no "HTS" [horizontal test stand] or "VTS" [vertical test stand] designations for test stands. Those are particularly dumb, as they contain unnecessary words. A "stand" at our test site is obviously a *test* stand. VTS-3 is four syllables compared with "Tripod," which is two, so the bloody acronym version actually takes longer to say than the name! The key test for an acronym is to ask whether it helps or hurts communication. An acronym that most engineers outside of SpaceX already know, such as GUI, is fine to use. It is also ok to make up a few acronyms/contractions every now and again, assuming I have approved them, eg MVac and M9 instead of Merlin 1C-Vacuum or Merlin 1C-Sea Level, but those need to be kept to a minimum.

About this email: Musk bans made-up acronyms at SpaceX unless he personally approves them, arguing that they impede communication and leave new employees sitting in ignorance during meetings. He singles out 'HTS' and 'VTS' test-stand names as especially dumb.

Notes: Text as printed in Vance's biography (bracketed glosses are Vance's). The Goodreads transcription runs the email as one block, so original paragraph breaks are not preserved. Musk later confirmed the email on Twitter.

Sources: [Ashlee Vance, 'Elon Musk: Tesla, SpaceX, and the Quest for a Fantastic Future' \(2015\), via Goodreads quote page](#); [IBTimes \(2015\)](#); [Manner of Speaking](#); [Verdict](#)

Tags: communication, acronyms, culture, policy

Musk asks supporters to call Congress against H.R. 5781

From: Elon Musk

Date: July 29, 2010 (Posted by NASA Watch on 29 July 2010; the message says H.R. 5781 'will be debated ... tomorrow' and reach the floor 'this Friday' (30 July 2010).)

To: SpaceX update mailing list

Subject: Your Help Urgently Needed to Save the Future of Human Spaceflight

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · SINGLE-SOURCE

Your Help Urgently Needed to Save the Future of Human Spaceflight

If you care about the future of American space exploration, your urgent help is necessary. The only hope for the average citizen to one day travel to space is in danger due to the actions of certain members of Congress. SpaceX does not have the enormous lobbying power of the big government contractors to stop them, however with your help the day can still be saved.

NASA's Authorization bill (H.R. 5781) will be debated on the floor of the US House of Representatives tomorrow. Despite the imminent retirement of the Space Shuttle, H.R. 5781 authorizes over five times as many taxpayer dollars to fly NASA astronauts on the Russian Soyuz than it invests in developing an American commercial alternative, moreover at a time when jobs are sorely needed in the United States. Quite simply, this bill represents the sort of senseless pork politics that has driven our national debt to the point where our economy can barely service it.

The bill is expected to be brought to the House floor this Friday under a special "suspension of the rules," which is a procedure that limits debate and amendments.

Telephone your Congressional representative right away via the House Switchboard at (202) 225-3121 and ask them to vote NO on H.R. 5781, and instead support the bill unanimously agreed to in the Senate last week.

Your five minutes will make a critical difference, ensuring an exciting and inspiring future in space travel! SpaceX rarely asks you to take action, so you know it really matters when we do.

-Elon-

Look up your representative here: <http://www.house.gov/zip/ZIP2Rep.html>

About this email: Musk asks SpaceX's supporters to phone their House representatives and urge a NO vote on the House NASA authorization bill, H.R. 5781, which he says funds Russian Soyuz seats at five times the level of an American commercial alternative.

Notes: Attribution: signed '-Elon-'. NASA Watch reproduced it under the headline 'SpaceX Organizes NO on H.R. 5781 Effort', credited it to SpaceX and linked SpaceRef's status-report copy. A short call-to-action message to SpaceX's supporters rather than a hardware progress update. The 'Look up your representative' line follows the signature in the NASA Watch copy.

Sources: [NASA Watch](#); [SpaceRef status report linked by NASA Watch \(legacy pid URL; no longer resolves to the original\)](#)

Tags: spacex-update, space-policy, congress, commercial-crew, nasa-authorization, call-to-action

Musk lays out SpaceX's costs and prices

From: Elon Musk

Date: May 4, 2011 (Posted on spacex.com on 4 May 2011 according to the NSS reprint ('quoted in full from the SpaceX website, dated May 4, 2011').)

To: SpaceX update mailing list

Subject: Why the US Can Beat China: The Facts About SpaceX Costs

FULL TEXT · NEWSLETTER · PUBLISHED BY MUSK · WIDELY REPORTED

WHY THE US CAN BEAT CHINA: THE FACTS ABOUT SPACEX COSTS

Whenever someone proposes to do something that has never been done before, there will always be skeptics.

So when I started SpaceX, it was not surprising when people said we wouldn't succeed. But now that we've successfully proven Falcon 1, Falcon 9 and Dragon, there's been a steady stream of misinformation and doubt expressed about SpaceX's actual launch costs and prices.

As noted last month by a Chinese government official, "SpaceX currently has the best launch prices in the world and they don't believe they can beat them." This is a clear case of American innovation trumping lower overseas labor rates.

I recognize that our prices shatter the historical cost models of government-led developments, but these prices are not arbitrary, premised on capturing a dominant share of the market, or "teaser" rates meant to lure in an eager market only to be increased later. These prices are based on known costs and a demonstrated track record, and they exemplify the potential of America's commercial space industry.

Here are the facts:

The price of a standard flight on a Falcon 9 rocket is \$54 million. We are the only launch company that publicly posts this information on our website (www.spacex.com). We have signed many legally binding contracts with both government and commercial customers for this price (or less). Because SpaceX is so vertically integrated, we know and can control the overwhelming majority of our costs. This is why I am so confident that our performance will increase and our prices will decline over time, as is the case with every other technology.

The average price of a full-up NASA Dragon cargo mission to the International Space Station is \$133 million including inflation, or roughly \$115m in today's dollars, and we have a firm, fixed price contract with NASA for 12 missions. This price includes the costs of the Falcon 9 launch, the Dragon spacecraft, all operations, maintenance and overhead, and all of the work required to integrate with the Space Station. If there are cost overruns, SpaceX will cover the difference. (This concept may be foreign to some traditional government space contractors that seem to believe that cost overruns should be the responsibility of the taxpayer.)

The total company expenditures since being founded in 2002 through the 2010 fiscal year were less than \$800 million, which includes all the development costs for the

Falcon 1, Falcon 9 and Dragon. Included in this \$800 million are the costs of building launch sites at Vandenberg, Cape Canaveral and Kwajalein, as well as the corporate manufacturing facility that can support up to 12 Falcon 9 and Dragon missions per year. This total also includes the cost of five flights of Falcon 1, two flights of Falcon 9, and one up and back flight of Dragon.

The Falcon 9 launch vehicle was developed from a blank sheet to first launch in four and half years for just over \$300 million. The Falcon 9 is an EELV class vehicle that generates roughly one million pounds of thrust (four times the maximum thrust of a Boeing 747) and carries more payload to orbit than a Delta IV Medium.

The Dragon spacecraft was developed from a blank sheet to the first demonstration flight in just over four years for about \$300 million. Last year, SpaceX became the first private company, in partnership with NASA, to successfully orbit and recover a spacecraft. The spacecraft and the Falcon 9 rocket that carried it were designed, manufactured and launched by American workers for an American company. The Falcon 9/Dragon system, with the addition of a launch escape system, seats and upgraded life support, can carry seven astronauts to orbit, more than double the capacity of the Russian Soyuz, but at less than a third of the price per seat.

SpaceX has been profitable every year since 2007, despite dramatic employee growth and major infrastructure and operations investments. We have over 40 flights on manifest representing over \$3 billion in revenues.

These are the objective facts, confirmed by external auditors. Moreover, SpaceX intends to make far more dramatic reductions in price in the long term when full launch vehicle reusability is achieved. We will not be satisfied with our progress until we have achieved this long sought goal of the space industry.

For the first time in more than three decades, America last year began taking back international market-share in commercial satellite launch. This remarkable turn-around was sparked by a small investment NASA made in SpaceX in 2006 as part of the Commercial Orbital Transportation Services (COTS) program. A unique public-private partnership, COTS has proven that under the right conditions, a properly incentivized contractor — even an all-American one — can develop extremely complex systems on rapid timelines and a fixed-price basis, significantly beating historical industry-standard costs.

China has the fastest growing economy in the world. But the American free enterprise system, which allows anyone with a better mouse-trap to compete, is what will ensure that the United States remains the world's greatest superpower of innovation.

-Elon-

About this email: Responding to doubts about SpaceX's pricing, Musk lists the company's audited figures: a \$54 million Falcon 9, \$133 million Dragon cargo missions, under \$800 million spent in total since 2002, and profitability every year since 2007.

Notes: Attribution: signed '-Elon-' and written in the first person ('when I started SpaceX'). NSS states it is 'quoted in full from the SpaceX website, dated May 4, 2011'. In the original the opening sentence of each 'fact' paragraph is in bold. Falls outside 2009-2010 but continues the Musk-signed update series; it is the only Musk-signed update found for 2011-2012.

Sources: [National Space Society](#); [NASA Watch](#), 4 May 2011, 'SpaceX Explains Why They Beat China on Cost' (quotes the opening paragraphs); [Universe Today](#), 4 May 2011; [Via Satellite](#), 5 May 2011 (analysis of the statement)

Tags: spacex-update, launch-costs, falcon-9, dragon, china, cots

Why SpaceX won't go public until Mars transport is flying

From: Elon Musk

Date: June 7, 2013 (Header as published: June 7, 2013, 12:43:06 AM PDT)

To: All (spacex-employees@spacex.com)

Subject: Going Public

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Per my recent comments, I am increasingly concerned about SpaceX going public before the Mars transport system is in place. Creating the technology needed to establish life on Mars is and always has been the fundamental goal of SpaceX. If being a public company diminishes that likelihood, then we should not do so until Mars is secure. This is something that I am open to reconsidering, but, given my experiences with Tesla and SolarCity, I am hesitant to foist being public on SpaceX, especially given the long term nature of our mission.

Some at SpaceX who have not been through a public company experience may think that being public is desirable. This is not so. Public company stocks, particularly if big step changes in technology are involved, go through extreme volatility, both for reasons of internal execution and for reasons that have nothing to do with anything except the economy. This causes people to be distracted by the manic-depressive nature of the stock instead of creating great products.

It is important to emphasize that Tesla and SolarCity are public because they didn't have any choice. Their private capital structure was becoming unwieldy and they needed to raise a lot of equity capital. SolarCity also needed to raise a huge amount of debt at the lowest possible interest rate to fund solar leases. The banks who provide that debt wanted SolarCity to have the additional painful scrutiny that comes with being public. Those rules, referred to as Sarbanes-Oxley, essentially result in a tax being levied on company execution by requiring detailed reporting right down to how your meal is expensed during travel and you can be penalized even for minor mistakes.

YES, BUT I COULD MAKE MORE MONEY IF WE WERE PUBLIC

For those who are under the impression that they are so clever that they can outsmart public market investors and would sell SpaceX stock at the "right time," let me relieve you of any such notion. If you really are better than most hedge fund managers, then there is no need to worry about the value of your SpaceX stock, as you can just invest in other public stocks and make billions of dollars in the market.

If you think: "Ah, but I know what's really going on at SpaceX and that will give me an edge," you are also wrong. Selling public company stock with insider knowledge is illegal. As a result, selling public stock is restricted to narrow time windows a few times per year. Even then, you can be prosecuted for insider trading. At Tesla, we had both an

employee and an investor go through a grand jury investigation for selling stock over a year ago, despite them doing everything right in both the letter and the spirit of the law. Not fun.

Another thing that happens to public companies is that you become a target of the trial lawyers who create a class action lawsuit by getting someone to buy a few hundred shares and then pretending to sue the company on behalf of all investors for any drop in the stock price. Tesla is going through that right now even though the stock price is relatively high, because the drop in question occurred last year.

It is also not correct to think that because Tesla and SolarCity share prices are on the lofty side right now, that SpaceX would be too. Public companies are judged on quarterly performance. Just because some companies are doing well, doesn't mean that all would. Both of those companies (Tesla in particular) had great first quarter results. SpaceX did not. In fact, financially speaking, we had an awful first quarter. If we were public, the short sellers would be hitting us over the head with a large stick.

We would also get beaten up every time there was an anomaly on the rocket or spacecraft, as occurred on flight 4 with the engine failure and flight 5 with the Dragon prelaunches. Delaying launch of V1.1, which is now over a year behind schedule, would result in particularly severe punishment, as that is our primary revenue driver. Even something as minor as pushing a launch back a few weeks from one quarter to the next gets you a spanking. Tesla vehicle production in Q4 last year was literally only three weeks behind and yet the market response was brutal.

BEST OF BOTH WORLDS

My goal at SpaceX is to give you the best aspects of a public and private company. When we do a financing round, the stock price is keyed off of approximately what we would be worth if publicly traded, excluding irrational exuberance or depression, but without the pressure and distraction of being under a hot public spotlight. Rather than have the stock up during one liquidity window and down during another, the goal is a steady upward trend and never to let the share price go below the last round. The end result for you (or an investor in SpaceX) financially will be the same as if we were public and you sold a steady amount of stock every year.

In case you are wondering about a specific number, I can say that I'm confident that our long term stock price will be over \$100 if we execute well on Falcon 9 and Dragon. For this to be the case, we must have a steady and rapid cadence of launch that is far better than what we have achieved in the past. We have more work ahead of us than you probably realize. Let me give you a sense of where things stand financially: SpaceX expenses this year will be roughly \$800 to \$900 million (which blows my mind btw). Since we get revenue of \$60M for every F9 flight or double that for a FH or F9-Dragon flight, we must have about twelve flights per year where four of those flights are either Dragon or Heavy merely in order to achieve 10% profitability!

For the next few years, we have NASA commercial crew funding that helps supplement those numbers, but, after that, we are on our own. That is not much time to finish F9, FH, Dragon V2 and achieve an average launch rate of at least one per month. And bear

in mind that is an average, so if we take an extra three weeks to launch a rocket for any reason (could even be due to the satellite), we have only one week to do the follow-on-flight.

MY RECOMMENDATION

Below is my advice about regarding selling SpaceX stock or options. No complicated analysis is required, as the rules of thumb are pretty simple. If you believe that SpaceX will execute better than the average public company, then our stock price will continue to appreciate at a rate greater than that of the stock market, which would be the next highest return place to invest money over the long term. Therefore, you should sell only the amount that you need to improve your standard of living in the short to medium term. I do actually recommend selling some amount of stock, even if you are certain it will appreciate, as life is short and a bit more cash can increase fun and reduce stress at home (so long as you don't ratchet up your ongoing personal expenditures proportionately).

To maximize your post tax return, you are probably best off exercising your options to convert them to stock (if you can afford to do this) and then holding the stock for a year before selling it at our roughly biannual liquidity events. This allows you to pay the capital gains tax rate, instead of the income tax rate.

On a final note, we are planning to do a liquidity event as soon as Falcon 9 qualification is complete in one to two months. I don't know exactly what the share price will be yet, but, based on initial conversations with investors, I would estimate probably between \$30 and \$35. This places the value of SpaceX at \$4 to \$5 billion, which is about what it would be if we were public right now and, frankly, an excellent number considering that the new F9, FH and Dragon V2 have yet to launch.

Elon

About this email: Musk tells all SpaceX employees he does not want the company public until the Mars transport system is in place, describing the distractions, volatility and litigation that Tesla and SolarCity faced as public companies. He explains the twice-yearly liquidity events, shares financial figures, and advises on selling stock and options.

Notes: Leaked within days of sending in June 2013; later reproduced in an appendix of Ashlee Vance's biography.

Sources: [ValueWalk \(reprint of the email\)](#); [NextBigFuture \(June 2013\)](#); [Wccfttech](#)

Tags: ipo, stock, mars, liquidity, finance

Blue Origin's 39A protest: 'unicorns dancing in the flame duct'

From: Elon Musk

Date: September 2013 (Sent 'Tuesday night' in response to a SpaceNews query during the September 2013 dispute over the Launch Complex 39A lease; exact day not shown on the retrieved page)

To: SpaceNews (reporter query)

FULL TEXT · PRESS & JOURNALISTS · PUBLISHED BY RECIPIENT · CONFIRMED

From a SpaceX standpoint, we view [Blue Origin] and [United Launch Alliance's] action as a phony blocking tactic and an obvious one at that. BO has not yet succeeded in creating a reliable suborbital spacecraft, despite spending over 10 years in development. It is therefore unlikely that they will succeed in developing an orbital vehicle that will meet NASA's exacting standards in the next 5 years, which is the length of the lease. That said, I can't say for sure whether [Blue Origin's] action stems from malice. No such doubt exists about ULA's motivation.

However, rather than fight this issue, there is an easy way to determine the truth, which is simply to call their bluff. If they do somehow show up in the next 5 years with a vehicle qualified to NASA's human rating standards that can dock with the Space Station, which is what Pad 39A is meant to do, we will gladly accommodate their needs. Frankly, I think we are more likely to discover unicorns dancing in the flame duct.

About this email: Replying to a SpaceNews query about Blue Origin and ULA's objections to SpaceX leasing Launch Complex 39A, Musk calls the move a phony blocking tactic, questions Blue Origin's ability to field an orbital vehicle within five years, and offers to accommodate them if they ever show up with one.

Notes: SpaceNews states 'Musk's email is reproduced below in its entirety.' Bracketed company names are SpaceNews's editorial substitutions as printed.

Sources: [SpaceNews](#)

Tags: blue-origin, ula, launch-pad-39a, nasa, competition

Musk asks Stanton about the cave's geometry

From: Elon Musk

Date: July 6, 2018 (approximate within July 6-8, 2018; thread mixes Pacific and Thailand timestamps)

To: Richard Stanton

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

Are you one of the divers who understands the cave geometry? Trying to help out, but need to know the details of the most problematic areas.

About this email: Musk's opening email to British cave diver Richard Stanton during the Tham Luang rescue, asking about the most problematic sections of the cave.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

Stanton: will review Musk's rescue concepts later

From: Richard Stanton

Date: July 6, 2018 (approximate within July 6-8, 2018; thread mixes Pacific and Thailand timestamps)

To: Elon Musk

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

Thanks for the chat just now, we're not going to be available for any communication for a short while. Will review any ideas or concepts you come up with and get back to you later.

About this email: Stanton thanks Musk for a phone call and says the divers will be out of contact for a while but will review his ideas.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018. The exhibit shows this message at Jul 6, 2018, 10:29 PM.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

Musk asks Stanton for video of hardest cave sections

From: Elon Musk

Date: July 7, 2018 (approximate within July 6-8, 2018; thread mixes Pacific and Thailand timestamps)

To: Richard Stanton

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

Would it be possible to send me a video segments or several pictures of the most difficult sections? This would be extremely helpful.

About this email: Musk asks the dive team for video or photos of the most difficult passages to guide the design of the rescue capsule.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

Stanton: a capsule that tightly fits a 15-year-old will pass

From: Richard Stanton

Date: July 7, 2018 (approximate within July 6-8, 2018; thread mixes Pacific and Thailand timestamps)

To: Elon Musk

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

We don't have this, it won't mean anything without scale. If you make a capsule which tightly encloses a 15 year old boy, and no bigger, it will fit through.

About this email: Stanton says no video is available and that a capsule tightly enclosing a 15-year-old boy would fit through the cave.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

Musk: building both an aluminium tube and rubber cocoon

From: Elon Musk

Date: July 7, 2018 (approximate within July 6-8, 2018; thread mixes Pacific and Thailand timestamps)

To: Richard Stanton

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

Ok

Building both a pressure resistant aluminium tube and a dry rubber cocoon. Baseline requirement is that whether the person is passed out or having a panic attack, they will still be fine. Also, no stray hoses that can get caught on anything or pulled out. Basically a straightjacket with an air feed. Is this roughly the right direction?

About this email: Musk describes the two designs his engineers are building and the baseline requirement that an unconscious or panicking child would still be safe.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018. One reading of the exhibit shows a further stand-alone 'Ok' from Musk after Stanton's reply about the rubber cocoon.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

Stanton: rigid capsule better than rubber cocoon

From: Richard Stanton

Date: July 7, 2018 (approximate within July 6-8, 2018; thread mixes Pacific and Thailand timestamps)

To: Elon Musk

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

I'm not sure about the rubber cocoon at it would transmit pressure onto the child. Rigid is better, plus O2 as a breathing gas rather than air.

About this email: Stanton advises against the rubber cocoon because it would transmit pressure to the child, and recommends oxygen as the breathing gas.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

Stanton: capsule only needs 5m depth rating

From: Richard Stanton

Date: July 7, 2018 (approximate within July 6-8, 2018; thread mixes Pacific and Thailand timestamps)

To: Elon Musk

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

It only needs to go to a water depth of 5m, plus some safety margin.

About this email: Stanton gives the depth requirement for the rescue capsule.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

Musk sends Stanton the mini-sub's specifications

From: Elon Musk

Date: July 7, 2018 (approximate within July 6-8, 2018; thread mixes Pacific and Thailand timestamps)

To: Richard Stanton

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

Will have more info for you soon. It is 35 cm in diameter and 160cm long. We can make more that are longer or shorter. Dual air ports on front and rear (so possible to attach up to four tanks at once). Front ports are recessed into metal, so not subject to front damage. All ports have secondary caps in case a leak develops. With the front caps on, this is strong enough to be used as a battering ram if need be. Nose is 2.5cm thick solid, high toughness 2219 alloy aluminium. It is made from very high strength rocket-grade aluminium. Cylinder section is what the Falcon rocket uses for high pressure liquid oxygen transfer. Weighs 35kg. Will send pics in a few hours. Anything else that could be helpful? I am talking to the Prime Minister later today. Is there anything you would like me to convey?

About this email: Musk gives the dimensions, materials and air-port layout of the aluminium 'mini-sub' built from Falcon rocket parts, and asks if Stanton wants anything conveyed to the Thai prime minister.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018. Paragraph breaks are not preserved in the retrieved transcription.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

Stanton: 'all I see is a tube'

From: Richard Stanton

Date: July 7, 2018 (approximate within July 6-8, 2018; thread mixes Pacific and Thailand timestamps)

To: Elon Musk

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

With respect all I see is a tube, albeit made of fancy materials. We're ferrying in food in such a thing. The devil is in the detail. Breathing systems, off board gas venting systems, ballast trimming arrangements etc adding additional gas via Quick Release SCUBA compatible fittings. Why not have a long high pressure cylinder between the legs? Or a stepped tube with wider body and thinner leg section.

About this email: Stanton tells Musk the capsule design lacks detail on breathing, gas venting and ballast systems and suggests alternative layouts.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

Musk to Stanton: 'Please do not be rude'

From: Elon Musk

Date: July 7, 2018 (approximate within July 6-8, 2018; thread mixes Pacific and Thailand timestamps)

To: Richard Stanton

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

With respect, I am trying to be helpful. Please do not be rude.

About this email: Musk's curt reply to Stanton's criticism of the capsule design.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

Stanton apologises for tone, backs tube concept

From: Richard Stanton

Date: July 7, 2018 (approximate within July 6-8, 2018; thread mixes Pacific and Thailand timestamps)

To: Elon Musk

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

Apologies, as the tone of that didn't come across well. We are all still very much on board with the concept of a tube for safe evacuation. Though there does need to be a much more thorough investigation of all the matters mentioned in my previous mail. I'm not going to be available for the rest of the day

About this email: Stanton apologises and says the dive team supports the tube concept but the engineering details need more investigation.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

Musk: engineers working 24 hours a day on capsule

From: Elon Musk

Date: July 8, 2018 (approximate within July 6-8, 2018; thread mixes Pacific and Thailand timestamps)

To: Richard Stanton

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

Understandable. Right now, I have one the world's best engineering teams who normally design spaceships and spacesuits working on this thing 24 hours a day. We are trying to get it right in a very short period of time. If it isn't needed or won't help, that would be great to know. Otherwise, it would be very helpful to have as much design direction as possible.

About this email: Musk says his spacecraft engineers are working around the clock and asks Stanton to say if the device is not needed, or otherwise to give more design direction.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018. 'one the world's best' is as written.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

Stanton: 'absolutely worth continuing' with the capsule

From: Richard Stanton

Date: July 8, 2018 (approximate within July 6-8, 2018; thread mixes Pacific and Thailand timestamps)

To: Elon Musk

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

It is absolutely worth continuing with the development of this system in as timely a manner as feasible. If the rain holds out it may well be used.

About this email: Stanton encourages Musk to keep developing the capsule, saying it may well be used if the rain holds off. Musk later cited this email publicly to rebut claims the mini-sub was impractical.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

Musk: capsule parts being assembled and tested

From: Elon Musk

Date: July 8, 2018 (approximate within July 6-8, 2018; thread mixes Pacific and Thailand timestamps)

To: Richard Stanton

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

Sounds good, will continue. Parts are being assembled and will undergo testing in water in a few hours. Will send pics and video. However, don't want to put it on a plane if you think there are important changes needed. Operating principle is same as spacecraft design - no loss of life even with two failures.

About this email: Musk tells Stanton the capsule will be water-tested within hours and that it follows the spacecraft principle of tolerating two failures.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

Stanton: 'worried about the smallest lad'

From: Richard Stanton

Date: July 8, 2018 (exhibit header: July 8, 2018 at 10:13:12 AM PDT)

To: Elon Musk

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

We're worried about the smallest lad please keep working on the capsule details

About this email: Stanton's last message in the chain urges Musk to keep working on the capsule because of concern for the smallest boy.

Notes: Musk Decl. Ex. C (email chain with British cave diver Richard Stanton, July 6-8, 2018). The exhibit prints the chain newest-first with timestamps from two mail clients (Pacific time and Thailand time), so the exact send time of individual messages is uncertain; the date is a best reading. Transcribed from a scanned exhibit; minor OCR variance possible. Musk posted screenshots of part of this chain on Twitter on July 10, 2018.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [Business Insider via Yahoo Finance \(Musk tweeted the thread\)](#); [Dezeen](#)

Tags: thai-cave-rescue, mini-sub, richard-stanton, unsworth-case

COVID-19 less risky than driving home

From: Elon Musk

Date: March 13, 2020 (BuzzFeed News: sent 'in the early hours of Friday morning' (Friday, March 13, 2020))

To: All SpaceX employees

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

The risk of death from C19 is **vastly** less than the risk of death from driving your car home.

[...]

suggests that this is **not** within the top 100 health risks in the United States.

[...]

There are about 36 thousand automotive deaths per deaths [sic], as compared to 36 so far this year for C19.

[...]

Isn't C19 growing so fast that it will soon become a top 100 health risk for people who are otherwise healthy and young to middle-aged?

[...]

The trends do not support this conclusion.

[...]

The 36 deaths were almost all in high risk groups, particularly older people who already had prior lung damage, diabetes or heart disease.

[...]

As said above, if you're feeling ill, especially with something contagious, it's always better to stay home and take care of yourself.

About this email: At the start of the pandemic Musk told SpaceX employees that the risk of dying from COVID-19 was vastly lower than the risk of dying in a car crash and that the virus was not among the top 100 U.S. health risks, while advising sick employees to stay home.

Notes: Email seen by BuzzFeed News; only quoted passages were published. '[sic]' is BuzzFeed's.

Sources: [BuzzFeed News](#); [Gulf Today](#); [Inverse](#)

Tags: covid-19, health, risk, workplace

Starship is the top SpaceX priority

From: Elon Musk

Date: June 6, 2020 (CNBC: 'writing Saturday in a company-wide email seen by CNBC' (Saturday June 6, 2020; article published June 7))

To: All SpaceX employees

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

dramatically and immediately

[...]

Please consider the top SpaceX priority (apart from anything that could reduce Dragon return risk) to be Starship

[...]

We need to accelerate Starship progress

[...]

For those considering moving, we will always offer a dedicated SpaceX aircraft to shuttle people

[...]

consider spending significant time

About this email: A week after SpaceX's first crewed Dragon launch, Musk told employees that Starship was now the company's top priority apart from Dragon return risk, urged them to accelerate progress 'dramatically and immediately' and to consider spending significant time at Boca Chica.

Notes: REPLACES: 2020-06-06 Starship is the top SpaceX priority. Adds one more quoted fragment ('dramatically and immediately') and lists fragments in the order CNBC printed them; CNBC says the last fragment asked employees to consider spending significant time in Boca Chica. Still only fragments: the full text was never published.

Sources: [CNBC](#); [Space Explored](#); [TechCrunch](#)

Tags: starship, priorities, boca-chica, urgency

Starship surge: 'fly, drive, or get here by any means possible'

From: Elon Musk

Date: 2021 (Isaacson places the email in his 'Starship Surge' chapter (summer 2021, ahead of stacking the first full Starship); the retrieved highlight page gives no exact date)

To: All SpaceX employees

EXCERPT · COMPANY-WIDE MEMO · BOOK/BIOGRAPHY · SINGLE-SOURCE

Anyone who is not working on other obviously critical path projects at SpaceX should shift immediately to work on the first Starship orbit. Please fly, drive, or get here by any means possible.

About this email: Musk ordered a surge at Starbase, telling everyone at SpaceX not on another critical-path project to shift immediately to the first Starship orbital attempt and get to Boca Chica by any means.

Notes: Quoted in Isaacson's biography; text retrieved from a Kindle highlight page, so only the highlighted sentences are available. Month not verified against a retrieved page.

Sources: [Walter Isaacson, 'Elon Musk' \(2023\), via Goodreads reader highlight](#)

Tags: starship, starbase, surge, urgency

Raptor production crisis and 'genuine risk of bankruptcy'

From: Elon Musk

Date: November 26, 2021 (Sent the day after Thanksgiving 2021 (Friday Nov 26); first reported by Space Explored on Nov 29 and CNBC on Nov 30)

To: All SpaceX employees

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Unfortunately, the Raptor production crisis is much worse than it had seemed a few weeks ago. As we have dug into the issues following the exiting of prior senior management, they have unfortunately turned out to be far more severe than was reported. There is no way to sugarcoat this.

I was going to take this weekend off, as my first weekend off in a long time, but instead, I will be on the Raptor line all night and through the weekend.

Unless you have critical family matters or cannot physically return to Hawthorne, we will need all hands on deck to recover from what is, quite frankly, a disaster.

The consequences for SpaceX if we can not get enough reliable Raptors made is that we then can't fly Starship, which means we then can't fly Starlink Satellite V2 (Falcon has neither the volume nor the mass to orbit needed for satellite V2). Satellite V1, by itself, is financially weak, while V2 is strong.

In addition, we are spooling up terminal production to several million units per year, which will consume massive capital, assuming that satellite V2 will be on orbit to handle the bandwidth demand. These terminals will be useless otherwise.

What it comes down to, is that we face a genuine risk of bankruptcy if we can't achieve a Starship flight rate of at least once every two weeks next year.

Thanks,

Elon

About this email: Musk tells SpaceX staff that the Raptor engine production problems are far worse than reported after senior management left, cancels his weekend off and calls for all hands on deck. He warns that without a Starship flight rate of once every two weeks in 2022, Starlink V2 cannot fly and the company faces genuine bankruptcy risk.

Notes: Musk publicly followed up on Twitter after the leak, saying bankruptcy was unlikely but not impossible.

Sources: [Space Explored](#); [CNBC](#); [Space.com](#); [Fortune](#)

Tags: raptor, starship, starlink, bankruptcy, urgency, holidays

SpaceX return-to-office: 40 hours a week in the office

From: Elon Musk

Date: May 31, 2022 (Sent Tuesday night, May 31, 2022, per reports of June 1, 2022 citing The New York Times)

To: SpaceX employees

EXCERPT • COMPANY-WIDE MEMO • LEAKED TO PRESS • WIDELY REPORTED

The more senior you are, the more visible must be your presence. That is why I spent so much time in the factory — so that those on the line could see me working alongside them.

About this email: In a memo paralleling the one sent to Tesla staff the same night, Musk told SpaceX employees they must spend at least 40 hours a week in the office and said senior staff in particular need to be visibly present.

Notes: The SpaceX version was obtained by The New York Times (not retrievable in this session); only this passage of the SpaceX email was quoted in accessible coverage. The fuller Tesla emails of the same night ('Remote work is no longer acceptable', 'To be super clear') belong to the Tesla slice.

Sources: [Space.com](#) (citing [The New York Times](#)); [Scripps/KENS5](#); [TechXplore/AFP](#)

Tags: remote-work, return-to-office, policy, work-ethic

SpaceX acquires xAI: 'the next book'

From: Elon Musk

Date: February 2, 2026 (Memo to staff posted on SpaceX's website Monday Feb 2, 2026)

To: SpaceX and xAI employees (also posted publicly on spacex.com)

FULL TEXT · COMPANY-WIDE MEMO · PUBLISHED BY MUSK · CONFIRMED

SpaceX has acquired xAI to form the most ambitious, vertically-integrated innovation engine on (and off) Earth, with AI, rockets, space-based internet, direct-to-mobile device communications and the world's foremost real-time information and free speech platform. This marks not just the next chapter, but the next book in SpaceX and xAI's mission: scaling to make a sentient sun to understand the Universe and extend the light of consciousness to the stars!

Current advances in AI are dependent on large terrestrial data centers, which require immense amounts of power and cooling. Global electricity demand for AI simply cannot be met with terrestrial solutions, even in the near term, without imposing hardship on communities and the environment.

In the long term, space-based AI is obviously the only way to scale. To harness even a millionth of our Sun's energy would require over a million times more energy than our civilization currently uses!

The only logical solution therefore is to transport these resource-intensive efforts to a location with vast power and space. I mean, space is called "space" for a reason. By directly harnessing near-constant solar power with little operating or maintenance costs, these satellites will transform our ability to scale compute. It's always sunny in space! Launching a constellation of a million satellites that operate as orbital data centers is a first step towards becoming a Kardashev II-level civilization, one that can harness the Sun's full power, while supporting AI-driven applications for billions of people today and ensuring humanity's multi-planetary future.

In the history of spaceflight, there has never been a vehicle capable of launching the megatons of mass that space-based data centers or permanent bases on the Moon and cities on Mars require. Even in 2025, the most prolific year in history in terms of the number of orbital launches, only about 3000 tons of payload was launched into orbit, primarily consisting of Starlink satellites carried by our Falcon rocket.

The requirement to launch thousands of satellites to orbit became a forcing function for the Falcon program, driving recursive improvements to reach the unprecedented flight rates necessary to make space-based internet a reality. This year, Starship will begin delivering the much more powerful V3 Starlink satellites to orbit, with each launch adding more than 20 times the capacity to the constellation as the current Falcon launches of the V2 Starlink satellites. Starship will also launch the next generation of direct-to-mobile satellites, which will deliver full cellular coverage everywhere on Earth.

While the need to launch these satellites will act as a similar forcing function to drive Starship improvements and launch rates, the sheer number of satellites that will be needed for space-based data centers will push Starship to even greater heights. With

launches every hour carrying 200 tons per flight, Starship will deliver millions of tons to orbit and beyond per year, enabling an exciting future where humanity is out exploring amongst the stars.

The basic math is that launching a million tons per year of satellites generating 100 kW of compute power per ton would add 100 gigawatts of AI compute capacity annually, with no ongoing operational or maintenance needs. Ultimately, there is a path to launching 1 TW/year from Earth.

My estimate is that within 2 to 3 years, the lowest cost way to generate AI compute will be in space. This cost-efficiency alone will enable innovative companies to forge ahead in training their AI models and processing data at unprecedented speeds and scales, accelerating breakthroughs in our understanding of physics and invention of technologies to benefit humanity.

This new constellation will build upon the well-established space sustainability design and operational strategies, including end-of-life disposal, that have proven successful for SpaceX's existing broadband satellite systems.

While launching AI satellites from Earth is the immediate focus, Starship's capabilities will also enable operations on other worlds. Thanks to advancements like in-space propellant transfer, Starship will be capable of landing massive amounts of cargo on the Moon. Once there, it will be possible to establish a permanent presence for scientific and manufacturing pursuits. Factories on the Moon can take advantage of lunar resources to manufacture satellites and deploy them further into space. By using an electromagnetic mass driver and lunar manufacturing, it is possible to put 500 to 1000 TW/year of AI satellites into deep space, meaningfully ascend the Kardashev scale and harness a non-trivial percentage of the Sun's power.

The capabilities we unlock by making space-based data centers a reality will fund and enable self-growing bases on the Moon, an entire civilization on Mars and ultimately expansion to the Universe.

Thank you for everything you have done and will do for the light cone of consciousness.

Ad Astra!

Elon

About this email: Musk announces that SpaceX has acquired xAI and lays out the rationale: terrestrial power cannot meet AI demand, so SpaceX will launch up to a million satellites as orbital data centers using Starship. He predicts space will be the cheapest place for AI compute within two to three years and sketches lunar manufacturing and Mars expansion.

Notes: Text retrieved in two passes from the AOL/Business Insider reprint (first four paragraphs, then the remainder); the SpaceX page itself is JavaScript-rendered and could not be read directly. Section sub-headings, if any in the original posting, were not returned. Described by press as a memo to employees that SpaceX posted publicly.

Sources: [SpaceX website \(spacex.com/updates\)](https://spacex.com/updates), as reproduced by [Business Insider/AOL](#); [SpaceX updates page](#); [Bloomberg](#); [Techstrong](#)

Tags: xai, merger, orbital-data-centers, starship, ai, kardashev

OpenAI

The founding-era and breakup correspondence, mostly from exhibits in Musk v. OpenAI. Includes replies from Altman, Sutskever, Brockman and others.

130 emails, May 25, 2015 to March 11, 2019

Altman proposes a 'Manhattan Project for AI'

From: Sam Altman

Date: May 25, 2015 (Sent 9:10 PM)

To: Elon Musk

Subject: question

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Been thinking a lot about whether it's possible to stop humanity from developing AI.

I think the answer is almost definitely not.

If it's going to happen anyway, it seems like it would be good for someone other than Google to do it first.

Any thoughts on whether it would be good for YC to start a Manhattan Project for AI? My sense is we could get many of the top ~50 to work on it, and we could structure it so that the tech belongs to the world via some sort of nonprofit but the people working on it get startup-like compensation if it works. Obviously we'd comply with/aggressively support all regulation.

Sam

About this email: Altman tells Musk that stopping AI development is almost certainly impossible and floats YC starting a 'Manhattan Project for AI' whose technology would belong to the world via a nonprofit.

Notes: First exhibited with Musk's 2024 complaint against OpenAI.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits](#), as transcribed in LessWrong's 'OpenAI Email Archives'; [Dkt. 32-2 \(Exhibit 1 to First Amended Complaint\)](#); [CourtListener docket](#), Musk v. Altman

Tags: founding, nonprofit, google, y-combinator, musk-v-openai

Musk: 'Probably worth a conversation'

From: Elon Musk

Date: May 25, 2015 (Sent 11:09 PM)

To: Sam Altman

Subject: Re: question

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Probably worth a conversation

About this email: Musk's one-line reply to Altman's proposal for a YC-backed 'Manhattan Project for AI'.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits](#), as transcribed in LessWrong's 'OpenAI Email Archives'; [Dkt. 32-2 \(Exhibit 1 to First Amended Complaint\)](#); [CourtListener docket](#), Musk v. Altman

Tags: founding, nonprofit, musk-v-openai

Altman's detailed plan: mission, team, governance

From: Sam Altman

Date: June 24, 2015 (Sent 10:24 AM)

To: Elon Musk

Subject: AI lab

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

- 1) The mission would be to create the first general AI and use it for individual empowerment—ie, the distributed version of the future that seems the safest. More generally, safety should be a first-class requirement.
- 2) I think we'd ideally start with a group of 7-10 people, and plan to expand from there. We have a nice extra building in Mountain View they can have.
- 3) I think for a governance structure, we should start with 5 people and I'd propose you, Bill Gates, Pierre Omidyar, Dustin Moskovitz, and me. The technology would be owned by the foundation and used "for the good of the world", and in cases where it's not obvious how that should be applied the 5 of us would decide. The researchers would have significant financial upside but it would be uncorrelated to what they build, which should eliminate some of the conflict (we'll pay them a competitive salary and give them YC equity for the upside). We'd have an ongoing conversation about what work should be open-sourced and what shouldn't. At some point we'd get someone to run the team, but he/she probably shouldn't be on the governance board.
- 4) Will you be involved somehow in addition to just governance? I think that would be really helpful for getting work pointed in the right direction getting the best people to be part of it. Ideally you'd come by and talk to them about progress once a month or whatever. We generically call people involved in some limited way in YC "part-time partners" (we do that with Peter Thiel for example, though at this point he's very involved) but we could call it whatever you want. Even if you can't really spend time on it but can be publicly supportive, that would still probably be really helpful for recruiting.
- 5) I think the right plan with the regulation letter is to wait for this to get going and then I can just release it with a message like "now that we are doing this, I've been thinking a lot about what sort of constraints the world needs for safety." I'm happy to leave you off as a signatory. I also suspect that after it's out more people will be willing to get behind it.

Sam

About this email: Altman lays out the proposed lab: a mission to build the first general AI for individual empowerment, a starting team of 7-10, a five-person governance board including Musk, and a plan for a regulation letter.

Notes: The court exhibit (Dkt. 32-3) shows the five paragraphs numbered 1) to 5) and the subject line 'AI lab'; body text is from the LessWrong transcription with the numbering restored per the exhibit. Spelling 'safety' is as published. Also shown at the 2026 trial (reported as trial Exhibit 5).

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-3 \(Exhibit 2 to First Amended Complaint\)](#); [CourtListener docket, Musk v. Altman](#)

Tags: founding, governance, mission, recruiting, regulation, musk-v-openai

Musk: 'Agree on all'

From: Elon Musk

Date: June 24, 2015 (Sent 11:05 PM)

To: Sam Altman

Subject: Re: AI lab

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Agree on all

About this email: Musk's reply accepting Altman's proposed mission, team, governance structure and regulation-letter plan.

Notes: Subject per the court exhibit (Dkt. 32-3): 'Re: AI lab'. Exhibit timestamp 11:05:44 PM.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-3 \(Exhibit 2 to First Amended Complaint\)](#); [CourtListener docket, Musk v. Altman](#)

Tags: founding, governance, musk-v-openai

Altman to Musk: \$100MM commitment and specific asks

From: Sam Altman

Date: October 18, 2015 (Header as published: Oct 18, 2015, 11:36 AM)

To: Elon Musk

Subject: followup

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Good seeing you a couple of weeks ago. As discussed I think starting with a \$100MM commitment (and leaving the time unspecified) is the way to go. Everyone we discussed but Gates is committed to donating. Hope to have him locked down next week. I am going to sit down with Zuck, but I still think it's probably too problematic given Facebook AI Research. Two big new recruits that we expect to sign offer letters this week: John Schulman and Wojciech Zaremba. Specific asks for you: 1) Can you donate \$30MM over the next 5 years? 2) Can we call you an "advisor" but leave unspecified what you'll do in detail and figure it out as we go? 3) Will you be on the Safety Board with me? I'd like it to eventually be 5 of us, adding the next 3 over the year or so. This will be the "second key" for releasing anything that could be dangerous.

About this email: Sam Altman proposes launching the AI lab with a \$100 million commitment, reports on donors and recruits, and asks Musk to donate \$30 million over five years, be named an advisor and join a safety board.

Notes: Email TO Musk in a thread he replied to. Outside the SpaceX slice (OpenAI); included because the coordinator asked for Tech Emails posts on any Musk company - check against the OpenAI slice for duplicates. Retrieved sentence by sentence, so paragraph breaks are not preserved; the retrieval tool reported this message as cut off by the post's paywall, so it may continue. Subject inferred from the reply header 'Re: followup'. Post listed in the Tech Emails archive under Dec 6, 2025.

Sources: [Internal Tech Emails \(document from Musk v. Altman\)](#)

Tags: openai, founding, funding, governance

Musk to Altman: 'Let's discuss governance. This is critical.'

From: Elon Musk

Date: October 19, 2015 (Header as published: Oct 19, 2015, 3:27 AM)

To: Sam Altman

Subject: followup

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Let's discuss governance. This is critical. I don't want to fund something that goes in what turns out to be the wrong direction.

Btw, great move on the continuity fund!

About this email: Musk answers Altman's funding asks by insisting on discussing governance first, saying he does not want to fund something that ends up going in the wrong direction, and praises YC's continuity fund.

Notes: Outside the SpaceX slice (OpenAI); included because the coordinator asked for Tech Emails posts on any Musk company - check against the OpenAI slice for duplicates. Retrieved sentence by sentence; the paragraph break before 'Btw' is an assumption about layout, the wording is as retrieved. Subject inferred from the next message's 'Re: followup' header.

Sources: [Internal Tech Emails \(document from Musk v. Altman\)](#)

Tags: openai, founding, governance, funding

Altman to Musk: 'very focused on getting this right'

From: Sam Altman

Date: October 19, 2015 (Header as published: Oct 19, 2015, 8:47 AM)

To: Elon Musk

Subject: Re: followup

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Happy to talk about it any time--very focused on getting this right!

Thanks--you were actually the main inspiration for that. We've had such a struggle getting our capital-intensive businesses funded (unless the founders were already rich from a previous startup) that we figured we should just do it ourselves.

About this email: Altman agrees to talk about governance any time and says Musk was the main inspiration for YC's continuity fund.

Notes: Email TO Musk closing the thread. Outside the SpaceX slice (OpenAI); check against the OpenAI slice for duplicates. Retrieved sentence by sentence; paragraph layout approximate.

Sources: [Internal Tech Emails \(document from Musk v. Altman\)](#)

Tags: openai, founding, governance, y-combinator

Musk sends Altman Lord Acton quotes on power

From: Elon Musk

Date: November 14, 2015 (Sent Saturday 5:19 PM)

To: Sam Altman

Subject: Great Acton quotes

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

"Liberty consists in the division of power. Absolutism, in concentration of power."

And

"There is no error so monstrous that it fails to find defenders among the ablest men"

About this email: Musk sends Altman two Lord Acton quotations about liberty as the division of power and absolutism as its concentration.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 379-61 (Exhibit 6); [Mother Jones](#), 'The Musk v. Altman court battle reveals how the oligarchy is afraid of itself' (Apr 2026); [Business Insider via Yahoo Finance](#), 'Musk pitched naming OpenAI Freemind in jab at Google, emails show' (Apr 29, 2026)

Tags: lord-acton, power, liberty, philosophy, musk-v-openai

Altman: second Acton quote an all-time favorite

From: Sam Altman

Date: November 14, 2015 (Sent Saturday 6:04 PM)

To: Elon Musk

Subject: Re: Great Acton quotes

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

the second is one of my all-time favorite quotes!

<https://www.youtube.com/watch?v=f-bwW1czuL4&feature=youtu.be&t=122>

About this email: Altman replies that the second Acton quote is one of his favorites and links a YouTube clip.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 379-61 (Exhibit 6); [Mother Jones](#), 'The Musk v. Altman court battle reveals how the oligarchy is afraid of itself' (Apr 2026); [Business Insider via Yahoo Finance](#), 'Musk pitched naming OpenAI Freemind in jab at Google, emails show' (Apr 29, 2026)

Tags: lord-acton, power, philosophy, musk-v-openai

Altman asks Musk to call Sutskever over Google counteroffer

From: Sam Altman

Date: November 18, 2015 (Sent Wednesday 1:06 PM)

To: Elon Musk (and Ilya Sutskever)

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Elon, as we discussed, Ilya is trying to decide between leaving YCAI and a massive counteroffer from Google. I think talking to you would help him make the right choice.

[...]

It sounds like he's free after 5:00 PM. Sam.

About this email: Altman tells Musk that Sutskever is weighing the new lab against a massive Google counteroffer and asks Musk to call him.

Notes: Text is as read aloud by counsel at Sutskever's deposition (deposition Exhibit 3, Bates 2024MUSK-0005305), not the document itself; a phone number between the two passages was omitted, and the transcript's wording ('leaving YCAI') may reflect a reading or transcription slip. Subject line not stated.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-46 (Ilya Sutskever deposition, Oct. 1, 2025; emails read into the record)

Tags: recruiting, ilya-sutskever, google, counteroffers, musk-v-openai

Sutskever asks Musk for a phone call

From: Ilya Sutskever

Date: November 18, 2015 (Sent 8:41 PM; the chain's top message is dated Nov. 19, so the date of this reply (Nov. 18 or 19) is not certain)

To: Elon Musk

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Hi, Elon. I would like to talk to you about this matter over the phone. Any chance you could call today before 4:00 PM., after 8:00 PM? Let me know what time works best for you. Ilya.

About this email: Sutskever asks Musk to call him to discuss his decision between the new lab and Google.

Notes: As read aloud by counsel at Sutskever's deposition (deposition Exhibit 3); punctuation is the court reporter's.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-46 (Ilya Sutskever deposition, Oct. 1, 2025; emails read into the record)

Tags: recruiting, ilya-sutskever, google, musk-v-openai

Musk to Sutskever: 'No problem. Will call today.'

From: Elon Musk

Date: November 19, 2015 (Sent 9:14 (PM), about half an hour after Sutskever's 8:41 PM message; exhibit header dated Nov. 19, 2015)

To: Ilya Sutskever

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

No problem. Will call today.

About this email: Musk agrees to call Sutskever, who was deciding between the new lab and a Google counteroffer.

Notes: As read aloud by counsel at Sutskever's deposition (deposition Exhibit 3, Bates 2024MUSK-0005305). Likely the complete message, but the document itself was not retrieved.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-46 (Ilya Sutskever deposition, Oct. 1, 2025; emails read into the record)

Tags: recruiting, ilya-sutskever, google, musk-v-openai

Altman: board plan for 'YC AI' Delaware nonprofit

From: Sam Altman

Date: November 20, 2015 (Sent 10:48 AM)

To: Elon Musk (cc: YC general counsel, name redacted)

Subject: Re: AI docs

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

Elon-

Plan is to have you, me, and Ilya on the Board of Directors for YC AI, which will be a Delaware non-profit. We will also state that we plan to elect two other outsiders by majority vote of the Board.

We will write into the bylaws that any technology that potentially compromises the safety of humanity has to get consent of the Board to be released, and we will reference this in the researchers' employment contracts.

At a high level, does that work for you?

I'm cc'ing our GC <redacted> here—is there someone in your office he can work with on the details?

Sam

About this email: Altman proposes that he, Musk and Sutskever sit on the board of 'YC AI', a Delaware nonprofit, with bylaws requiring board consent to release technology that could compromise humanity's safety.

Notes: Published by OpenAI (Altman's side) in its December 2024 blog post; LessWrong marks it as sourced from the OpenAI blog.

Sources: [OpenAI blog](#), 'Elon Musk wanted an OpenAI for-profit' (Dec 13, 2024); transcribed in LessWrong's 'OpenAI Email Archives'; [LessWrong](#), [OpenAI Email Archives](#)

Tags: governance, board, nonprofit, y-combinator, safety

Musk: use a standard C corp with parallel nonprofit

From: Elon Musk

Date: November 20, 2015 (Sent 12:29 PM)

To: Sam Altman

Subject: Re: AI docs

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

I think this should be independent from (but supported by) YC, not what sounds like a subsidiary.

Also, the structure doesn't seem optimal. In particular, the YC stock along with a salary from the nonprofit muddies the alignment of incentives. Probably better to have a standard C corp with a parallel nonprofit.

About this email: Musk says the lab should be independent from YC rather than a subsidiary, and that YC stock plus a nonprofit salary muddies incentives; he suggests a standard C corp with a parallel nonprofit.

Notes: Published by OpenAI in its December 2024 blog post.

Sources: [OpenAI blog](#), 'Elon Musk wanted an OpenAI for-profit' (Dec 13, 2024); transcribed in LessWrong's 'OpenAI Email Archives'; [LessWrong](#), [OpenAI Email Archives](#)

Tags: structure, c-corp, nonprofit, y-combinator, incentives

Musk: good talk with Ilya; 'need to do more in comp'

From: Elon Musk

Date: November 20, 2015 (Time not stated)

To: Sam Altman

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

I had a good conversation with Ilya yesterday. We definitely need to do more in comp and he asked for a commitment on my part to be involved which I agreed to.

About this email: Musk reports to Altman that he spoke with Sutskever, that compensation must be improved, and that he agreed to Sutskever's request for a commitment that Musk would be involved.

Notes: As read aloud by counsel at Sutskever's deposition (deposition Exhibit 5, Bates 2024MUSK-0005266); the email may contain more text than was read. Subject line not stated.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-46 ([Ilya Sutskever deposition, Oct. 1, 2025; emails read into the record](#))

Tags: recruiting, ilya-sutskever, compensation, involvement, musk-v-openai

Brockman sends draft launch blog and offer template

From: Greg Brockman

Date: November 22, 2015 (Sent 6:11 PM)

To: Elon Musk (cc: Sam Altman)

Subject: follow up from call

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Hey Elon,

Nice chatting earlier.

As I mentioned on the phone, here's the latest early draft of the blog post:

<<https://quip.com/6YnqA26RJgKr>>. (Sam, Ilya, and I are thinking about new names; would love any input from you.)

Obviously, there's a lot of other detail to change too, but I'm curious what you think of that kind of messaging. I don't want to pull any punches, and would feel comfortable broadcasting a stronger message if it feels right. I think it's mostly important that our messaging appeals to the research community (or at least the subset we want to hire). I hope for us to enter the field as a neutral group, looking to collaborate widely and shift the dialog towards being about humanity winning rather than any particular group or company. (I think that's the best way to bootstrap ourselves into being a leading research institution.)

I've attached the offer letter template we've been using, with a salary of \$175k. Here's the email template I've been sending people:

Attached is your official YCR offer letter! Please sign and date the your convenience. There will be two more documents coming:

- A separate letter offering you 0.25% of each YC batch you are present for (as compensation for being an Advisor to YC).
- The At-Will Employment, Confidential Information, Invention Assignment and Arbitration Agreement

(As this is the first batch of official offers we've done, please forgive any bumpiness along the way, and please let me know if anything looks weird!)

We plan to offer the following benefits:

- Health, dental, and vision insurance
- Unlimited vacation days with a recommendation of four weeks per year
- Paid parental leave

- Paid conference attendance when you are presenting YC AI work or asked to attend by YC AI

We're also happy to provide visa support. When you're ready to talk about visa-related questions, I'm happy to put you in touch with Kirsty from YC.

Please let me know if you have any questions — I'm available to chat any time! Looking forward to working together :).

- gdb

About this email: Brockman shares the early draft of the launch blog post, asks for input on naming and messaging, and attaches the \$175k offer letter template and benefits email being sent to recruits.

Notes: The passage from 'Attached is your official YCR offer letter!' to 'Looking forward to working together :).' is a quoted template inside the email.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-4 \(Exhibit 3 to First Amended Complaint\)](#); [CourtListener docket, Musk v. Altman](#)

Tags: launch, blog-draft, offers, compensation, recruiting, musk-v-openai

Musk: announce a \$1B commitment, not \$100M

From: Elon Musk

Date: November 22, 2015 (Sent 7:48 PM)

To: Greg Brockman (cc: Sam Altman)

Subject: follow up from call

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Blog sounds good, assuming adjustments for neutrality vs being YC-centric.

I'd favor positioning the blog to appeal a bit more to the general public -- there is a lot of value to having the public root for us to succeed -- and then having a longer, more detailed and inside-baseball version for recruiting, with a link to it at the end of the general public version.

We need to go with a much bigger number than \$100M to avoid sounding hopeless relative to what Google or Facebook are spending. I think we should say that we are starting with a \$1B funding commitment. This is real. I will cover whatever anyone else doesn't provide.

Template seems fine, apart from shifting to a vesting cash bonus as default, which can optionally be turned into YC or potentially SpaceX (need to understand how much this will be) stock.

About this email: Musk approves the blog direction with less YC-centric framing, urges a general-public version, insists on announcing a \$1B funding commitment that he will backstop, and suggests a vesting cash bonus convertible to YC or SpaceX stock.

Notes: The \$1B passage was first published in excerpt by OpenAI in March 2024; the full email appeared later in court exhibits.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'](#); [CourtListener docket, Musk v. Altman](#); [OpenAI, 'OpenAI and Elon Musk' \(Mar 5, 2024\)](#)

Tags: funding, 1-billion, launch, blog-draft, compensation, musk-v-openai

Musk: pure-play 501c3, 'Freemind' name, time commitment

From: Elon Musk

Date: November 22, 2015 (Sent Sunday 6:54 PM)

To: Sam Altman

Subject: Re: Thought Summary

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Had a great call with Greg. I'm super impressed with everyone so far. This is a great team.

Doing this as an independent, pure play 501c3, but with a crystal clear focus on the positive advent of strong AI distributed widely to humanity does seem to make the most sense to me. Please correct me if we aren't on the same page. The company would still aim to bring in revenue in excess of costs at some point, but positive net revenue would just flow to cash reserves.

On the comp front, I think having the default approach be a cash salary and a cash bonus that vests over four years and stacks with refresher vesting bonuses every year is the way to go. People could, at their option, if you are amenable, convert that cash amount to YC stock. I'm also fine if they want to convert some or all to SpaceX stock. I can pretty much do what I want on the SpaceX side, as it is private (thank goodness).

Also, we will have insane amounts of real world sensor data -- several orders of magnitude greater than any other company -- at Tesla that the team could use when helpful.

Was thinking a bit about names. It is always super hard to come up with one that doesn't suck and isn't already taken by someone that won't give it up. Least bad name I have thought of so far is Freemind. Conveys the sense that we are trying to create digital intelligence that will be freely available to all -- the opposite of Deepmind's one-ring-to-rule-them-all approach. Having "mind" in there isn't great, but the synonyms suck and their name has been subsumed in the public sphere by Google/Alphabet. You come up with anything?

On the time allocation front, I've decided to dedicate whatever amount of my time that is materially useful, even though it will come at some cost to SpaceX and Tesla. If I really believe that this is potentially the biggest near-term existential threat, then action should follow belief. That said, this probably amounts to half a day per week (higher in the beginning), with more time allocated when we hit a key decision point or crisis.

About this email: Musk backs an independent, pure-play 501c3 focused on widely distributed strong AI, outlines cash-plus-vesting-bonus pay convertible to YC or SpaceX stock, offers Tesla sensor data, proposes the name 'Freemind', and commits roughly half a day a week.

Notes: Also shown at the 2026 jury trial (reported as trial Exhibit 12). The earlier 'Thought Summary' message this replies to is not in the exhibit as retrieved.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 379-61 (Exhibit 6); [Mother Jones, 'The Musk v. Altman court battle reveals how the oligarchy is afraid of itself' \(Apr 2026\)](#); [Business Insider via Yahoo Finance, 'Musk pitched naming OpenAI Freemind in jab at Google, emails show' \(Apr 29, 2026\)](#)

Tags: structure, 501c3, naming, freemind, compensation, tesla-data, musk-v-openai

Altman agrees on 501c3; floats 'Axon'; asks Musk's title

From: Sam Altman

Date: November 22, 2015 (Sent Sunday 7:08 PM)

To: Elon Musk

Subject: Re: Thought Summary

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Yes--I think an independent 501c3 is correct. Are you ok if, initially, it shares a building with YC and YC helps with legal stuff etc? I think that will save us a few months while we hire dedicated people.

I like the idea of defaulting to a cash bonus that stacks each year--I'd independently come to that same conclusion after talking to Greg yesterday. Offering them the option to convert it to SpaceX or YC stock sounds great to me.

Making Tesla data available would be incredible. I think most YC companies will offer to do the same.

Freemind seems pretty good to me but I'd to keep thinking--it may be a little too close to Deepmind. I really like the word free here thought--freethink? freemerge?

Best idea I've had so far is Axon, which I think we could get .com for from TPG-Axon. Intelligence.com looks getable, but perhaps too generic.

I'll keep thinking.

Half a day per week sounds awesome. I assume that if gets to the point where AGI looks like it's about to happen, you and I will both find a way to spend a lot of time on it. And if it seems further off, then I expect you'll be very busy getting to Mars.

Have you given any thought to how you want to describe your involvement? I'm fine with anything from "advisor" to "co-CEO" but think it's probably good to have a clear answer.

About this email: Altman agrees an independent 501c3 is right, asks if it can initially share a building with YC, suggests names including Axon, and asks how Musk wants to describe his involvement, from 'advisor' to 'co-CEO'.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\), Dkt. 379-61 \(Exhibit 6\)](#); [Mother Jones, 'The Musk v. Altman court battle reveals how the oligarchy is afraid of itself' \(Apr 2026\)](#); [Business Insider via Yahoo Finance, 'Musk pitched naming OpenAI Freemind in jab at Google, emails show' \(Apr 29, 2026\)](#)

Tags: structure, 501c3, naming, axon, compensation, titles, musk-v-openai

Musk: 'Can't be lukewarm'; proposes co-chairs

From: Elon Musk

Date: November 22, 2015 (Sent Sunday 7:32 PM)

To: Sam Altman

Subject: Re: Thought Summary

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Great, sounds like we are independently following the same thought tracks.

No problem with YC support in the beginning. That's normal for a lot of companies, so I don't think it materially affects the neutrality position.

The "mind" part of Freemind is partly an intentional philosophical counter to Deepmind and also, as mentioned, I couldn't find a synonym for "mind" that wasn't terrible.

I like Axon as an option, although it does slightly sound like Google Brain or, more generally, that we think digital intelligence consists of brain emulation. Pretty much all names suck in the beginning though. Given several choices, I tend to favor names that convey the mission of the company and hopefully have a positive impact on recruitment.

Yeah, my role should be clear and I have to bite the bullet on admitting real involvement. This will come as a shocker to many, but so be it. Can't be lukewarm about this.

In terms of title, what about me as chair and you as CEO or us both as co-chairs? My main concern about being called a co-CEO, even if that is de facto true, is that it might cause problems at SpaceX and Tesla, where they know I'm already overbooked.

About this email: Musk accepts early YC support, explains the 'mind' in Freemind as a counter to DeepMind, comments on Axon, says he must admit real involvement, and proposes being chair with Altman as CEO or both as co-chairs.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\), Dkt. 379-61 \(Exhibit 6\)](#); [Mother Jones, 'The Musk v. Altman court battle reveals how the oligarchy is afraid of itself' \(Apr 2026\)](#); [Business Insider via Yahoo Finance, 'Musk pitched naming OpenAI Freemind in jab at Google, emails show' \(Apr 29, 2026\)](#)

Tags: naming, freemind, titles, co-chair, involvement, musk-v-openai

Altman: warming to 'freemind'; co-chairs works best

From: Sam Altman

Date: November 22, 2015 (Sent Sunday 7:36 PM)

To: Elon Musk

Subject: Re: Thought Summary

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Ok, let me work on a draft of how to step up the bonuses for the senior positions.

I'm warming up fast to freemind. It definitely conveys the right spirit. Also thinking about names related to Turing somehow.

I think both as co-chairs works best. We can evolve from there as needed.

About this email: Altman says he will draft stepped-up bonuses for senior roles, is warming to 'freemind', is considering Turing-related names, and favors both men as co-chairs.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\), Dkt. 379-61 \(Exhibit 6\)](#); [Mother Jones, 'The Musk v. Altman court battle reveals how the oligarchy is afraid of itself' \(Apr 2026\)](#); [Business Insider via Yahoo Finance, 'Musk pitched naming OpenAI Freemind in jab at Google, emails show' \(Apr 29, 2026\)](#)

Tags: naming, freemind, titles, co-chair, compensation, musk-v-openai

Altman introduces Brockman to Musk

From: Sam Altman

Date: November 22, 2015 (Time not stated)

To: Elon Musk, Greg Brockman

Subject: One More AI intro

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Elon, please meet Greg, former CTO of Stripe. You guys met at the dinner we had but would be great if you could talk more.

About this email: Altman introduces Greg Brockman, former Stripe CTO, to Musk and suggests they talk more.

Notes: As read aloud by counsel at Brockman's deposition (deposition Exhibit 4, Bates SPX5733). The introduction led to the Nov. 22, 2015 Musk-Brockman call referenced in 'follow up from call'.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 379-67 (Greg Brockman deposition, Sept. 23, 2025; emails read into the record)

Tags: introductions, greg-brockman, founding, musk-v-openai

Brockman to Musk: 'Nice to virtually connect'

From: Greg Brockman

Date: November 22, 2015 (Time not stated)

To: Elon Musk

Subject: Re: One More AI intro

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Hi, Elon. Nice to virtually connect. Would there be a good [...]

About this email: Brockman follows Altman's introduction by asking Musk for a time to chat.

Notes: As read aloud by counsel at Brockman's deposition; the transcript excerpt filed with the court cuts off mid-sentence. Subject line inferred from the thread.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 379-67 (Greg Brockman deposition, Sept. 23, 2025; emails read into the record)

Tags: introductions, greg-brockman, founding, musk-v-openai

Musk: avoid a Turing Test association in the name

From: Elon Musk

Date: November 23, 2015 (Timestamp in exhibit: Monday 03:50:06 AM (time zone not stated))

To: Sam Altman

Subject: Re: Thought Summary

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Great.

Something Turing-related that doesn't sound too ominous might be good. Want to avoid the Turing Test association though, as that sounds too much like we are replacing humans.

About this email: Musk says a Turing-related name that is not ominous could work but warns against evoking the Turing Test, which sounds like replacing humans.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 379-61 (Exhibit 6); Mother Jones, 'The Musk v. Altman court battle reveals how the oligarchy is afraid of itself' (Apr 2026); Business Insider via Yahoo Finance, 'Musk pitched naming OpenAI Freemind in jab at Google, emails show' (Apr 29, 2026)

Tags: naming, turing, messaging, musk-v-openai

Musk drafts OpenAI's opening paragraphs

From: Elon Musk

Date: December 8, 2015 (Sent 9:29 AM)

To: Sam Altman

Subject: Draft opening paragraphs

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

It is super important to get the opening summary section right. This will be what everyone reads and what the press mostly quotes. The whole point of this release is to attract top talent. Not sure Greg totally gets that.

---- OpenAI is a non-profit artificial intelligence research company with the goal of advancing digital intelligence in the way that is most likely to benefit humanity as a whole, unencumbered by an obligation to generate financial returns.

The underlying philosophy of our company is to disseminate AI technology as broadly as possible as an extension of all individual human wills, ensuring, in the spirit of liberty, that the power of digital intelligence is not overly concentrated and evolves toward the future desired by the sum of humanity.

The outcome of this venture is uncertain and the pay is low compared to what others will offer, but we believe the goal and the structure are right. We hope this is what matters most to the best in the field.

About this email: Musk stresses that the launch post's opening summary is what the press will quote and that its purpose is to attract top talent, then supplies his own three-paragraph draft describing OpenAI as a nonprofit.

Notes: Text matches the court exhibit (Dkt. 32-5). Also shown at the 2026 trial (reported as trial Exhibit 14).

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\)](#) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'; Dkt. 32-5 (Exhibit 4 to First Amended Complaint); CourtListener docket, Musk v. Altman

Tags: launch, blog-draft, mission, recruiting, nonprofit, musk-v-openai

Altman's revision of the opening paragraphs

From: Sam Altman

Date: December 8, 2015 (Sent 10:34 AM)

To: Elon Musk

Subject: Draft opening paragraphs

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

how is this?

—

OpenAI is a non-profit artificial intelligence research company with the goal of advancing digital intelligence in the way that is most likely to benefit humanity as a

whole, unencumbered by an obligation to generate financial returns.

Because we don't have any financial obligations, we can focus on the maximal positive human impact and disseminating AI technology as broadly as possible. We believe AI should be an extension of individual human wills and, in the spirit of liberty, not be concentrated in the hands of the few.

The outcome of this venture is uncertain and the pay is low compared to what others will offer, but we believe the goal and the structure are right. We hope this is what matters most to the best in the field.

About this email: Altman returns an edited version of Musk's draft opening, tightening the second paragraph about broad dissemination and AI as an extension of individual human wills.

Notes: Text matches the court exhibit (Dkt. 32-5). Also shown at the 2026 trial (reported as trial Exhibit 14).

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-5 \(Exhibit 4 to First Amended Complaint\)](#); [CourtListener docket, Musk v. Altman](#)

Tags: launch, blog-draft, mission, nonprofit, musk-v-openai

Altman: DeepMind planning massive counteroffers

From: Sam Altman

Date: December 11, 2015 (Sent 11:30 AM; the subject line runs into the first sentence)

To: Elon Musk

Subject: just got word...

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

that deepmind is going to give everyone in openAI massive counteroffers tomorrow to try to kill it.

do you have any objection to me proactively increasing everyone's comp by 100-200k per year? i think they're all motivated by the mission here but it would be a good signal to everyone we are going to take care of them over time.

sounds like deepmind is planning to go to war over this, they've been literally cornering people at NIPS.

About this email: On launch day Altman warns Musk that DeepMind will make large counteroffers to OpenAI's recruits and asks to raise everyone's pay by \$100-200k a year.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-6 \(Exhibit 5 to First Amended Complaint\)](#); [CourtListener docket, Musk v. Altman](#)

Tags: deepmind, recruiting, compensation, counteroffers, launch, musk-v-openai

Musk: 'Has Ilya come back with a solid yes?'

From: Elon Musk

Date: December 11, 2015 (Time not shown in the exhibit; between 11:30 AM and 12:15 PM)

To: Sam Altman

Subject: just got word...

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Has Ilya come back with a solid yes?

If anyone seems at all uncertain, I'm happy to call them personally too. Have told Emma this is my absolute top priority 24/7.

About this email: Musk asks whether Sutskever has firmly committed and offers to personally call anyone wavering, calling recruiting his top priority.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits](#), as transcribed in [LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-6 \(Exhibit 5 to First Amended Complaint\)](#); [CourtListener docket, Musk v. Altman](#)

Tags: deepmind, recruiting, ilya-sutskever, counteroffers, musk-v-openai

Altman: Ilya 'just gave his word'

From: Sam Altman

Date: December 11, 2015 (Sent 12:15 PM)

To: Elon Musk

Subject: just got word...

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

yes committed committed. just gave his word.

About this email: Altman confirms Sutskever has committed to OpenAI.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits](#), as transcribed in [LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-6 \(Exhibit 5 to First Amended Complaint\)](#); [CourtListener docket, Musk v. Altman](#)

Tags: recruiting, ilya-sutskever, deepmind, musk-v-openai

Musk: 'awesome'

From: Elon Musk

Date: December 11, 2015 (Sent 12:32 PM)

To: Sam Altman

Subject: just got word...

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

awesome

About this email: Musk's one-word reaction to news that Sutskever committed.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits](#), as transcribed in [LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-6 \(Exhibit 5 to First Amended Complaint\)](#); [CourtListener docket, Musk v. Altman](#)

Tags: recruiting, ilya-sutskever, musk-v-openai

Altman: team ready for DeepMind offers; news at 1:30

From: Sam Altman

Date: December 11, 2015 (Sent 12:35 PM)

To: Elon Musk

Subject: just got word...

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

everyone feels great, saying stuff like "bring on the deepmind offers, they unfortunately dont have 'do the right thing' on their side"

news out at 130 pm pst

About this email: Altman reports that the founding team is upbeat about resisting DeepMind's counteroffers and that the announcement goes out at 1:30 pm PST.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'; Dkt. 32-6 \(Exhibit 5 to First Amended Complaint\); CourtListener docket, Musk v. Altman](#)

Tags: recruiting, deepmind, launch, musk-v-openai

Musk's launch-day note to the founding team

From: Elon Musk

Date: December 11, 2015 (Sent 4:41 PM)

To: Ilya Sutskever, Pamela Vagata, Vicki Cheung, Diederik Kingma, Andrej Karpathy, John D. Schulman, Trevor Blackwell, Greg Brockman (cc: Sam Altman)

Subject: The OpenAI Company

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Congratulations on a great beginning!

We are outmanned and outgunned by a ridiculous margin by organizations you know well, but we have right on our side and that counts for a lot. I like the odds.

Our most important consideration is recruitment of the best people. The output of any company is the vector sum of the people within it. If we are able to attract the most talented people over time and our direction is correctly aligned, then OpenAI will prevail.

To this end, please give a lot of thought to who should join. If I can be helpful with recruitment or anything else, I am at your disposal. I would recommend paying close attention to people who haven't completed their grad or even undergrad, but are obviously brilliant. Better to have them join before they achieve a breakthrough.

Looking forward to working together,

Elon

About this email: On the day OpenAI was announced, Musk congratulates the founding team, says they are outmanned and outgunned but have right on their side, and makes recruiting the best people the top priority.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'; Dkt. 32-7 \(Exhibit 6 to First Amended Complaint\); CourtListener docket, Musk v. Altman](#)

Tags: launch, recruiting, founding-team, talent, musk-v-openai

Musk: 'Great' (launch minutes away)

From: Elon Musk

Date: December 11, 2015 (Sent 1:27:14 PM)

To: Sam Altman

Subject: Re: just got word...

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Great

About this email: Musk's one-word reply to Altman's note that the team feels great and the announcement goes out at 1:30 pm.

Notes: Top message of the exhibit; not included in the LessWrong transcription of this thread.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 32-6 (Exhibit 5 to First Amended Complaint, filed Nov. 14, 2024); [LessWrong](#), [OpenAI Email Archives](#)

Tags: launch, recruiting, deepmind, musk-v-openai

Redacted correspondent challenges open-sourcing AI

From: [redacted]

Date: January 2, 2016 (Sent 10:12 AM CST)

To: Elon Musk

Subject: congrats on the falcon 9

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

Hi Elon Happy new year to you, [REDACTED]!

Congratulations on landing the Falcon 9, what an amazing achievement. Time to build out the fleet now!

I've seen you (and Sam and other OpenAI people) doing a lot of interviews recently extolling the virtues of open sourcing AI, but I presume you realise that this is not some sort of panacea that will somehow magically solve the safety problem? There are many good arguments as to why the approach you are taking is actually very dangerous and in fact may increase the risk to the world. Some of the more obvious points are well articulated in this blog post, that I'm sure you've seen, but there are also other important considerations: <http://slatestarcodex.com/2015/12/17/should-ai-be-open/>

I'd be interested to hear your counter-arguments to these points.

Best,

[REDACTED]

About this email: A correspondent whose name is redacted congratulates Musk on the Falcon 9 landing and argues that OpenAI's open-sourcing approach is not a safety panacea and may increase risk, citing a Slate Star Codex post.

Notes: Sender name redacted in the published version. Musk forwarded this email, without recorded comment, to Sam Altman, Ilya Sutskever and Greg Brockman on Jan 2, 2016 at 8:18 AM (Pacific); that bare forward is not listed as a separate entry.

Sources: [OpenAI blog](#), 'OpenAI and Elon Musk' (Mar 5, 2024); transcribed in LessWrong's 'OpenAI Email Archives'; [LessWrong](#), [OpenAI Email Archives](#)

Tags: open-source, safety, slate-star-codex, criticism

Sutskever: 'Open' means sharing benefits, not science

From: Ilya Sutskever

Date: January 2, 2016 (Sent 9:06 AM)

To: Elon Musk, Sam Altman, Greg Brockman

Subject: Fwd: congrats on the falcon 9

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

The article is concerned with a hard takeoff scenario: if a hard takeoff occurs, and a safe AI is harder to build than an unsafe one, then by opensorucing everything, we make it easy for someone unscrupulous with access to overwhelming amount of hardware to build an unsafe AI, which will experience a hard takeoff. As we get closer to building AI, it will make sense to start being less open. The Open in openAI means that everyone should benefit from the fruits of AI after its built, but it's totally OK to not share the science (even though sharing everything is definitely the right strategy in the short and possibly medium term for recruitment purposes).

About this email: Replying to Musk's forward, Sutskever says it will make sense to be less open as AI gets closer, and that the 'Open' in OpenAI means everyone benefits from AI's fruits, not that the science must be shared.

Notes: Typo 'opensorucing' is in the published text. Paragraphing follows the LessWrong transcription as retrieved (single block).

Sources: [OpenAI blog, 'OpenAI and Elon Musk' \(Mar 5, 2024\)](#); transcribed in LessWrong's 'OpenAI Email Archives'; [LessWrong, OpenAI Email Archives](#)

Tags: open-source, safety, hard-takeoff, recruiting

Musk: 'Yup'

From: Elon Musk

Date: January 2, 2016 (Sent 9:11 AM)

To: Ilya Sutskever

Subject: Fwd: congrats on the falcon 9

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

Yup

About this email: Musk's one-word agreement with Sutskever's view that OpenAI need not share its science as AI gets closer.

Sources: [OpenAI blog, 'OpenAI and Elon Musk' \(Mar 5, 2024\)](#); transcribed in LessWrong's 'OpenAI Email Archives'; [LessWrong, OpenAI Email Archives](#)

Tags: open-source, safety

Sutskever requests biweekly 30-minute meetings with Musk

From: Ilya Sutskever

Date: February 18, 2016 (Time not stated)

To: Sam Teller, Elon Musk, Greg Brockman

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Greg and I think it would be useful for OpenAI's long-term success if the two of us had a 30-minute meeting with Elon every two weeks.

About this email: Sutskever proposes that he and Brockman meet with Musk for 30 minutes every two weeks.

Notes: As read aloud by counsel at Sutskever's deposition (deposition Exhibit 9). Subject line not stated.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\), Dkt. 391-46 \(Ilya Sutskever deposition, Oct. 1, 2025; emails read into the record\)](#)

Tags: meetings, management, ilya-sutskever, musk-v-openai

Musk: DeepMind is 'playing the Super Bowl'

From: Elon Musk

Date: February 18, 2016 (Sent the same day, 'a few hours later' per counsel)

To: Ilya Sutskever (and others on the chain)

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

I think I should probably do this every week.

[...]

My dinner with Dennis was extremely alarming. I feel like they are playing the Super Bowl and we are playing the Puppy Bowl. Unless want to have our ass handed to us, we need to step up our game dramatically.

[...]

Let's have a conference call later today with the key people of OpenAI. Later today, if possible; otherwise, tomorrow.

About this email: Musk says he should meet OpenAI's leaders weekly, calls a dinner with DeepMind's chief extremely alarming, says OpenAI must dramatically step up its game, and asks for a conference call with key people.

Notes: As read aloud by counsel at Sutskever's deposition (deposition Exhibit 9) in three passages; 'Dennis' is the court reporter's rendering (apparently Demis Hassabis of DeepMind) and 'Unless want' is as transcribed. This exchange immediately precedes the 'Re: Followup thoughts' emails of Feb. 19, 2016. Subject line not stated.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\), Dkt. 391-46 \(Ilya Sutskever deposition, Oct. 1, 2025; emails read into the record\)](#); [Big Technology, 'Satya, Sam To Take The Stand This Week + Highlights From Musk v. Altman' \(May 2026\)](#)

Tags: deepmind, demis-hassabis, urgency, meetings, competition, musk-v-openai

Musk: DeepMind odds rising; OpenAI needs urgency

From: Elon Musk

Date: February 19, 2016 (Sent 12:05 AM)

To: Ilya Sutskever, Greg Brockman, Sam Altman

Subject: Re: Followup thoughts

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Frankly, what surprises me is that the AI community is taking this long to figure out concepts. It doesn't sound super hard. High-level linking of a large number of deep nets sounds like the right approach or at least a key part of the right approach. [REDACTED]

The probability of DeepMind creating a deep mind increases every year. Maybe it doesn't get past 50% in 2 to 3 years, but it likely moves past 10%. That doesn't sound

crazy to me, given their resources.

In any event, I have found that it is far better to overestimate than underestimate competitors.

This doesn't mean we should rush out and hire weak talent. I agree that nothing good would be achieved by that. What we need to do is redouble our efforts to seek out the best people in the world, do whatever it takes to bring them on board and imbue the company with a high sense of urgency.

It will be important for OpenAI to achieve something significant in the next 6 to 9 months to show that we are for real. Doesn't need to be a whopper breakthrough, but it should be enough for key talent around the world to sit up and take notice.

[REDACTED]

About this email: Musk says he is surprised the AI community is slow to figure out concepts, estimates DeepMind's odds of success are growing each year, and urges redoubled recruiting and a significant result within 6 to 9 months.

Notes: Two passages are redacted in the exhibit. The earlier message(s) in the 'Followup thoughts' thread that Musk is replying to were not published.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'](#); [CourtListener docket, Musk v. Altman](#); [OpenAI, 'Elon Musk wanted an OpenAI for-profit' \(Dec 13, 2024\)](#)

Tags: deepmind, urgency, recruiting, competition, timelines, musk-v-openai

Sutskever: more than one problem stands before AGI

From: Ilya Sutskever

Date: February 19, 2016 (Sent 10:28 AM)

To: Elon Musk (cc: Greg Brockman, Sam Altman)

Subject: Re: Followup thoughts

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Several points:

- It is not the case that once we solve "concepts," we get AI. Other problems that will have to be solved include unsupervised learning, transfer learning, and lifetime learning. We're also doing pretty badly with language right now. It does not mean that these problems will not see significant progress in the coming years, but it is not the case that there is only one problem that stands between us and full human level AI.
- We can't build AI today because we lack key ideas (computers may be too slow, too, but we can't tell). Powerful ideas are produced by top people. Massive clusters help, and are very worth getting, but they play a less important role.
- We will be able to achieve a conventionally significant result in the next 6 to 9 months, simply because the people we already have are very good. Achieving a field-altering result will be harder, riskier, and take longer. But we have a not unreasonable plan for that as well.

About this email: Sutskever pushes back that solving 'concepts' does not yield AI, says key ideas from top people matter more than clusters, and predicts a conventionally significant result within 6 to 9 months.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\)](#) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'; [CourtListener](#) docket, [Musk v. Altman](#); OpenAI, 'Elon Musk wanted an OpenAI for-profit' (Dec 13, 2024)

Tags: research, timelines, deepmind, compute, musk-v-openai

Brockman asks for guidance on pay and offers

From: Greg Brockman

Date: February 21, 2016 (Sent 11:34 AM)

To: Elon Musk (cc: Sam Altman)

Subject: compensation framework

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Hi all,

We're currently doing our first round of full-time offers post-founding. It's obviously super important to get these right, as the implications are very long-term. I don't yet feel comfortable making decisions here on my own, and would love any guidance.

Here's what we're currently doing:

Founding team: \$275k salary + 25bps of YC stock

- Also have option of switching permanently to \$125k annual bonus or equivalent in YC or SpaceX stock. I don't know if anyone's taken us up on this.

New offers: \$175k annual salary + \$125k annual bonus || equivalent in YC or SpaceX stock. Bonus is subject to performance review, where you may get 0% or significantly greater than 100%.

Special cases: gdb + Ilya + Trevor

The plan is to keep a mostly flat salary, and use the bonus multiple as a way to reward strong performers.

Some notes:

- We use a 20% annualized discount for the 8 years until the stock becomes liquid, the \$125k bonus equates to 12bps in YC. So the terminal value is more like \$750k. This number sounds a lot more impressive, though obviously it's hard to value exactly.
- The founding team was initially offered \$175k each. The day after the lab launched, we proactively increased everyone's salary by \$100k, telling them that we are financially committed to them as the lab becomes successful, and asking for a personal promise to ignore all counteroffers and trust we'll take care of them.
- We're currently interviewing Ian Goodfellow from Brain, who is one of the top 2 scientists in the field we don't have (the other being Alex Graves, who is a DeepMind loyalist). He's the best person on Brain, so Google will fight for him. We're grandfathering him into the founding team offer.

Some salary datapoints:

- John was offered \$250k all-in annualized at DeepMind, thought he could negotiate to \$300k easily.
- Wojciech was verbally offered ~\$1.25M/year at FAIR (no concrete letter though)
- Andrew Tulloch is getting \$800k/year at FB. (A lot is stock which is vesting.)
- Ian Goodfellow is currently getting \$165k cash + \$600k stock/year at Google.
- Apple is a bit desperate and offering people \$550k cash (plus stock, presumably). I don't think anyone good is saying yes.

Two concrete candidates that are on my mind:

- Andrew is very close to saying yes. However, he's concerned about taking such a large paycut.
- Ian has stated he's not primarily concerned with money, but the Bay Area is expensive / wants to make sure he can buy a house. I don't know what will happen if/when Google starts throwing around the numbers they threw at Ilya.

My immediate questions:

1. I expect Andrew will try to negotiate up. Should we stick to his offer, and tell him to only join if he's excited enough to take that kind of paycut (and that others have left more behind)?
2. Ian will be interviewing + (I'm sure) getting an offer on Wednesday. Should we consider his offer final, or be willing to slide depending on what Google offers?
3. Depending on the answers to 1 + 2, I'm wondering if this flat strategy makes sense. If we keep it, I feel we'll have to really sell people on the bonus multiplier. Maybe one option would be using a signing bonus as a lever to get people to sign?
4. Very secondary, but our intern comp is also below market: \$9k/mo. (FB offers \$9k + free housing, Google offers like \$11k/mo all-in.) Comp is much less important to interns than to FT people, since the experience is primary. But I think we may have lost a candidate who was on the edge to this. Given the dollar/hour is so much lower than for FT, should we consider increasing the amount?

I'm happy to chat about this at any time.

- gdb

About this email: Brockman lays out OpenAI's compensation structure for the founding team and new hires, lists competing offers from DeepMind, Facebook, Google and Apple, and asks how to handle candidates such as Andrew Tulloch and Ian Goodfellow.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-8 \(Exhibit 7 to First Amended Complaint\)](#); [CourtListener docket, Musk v. Altman](#)

Tags: compensation, salaries, offers, recruiting, ian-goodfellow, musk-v-openai

Musk: pay whatever it takes or lose to DeepMind

From: Elon Musk

Date: February 22, 2016 (Sent 12:09 AM)

To: Greg Brockman (cc: Sam Altman)

Subject: compensation framework

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

We need to do what it takes to get the top talent. Let's go higher. If, at some point, we need to revisit what existing people are getting paid, that's fine.

Either we get the best people in the world or we will get whipped by Deepmind.

Whatever it takes to bring on ace talent is fine by me.

Deepmind is causing me extreme mental stress. If they win, it will be really bad news with their one mind to rule the world philosophy. They are obviously making major progress and well they should, given the talent level over there.

About this email: Musk tells Brockman to raise offers and do whatever it takes to hire top talent, saying DeepMind is causing him extreme mental stress and that its winning would be very bad.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-8 \(Exhibit 7 to First Amended Complaint\)](#); [CourtListener docket, Musk v. Altman](#)

Tags: compensation, deepmind, recruiting, talent, competition, musk-v-openai

Brockman: 'Read you loud and clear'

From: Greg Brockman

Date: February 22, 2016 (Sent 12:21 AM)

To: Elon Musk (cc: Sam Altman)

Subject: compensation framework

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Read you loud and clear. Sounds like a plan. Will plan to continue working with sama on specifics, but let me know if you'd like to be kept in the loop.

- gdb

About this email: Brockman acknowledges Musk's direction on compensation and says he will work out specifics with Altman.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-8 \(Exhibit 7 to First Amended Complaint\)](#); [CourtListener docket, Musk v. Altman](#)

Tags: compensation, recruiting, musk-v-openai

Altman asks Musk for \$20M a year for three years

From: Sam Altman

Date: February 25, 2016 (Sent 6:13 PM)

To: Elon Musk

Subject: cash

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

I think we're going to need more than I was originally budgeting given a) the salaries in the field and b) the speed at which you want to grow.

Can you do \$20MM a year for the each of the next 3 years? I can do \$10MM/year, and we'll have brought \$5MM a year from other donors.

Sam

About this email: Altman says OpenAI will need more than originally budgeted given salaries and Musk's desired growth rate, and asks Musk for \$20M a year for three years alongside \$10M a year from Altman and \$5M from other donors.

Notes: Subject line inferred from the replies ('Re: cash'). Email addresses are redacted in the public version.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-13 (Exhibit 55, public redacted version, filed Jan. 16, 2026); Dkt. 391-22 (Exhibit 74) and Dkt. 391-54 (Exhibit 69), longer versions of the same chain; Dkt. 379-64, Jared Birchall deposition excerpts discussing this chain

Tags: funding, donations, budget, commitment, musk-v-openai

Musk agrees to \$20M a year: 'Agreed / Yes'

From: Elon Musk

Date: February 25, 2016 (Sent Thursday 6:30 PM)

To: Sam Altman

Subject: Re: cash

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Agreed

Yes

About this email: Musk agrees to Altman's request for \$20M a year for three years.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-13 (Exhibit 55, public redacted version, filed Jan. 16, 2026); Dkt. 391-22 (Exhibit 74) and Dkt. 391-54 (Exhibit 69), longer versions of the same chain; Dkt. 379-64, Jared Birchall deposition excerpts discussing this chain

Tags: funding, donations, commitment, musk-v-openai

Altman asks who to coordinate funding with

From: Sam Altman

Date: February 25, 2016 (Sent 6:32 PM)

To: Elon Musk

Subject: Re: cash

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

ok thanks, do you have an assistant or finance person or something i can coordinate with?

About this email: Altman thanks Musk and asks for an assistant or finance contact to coordinate the transfers.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-13 (Exhibit 55, public redacted version, filed Jan. 16, 2026); Dkt. 391-22 (Exhibit 74) and Dkt. 391-54 (Exhibit 69), longer versions of the same chain; Dkt. 379-64, Jared Birchall deposition excerpts discussing this chain

Tags: funding, donations, logistics, musk-v-openai

Musk: wire \$5M per quarter to OpenAI; 'a non-profit'

From: Elon Musk

Date: February 25, 2016 (Sent Thursday 6:36 PM)

To: Sam Altman; Ronald Gong (myCFO / BMO family office)

Subject: Re: cash

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Ron, we need to wire \$5M per quarter to OpenAI, starting April 1. This is a non-profit.

About this email: Musk loops in his family-office adviser Ron Gong and instructs him to wire \$5M per quarter to OpenAI starting April 1, noting it is a nonprofit.

Notes: Later messages in the chain (Feb 29 - Mar 11, 2016) are between Ron Gong, Sam Altman and Chris Clark about tax-exemption paperwork and do not involve Musk.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-13 (Exhibit 55, public redacted version, filed Jan. 16, 2026); Dkt. 391-22 (Exhibit 74) and Dkt. 391-54 (Exhibit 69), longer versions of the same chain; Dkt. 379-64, Jared Birchall deposition excerpts discussing this chain

Tags: funding, donations, nonprofit, family-office, musk-v-openai

Brockman checks Wired fact-check answers with Musk

From: Greg Brockman

Date: March 21, 2016 (Sent 12:53 AM)

To: Elon Musk (cc: Sam Teller)

Subject: wired article

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Hi Elon,

I was interviewed for a Wired article on OpenAI, and the fact checker sent me some questions. Wanted to sync with you on two in particular to make sure they sound reasonable / aligned with what you'd say:

> Would it be accurate to say that OpenAI is giving away ALL of its research?

At any given time, we will take the action that is likely to most strongly benefit the world. In the short term, we believe the best approach is giving away our research. But longer-term, this might not be the best approach: for example, it might be better not to immediately share a potentially dangerous technology. In all cases, we will be giving away all the benefits of all of our research, and want those to accrue to the world rather than any one institution.

> Does OpenAI believe that getting the most sophisticated AI possible in as many hands as possible is humanity's best chance at preventing a too-smart AI in private hands that could find a way to unleash itself on the world for malicious ends?

We believe that using AI to extend individual human wills is the most promising path to ensuring AI remains beneficial. This is appealing because if there are many agents with about the same capabilities they could keep any one bad actor in check. But I wouldn't claim we have all the answers: instead, we're building an organization that can both seek those answers, and take the best possible action regardless of what the answer turns out to be.

Thanks!

- gdb

About this email: Brockman runs past Musk his proposed answers to a Wired fact-checker on whether OpenAI gives away all its research and whether widely distributed AI is the best safeguard.

Notes: Lines beginning '>' are the fact-checker's questions quoted in the email.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-9 \(Exhibit 8 to First Amended Complaint\)](#); [CourtListener docket, Musk v. Altman](#)

Tags: press, wired, open-source, messaging, safety, musk-v-openai

Musk: 'Sounds good'

From: Elon Musk

Date: March 21, 2016 (Sent 6:53:47 AM)

To: Greg Brockman (cc: Sam Teller)

Subject: Re: wired article

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Sounds good

About this email: Musk approves Brockman's proposed answers to the Wired fact-checker.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-9 \(Exhibit 8 to First Amended Complaint\)](#); [CourtListener docket, Musk v. Altman](#)

Tags: press, wired, open-source, musk-v-openai

Musk asks Nvidia's Huang for an early DGX-1

From: Elon Musk

Date: April 13, 2016 (Sent Wednesday 12:01 AM)

To: Jensen Huang (Nvidia CEO)

Subject: DGX-1

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

can OpenAI buy one of the early units? The team was asking me about this earlier today.

OpenAI is unaffiliated with Tesla. It is a non-profit funded by me and a few others with the goal of developing safe AGI (and hopefully not paving the road to hell with good intentions).

About this email: Musk asks Nvidia CEO Jensen Huang whether OpenAI can buy one of the first DGX-1 supercomputers, describing OpenAI as a nonprofit unaffiliated with Tesla and funded by him and a few others.

Notes: Sent from Musk's Tesla address. Also shown at the 2026 jury trial (reported as trial Exhibit 70).

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 379-74 (Exhibit 34, filed Jan. 6, 2026); [The Verge](#), 'All the evidence unveiled so far in Musk v. Altman' (Apr 2026); [Olhar Digital summary of trial exhibits \(Apr 30, 2026\)](#)

Tags: nvidia, jensen-huang, compute, dgx-1, nonprofit, musk-v-openai

Huang: OpenAI will get one of the first DGX-1s

From: Jensen Huang

Date: April 19, 2016 (Sent 10:11 AM)

To: Elon Musk

Subject: Re: DGX-1

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

I've not forgotten.

This is the first supercomputer that's selling itself off the web! Demand coming from all over.

First shipment starts end next month. And already taking orders for delivery in September.

I will make sure OpenAI gets one of the first ones.

About this email: Huang replies that demand for the DGX-1 is strong, first shipments start the following month, and he will make sure OpenAI gets one of the first units.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 379-74 (Exhibit 34, filed Jan. 6, 2026); [The Verge](#), 'All the evidence unveiled so far in Musk v. Altman' (Apr 2026); [Olhar Digital summary of trial exhibits \(Apr 30, 2026\)](#)

Tags: nvidia, jensen-huang, compute, dgx-1, musk-v-openai

Musk to Huang: 'Much appreciated'

From: Elon Musk

Date: April 20, 2016 (Sent 8:20:42 AM)

To: Jensen Huang

Subject: Re: DGX-1

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

Much appreciated

About this email: Musk thanks Huang for promising OpenAI an early DGX-1.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 379-74 (Exhibit 34, filed Jan. 6, 2026); [The Verge](#), 'All the evidence unveiled so far in Musk v. Altman' (Apr 2026); [Olhar Digital](#) summary of trial exhibits (Apr 30, 2026)

Tags: nvidia, jensen-huang, compute, dgx-1, musk-v-openai

NYT asks for Musk's reaction to Zuckerberg

From: Sam Teller (forwarding Alex Thompson, The New York Times)

Date: April 27, 2016 (Alex Thompson emailed Sam Teller at 7:25 AM; Teller forwarded it to Musk at 11:03 AM)

To: Elon Musk

Subject: Re: Maureen Dowd

FULL TEXT · PRESS & JOURNALISTS · COURT RECORD · CONFIRMED

Hi Sam,

I hope you are having a great day and I apologize for interrupting it with another question. Maureen wanted to see if Mr. Musk had any reaction to some of Mr. Zuckerberg's public comments since their interview. In particular, his labelling of Mr. Musk as "hysterical" for his A.I. fears and lectured those who "fearmonger" about the dangers of A.I.. I have included more details below of Mr. Zuckerberg's comments.

Asked in Germany recently about Musk's forebodings, Zuckerberg called them "hysterical" and praised A.I. breakthroughs, including one system he claims can make cancer diagnoses for skin lesions on a mobile phone with the accuracy of "the best dermatologist."

"Unless we really mess something up," he said, the machines will always be subservient, not "superhuman."

"I think we can build A.I. so it works for us and helps us...Some people fearmonger about how A.I. is a huge danger, but that seems farfetched to me and much less likely than disasters due to widespread disease, violence, etc." Or as he put his philosophy at an April Facebook developers conference: "Choose hope over fear."

--

Alex Thompson

The New York Times

About this email: Musk's chief of staff forwards a query from a New York Times researcher working with Maureen Dowd asking for Musk's response to Mark Zuckerberg calling his AI fears 'hysterical'.

Notes: Text is the journalist's message as forwarded by Sam Teller; the exhibit (Dkt. 32-10) shows no covering note from Teller.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\)](#) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'; [Dkt. 32-10 \(Exhibit 9 to First Amended Complaint\)](#); [CourtListener](#) docket, [Musk v. Altman](#)

Tags: press, maureen-dowd, mark-zuckerberg, ai-risk, musk-v-openai

Musk's statement on Zuckerberg and why OpenAI exists

From: Elon Musk

Date: April 27, 2016 (Sent 12:24 PM)

To: Sam Teller

Subject: Re: Maureen Dowd

FULL TEXT · PRESS & JOURNALISTS · COURT RECORD · CONFIRMED

History unequivocally illustrates that a powerful technology is a double-edged sword. It would be foolish to assume that AI, arguably the most powerful of all technologies, only has a single edge.

The recent example of Microsoft's AI chatbot shows how quickly it can turn incredibly negative. The wise course of action is to approach the advent of AI with caution and ensure that its power is widely distributed and not controlled by any one company or person.

That is why we created OpenAI.

About this email: Musk drafts a response for the New York Times: powerful technology is a double-edged sword, Microsoft's chatbot shows how fast AI can turn negative, and AI's power should be widely distributed, which is why OpenAI was created.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\)](#) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'; [Dkt. 32-10 \(Exhibit 9 to First Amended Complaint\)](#); [CourtListener](#) docket, [Musk v. Altman](#)

Tags: press, maureen-dowd, mark-zuckerberg, ai-risk, mission, musk-v-openai

Sutskever asks how Musk's recruiting calls went

From: Ilya Sutskever

Date: May 3, 2016 (Time not stated)

To: Elon Musk

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

How did your calls with Roger and Igor go? Anything notable? Ilya.

About this email: Sutskever asks Musk how his calls with two OpenAI recruiting targets, 'Roger and Igor', went.

Notes: As read aloud by counsel at Sutskever's deposition (deposition Exhibit 8, described as an email exchange between Sutskever and Musk); Musk's side of the exchange was not read into the record. Sutskever testified these were people being recruited to OpenAI.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), [Dkt. 391-46 \(Ilya Sutskever deposition, Oct. 1, 2025; emails read into the record\)](#)

Tags: recruiting, ilya-sutskever, musk-v-openai

Landlord sends Musk the Pioneer Building lease draft

From: Atit Jariwala

Date: June 3, 2016 (Sent 4:41:38 AM PDT)

To: Elon Musk

Subject: RE: Pioneer Building

FULL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

Good morning - see attached lease draft. Please let me know if you have any comments/questions.

About this email: The Pioneer Building's landlord representative sends Musk a draft lease for the San Francisco building that would house OpenAI.

Notes: Earlier messages in the 'Pioneer Building' thread to which this replies are not shown in the exhibit as retrieved.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-56 (Exhibit 71, public redacted version, filed Jan. 16, 2026)

Tags: pioneer-building, lease, real-estate, musk-v-openai

Musk forwards lease draft: 'Please review'

From: Elon Musk

Date: June 3, 2016 (Sent Friday 12:51 PM)

To: Jared Birchall (recipient inferred from the chain)

Subject: Fwd: Pioneer Building

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Please review

About this email: Musk forwards the Pioneer Building lease draft to his family-office head for review.

Notes: Recipient and subject line are inferred from the surrounding chain; the exhibit header for this message is not fully shown.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-56 (Exhibit 71, public redacted version, filed Jan. 16, 2026)

Tags: pioneer-building, lease, real-estate, musk-v-openai

Birchall flags self-dealing risk on OpenAI building lease

From: Jared Birchall

Date: June 13, 2016 (Sent 2:19 PM)

To: Elon Musk

Subject: Pioneer building lease

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

The myCFO team suggests that having the Musk foundation lease a building for OpenAI is a problem because you are a director/chair of OpenAI. Because the foundation now has assets in excess of \$2 million, there will be an obligatory audit of the foundation. The lease would need to be disclosed. This could become a 'self dealing' issue in the eyes of the IRS because of your fiduciary position with both organizations. The way around it is to have the lease cosigned with OpenAI and another one of your entities - like Musk industries. We can still make the payments through the donor advised fund or the Musk foundation, but the lease wouldn't have to be disclosed as a Musk foundation transaction. Are you OK proceeding as outlined?

Jared J Birchall
Managing Director
Musk Family Office

About this email: Musk's family-office head warns that having the Musk Foundation lease a building for OpenAI could be seen as self-dealing because Musk chairs OpenAI, and proposes co-signing the lease with OpenAI and another Musk entity.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-8 (Exhibit 38, public redacted version, filed Jan. 16, 2026)

Tags: pioneer-building, lease, musk-foundation, self-dealing, tax, musk-v-openai

Musk: 'My control of OpenAI is limited'

From: Elon Musk

Date: June 13, 2016 (Sent between 2:19 PM and 4:08 PM, when Birchall forwarded it to myCFO; exact time not shown)

To: Jared Birchall

Subject: Pioneer building lease

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

This building will be used for more than OpenAI. My control of OpenAI is limited. There is more than one donor and there will ultimately be many donors.

I'm personally on the hook for the building, so it needs to be a building over which I have full control.

About this email: Musk says the Pioneer Building will be used for more than OpenAI, that his control of OpenAI is limited and it will have many donors, and that since he is personally liable for the building he needs full control of it.

Notes: Appears in the exhibit as a message forwarded by Jared Birchall to Musk's myCFO advisers on June 13, 2016 at 4:08 PM; the remainder of the chain is between Birchall and Ron Gong.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-8 (Exhibit 38, public redacted version, filed Jan. 16, 2026)

Tags: pioneer-building, lease, control, donors, musk-v-openai

Birchall weekly update: OpenAI's \$5M quarterly request

From: Jared Birchall

Date: July 9, 2016 (Sent 12:03 AM)

To: Elon Musk

Subject: Weekly Update

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Elon - I hope the trip has gone well! Attached is your weekly balance sheet update. You can 'ignore links' when it asks.

Confidential

- I spoke with Chris Clark yesterday and he requested that we send the quarterly \$5MM commitment at the beginning of the quarter - which is now. The only problem is we don't have sufficient cash to make that happen. We have \$150k in the Foundation and \$1.8 mil in the DAF. The foundation would need to sell some shares in order to generate the necessary cash. I share your desire to hold off for better valuations, but if we want to accommodate this request we'd need to gift 23k shares to the DAF where they would provide the payment. Thoughts?
- We will also have the ~\$150k payable each month for rent. That's another \$600k this year, in addition to the \$600k we'll need to support the USC project (among a few other commitments). Plus another \$10MM committed in January.
- Chris asked me about using the extra space in the building for some of the Y Combinator companies. I told him to hold off on any of those plans until I talk it through with you.
- As suggested, I have requested a copy of all the OpenAI formation documents and am awaiting these from Chris.

Jared J Birchall
Managing Director
Musk Family Office

About this email: Musk's family-office head reports that OpenAI's Chris Clark wants the quarterly \$5M commitment sent now but cash is short, lists rent and other commitments, and says Clark asked about using spare building space for YC companies.

Notes: Public redacted version; parts of the weekly update may be redacted.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-42 (Exhibit 18, [public redacted version](#), filed Jan. 16, 2026)

Tags: donations, funding, pioneer-building, family-office, y-combinator, musk-v-openai

Musk worries OpenAI is run 'as an extension of YCombinator'

From: Elon Musk

Date: July 9, 2016 (Sent Saturday, July 9, 2016; time not shown (between 12:03 AM and 9:03 PM))

To: Jared Birchall

Subject: Re: Weekly Update

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Did we pay the last OpenAI installment at beginning of quarter? If so, we should continue on that track. I committed to \$5M per quarter. It's fine to take the actions necessary for that support. Just need to know in advance if anything is going to generate press.

Please ask for (ideally electronic) copies of everything at OpenAI, including all offer letters plus anything else and lmk if there is anything concerning. Who is responsible for administrative work there? I have had very little bandwidth to think about the company and am a little worried that it is being managed as an extension of YCombinator. The YC stock as comp for many who joined obviously incents them in that direction. Does everyone have YC stock? That would be weird.

When exactly does the building become available and the rent start? I would like to use part of the building for Neuralink at some point, so no YC stuff.

About this email: Musk reaffirms his \$5M-per-quarter commitment, asks for copies of all OpenAI documents including offer letters, worries the company is being managed as an extension of Y Combinator, and says he wants part of the building for Neuralink.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-42 (Exhibit 18, [public redacted version](#), filed Jan. 16, 2026)

Tags: donations, y-combinator, governance, oversight, neuralink, pioneer-building, musk-v-openai

Birchall: no OpenAI entity existed to receive April payment

From: Jared Birchall

Date: July 9, 2016 (Sent 9:03 PM)

To: Elon Musk

Subject: Re: Weekly Update

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

We were supposed to begin the quarterly contributions to OpenAI in April, but because they didn't have an entity in place to even make a contribution we didn't pay. They scrambled and in June started using another 501(c)3 as a conduit until OpenAI is officially granted 501(c)3 status. We verified the conduit and just made the first payment in June. The application process is long and tedious. I'm not sure why they have taken so long to apply, but my inquiry lit a fire under them. Earlier today I received all the electronic formation documents for OpenAI, as well as for the conduit organization and have started going through them. I will request the other docs as well.

The plan is to always give through the DAF so no press is generated on that front. I'll keep tabs on other items that could potentially generate press.

The building rent starts August 15th. Atit already has our first month's rent. He'll make the building available as needed for planning and prep.

About this email: Birchall explains that quarterly contributions were due to start in April but OpenAI had no entity able to receive them, so the first payment went in June through another 501(c)(3) acting as a conduit; rent on the building starts August 15.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-42 (Exhibit 18, public redacted version, filed Jan. 16, 2026)

Tags: donations, funding, 501c3, fiscal-sponsor, pioneer-building, musk-v-openai

Musk: 'My credibility is at stake here'

From: Elon Musk

Date: July 10, 2016 (Sent Sunday, July 10, 2016; time not shown (before 5:42 AM PDT))

To: Jared Birchall

Subject: Re: Weekly Update

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

So I haven't sent anything to OpenAI? That's a really big deal. My credibility is at stake here.

About this email: Musk reacts with alarm to the idea that he has not yet sent money directly to OpenAI.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-42 (Exhibit 18, public redacted version, filed Jan. 16, 2026)

Tags: donations, funding, credibility, musk-v-openai

Birchall: OpenAI has received \$5.5M via the DAF

From: Jared Birchall

Date: July 10, 2016 (Sent 5:42:53 AM PDT)

To: Elon Musk

Subject: Re: Weekly Update

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

OpenAI has received \$5.5MM from you. But it was sent from the DAF and was channeled through the temporary 501(c)3 that they are using. Anyone who has given any money to OpenAI has gone the same route at this point.

About this email: Birchall reassures Musk that OpenAI has received \$5.5M from him, sent from his donor-advised fund through the temporary 501(c)(3) all donors are using.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 391-42 (Exhibit 18, public redacted version, filed Jan. 16, 2026)

Tags: donations, funding, fiscal-sponsor, musk-v-openai

Altman: negotiated \$50M Microsoft compute donation

From: Sam Altman

Date: August 29, 2016 (Sent 10:00 AM)

To: Elon Musk

Subject: microsoft/openAI

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

I have negotiated a \$50 million compute donation from them over the next 3 years!

Do you have any reason not to like them, or care about us switching over from Amazon?

About this email: Altman tells Musk he has negotiated a \$50 million compute donation from Microsoft over three years and asks if Musk objects to switching from Amazon.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\), Dkt. 379-38 \(Exhibit 4 to Microsoft's summary judgment motion, filed Jan. 6, 2026\)](#)

Tags: microsoft, compute, amazon, donation, musk-v-openai

Musk: OK only if Microsoft doesn't use it in marketing

From: Elon Musk

Date: August 29, 2016 (Sent Monday 10:15 AM)

To: Sam Altman

Subject: Re: microsoft/openAI

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

I'm ok with this only if they don't use it in marketing. I would also like to see the exact terms and conditions.

Gifts are only as good as the T&C.

What happened to Amazon?

I think Jeff is a bit of a tool and Satya is not, so I slightly prefer Microsoft, but I hate their marketing dept.

About this email: Musk says he accepts the Microsoft compute donation only if it is not used in marketing, asks for the exact terms, asks what happened with Amazon, and says he slightly prefers Microsoft but hates its marketing department.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\), Dkt. 379-38 \(Exhibit 4 to Microsoft's summary judgment motion, filed Jan. 6, 2026\)](#)

Tags: microsoft, compute, amazon, marketing, jeff-bezos, satya-nadella, musk-v-openai

Altman: Amazon balked on terms; details coming

From: Sam Altman

Date: August 29, 2016 (Sent 10:18 AM)

To: Elon Musk

Subject: Re: microsoft/openAI

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Amazon started really kicking us around on the T+C, especially on marketing commits.

And their offering wasn't that good technically anyway.

Will send you detailed terms by end of this week or so, still going back on forth on that part. So far looking clean though.

About this email: Altman explains that Amazon was difficult on terms, especially marketing commitments, that its offering was technically weaker, and that detailed Microsoft terms will follow.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\), Dkt. 379-38 \(Exhibit 4 to Microsoft's summary judgment motion, filed Jan. 6, 2026\)](#)

Tags: microsoft, compute, amazon, terms, musk-v-openai

Musk: will call Satya if terms are decent

From: Elon Musk

Date: August 30, 2016 (Header timestamp Tue, 30 Aug 2016 06:25:19 -0000 (UTC); late evening Aug 29 Pacific time)

To: Sam Altman

Subject: Re: microsoft/openAI

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Sounds good

I will call Satya if we get to decent terms. A simple text blog expressing appreciation of Microsoft's donation on our website will help them a lot. They can always point people to that.

About this email: Musk agrees, says he will call Satya Nadella if the terms are decent, and suggests a simple blog post thanking Microsoft for the donation.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\), Dkt. 379-38 \(Exhibit 4 to Microsoft's summary judgment motion, filed Jan. 6, 2026\)](#)

Tags: microsoft, compute, satya-nadella, terms, musk-v-openai

Altman sends Microsoft's Azure term sheet

From: Sam Altman

Date: September 16, 2016 (Sent 2:37 PM)

To: Elon Musk (cc: Sam Teller)

Subject: MSFT hosting deal

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Here are the MSFT terms. \$60MM of compute for \$10MM, and input from us on what they deploy in the cloud. LMK if you have any feedback.

Sam

Microsoft/OpenAI Terms

Microsoft and OpenAI: Accelerate the development of deep learning on Azure and CNTK

This non-binding term sheet ("Term Sheet") between Microsoft Corporation ("Microsoft") and OpenAI ("OpenAI") sets forth the terms for a potential business relationship between the parties. This Term Sheet is intended to form a basis of

discussion and does not state all matters upon which agreement must be reached before executing a legally binding commercial agreement ("Commercial Agreement"). The existence and terms of this Term Sheet, and all discussions related thereto or to a Commercial Agreement, are Confidential Information as defined and governed by the Non-Disclosure Agreement between the parties dated 17 March, 2016 ("NDA"). Except for the binding nature of the foregoing confidentiality obligations, this Term Sheet is non-binding.

Deal Purpose

OpenAI is focused on deep learning in such a way as to benefit humanity. Microsoft and OpenAI desire to partner to enable the acceleration of deep learning on Microsoft Azure. Towards this goal, Microsoft will provide OpenAI with Azure compute capabilities at a favorable price that would enable OpenAI to continue their mission effectively.

Deal Business Goal

Microsoft

- Accelerate deep learning environment on Azure
- Attract a net new audience of next generation developers
- Joint PR and evangelism of deep learning on Azure

OpenAI

- Deeply discounted GPU compute offering over the deal term (3 years) for use in their nonprofit research: \$60m of Compute for \$10m
- Joint PR and evangelism of OpenAI on Azure

Parties (Legal entities) Microsoft OpenAI

Proposed Deal Execution Date September 19, 2016

Proposed Deal Commencement Date Same as deal execution date

Legal Authoring Microsoft holds the pen.

Deal Term 3 years

Engineering Terms

- Compute: Microsoft will provide OpenAI GPU core hours of compute at the agreed upon price for OpenAI's workloads to run in Azure.
- Geographic Location: Geographic location decisions will be at Microsoft discretion depending on capacity and availability. Microsoft will also be responsible for sharing the deployment strategy and timelines with OpenAI.
- SLA: For all Virtual Machines that have two or more instances deployed in the same availability set, Microsoft guarantee OpenAI will have virtual machine connectivity to at least one instance at least 99.95% of the time. Microsoft will be held accountable to the SLA's provided on https://azure.microsoft.com/enus/support/legal/sla/virtual-machines/v1_2/
- Evaluation, Evangelization, and Usage of CNTK v2, Azure Batch and HD-Insight: OpenAI will evaluate CNTK v2, Azure Batch, and HDInsight for their research, provide feedback on how Microsoft can improve these products. OpenAI will work with

Microsoft to evangelize these products to their research and developer ecosystems, and evangelize Microsoft Azure as their preferred public cloud provider. At their sole discretion, and as it makes sense for their research, OpenAI will adopt these products

- Ramp: Microsoft and OpenAI will work together for creating a ramp plan that balances capacity per clusters. The initial timeline for ramp is a minimum of 30 days that will be augmented by Microsoft's capacity expansion plans in the coming months.
- Capacity: OpenAI will be given an allocation of capacity in the preview cluster (located in US South Central) for short term requirements and Microsoft will provide quota access to the subsequent K80 GPU clusters that go live in the 4th quarter of 2016 with the intention of more capacity in Q1 2017 (calendar year).

Financial Terms

- Financial Terms: Microsoft will offer \$60m worth of List Compute (including GPU) at a deep discount which results in a price of \$10m to be paid by OpenAI over the course of the deal. In the event OpenAI consumes less than \$10m worth of Azure compute, OpenAI will be responsible for paying the balance between the used amount and \$10m at the end of the deal term to Microsoft.

Marketing & PR Terms

Microsoft and OpenAI commit to jointly evangelizing deep learning capabilities on Azure as agreed upon by both parties.

- Ignite: Announce the partnership at Microsoft's Ignite event with executives (Sam Altman from OpenAI and Satya Nadella from Microsoft) from both parties inaugurating the collaboration
- PR: Microsoft and OpenAI will work together to issue a joint press release about the partnership including any materials such as blog posts and videos.

About this email: Altman forwards Microsoft's proposed terms, \$60M of Azure compute for \$10M plus OpenAI feedback and joint evangelism, and asks Musk for feedback.

Notes: The Microsoft/OpenAI non-binding term sheet is reproduced in the email as published; its original table layout is flattened in the transcription.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits, as transcribed in LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-11 \(Exhibit 10 to First Amended Complaint\)](#); [CourtListener docket, Musk v. Altman](#)

Tags: microsoft, compute, azure, term-sheet, musk-v-openai

Musk on Microsoft terms: 'made me feel nauseous'

From: Elon Musk

Date: September 16, 2016 (Sent 3:10 PM)

To: Sam Altman (cc: Sam Teller)

Subject: MSFT hosting deal

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

This actually made me feel nauseous. It sucks and is exactly what I would expect from them.

Let's just say that we are willing to have Microsoft donate spare computing time to OpenAI and have that be known, but we want do any contract or agree to "evangelize".

They can turn us off at any time and we can leave at any time.

About this email: Musk rejects Microsoft's proposed terms, saying OpenAI should accept only donated spare compute with no contract and no obligation to 'evangelize' Microsoft products.

Notes: 'we want do any contract' is as transcribed (apparently 'won't').

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits](#), as transcribed in [LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-11 \(Exhibit 10 to First Amended Complaint\)](#); [CourtListener docket](#), [Musk v. Altman](#)

Tags: microsoft, compute, azure, independence, musk-v-openai

Altman: Microsoft already agreed to drop the clause

From: Sam Altman

Date: September 16, 2016 (Sent 3:33 PM)

To: Elon Musk (cc: Sam Teller)

Subject: MSFT hosting deal

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

I had the same reaction after reading that section and they've already agreed to drop.

We had originally just wanted space cycles donated but the team wanted more certainty that capacity will be available. But I'll work with MSFT to make sure there are no strings attached

About this email: Altman says he had the same reaction, that Microsoft has agreed to drop the evangelism section, and that he will ensure there are no strings attached.

Notes: 'space cycles' is as transcribed (apparently 'spare').

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits](#), as transcribed in [LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-11 \(Exhibit 10 to First Amended Complaint\)](#); [CourtListener docket](#), [Musk v. Altman](#)

Tags: microsoft, compute, azure, musk-v-openai

Musk: 'No certainty either way. No contract.'

From: Elon Musk

Date: September 16, 2016 (Time not shown in the exhibit; between 3:33 PM and 6:45 PM)

To: Sam Altman (cc: Sam Teller)

Subject: MSFT hosting deal

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

We should just do this low key. No certainty either way. No contract.

About this email: Musk says the Microsoft arrangement should be low key with no contract.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits](#), as transcribed in [LessWrong's 'OpenAI Email Archives'](#); [Dkt. 32-11 \(Exhibit 10 to First Amended Complaint\)](#); [CourtListener docket](#), [Musk v. Altman](#)

Tags: microsoft, compute, independence, musk-v-openai

Altman: will see how much he can get

From: Sam Altman

Date: September 16, 2016 (Sent 6:45 PM)

To: Elon Musk (cc: Sam Teller)

Subject: MSFT hosting deal

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

ok will see how much \$ I can get in that direction.

About this email: Altman agrees to pursue a looser arrangement with Microsoft.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits](#), as transcribed in LessWrong's 'OpenAI Email Archives'; [Dkt. 32-11 \(Exhibit 10 to First Amended Complaint\)](#); [CourtListener docket](#), Musk v. Altman

Tags: microsoft, compute, musk-v-openai

Teller: Microsoft offers \$50M with no strings

From: Sam Teller

Date: September 20, 2016 (Sent 8:05 PM)

To: Elon Musk

Subject: MSFT hosting deal

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Microsoft is now willing to do the agreement for a full \$50m with "good faith effort at OpenAI's sole discretion" and full mutual termination rights at any time. No evangelizing. No strings attached. No looking like lame Microsoft marketing pawns. Ok to move ahead?

About this email: Teller reports Microsoft will do the agreement for \$50M with a good-faith-effort clause at OpenAI's discretion, mutual termination rights and no evangelizing, and asks if it is OK to proceed.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits](#), as transcribed in LessWrong's 'OpenAI Email Archives'; [Dkt. 32-11 \(Exhibit 10 to First Amended Complaint\)](#); [CourtListener docket](#), Musk v. Altman

Tags: microsoft, compute, azure, musk-v-openai

Musk: don't seem like 'Microsoft's marketing bitch'

From: Elon Musk

Date: September 21, 2016 (Sent 12:09 AM)

To: Sam Teller

Subject: MSFT hosting deal

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Fine by me if they don't use this in active messaging. Would be worth way more than \$50M not to seem like Microsoft's marketing bitch.

About this email: Musk approves the revised Microsoft deal provided Microsoft does not use it in active messaging.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722\) court exhibits](#), as transcribed in LessWrong's 'OpenAI Email Archives'; [Dkt. 32-11 \(Exhibit 10 to First Amended Complaint\)](#); [CourtListener docket](#), Musk v. Altman

Tags: microsoft, compute, independence, pr, musk-v-openai

Sutskever: meetings with Musk help run OpenAI

From: Ilya Sutskever

Date: September 24, 2016 (Sent Saturday 11:12 AM)

To: Elon Musk

Subject: Re: BTW

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

I find our meetings to be very helpful for running OpenAI and for figuring out the right medium term goals.

About this email: Sutskever tells Musk their meetings are very helpful for running OpenAI and setting medium-term goals.

Notes: Earliest message shown in the exhibit; any earlier 'BTW' message it replies to is not reproduced.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 379-72 (Exhibit 31, filed Jan. 6, 2026)

Tags: ilya-sutskever, meetings, management, musk-v-openai

Musk wants to learn OpenAI's technical issues in detail

From: Elon Musk

Date: September 24, 2016 (Sent Saturday 11:16 AM)

To: Ilya Sutskever

Subject: Re: BTW

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Cool, I certainly enjoy them too. Also, really want to learn more at a detailed level about the technical issues that we are wrestling with. Really interesting.

About this email: Musk says he enjoys the meetings too and wants to learn in more detail about the technical problems OpenAI is working on.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 379-72 (Exhibit 31, filed Jan. 6, 2026)

Tags: ilya-sutskever, meetings, technical, musk-v-openai

Sutskever: longer visits; primer for non-experts coming

From: Ilya Sutskever

Date: September 24, 2016 (Sent Saturday 11:47 AM)

To: Elon Musk

Subject: Re: BTW

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

For that you'd need to take the expensive move of lengthening your visits ...

Also, some of us have been thinking of writing a condensed introduction into our subject matter for people who not experts in the field -- I'll send you this document once its ready. Expect a few weeks to a month.

About this email: Sutskever says learning the technical detail would require longer visits and promises a condensed introduction to the field for non-experts within a few weeks to a month.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 379-72 (Exhibit 31, filed Jan. 6, 2026)

Tags: ilya-sutskever, meetings, technical, musk-v-openai

Musk: more time after SolarCity merger, rocket flying

From: Elon Musk

Date: September 24, 2016 (Sent Saturday 11:55 AM)

To: Ilya Sutskever

Subject: Re: BTW

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Once the SolarCity merger is done and the rocket is flying again, I will should be able to do so.

About this email: Musk says he should be able to spend longer at OpenAI once the SolarCity merger is done and SpaceX's rocket is flying again.

Notes: 'I will should' is as printed in the exhibit.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 379-72 (Exhibit 31, filed Jan. 6, 2026)

Tags: ilya-sutskever, time-commitment, solarcity, spacex, musk-v-openai

Sutskever: 'Good luck with both.'

From: Ilya Sutskever

Date: September 24, 2016 (Sent 6:59:56 PM)

To: Elon Musk

Subject: Re: BTW

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Good luck with both.

About this email: Sutskever wishes Musk luck with the SolarCity merger and SpaceX's return to flight.

Sources: [Musk v. Altman \(N.D. Cal., No. 4:24-cv-04722-YGR\)](#), Dkt. 379-72 (Exhibit 31, filed Jan. 6, 2026)

Tags: ilya-sutskever, solarcity, spacex, musk-v-openai

Sutskever's first bi-weekly update: compute, Dota, robotics

From: Ilya Sutskever

Date: June 12, 2017

To: Greg Brockman, [redacted], Elon Musk

Subject: Bi-weekly updates

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

This is the first of our bi-weekly updates. The goal is to keep you up to date, and to help us make greater use from your visits.

Compute:

- Compute is used in two ways: it is used to run a big experiment quickly, and it is used to run many experiments in parallel.
- 95% of progress comes from the ability to run big experiments quickly. The utility of running many experiments is much less useful.
- In the old days, a large cluster could help you run more experiments, but it could not

help with running a single large experiment quickly.

- For this reason, an academic lab could compete with Google, because Google's only advantage was the ability to run many experiments. This is not a great advantage.
- Recently, it has become possible to combine 100s of GPUs and 100s of CPUs to run an experiment that's 100x bigger than what is possible on a single machine while requiring comparable time. This has become possible due to the work of many different groups. As a result, the minimum necessary cluster for being competitive is now 10-100x larger than it was before.
- Currently, every Dota experiment uses 1000+ cores, and it is only for the small 1v1 variant, and on extremely small neural network policies. We will need more compute to just win the 1v1 variant. To win the full 5v5 game, we will need to run fewer experiments, where each experiment is at least 1 order of magnitude larger (possibly more!).
- TLDR: What matters is the size and speed of our experiments. In the old days, a big cluster could not let anyone run a larger experiment quickly. Today, a big cluster lets us run a large experiment 100x faster.
- In order to be capable of accomplishing our projects even in theory, we need to increase the number of our GPUs by a factor of 10x in the next 1-2 months (we have enough CPUs). We will discuss the specifics in our in-person meeting.

Dota 2:

- We will solve the 1v1 version of the game in 1 month. Fans of the game care about 1v1 a fair bit.
- We are now at a point where a single experiment consumes 1000s of cores, and where adding more distributed compute increases performance.
- Here is a cool video of our bot doing something rather clever:
<https://www.youtube.com/watch?v=Y-vxbREX5ck&feature=youtu.be&t=99>.

Rapid learning of new games:

- Infra work is underway
- We implemented several baselines
- Fundamentally, we're not where we want to be, and are taking action to correct this.

Robotics:

- Current status: The HER algorithm (https://www.youtube.com/watch?v=Dz_HuzgMzxo) can learn to solve many low-dimensional robotics tasks that were previously unsolvable very rapidly. It is non-obvious, simple, and effective.
- In 6 months, we will accomplish at least one of: single-handed Rubik's cube, pen spinning (<https://www.youtube.com/watch?v=dDavyRnEPrI>), Chinese balls spinning (<https://www.youtube.com/watch?v=M9N1duIl4Fc>) using the HER algorithm and using a sim2real method [such as <https://blog.openai.com/spam-detection-in-the-physical-world/>].
- The above will be deployed on the robotic hand: [Link to Google Drive] [this video is human controlled, not algorithmic controlled. Need to be logged in to the OpenAI account to see the video].

Self play as a key path to AGI:

- Self play in multiagent environments is magical: if you place agents into an environment, then no matter how smart (or not smart) they are, the environment will provide them with the exact level of challenge, which can be faced only by outsmarting the competition. So for example, if you have a group of children, they will find each other's company to be challenging; likewise for a collection of super intelligences of comparable intelligence. So the "solution" to self-play is to become more and more intelligent, without bound.
- Self-play lets us get "something out of nothing." The rules of a competitive game can be simple, but the best strategy for playing this game can be immensely complex. [motivating example: <https://www.youtube.com/watch?v=u2T77mQmJYI>].
- Training agents in simulation to develop very good dexterity via competitive fighting, such as wrestling. Here is a video of ant-shaped robots that we trained to struggle: <redacted>
- Current work on self-play: getting agents to learn to develop a language [gifs in <https://blog.openai.com/learning-to-cooperate-compete-and-communicate/>]. Agents are doing "stuff," but it's still work in progress.

We have a few more cool smaller projects. Updates to be presented as they produce significant results.

About this email: Sutskever sends Musk the first of a planned series of bi-weekly progress updates, arguing that large fast experiments drive progress and that OpenAI needs 10x more GPUs within one to two months, and reporting on Dota 2, robotics and self-play.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Published by OpenAI in its Dec 2024 blog post (marked as blog-sourced in the LessWrong archive). One Google Drive link was rendered by the retrieval tool as '[Link to Google Drive]' rather than the full URL; one link is redacted in the source.

Sources: [OpenAI blog: Elon Musk wanted an OpenAI for-profit \(Dec 13, 2024\)](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: compute, dota, robotics, self-play, status-update

Musk: 'Thanks, this is a great update'

From: Elon Musk

Date: June 12, 2017

To: Ilya Sutskever; cc Greg Brockman, [redacted]

Subject: Re: Bi-weekly updates

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

Thanks, this is a great update.

About this email: Musk's one-line acknowledgement of Sutskever's first bi-weekly progress update.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Jun 12, 2017 10:52 PM. The retrieval tool listed recipients as Ilya Sutskever, Greg Brockman and a redacted party; exact To/CC split not verified.

Sources: [OpenAI blog: Elon Musk wanted an OpenAI for-profit \(Dec 13, 2024\)](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: status-update, compute, short-reply

Musk: make sure compute is not a constraint

From: Elon Musk

Date: June 13, 2017

To: Ilya Sutskever; cc Greg Brockman, [redacted]

Subject: Re: Bi-weekly updates

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

Ok. Let's figure out the least expensive way to ensure compute power is not a constraint...

About this email: Replying the next morning to Sutskever's request for 10x more GPUs, Musk says to find the cheapest way to ensure compute does not limit OpenAI.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Jun 13, 2017 10:24 AM. Published in OpenAI's Dec 2024 blog post.

Sources: [OpenAI blog: Elon Musk wanted an OpenAI for-profit \(Dec 13, 2024\)](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: compute, funding, short-reply

Musk pitches Nadella on 10,000 GPU servers for OpenAI

From: Elon Musk

Date: June 28, 2017

To: Ilya Sutskever, Greg Brockman; cc Sam Altman

Subject: The 10,000

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Just spoke to Satya and talked about OpenAI needing one the order of 10,000 servers with the latest Nvidia GPUs to beat the best human players at competitive eSports games.

This would obviously be a major opportunity for Microsoft to promote Azure relative to other cloud systems.

He said he'd talk internally and get back to me soon. Please send me the ideal spec for the individual servers and how they are connected. What ideally do we want Microsoft to do? Sounds like there is a good chance they will do it.

About this email: Musk reports that he has just spoken to Satya Nadella about OpenAI needing on the order of 10,000 servers with the latest Nvidia GPUs to beat top human esports players, pitched as an Azure marketing opportunity, and asks for the ideal server spec.

Notes: Text taken from the PDF text layer of the court exhibit as filed on the Musk v. Altman docket (N.D. Cal. 4:24-cv-04722), via CourtListener/RECAP. Sent Wed, 28 Jun 2017 14:10:58 -0700. 'one the order' is as written in the exhibit. Paragraph breaks inferred from the line layout. The same email is Ex. 11 to the Eynon Declaration (Dkt. 353).

Sources: [Musk v. Altman, Dkt. 351-36 \(Ex. 33 to Hawes Declaration, Bates 2024MUSK-0006691\)](#)

Tags: microsoft, compute, azure, dota, satya-nadella, musk-v-openai

Sutskever: 'The business of building AGI'

From: Ilya Sutskever

Date: July 12, 2017

To: Elon Musk, Greg Brockman

Subject: The business of building AGI

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

We usually decide that problems are hard because smart people have worked on them unsuccessfully for a long time. It's easy to think that this is true about AI. However, the past five years of progress have shown that the earliest and simplest ideas about AI — neural networks — were right all along, and we needed modern hardware to get them working.

Historically, AI breakthroughs have consistently happened with models that take between 7-10 days to train. This means that hardware defines the surface of potential AI breakthroughs. This is a statement about human psychology more than about AI. If experiments take longer than this, it's hard to keep all the state in your head and iterate and improve. If experiments are shorter, you'll just use a bigger model.

It's not so much that AI progress is a hardware game, any more than physics is a particle accelerator game. But if our computers are too slow, no amount of cleverness will result in AGI, just like if a particle accelerator is too small, we have no shot at figuring out how the universe works. Fast enough computers are a necessary ingredient, and all past failures may have been caused by computers being too slow for AGI.

Until very recently, there was no way to use many GPUs together to run faster experiments, so academia had the same "effective compute" as industry. But earlier this year, Google used two orders of magnitude more compute than is typical to optimize the architecture of a classifier, something that usually requires lots of researcher time. And a few months ago, Facebook released a paper showing how to train a large ImageNet model with near-linear speedup to 256 GPUs (given a specially-configured cluster with high-bandwidth interconnects).

Over the past year, Google Brain produced impressive results because they have an order of magnitude or two more GPUs than anyone. We estimate that Brain has around 100k GPUs, FAIR has around 15-20k, and DeepMind allocates 50 per researcher on question asking, and rented 5k GPUs from Brain for AlphaGo. Apparently, when people run neural networks at Google Brain, it eats up everyone's quotas at DeepMind.

We're still missing several key ideas necessary for building AGI. How can we use a system's understanding of "thing A" to learn "thing B" (e.g. can I teach a system to count, then to multiply, then to solve word problems)? How do we build curious systems? How do we train a system to discover the deep underlying causes of all types of phenomena — to act as a scientist? How can we build a system that adapts to new situations on which it hasn't been trained on precisely (e.g. being asked to apply familiar concepts in an unfamiliar situation)? But given enough hardware to run the relevant experiments in 7-10 days, history indicates that the right algorithms will be found, just like physicists would quickly figure out how the universe works if only they had a big

enough particle accelerator.

There is good reason to believe that deep learning hardware will speed up 10x each year for the next four to five years. The world is used to the comparatively leisurely pace of Moore's Law, and is not prepared for the drastic changes in capability this hardware acceleration will bring. This speedup will happen not because of smaller transistors or faster clock cycles; it will happen because like the brain, neural networks are intrinsically parallelizable, and new highly parallel hardware is being built to exploit this.

Within the next three years, robotics should be completely solved, AI should solve a long-standing unproven theorem, programming competitions should be won consistently by AIs, and there should be convincing chatbots (though no one should pass the Turing test). In as little as four years, each overnight experiment will feasibly use so much compute capacity that there's an actual chance of waking up to AGI, given the right algorithm — and figuring out the algorithm will actually happen within 2-4 further years of experimenting with this compute in a competitive multiagent simulation.

To be in the business of building safe AGI, OpenAI needs to:

- Have the best AI results each year. In particular, as hardware gets exponentially better, we'll have dramatically better results. Our DOTA and Rubik's cube projects will have impressive results for the current level of compute. Next year's projects will be even more extreme, and what's realistic depends primarily on what compute we can access.
- Increase our GPU cluster from 600 GPUs to 5000 GPUs ASAP. As an upper bound, this will require a capex of \$12M and an opex of \$5-6M over the next year. Each year, we'll need to exponentially increase our hardware spend, but we have reason to believe AGI can ultimately be built with less than \$10B in hardware.
- Increase our headcount: from 55 (July 2017) to 80 (January 2018) to 120 (January 2019) to 200 (January 2020). We've learned how to organize our current team, and we're now bottlenecked by number of smart people trying out ideas.
- Lock down an overwhelming hardware advantage. The 4-chip card that Jim Keller says he can build in 2 years is effectively TPU 3.0 and (given enough quantity) would allow us to be on an almost equal footing with Google on compute. The Cerebras design is far ahead of both of these, and if they're real then having exclusive access to them would put us far ahead of the competition. We have a structural idea for how to do this given more due diligence, best to discuss on a call.

2/3/4 will ultimately require large amounts of capital. If we can secure the funding, we have a real chance at setting the initial conditions under which AGI is born. Increased funding needs will come lockstep with increased magnitude of results. We should discuss options to obtain the relevant funding, as that's the biggest piece that's outside of our direct control.

Progress this week:

- We've beat our top 1v1 test player (he's top 30 in North America at 1v1, and beats the top 1v1 player about 30% of the time), but the bot can also be exploited by playing

weirdly. We're working on understanding these exploits and cracking down on them.

* Repeated from Saturday, here's the first match where we beat our top test player:

<https://www.youtube.com/watch?v=FBoUHay7XBI&feature=youtu.be&t=345>

* Every additional day of training makes the bot stronger and harder to exploit.

- Robot getting closer to solving Rubik's cube.

* The improved cube simulation teleoperated by a human: <redacted>.

- Our defense against adversarial examples is starting to work on ImageNet.

* We will completely solve the problem of adversarial examples by the end of August.

We also have four strong researchers leaving OpenAI:

- Peter Xi Chen, Rocky Yan Duan, and Jonathan Ho are leaving to start a robotics startup to solve all robotics manufacturing in China. They're highly motivated to start a company (Peter and Rocky had an unsuccessful startup before OpenAI).

- Alec Radford is leaving due to a mismatch in research interests (he doesn't believe AGI is near, and wants to work on what he views as shorter-term problems).

About this email: Sutskever lays out the thesis that hardware is the binding constraint on AGI, predicts 10x yearly hardware speedups, and lists what OpenAI needs: best results each year, 5,000 GPUs, headcount growth to 200 by 2020, and an overwhelming hardware advantage, all requiring large amounts of capital.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Published by OpenAI in its Dec 2024 blog post. The email ends with three blacked-out passages in the source. No reply from Musk is reproduced. Redacted passages restored from the unsealed court exhibit (Dkt. 379-77).

Sources: [OpenAI blog: Elon Musk wanted an OpenAI for-profit \(Dec 13, 2024\)](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [Musk v. Altman, Dkt. 379-77 \(Ex. 38 to Hawes Declaration, Bates 2024MUSK-0006991; unsealed Jan 6, 2026\)](#)

Tags: agi, compute, hardware, funding, cerebras, strategy

Zilis preps Musk: 'build AGI before anyone else', bet on hardware

From: Shivon Zilis

Date: July 18, 2017

To: Elon Musk, Sam Teller

Subject: OpenAI notes in advance of tonight's meeting

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

A quick heads up what the conversation with Greg and Ilya will look like:

OpenAI's Mission: Want to confirm what it should be. Given the AGI timelines keeps compressing, they think it should be "build AGI before anyone else" (thinking of Google and China, specifically). This isn't a departure, per se, but want to confirm that you still think the best possible future would be brought about this way? This would of course involve being much less "open" and require decisions that would prioritize speed over other considerations (safety, government involvement, etc).

Hardware: if it's true that OpenAI should dead set on building AGI before anyone else, the critical factor is hardware. They think if all resources exhausted, best you can get in terms of human capital/ ingenuity over DeepMind 2-3x, whereas if you buy that HW is

really going to depart from Moore's law and hit 10x per year it's not even a contest as to which battlefield more critical right now. They are at the point where they want to bet the farm on the right hardware strategy.

Cerebras v A Billion Cores: In an ideal world both would be pursued. Cerebras is a high beta experiment, they understand that, but think that if it succeeds it's so game changing that if Google gets it, it's game over re: race for AGI. Reason is that even if we started on a billion core system today, it's still 18 months behind and that's enough to lose the battle. Want to keep working on billion core idea since higher probability of eventual success but think it's prudent to figure out the right strategy to still merge or exclusively partner with Cerebras as well. They believe the combined entity, assuming OpenAI converts from being a non-profit, would be more valuable?? (Note: Andrew Feldman reached out to them to discuss next steps. That meeting is planned for next week)

Dota supremacy ... but at what cost: The compute for Dota 5v5 is hard to estimate but would get into the \$50M range on the high side. They don't have perfect data but estimate that AlphaGo was somewhere in this ballpark. That begs the question of is this worth the price tag? They think yes, since it's likely to have an equal or better outcome to AlphaGo but want to both confirm that you agree and also make sure there is a path to funding should it come to that.

About this email: Before a meeting with Brockman and Sutskever, Zilis briefs Musk that they want to confirm OpenAI's mission should be to build AGI before anyone else (meaning being much less open), bet the farm on hardware, pursue a merger or exclusive partnership with Cerebras assuming OpenAI converts from a non-profit, and spend up to about \$50M on compute for Dota 5v5.

Notes: Text taken from the PDF text layer of the court exhibit as filed on the Musk v. Altman docket (N.D. Cal. 4:24-cv-04722), via CourtListener/RECAP. Sent Tuesday, July 18, 2017 1:24:27 PM. Three obvious OCR artifacts in the PDF text layer were normalised ('l Ox' to '10x', 'AGL' to 'AGI.', 'ifwe' to 'if we'); 'should dead set' is as in the exhibit. Originally filed under seal with OpenAI's summary judgment motion.

Sources: [Musk v. Altman, Dkt. 379-27 \(Ex. 58 to Wiener Declaration, Bates 2024MUSK-0010721; unsealed Jan 6, 2026\)](#)

Tags: mission, hardware, cerebras, for-profit, dota, agi-race, musk-v-openai

Sutskever biweekly update: for-profit structure 'coming up'

From: Ilya Sutskever

Date: July 20, 2017

To: Elon Musk, Greg Brockman

Subject: biweekly update

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

- The robot hand can now solve a Rubik's cube in simulation:

<https://drive.google.com/a/openai.com/file/d/0B60rCy4P2FOIenlLdzN2LXdiOTQ/view?usp=sharing> (needs OpenAI login)

Physical robot will do same in September

- 1v1 bot is no longer exploitable

It can no longer be beaten using "unconventional" strategies

On track to beat all humans in 1 month

- Athletic competitive robots:

<https://drive.google.com/a/openai.com/file/d/0B60rCy4P2FOIZE4wNVdlbkx6U2M/view?usp=sharing> (needs OpenAI login)

- Released an adversarial example that fools a camera from all angles simultaneously:

<https://blog.openai.com/robust-adversarial-inputs/>

- DeepMind's directly used one of our algorithms to produce their parkour results:

DeepMind's results:

<https://deepmind.com/blog/producing-flexible-behaviourssimulated-environments/>

DeepMind's technical papers explicitly state they directly used our algorithms

Our blogpost about our algorithm: <https://blog.openai.com/openai-baselines-ppo/> (DeepMind used an older version).

- Coming up:

Designing the for-profit structure

Negotiate merger terms with Cerebras

More due diligence with Cerebras

About this email: A short progress update to Musk on the robot hand, the Dota 1v1 bot and adversarial examples, ending with a 'coming up' list that includes designing the for-profit structure and negotiating a merger with Cerebras.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Indentation reconstructed from the nested bullets in the transcription.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: status-update, robotics, dota, for-profit, cerebras, musk-v-openai

Musk on China's AI plan: 'another reason to change course'

From: Elon Musk

Date: July 21, 2017

To: Greg Brockman, Ilya Sutskever

Subject: Beijing Wants A.I. to Be Made in China by 2030

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

They will do whatever it takes to obtain what we develop. Maybe another reason to change course.

<https://mobile.nytimes.com/2017/07/20/business/china-artificialintelligence.html?smprod=nytcore-iphone&smid=nytcore-iphone-share&referrer=>

About this email: Forwarding a news article about China's 2030 AI ambitions, Musk says China will do whatever it takes to obtain what OpenAI develops and suggests it is another reason to change course.

Notes: Sent Fri, Jul 21, 2017 at 3:34 AM. Body from the LessWrong transcription of OpenAI's Dec 2024 blog post; the article link is taken from the unredacted chain filed as Ex. 51 to the Wiener Declaration (Dkt. 328-55), whose OCR text layer garbles the URL slightly (it is shown here as the retrieval tool normalised it).

Sources: [OpenAI blog: Elon Musk wanted an OpenAI for-profit \(Dec 13, 2024\)](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [Musk v. Altman, Dkt. 328-55 \(Ex. 51 to Wiener Declaration; full unredacted chain\)](#)

Tags: china, for-profit, strategy, national-security

Brockman: non-profit, then for-profit, then government project

From: Greg Brockman

Date: July 21, 2017

To: Elon Musk; cc Ilya Sutskever

Subject: Re: Beijing Wants A.I. to Be Made in China by 2030

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

100% agreed. We think the path must be:

1. AI research non-profit (through end of 2017)
2. AI research + hardware for-profit (starting 2018)
3. Government project (when: ??)

Government project is helpful for financial resources, but crucial for security - don't know how to defend against Putin or North Korea otherwise. When ready, we must proactively form the project around us, rather than being slurped in against our will.

-gdb

About this email: Brockman agrees with Musk and sketches a three-stage path: research non-profit through 2017, a research-plus-hardware for-profit from 2018, and eventually a government project, which he calls crucial for security against Putin or North Korea.

Notes: The paragraph about a government project was blacked out in OpenAI's Dec 2024 blog post and in the LessWrong archive; it appears unredacted in the chain filed as Ex. 51 to the Wiener Declaration (Dkt. 328-55, Oct 17, 2025), from which this text is taken. The exhibit quotes the message as sent 'Jul 22, 2017, at 4:32 AM' (time zone of the quoting client); the blog post gives Jul 21, 2017 1:18 PM.

Sources: [OpenAI blog: Elon Musk wanted an OpenAI for-profit \(Dec 13, 2024\)](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [Musk v. Altman, Dkt. 328-55 \(Ex. 51 to Wiener Declaration; full unredacted chain\)](#)

Tags: china, for-profit, strategy, government, security

Musk: 'I have a tentative game plan'

From: Elon Musk

Date: July 21, 2017

To: Greg Brockman; cc Ilya Sutskever, [redacted]

Subject: Re: Beijing Wants A.I. to Be Made in China by 2030

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

Let's talk Sat or Sun. I have a tentative game plan that I'd like to run by you.

About this email: Musk proposes talking over the weekend about a tentative game plan for OpenAI's change of course.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Timestamp given as Jul 21, 2017 1:18 PM in the transcription (same minute as Brockman's message).

Sources: [OpenAI blog: Elon Musk wanted an OpenAI for-profit \(Dec 13, 2024\)](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: for-profit, strategy, short-reply

Musk offers OpenAI leaders Founder Series Model 3s

From: Elon Musk

Date: July 21, 2017

To: Ilya Sutskever, Greg Brockman (reply on the 'Beijing' thread)

Subject: Re: Beijing Wants A.I. to Be Made in China by 2030 - NYTimes.com

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

By the way, in appreciation for what you've done to get OpenAI to where it is today, I would like to give you each a Founder Series Model 3. These are the earliest cars produced and are not available to the public.

If there are a few others at OpenAI who you think also really deserve one (maybe Wojciech?), I am happy to do the same for them.

Hope you like it!

Elon

About this email: Minutes after agreeing to discuss a change of course for OpenAI, Musk tells Sutskever and Brockman he wants to give each of them a Founder Series Tesla Model 3 in appreciation, and offers the same for a few other deserving staff.

Notes: Text taken from the PDF text layer of the court exhibit as filed on the Musk v. Altman docket (N.D. Cal. 4:24-cv-04722), via CourtListener/RECAP. Quoted in the chain as sent 'Fri, Jul 21, 2017 at 1:25 PM'. This part of the thread was not included in OpenAI's Dec 2024 blog post. OpenAI's summary judgment filing says Musk donated the four vehicles to OpenAI to be given to four employees as bonuses.

Sources: [Musk v. Altman, Dkt. 328-55 \(Ex. 51 to Wiener Declaration, Bates OPENAI_MUSK00020777\)](#)

Tags: tesla, model-3, gift, compensation, founders, musk-v-openai

Sutskever thanks Musk for the Model 3 offer

From: Ilya Sutskever

Date: July 21, 2017 (Quoted in the chain as 'Jul 22, 2017, at 6:48 AM' in the sender-side time zone shown; it falls between Musk's 1:25 PM and 8:53 PM (header time) messages of Jul 21.)

To: Elon Musk; cc Greg Brockman, John Schulman, Wojciech Zaremba

Subject: Re: Beijing Wants A.I. to Be Made in China by 2030 - NYTimes.com

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Wow, thank you! This is a very kind gesture that we greatly appreciate. Yes, Wojciech definitely deserves it as much as we do, as does + John Schulman.

About this email: Sutskever thanks Musk for the offer of Founder Series Model 3s and says Wojciech Zaremba and John Schulman deserve one as well.

Notes: Text taken from the PDF text layer of the court exhibit as filed on the Musk v. Altman docket (N.D. Cal. 4:24-cv-04722), via CourtListener/RECAP. Recipients inferred from the header of Musk's reply that follows.

Sources: [Musk v. Altman, Dkt. 328-55 \(Ex. 51 to Wiener Declaration, Bates OPENAI_MUSK00020777\)](#)

Tags: tesla, model-3, gift, founders, musk-v-openai

Musk: 'Consider it done!' on four Model 3s

From: Elon Musk

Date: July 21, 2017

To: Ilya Sutskever; cc Greg Brockman, John Schulman, Wojciech Zaremba, Jared Birchall

Subject: Re: Beijing Wants A.I. to Be Made in China by 2030 - NYTimes.com

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Consider it done!

Jared, please facilitate. Let's try to slot these four cars into next month's production.

About this email: Musk agrees to give four Founder Series Model 3s to OpenAI staff and asks Jared Birchall to arrange for them to be slotted into the next month's production.

Notes: Text taken from the PDF text layer of the court exhibit as filed on the Musk v. Altman docket (N.D. Cal. 4:24-cv-04722), via CourtListener/RECAP. Header timestamp 7/21/2017 8:53:24 PM (time zone not stated).

Sources: [Musk v. Altman, Dkt. 328-55 \(Ex. 51 to Wiener Declaration, Bates OPENAI_MUSK00020777\)](#)

Tags: tesla, model-3, gift, jared-birchall, musk-v-openai

Musk: 'Time to make the next step for OpenAI'

From: Elon Musk

Date: August 11, 2017

To: Greg Brockman, Ilya Sutskever, Sam Altman; cc [redacted], Shivon Zilis

Subject: Tomorrow afternoon

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

[redacted] Are you guys able to meet or do a conf call tomorrow afternoon? Time to make the next step for OpenAI. This is the triggering event.

About this email: On the night OpenAI's Dota bot beat a top professional, Musk asks the founders to meet or call the next afternoon, calling it the triggering event for OpenAI's next step.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Aug 11, 2017 9:17 PM. The opening of the message is redacted. Published in OpenAI's Dec 2024 blog post, which ties it to the Dota 2 1v1 win at The International.

Sources: [OpenAI blog: Elon Musk wanted an OpenAI for-profit \(Dec 13, 2024\)](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: for-profit, dota, triggering-event, meeting

Zilis relays Brockman and Sutskever's seven open questions

From: Shivon Zilis

Date: August 28, 2017

To: Elon Musk; cc Sam Teller

Subject: OpenAI notes

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Elon,

As I'd mentioned, Greg had asked to talk through a few things this weekend. Ilya ended up joining, and they pretty much just shared all of what they are still trying to think through. This is the distillation of that random walk of a conversation... came down to 7

unanswered questions with their commentary below. Please note that I'm not advocating for any of this, just structuring and sharing the information I heard.

1. Short-term control structure?

-Is the requirement for absolute control? They wonder if there is a scenario where there could be some sort of creative overrule provision if literally everyone else disagreed on direction (not just the three of them, but perhaps a broader board)?

2. Duration of control and transition?

-The non-negotiable seems to be an ironclad agreement to not have any one person have absolute control of AGI if it's created. Satisfying this means a situation where, regardless of what happens to the three of them, it's guaranteed that power over the company is distributed after the 2-3 year initial period.

3. Time spent?

-How much time does Elon want to spend on this, and how much time can he actually afford to spend on this? In what timeframe? Is this an hour a week, ten hours a week, something in between?

4. What to do with time spent?

-They don't really know how he prefers to spend time at his other companies and how he'd want to spend his time on this. Greg and Ilya are confident they could build out SW / ML side of things pretty well. They are not confident on the hardware front. They seemed hopeful Elon could spend some time on that since that's where they are weak, but did want his help in all domains he was interested in.

5. Ratio of time spent to amount of control?

-They are cool with less time / less control, more time / more control, but not less time / more control. Their fear is that there won't be enough time to discuss relevant contextual information to make correct decisions if too little time is spent.

6. Equity split?

-Greg still instinctually anchored on equal split. I personally disagree with him on that instinct and he asked for and was receptive to hearing other things he could use to recalibrate his mental model.

-Greg noted that Ilya in some ways has contributed millions by leaving his earning potential on the table at Google.

-One concern they had was the proposed employee pool was too small.

7. Capitalization strategy?

-Their instinct is to raise much more than \$100M out of the gate. They are of the opinion that the datacenter they need alone would cost that so they feel more comfortable raising more.

Takeaways:

Unsure if any of this is amenable but just from listening to all of the data points they threw out, the following would satisfy their current sticky points:

- Spending 5-10 hours a week with near full control, or spend less time and have less control.
- Having a creative short-term override just for extreme scenarios that was not just Greg / Sam / Ilya.
- An ironclad 2-3yr minority control agreement, regardless of the fates of Greg / Sam / Ilya.
- \$200M-\$1B initial raise.
- Greg and Ilya's stakes end up higher than 1/10 of Elon's but not significantly (this remains the most ambiguous).
- Increasing employee pool.

About this email: Zilis reports to Musk on a weekend conversation with Brockman and Sutskever, distilling seven unresolved questions about the proposed for-profit: control, its duration, Musk's time commitment, equity split and capitalization, noting their non-negotiable that no one person have absolute control of AGI.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Aug 28, 2017 12:01 AM. Bold/italic emphasis in the transcription (section headings; 'The' in item 2) is not preserved.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: for-profit, control, equity, negotiation, governance, musk-v-openai

Musk: 'encourage them to go start a company. I've had enough'

From: Elon Musk

Date: August 28, 2017

To: Shivon Zilis; cc Sam Teller

Subject: Re: OpenAI notes

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

This is very annoying. Please encourage them to go start a company. I've had enough.

About this email: Seven minutes after Zilis relays Brockman and Sutskever's conditions on control and equity, Musk replies that it is very annoying and that she should encourage them to go start their own company.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Aug 28, 2017 12:08 AM.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: for-profit, control, negotiation, frustration, musk-v-openai

Birchall: holding the quarterly \$5M to OpenAI — 'Continue to hold?'

From: Jared Birchall

Date: August 29, 2017

To: Elon Musk

Subject: Re: Q3 2017 Grant

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

For now I have held off on the quarterly \$5M. Continue to hold?

[Forwarded message from Chris Clark (OpenAI) to Jared Birchall, Mon, Aug 28, 2017 at 12:32 PM, subject "Q3 2017 Grant":]

Hi Jared,

I just wanted to check in re: Q3 2017 grant. The compute payment covering what we used for the Dota project (the one that got all the press a couple weeks ago) will likely be due in Sept, so I wanted to double check on timing.

Thanks,
Chris

About this email: Forwarding an OpenAI request about the Q3 2017 grant, Musk's family-office head Jared Birchall says he has held off on the quarterly \$5 million and asks whether to keep holding it.

Notes: Text taken from the PDF text layer of the court exhibit as filed on the Musk v. Altman docket (N.D. Cal. 4:24-cv-04722), via CourtListener/RECAP. Sent Aug 29, 2017 at 12:07 AM, the day after Musk told Zilis 'I've had enough'. The bracketed line is an editorial label for the forwarded message quoted beneath Birchall's note.

Sources: [Musk v. Altman, Dkt. 379-29 \(Ex. 71 to Wiener Declaration, Bates 2024MUSK-0009625; unsealed Jan 6, 2026\)](#); [Neura News summary of trial exhibits \(Ex. 662\)](#)

Tags: funding, donations, jared-birchall, leverage, musk-v-openai

Musk confirms hold on OpenAI's \$5M quarterly grant: 'Yes'

From: Elon Musk

Date: August 29, 2017

To: Jared Birchall

Subject: Re: Q3 2017 Grant

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Yes

About this email: Musk's one-word instruction to keep withholding the quarterly \$5 million donation to OpenAI during the for-profit control dispute.

Notes: Text taken from the PDF text layer of the court exhibit as filed on the Musk v. Altman docket (N.D. Cal. 4:24-cv-04722), via CourtListener/RECAP. Header timestamp Tue, 29 Aug 2017 07:18:08 -0000 (UTC), i.e. about eleven minutes after Birchall's question. OpenAI cites this exhibit for the statement that Musk directed his team to stop the \$5 million quarterly donations in August 2017.

Sources: [Musk v. Altman, Dkt. 379-29 \(Ex. 71 to Wiener Declaration, Bates 2024MUSK-0009625; unsealed Jan 6, 2026\)](#); [Neura News summary of trial exhibits \(Ex. 662\)](#)

Tags: funding, donations, jared-birchall, leverage, musk-v-openai

Musk's board and control proposal for the for-profit

From: Elon Musk

Date: September 13, 2017

To: Ilya Sutskever; cc Greg Brockman

Subject: Re: Current State

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Sounds good. The three common stock seats (you, Greg and Sam) should be elected by common shareholders. They will de facto be yours, but not in the unlikely event that you lose the faith of a huge percentage of common stockholders over time or step away from the company by choice.

I think that the Preferred A investment round (supermajority me) should have the right to appoint four (not three) seats. I would not expect to appoint them immediately, but, like I said I would unequivocally have initial control of the company, but this will change quickly.

The rough target would be to get to a 12 person board (probably more like 16 if this board really ends up deciding the fate of the world) where each board member has a deep understanding of technology, at least a basic understanding of AI and strong & sensible morals.

Apart from the Series A four and the Common three, there would likely be a board member with each new lead investor/ally. However, the specific individual new board members can only be added if all but one existing board member agrees. Same for removing board members.

There will also be independent board members we want to add who aren't associated with an investor. Same rules apply: requires all but one of existing directors to add or remove.

I'm super tired and don't want to overcomplicate things, but this seems approx right. At the sixteen person board level, we would have 7/16 votes and I'd have a 25% influence, which is my min comfort level. That sounds about right to me. If everyone else we asked to join our board is truly against us, we should probably lose.

As mentioned, my experience with boards (assuming they consist of good, smart people) is that they are rational and reasonable. There is basically never a real hardcore battle where an individual board vote is pivotal, so this is almost certainly (sure hope so) going to be a moot point.

As a closing note, I've been really impressed with the quality of discussion with you guys on the equity and board stuff. I have a really good feeling about this.

Lmk if above seems reasonable.

Elon

About this email: Musk sets out his terms for the planned for-profit: three common-stock seats for Sutskever, Brockman and Altman, four seats for his Preferred A round, and 'unequivocal' initial control that he says would dilute as the board grew to 12-16 members, leaving him about 25% influence.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Sep 13, 2017 12:40 AM. The message it replies to ('Current State') is not reproduced in the archive. Also shown in OpenAI's Dec 2024 blog post.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits, CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [OpenAI blog: Elon Musk wanted an OpenAI for-profit](#)

Tags: for-profit, board, control, governance, equity, musk-v-openai

'Honest Thoughts': Sutskever and Brockman on control of AGI

From: Ilya Sutskever

Date: September 20, 2017

To: Elon Musk, Sam Altman; cc Greg Brockman, Sam Teller, Shivon Zilis

Subject: Honest Thoughts

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Elon, Sam,

This process has been the highest stakes conversation that Greg and I have ever participated in, and if the project succeeds, it'll turn out to have been the highest stakes conversation the world has seen. It's also been a deeply personal conversation for all of us.

Yesterday while we were considering making our final commitment given the non-solicit agreement, we realized we'd made a mistake. We have several important concerns that we haven't raised with either of you. We didn't raise them because we were afraid to: we were afraid of harming the relationship, having you think less of us, or losing you as partners.

There is some chance that our concerns will prove to be unresolvable. We really hope it's not the case, but we know we will fail for sure if we don't all discuss them now. And we have hope that we can work through them and all continue working together.

Elon:

We really want to work with you. We believe that if we join forces, our chance of success in the mission is the greatest. Our upside is the highest. There is no doubt about that. Our desire to work with you is so great that we are happy to give up on the equity, personal control, make ourselves easily firable — whatever it takes to work with you.

But we realized that we were careless in our thinking about the implications of control for the world. Because it seemed so hubristic, we have not been seriously considering the implications of success.

The current structure provides you with a path where you end up with unilateral absolute control over the AGI. You stated that you don't want to control the final AGI, but during this negotiation, you've shown to us that absolute control is extremely important to you.

As an example, you said that you needed to be CEO of the new company so that everyone will know that you are the one who is in charge, even though you also stated that you hate being CEO and would much rather not be CEO.

Thus, we are concerned that as the company makes genuine progress towards AGI, you will choose to retain your absolute control of the company despite current intent to the contrary. We disagree with your statement that our ability to leave is our greatest power, because once the company is actually on track to AGI, the company will be much more important than any individual.

The goal of OpenAI is to make the future good and to avoid an AGI dictatorship. You are concerned that Demis could create an AGI dictatorship. So do we. So it is a bad idea to create a structure where you could become a dictator if you chose to, especially given that we can create some other structure that avoids this possibility.

We have a few smaller concerns, but we think it's useful to mention it here:

In the event we decide to buy Cerebras, my strong sense is that it'll be done through Tesla. But why do it this way if we could also do it from within OpenAI? Specifically, the concern is that Tesla has a duty to shareholders to maximize shareholder return, which is not aligned with OpenAI's mission. So the overall result may not end up being optimal for OpenAI.

We believe that OpenAI the non-profit was successful because both you and Sam were in it. Sam acted as a genuine counterbalance to you, which has been extremely fruitful. Greg and I, at least so far, are much worse at being a counterbalance to you. We feel this is evidenced even by this negotiation, where we were ready to sweep the long-term AGI control questions under the rug while Sam stood his ground.

Sam:

When Greg and I are stuck, you've always had an answer that turned out to be deep and correct. You've been thinking about the ways forward on this problem extremely deeply and thoroughly. Greg and I understand technical execution, but we don't know how structure decisions will play out over the next month, year, or five years.

But we haven't been able to fully trust your judgements throughout this process, because we don't understand your cost function.

We don't understand why the CEO title is so important to you. Your stated reasons have changed, and it's hard to really understand what's driving it.

Is AGI truly your primary motivation? How does it connect to your political goals? How has your thought process changed over time?

Greg and Ilya:

We had a fair share of our own failings during this negotiation, and we'll list some of them here (Elon and Sam, I'm sure you'll have plenty to add...):

During this negotiation, we realized that we have allowed the idea of financial return 2-3 years down the line to drive our decisions. This is why we didn't push on the control — we thought that our equity is good enough, so why worry? But this attitude is wrong, just like the attitude of AI experts who don't think that AI safety is an issue because they don't really believe that they'll build AGI.

We did not speak our full truth during the negotiation. We have our excuses, but it was damaging to the process, and we may lose both Sam and Elon as a result.

There's enough baggage here that we think it's very important for us to meet and talk it out. Our collaboration will not succeed if we don't. Can all four of us meet today? If all of us say the truth, and resolve the issues, the company that we'll create will be much more likely to withstand the very strong forces it'll experience.

- Greg & Ilya

About this email: Sutskever and Brockman tell Musk and Altman they have unraised concerns: that the proposed structure would give Musk unilateral absolute control over AGI, that they do not understand Altman's motives for wanting to be CEO, and that they themselves let financial upside drive their negotiating. They ask for all four to meet that day.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Sep 20, 2017 2:08 PM from Sutskever's account, signed by both. Retrieved in two halves and joined at the 'Sam:' heading.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [OpenAI blog: Elon Musk wanted an OpenAI for-profit](#)

Tags: control, for-profit, agi-dictatorship, negotiation, governance, trust, musk-v-openai

Musk: 'Guys, I've had enough. This is the final straw'

From: Elon Musk

Date: September 20, 2017

To: Ilya Sutskever, Sam Altman; cc Greg Brockman, Sam Teller, Shivon Zilis

Subject: Re: Honest Thoughts

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Guys, I've had enough. This is the final straw.

Either go do something on your own or continue with OpenAI as a nonprofit. I will no longer fund OpenAI until you have made a firm commitment to stay or I'm just being a fool who is essentially providing free funding for you to create a startup.

Discussions are over

About this email: Nine minutes after the 'Honest Thoughts' email, Musk ends the for-profit negotiations and says he will stop funding OpenAI until the founders firmly commit to stay as a nonprofit or go off on their own.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Sep 20, 2017 2:17 PM. Paragraph breaks follow the commonly published layout; the retrieval tool returned the text as a single run.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [OpenAI blog: Elon Musk wanted an OpenAI for-profit](#)

Tags: control, for-profit, funding, ultimatum, negotiation, musk-v-openai

Musk: earlier terms are 'no longer on the table'

From: Elon Musk

Date: September 20, 2017

To: Ilya Sutskever, Sam Altman; cc Greg Brockman, Sam Teller, Shivon Zilis

Subject: Re: Honest Thoughts

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

To be clear, this is not an ultimatum to accept what was discussed before. That is no longer on the table.

About this email: Musk clarifies that his previous message is not an ultimatum to accept the earlier deal; that offer is withdrawn.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Sep 20, 2017 3:08 PM.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [OpenAI blog: Elon Musk wanted an OpenAI for-profit](#)

Tags: control, for-profit, negotiation, ultimatum, musk-v-openai

Altman: 'i remain enthusiastic about the non-profit structure!'

From: Sam Altman

Date: September 21, 2017

To: Elon Musk, Ilya Sutskever; cc Greg Brockman, Sam Teller, Shivon Zilis

Subject: Re: Honest Thoughts

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

i remain enthusiastic about the non-profit structure!

About this email: Altman's one-line response to Musk's ultimatum, affirming the non-profit structure.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Sep 21, 2017 9:17 AM.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [OpenAI blog: Elon Musk wanted an OpenAI for-profit](#)

Tags: nonprofit, negotiation, short-reply, musk-v-openai

Zilis: Brockman and Sutskever will stay non-profit

From: Shivon Zilis

Date: September 22, 2017

To: Elon Musk; cc Sam Teller

Subject: Non-profit

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Hi Elon,

Quick FYI that Greg and Ilya said they would like to continue with the non-profit structure. They know they would need to provide a guarantee that they won't go off doing something else to make it work.

About this email: Zilis tells Musk that Brockman and Sutskever want to continue with the non-profit structure and know they must guarantee they will not go off and do something else.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Sep 22, 2017 9:50 AM. Line break after the greeting is assumed; the retrieval tool returned the text on one line.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: nonprofit, negotiation, commitment, musk-v-openai

Musk: 'Ok'

From: Elon Musk

Date: September 22, 2017

To: Shivon Zilis; cc Sam Teller

Subject: Re: Non-profit

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Ok

About this email: Musk's one-word acknowledgement that Brockman and Sutskever will stay with the non-profit.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Sep 22, 2017 10:01 AM.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: nonprofit, short-reply, musk-v-openai

Zilis relays Altman's position after the blow-up

From: Shivon Zilis

Date: September 22, 2017

To: Elon Musk; cc Sam Teller

Subject: Re: Non-profit

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

From Altman:

Structure: Great with keeping non-profit and continuing to support it.

Trust: Admitted that he lost a lot of trust with Greg and Ilya through this process. Felt their messaging was inconsistent and felt childish at times.

Hiatus: Sam told Greg and Ilya he needs to step away for 10 days to think. Needs to figure out how much he can trust them and how much he wants to work with them. Said he will come back after that and figure out how much time he wants to spend.

Fundraising: Greg and Ilya have the belief that 100's of millions can be achieved with donations if there is a definitive effort. Sam thinks there is a definite path to 10's of millions but TBD on more. He did mention that Holden was irked by the move to for-profit and potentially offered more substantial amount of money if OpenAI stayed a non-profit, but hasn't firmly committed. Sam threw out a \$100M figure for this if it were to happen.

Communications: Sam was bothered by how much Greg and Ilya keep the whole team in the loop with happenings as the process unfolded. Felt like it distracted the team. On the

other hand, apparently in the last day almost everyone has been told that the for-profit structure is not happening and he is happy about this at least since he just wants the team to be heads down again.

About this email: Zilis passes along Altman's view: he is fine keeping the non-profit, has lost trust in Brockman and Sutskever, is taking a ten-day hiatus, and sees a path to tens of millions in donations, with Holden Karnofsky possibly offering more if OpenAI stays non-profit.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Sep 22, 2017 5:54 PM.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: nonprofit, trust, fundraising, altman, negotiation, musk-v-openai

Brockman's New Year note: 'It's an honor to work alongside you'

From: Greg Brockman

Date: January 1, 2018

To: Elon Musk; bcc Ilya Sutskever

Subject: thank you

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Hi Elon,

We've been working together for 2 years now, and in every meeting with you I continue to learn, grow, and see the world in a new way. I particularly admire your clarity of purpose, the simple and consistent heuristics you use to evaluate decisions, and that you stick to what's right rather than what's easy.

I am grateful and humbled that you see potential in Ilya and me and are willing to give us a chance. 2017 was a hard year, and we've made mistakes, but I believe the lessons we've learned will ultimately make us successful in our mission.

It's an honor to work alongside you. Let's make sure that the post-AGI world is one that is good for humans.

- gdb

About this email: On New Year's Day 2018, Brockman thanks Musk for two years of working together, praises his clarity of purpose, and says he is grateful Musk is willing to give him and Sutskever a chance after a hard 2017.

Notes: Text taken from the PDF text layer of the court exhibit as filed on the Musk v. Altman docket (N.D. Cal. 4:24-cv-04722), via CourtListener/RECAP. Header timestamp 1/1/2018 7:56:57 AM. Offered by Musk's side as evidence of the relationship weeks before he left the board. No reply from Musk is in the exhibit.

Sources: [Musk v. Altman, Dkt. 379-71 \(Ex. 29 to Hawes Declaration, Bates OPENAI_MUSK00000814; unsealed Jan 6, 2026\)](#); [Neura News summary of trial exhibits \(Ex. 99\)](#)

Tags: gratitude, new-year, relationship, agi, musk-v-openai

Sutskever's New Year note: 'most overwhelmingly competent person in the world'

From: Ilya Sutskever

Date: January 1, 2018

To: Elon Musk; bcc Greg Brockman

Subject: Note

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Hi Elon,

We've been working together for almost exactly two years, so it felt appropriate to send this note.

I enjoy working together. You quickly pushed me out of my academic comfort zone. With time, I grew to appreciate the vast depth of your strategic insight -- I can think of many examples, the biggest one is the decision to build the custom hardware. I've made my share of non-negligible mistakes in 2017, but their lessons will help us overcome the harder challenges of the future.

We're trying to accomplish a difficult task, but with effort, we may be able to do it. It helps that we have the most overwhelmingly competent person in the world helping us.

Happy 2018. Let's make 2017 pale in comparison.

About this email: Sutskever marks two years of working with Musk, credits his strategic insight (above all the decision to build custom hardware), admits mistakes in 2017 and calls Musk the most overwhelmingly competent person in the world.

Notes: Text taken from the PDF text layer of the court exhibit as filed on the Musk v. Altman docket (N.D. Cal. 4:24-cv-04722), via CourtListener/RECAP. Header timestamp 1/1/2018 8:45:15 AM. No reply from Musk is in the exhibit.

Sources: [Musk v. Altman, Dkt. 379-70 \(Ex. 28 to Hawes Declaration, Bates OPENAI_MUSK00007198; unsealed Jan 6, 2026\)](#); [Neura News summary of trial exhibits \(Ex. 98\)](#)

Tags: gratitude, new-year, relationship, hardware, musk-v-openai

Altman flags safety team's concerns about the ICO

From: Sam Altman

Date: January 21, 2018

To: Elon Musk; cc Greg Brockman, Ilya Sutskever, Sam Teller, Shivon Zilis

Subject: ICO

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Elon—

Heads up, spoke to some of the safety team and there were a lot of concerns about the ICO and possible unintended effects in the future.

Planning to talk to the whole team tomorrow and invite input. Going to emphasize the need to keep this confidential, but I think it's really important we get buy-in and give people the chance to weigh in early.

Sam

About this email: Altman warns Musk that members of the safety team have many concerns about the proposed initial coin offering and that he plans to brief the whole team and invite input while stressing confidentiality.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Jan 21, 2018 5:08 PM. Paragraph breaks are approximate; the retrieval tool returned the body as one run of text.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: ico, fundraising, safety-team, confidentiality, musk-v-openai

Musk: 'Absolutely'

From: Elon Musk

Date: January 21, 2018

To: Sam Altman; cc Greg Brockman, Ilya Sutskever, Sam Teller, Shivon Zilis

Subject: Re: ICO

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Absolutely

About this email: Musk's one-word agreement that the team should be consulted about the ICO idea.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Jan 21, 2018 5:56 PM.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: ico, fundraising, short-reply, musk-v-openai

Karpathy: Google dominates ICLR submissions

From: Andrej Karpathy

Date: January 31, 2018

To: Elon Musk; cc Shivon Zilis

Subject: Top AI institutions today

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

The ICLR conference (which is the top deep learning - specific conference (NIPS is larger, but more diffuse)) released their decisions for accepted/rejected papers, and someone made some nice plots that show where the current deep learning / AI research happens at. It's an imperfect measure because not every company might prioritize paper publications, but it's indicative.

Here's a plot that shows the total number of papers (broken down by oral/poster/workshop/rejected) from any institution:

Long story short, Google is dominating with 83 paper submissions. The academic institutions (Berkeley / Stanford / CMU / MIT) are next, in 20-30 ranges each.

Just thought it was an interesting snapshot of where all the action is today. The full data is here: <http://webia.lip6.fr/~pajot/dataviz.html>

-Andrej

About this email: Karpathy sends Musk a breakdown of ICLR paper submissions by institution, showing Google far ahead with 83 submissions and the top universities at 20-30 each.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Jan 31, 2018 1:20 PM. The original included an embedded plot.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [Musk v. Altman, Dkt. 328-63 \(Ex. 59 to Wiener Declaration, Bates OPENAI_MUSK00000837\)](#)

Tags: google, research, iclr, competition, musk-v-openai

Musk: 'OpenAI is on a path of certain failure relative to Google'

From: Elon Musk

Date: January 31, 2018

To: Greg Brockman, Ilya Sutskever, Sam Altman; cc Sam Teller, Shivon Zilis; fwd Andrej Karpathy's email

Subject: Fwd: Top AI institutions today

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

OpenAI is on a path of certain failure relative to Google. There obviously needs to be immediate and dramatic action or everyone except for Google will be consigned to irrelevance.

I have considered the ICO approach and will not support it. In my opinion, that would simply result in a massive loss of credibility for OpenAI and everyone associated with the ICO. If something seems too good to be true, it is. This was, in my opinion, an unwise diversion.

The only paths I can think of are a major expansion of OpenAI and a major expansion of Tesla AI. Perhaps both simultaneously. The former would require a major increase in funds donated and highly credible people joining our board. The current board situation is very weak.

I will set up a time for us to talk tomorrow. To be clear, I have a lot of respect for your abilities and accomplishments, but I am not happy with how things have been managed. That is why I have had trouble engaging with OpenAI in recent months. Either we fix things and my engagement increases a lot or we don't and I will drop to near zero and publicly reduce my association. I will not be in a situation where the perception of my influence and time doesn't match the reality.

About this email: Forwarding Karpathy's chart, Musk tells OpenAI's leaders the organization is on a path of certain failure relative to Google, rejects the ICO, says the only paths are a major expansion of OpenAI and/or Tesla AI, and threatens to cut his involvement to near zero and publicly reduce his association.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Jan 31, 2018 2:02 PM. Musk left the OpenAI board three weeks later.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [OpenAI blog: Elon Musk wanted an OpenAI for-profit](#); [Musk v. Altman, Dkt. 328-63 \(Ex. 59 to Wiener Declaration, Bates OPENAI_MUSK00000837\)](#)

Tags: google, ico, funding, board, ultimatum, tesla-ai, musk-v-openai

Musk asks Karpathy: 'What do you think makes sense?'

From: Elon Musk

Date: January 31, 2018

To: Andrej Karpathy

Subject: Fwd: Top AI institutions today

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

fyi

What do you think makes sense? Happy to talk by phone if that's better.

About this email: Musk forwards his 'certain failure' email to Karpathy, then at Tesla, and asks for his view.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Jan 31, 2018 2:07 PM.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [Musk v. Altman, Dkt. 328-63 \(Ex. 59 to Wiener Declaration, Bates OPENAI_MUSK00000837\)](#)

Tags: google, tesla-ai, strategy, short-reply, musk-v-openai

Brockman's reply: expand OpenAI, keep the moral high ground

From: Greg Brockman

Date: January 31, 2018

To: Elon Musk; cc Ilya Sutskever, Sam Altman, [redacted], Shivon Zilis

Subject: Re: Top AI institutions today

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Hi Elon,

Thank you for the thoughtful note. I have always been impressed by your focus on the big picture, and agree completely we must change trajectory to achieve our goals. Let's speak tomorrow, any time 4p or later will work.

My view is that the best future will come from a major expansion of OpenAI. Our goal and mission are fundamentally correct, and that will increasingly be a superpower as AGI grows near.

Fundraising

Our fundraising conversations show that:

- Ilya and I are able to convince reputable people that AGI can really happen in the next ≤ 10 years
- There's appetite for donations from those people
- There's very large appetite for investments from those people

I respect your decision on the ICO idea, which matches the evolution of our own thinking. Sam Altman has been working on a fundraising structure that does not rely on a public offering, and we will be curious to hear your feedback.

Of the people we've been talking to, the following people are currently my top suggestions for board members. Would also love suggestions for your top picks not on this list, and we can figure out how to approach them.

- <redacted>
- <redacted>
- <redacted>
- <redacted>
- <redacted>
- <redacted> (she heads Partnership on AI, originally created by Demis to steal OpenAI's thunder - would bring a lot of outside credibility)

The next 3 years

Over the next 3 years, we must build 3 things:

- Custom AI hardware (such as <redacted> computer)
- Massive AI data center (likely multiple revs thereof)
- Best software team, mixing between algorithm development, public demonstrations, and safety

We've talked the most about the custom AI hardware and AI data center. On the software front, we have a credible path (self-play in a competitive multiagent environment) which has been validated by Dota and AlphaGo. We also have identified a small but finite number of limitations in today's deep learning which are barriers to learning from human levels of experience. And we believe we uniquely are on trajectory to solving safety (at least in broad strokes) in the next three years.

We would like to scale headcount in this way:

- Beginning of 2017: ~40
- End of 2018: 100
- End of 2019: 300
- End of 2020: 900

[redacted passage]

Moral high ground

Our biggest tool is the moral high ground. To retain this, we must:

- Try our best to remain a non-profit. AI is going to shake up the fabric of society, and our fiduciary duty should be to humanity.
- Put increasing effort into the safety/control problem, rather than the fig leaf you've noted in other institutions. It doesn't matter who wins if everyone dies. Related to this, we need to communicate a "better red than dead" outlook — we're trying to build safe AGI, and we're not willing to destroy the world in a down-to-the-wire race to do so.
- Engage with government to provide trusted, unbiased policy advice — we often hear that they mistrust recommendations from companies such as <redacted>.
- Be perceived as a place that provides public good to the research community, and keeps the other actors honest and open via leading by example.

The past 2 years

I would be curious to hear how you rate our execution over the past two years, relative to resources. In my view:

- Over the past five years, there have been two major demonstrations of working systems: AlphaZero [DeepMind] and Dota 1v1 [OpenAI]. (There are a larger number of breakthroughs of "capabilities" popular among practitioners, the top of which I'd say are: ProgressiveGAN [NVIDIA], unsupervised translation [Facebook], WaveNet [DeepMind], Atari/DQN [DeepMind], machine translation [Ilya at Google — now at OpenAI], generative adversarial network [Ian Goodfellow at grad school — now at Google], variational autoencoder [Durk at grad school — now at OpenAI], AlexNet [Ilya at grad school — now at OpenAI].) We benchmark well on this axis.
- We grew very rapidly in 2016, and in 2017 iterated to a working management structure. We're now ready to scale massively, given the resources. We lose people on comp currently, but pretty much only on comp. I've been resuming the style of recruiting I did in the early days, and believe I can exceed those results.
- We have the most talent dense team in the field, and we have the reputation for it as well.
- We don't encourage paper writing, and so paper acceptance isn't a measure we optimize. For the ICLR chart Andrej sent, I'd expect our (accepted papers)/(people submitting papers) to be the highest in the field.

- gdb

About this email: Brockman agrees OpenAI must change trajectory, accepts Musk's rejection of the ICO, proposes board candidates, a three-year plan for custom hardware, a massive data center and headcount growth to 900, and argues OpenAI should try its best to remain a non-profit to keep the moral high ground.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Jan 31, 2018 10:56 PM. Names of proposed board members and one passage are redacted in the exhibit.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits, CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [OpenAI blog: Elon Musk wanted an OpenAI for-profit](#); [Musk v. Altman, Dkt. 328-63 \(Ex. 59 to Wiener Declaration, Bates OPENAI_MUSK00000837\)](#)

Tags: fundraising, board, nonprofit, strategy, safety, headcount, musk-v-openai

Karpathy: OpenAI should 'attach to Tesla as its cash cow'

From: Andrej Karpathy

Date: January 31, 2018

To: Elon Musk

Subject: Re: Top AI institutions today

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Working at the cutting edge of AI is unfortunately expensive. For example, DeepMind's operating expenses in 2016 were at around \$250M USD (does not include compute). With their growing team today it might be ~0.5B/yr. But then Alphabet in 2016 reported ~20B net income so it's still fairly cheap even if DeepMind had no revenue of its own. In addition to DeepMind, Google also has Google Brain, Research, and Cloud. And TensorFlow, TPUs, and they own about a third of all research (in fact, they hold their own AI conferences).

I also strongly suspect that compute horsepower will be necessary (and possibly even sufficient) to reach AGI. If historical trends are any indication, progress in AI is primarily driven by systems - compute, data, infrastructure. The core algorithms we use today have remained largely unchanged from the ~90s. Not only that, but any algorithmic advances published in a paper somewhere can be almost immediately re-implemented and incorporated. Conversely, algorithmic advances alone are inert without the scale to also make them scary.

It seems to me that OpenAI today is burning cash and that the funding model cannot reach the scale to seriously compete with Google (an 800B company). If you can't seriously compete but continue to do research in open, you might in fact be making things worse and helping them out "for free", because any advances are fairly easy for them to copy and immediately incorporate, at scale.

A for-profit pivot might create a more sustainable revenue stream over time and would, with the current team, likely bring in a lot of investment. However, building out a product from scratch would steal focus from AI research, it would take a long time and it's unclear if a company could "catch up" to Google scale, and the investors might exert too much pressure in the wrong directions.

The most promising option I can think of, as I mentioned earlier, would be for OpenAI to attach to Tesla as its cash cow. I believe attachments to other large suspects (e.g. Apple? Amazon?) would fail due to an incompatible company DNA. Using a rocket analogy, Tesla already built the "first stage" of the rocket with the whole supply chain of Model 3 and its onboard computer and a persistent internet connection. The "second stage" would be a full self driving solution based on large-scale neural network training, which OpenAI expertise could significantly help accelerate. With a functioning full self-driving solution in ~2-3 years we could sell a lot of cars/trucks. If we do this really well, the transportation industry is large enough that we could increase Tesla's market cap to high O(~100K), and use that revenue to fund the AI work at the appropriate scale.

I cannot see anything else that has the potential to reach sustainable Google-scale capital within a decade.

-Andrej

About this email: Karpathy argues that cutting-edge AI is too expensive for OpenAI's funding model, that open research may just help Google, that a for-profit pivot has drawbacks, and that the most promising option is for OpenAI to attach to Tesla as its cash cow, funding AI through full self-driving revenue.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Jan 31, 2018 11:54 PM. First published (with the author's name redacted) in OpenAI's March 2024 blog post; Karpathy was identified as the author in the Nov 2024 court exhibits.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [OpenAI blog: OpenAI and Elon Musk \(Mar 5, 2024\)](#); [Musk v. Altman, Dkt. 328-63 \(Ex. 59 to Wiener Declaration, Bates OPENAI_MUSK00000837\)](#)

Tags: tesla, cash-cow, google, funding, for-profit, self-driving, musk-v-openai

Musk: Tesla is 'the only path that could even hope to hold a candle to Google'

From: Elon Musk

Date: February 1, 2018

To: Ilya Sutskever, Greg Brockman

Subject: Fwd: Top AI institutions today

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Andrej is exactly right. We may wish it otherwise, but, in my and Andrej's opinion, Tesla is the only path that could even hope to hold a candle to Google. Even then, the probability of being a counterweight to Google is small. It just isn't zero.

About this email: Musk forwards Karpathy's analysis to Sutskever and Brockman and endorses it, saying Tesla is the only path that could hope to rival Google, and even then the odds are small.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Feb 1, 2018 3:52 AM, above the forwarded Karpathy email. First published in OpenAI's March 2024 blog post with Karpathy's name redacted.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [OpenAI blog: OpenAI and Elon Musk \(Mar 5, 2024\)](#); [Musk v. Altman, Dkt. 328-63 \(Ex. 59 to Wiener Declaration, Bates OPENAI_MUSK00000837\)](#)

Tags: tesla, google, cash-cow, funding, strategy, musk-v-openai

Zilis: no ICO, capped-return idea, Musk's formal board resignation

From: Shivon Zilis

Date: March 25, 2018

To: Elon Musk; cc Sam Teller

Subject: AI updates

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

OpenAI

Fundraising:

-No longer doing the ICO / "instrument to purchase compute in advance" type structure. Altman is thinking through an instrument where the 4-5 large corporates who are interested can invest with a return capped at 50x if OpenAI does get to some semblance

of money-making AGI. They apparently seem willing just for access reasons. He wants to discuss with you in more detail.

Formal Board Resignation:

-You're still technically on the board so need to send a quick one liner to Sam Altman saying something like "With this email I hereby resign as a director of OpenAI, effective Feb 20th 2018".

Future Board:

-Altman said he is cool with me joining then having to step off if I become conflicted, but is concerned that others would consider it a burned bridge if I had to step off. I think best bet is not to join for now and be an ambiguous advisor but let me know if you feel differently. They have Adam D'Angelo as the potential fifth to take your place, which seems great?

TeslaAI

Andrej has three candidates in pipeline, may have 1-2 come in to meet you on Tuesday. He will send you a briefing note about them. Also, he's working on starter language for a potential release that will be ready to discuss Tuesday. It will follow the "full-stack AI lab" angle we talked about but, if that doesn't feel right, please course correct... is tricky messaging.

Cerebras

Chip should be available in August for them to test, and they plan to let others have remote access in September. The Cerebras guy also mentioned that a lot of their recent customer interest has been from companies upset about the Nvidia change in terms of service (the one that forces companies away from consumer grade GPUs to enterprise Pascals / Voltas). Scott Gray and Ilya continue to spend a bunch of time with them

About this email: Zilis briefs Musk that OpenAI has dropped the ICO and Altman is designing an instrument with returns capped at 50x, reminds him he must formally resign as a director effective Feb 20, 2018, discusses whether she should join the board, and updates him on Tesla AI hiring and Cerebras.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Mar 25, 2018 11:03 AM. Section headings were bold in the transcription.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: capped-profit, board-resignation, fundraising, tesla-ai, cerebras, musk-v-openai

Altman sends Musk the draft OpenAI Charter

From: Sam Altman

Date: April 2, 2018

To: Elon Musk; cc Shvon Zilis

Subject: The OpenAI Charter

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

We are planning to release this next week--any thoughts?

The OpenAI Charter

OpenAI's mission is to ensure that artificial general intelligence (AGI) — by which we mean highly autonomous systems that outperform humans at most economically-valuable creative work — benefits all of humanity. We will attempt to directly build safe and beneficial AGI, but will also consider our mission fulfilled if our work aids others to achieve this outcome. To that end, we commit to the following principles:

Broadly Distributed Benefits

We commit to use any influence we obtain over AGI's deployment to ensure it is used for the benefit of all, and to avoid enabling uses of AI or AGI that harm humanity or unduly concentrate power.

Our primary fiduciary duty is to humanity. We anticipate needing to marshal substantial resources to fulfill our mission, but will always assiduously act to minimize conflicts of interest among our employees and stakeholders that could compromise broad benefit.

Long-Term Safety

We are committed to doing the research required to make AGI safe, and to driving the broad adoption of such research across the AI community.

We are concerned about late-stage AGI development becoming a competitive race without time for adequate safety precautions. Therefore, if a value-aligned, safety-conscious project comes close to building AGI before we do, we commit to stop competing with and start assisting this project. We will work out specifics in case by case agreements, but a typical triggering condition might be "a better-than-even chance of success in the next 2 years".

Technical Leadership

To be effective at addressing AGI's impact on society, OpenAI must be on the cutting edge of AI capabilities — policy and safety advocacy alone would be insufficient.

We believe that AI will have broad societal impact before AGI, and we'll strive to lead in those areas that are directly aligned with our mission and expertise.

Cooperative Orientation

We will actively cooperate with other research and policy institutions; we seek to create a global community working together to address AGI's global challenges.

We are committed to providing public goods that help society navigate the path to AGI. Today this includes publishing most of our AI research, but we expect that safety and security concerns will reduce our traditional publishing in the future, while increasing the importance of sharing safety, policy, and standards research.

About this email: Altman sends Musk the text of the OpenAI Charter a week before its release and asks for any thoughts.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Apr 2, 2018 1:54 PM.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: charter, mission, safety, governance, musk-v-openai

Musk on the OpenAI Charter: 'Sounds fine'

From: Elon Musk

Date: April 2, 2018

To: Sam Altman

Subject: Re: The OpenAI Charter

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Sounds fine

About this email: Musk's two-word approval of the draft OpenAI Charter.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Apr 2, 2018 2:45 PM.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: charter, short-reply, governance, musk-v-openai

Zilis: OpenAI raising equity with a fixed maximum return

From: Shivon Zilis

Date: April 23, 2018

To: Elon Musk; cc Sam Teller

Subject: Re: AI updates

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Updated info per a conversation with Altman. You're tentatively set to speak with him on Tuesday.

Financing:

- He confirmed again that they are definitely not doing an ICO but rather equity that has a fixed maximum return.

- Would be a rather unique subsidiary structure for the raise which he wants to walk you through.

- Wants to move within 4-6 week on first round (probably largely Reid money, potentially some corporates).

Tech:

Says Dota 5v5 looking better than anticipated.

- The sharp rise in Dota bot performance is apparently causing people internally to worry that the timeline to AGI is sooner than they'd thought before.

- Thinks they are on track to beat Montezuma's Revenge shortly.

Time allocation:

- I've reallocated most of the hours I used to spend with OpenAI to Neuralink and Tesla. This naturally happened with you stepping off the board and related factors — but if you'd prefer I pull more hours back to OpenAI oversight please let me know.

-Sam and Greg asked if I'd be on their informal advisory board (just Gabe Newell so far), which seems fine and better than the formal board given potential conflicts? If that doesn't feel right let me know what you'd prefer.

About this email: Zilis updates Musk after talking to Altman: OpenAI is definitely not doing an ICO but equity with a fixed maximum return through a unique subsidiary structure, with a first round probably largely from Reid Hoffman; Dota progress is making staff think AGI is nearer; and she has shifted her hours to Neuralink and Tesla.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Apr 23, 2018 1:49 AM as a continuation of the Mar 25 'AI updates' thread. No reply from Musk is reproduced.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: capped-profit, fundraising, dota, agi-timelines, advisory-board, musk-v-openai

Musk on OpenAI's capped-return raise and Zilis's advisory role: 'Ok by me'

From: Elon Musk

Date: April 23, 2018

To: Shivon Zilis; cc Sam Teller

Subject: Re: AI updates

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Ok by me

About this email: Seventeen minutes after Zilis reports that OpenAI is raising equity with a fixed maximum return through a subsidiary and asks about her time allocation and joining an informal advisory board, Musk replies that it is OK by him.

Notes: Text taken from the PDF text layer of the court exhibit as filed on the Musk v. Altman docket (N.D. Cal. 4:24-cv-04722), via CourtListener/RECAP. Header timestamp Mon, 23 Apr 2018 02:06:23 -0700. The PDF text layer runs the words together as 'Okbyme'. The exhibit quotes Zilis's Apr 23 and Mar 25 'AI updates' emails beneath the reply. Also Ex. 60 to the Wiener Declaration; OpenAI cites it to show Musk knew of the planned for-profit subsidiary by April 2018.

Sources: [Musk v. Altman, Dkt. 379-87 \(Ex. 62 to Hawes Declaration, Bates 2024MUSK-0005604; unsealed Jan 6, 2026\)](#); [Neura News summary of trial exhibits \(Ex. 233\)](#)

Tags: capped-profit, fundraising, short-reply, advisory-board, musk-v-openai

Zilis asks Musk to formally resign from the OpenAI board

From: Shivon Zilis

Date: April 27, 2018

To: Elon Musk

Subject: Re: OpenAI Board

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Would you prefer to formally resign from the OpenAI board or have Sam and Chris do their workaround option outlined below?

If the former, you can just copy/paste this line to Altman:

I hereby resign as a director of OpenAI, Inc., effective February 21, 2018.

FYI, noticed that Adam has already shared his joining the board:

<https://twitter.com/adamdangelo/status/988859015315701760>

[Forwarded message from Chris Clark to Shivon Zilis, cc Sam Altman, April 27, 2018 at 2:51:17 PM PDT, subject "Board Resignation":]

Hi Shivon,

We need to add Adam to the the board, but to do that we need a signed resignation letter or even a simple one line email to Sam with:

"I hereby resign as a director of OpenAI, Inc., effective February 21, 2018."

If you don't think that's doable then I'll add language to the board resolution acknowledging that Elon made a public statement that he's leaving the OpenAI board as of Feb 21, 2018 and electing his replacement based on that statement.

Let me know if I can help with anything.

Thanks,
Chris

About this email: Zilis forwards OpenAI's request for a signed resignation so Adam D'Angelo can be added to the board, and asks Musk whether he prefers to resign formally or let OpenAI use a board-resolution workaround; she supplies the one-line text to send Altman.

Notes: Text taken from the PDF text layer of the court exhibit as filed on the Musk v. Altman docket (N.D. Cal. 4:24-cv-04722), via CourtListener/RECAP. Sent April 27, 2018 at 8:25 PM. The bracketed line is an editorial label for the forwarded message included beneath Zilis's note; 'the the' is as in the exhibit.

Sources: [Musk v. Altman, Dkt. 328-58 \(Ex. 54 to Wiener Declaration, Bates OPENAI_MUSK00000865\)](#)

Tags: board-resignation, governance, adam-dangelo, musk-v-openai

Musk's formal resignation as an OpenAI director

From: Elon Musk

Date: April 28, 2018

To: Sam Altman; cc Shivon Zilis

Subject: Re: OpenAI Board

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

I hereby resign as a director of OpenAI, Inc., effective February 21, 2018.

About this email: Musk's one-line email formally resigning from the board of OpenAI, Inc., backdated to February 21, 2018, the date his departure was publicly announced.

Notes: Text taken from the PDF text layer of the court exhibit as filed on the Musk v. Altman docket (N.D. Cal. 4:24-cv-04722), via CourtListener/RECAP. Header timestamp 4/28/2018 5:54:44 AM (time zone not stated). The wording was supplied by OpenAI's Chris Clark via Shivon Zilis.

Sources: [Musk v. Altman, Dkt. 328-58 \(Ex. 54 to Wiener Declaration, Bates OPENAI_MUSK00000865\)](#)

Tags: board-resignation, governance, departure, musk-v-openai

Altman sends Musk the OpenAI LP term sheet: 'I won't take any equity'

From: Sam Altman

Date: August 31, 2018

To: Elon Musk; cc Sam Teller, Shivon Zilis

Subject: OpenAI term sheet

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Elon--

Please see attached, look forward to feedback.

Also, my current thought is that I won't take any equity. I'm not doing this for the money anyway, and I like the idea of being completely unconflicted and just focused on the best outcome for the world. If it appeared at some point we weren't going to build AGI but were going to build something valuable, then maybe I'd want equity then.

Sam

[Attachment: "OpenAI LP - Summary of Principal Terms" (PDF), reproduced from the exhibit:]

OpenAI

Summary Term Sheet

Mission

OpenAI's mission is to ensure that artificial general intelligence - by which we mean highly autonomous systems that outperform humans at most economically valuable work - (AGI), when developed, is safe and benefits all of humanity. We will attempt to directly build safe and beneficial AGI, but will also consider our mission fulfilled if our work aids others to achieve this outcome.

Corporate Structure

OpenAI LP will be a for-profit Delaware Limited Partnership managed by its General Partner, a single-member Delaware LLC controlled by OpenAI, Inc. (the Nonprofit)'s Board of Directors. At all times, no more than a minority of the Nonprofit's Board of Directors will be holders of any economic interest in OpenAI LP or in the employee holdings entity (described below).

Funding/ Revenue

Initial Capitalization

OpenAI LP will initially be capitalized by a contribution of assets from the Nonprofit. The Nonprofit will get an interest equivalent to that of a Limited Partner in the initial raise that is consistent with the value of its capital contribution.

Limited Partner Interests

We will sell preferred capped Limited Partner Interests redeemable (on a first in, first out basis) for a multiple of the purchase price (adjusted for inflation as described below). Redemption will commence either (a) if and when we successfully create AGI capable of generating the requisite returns and the General Partner, in its sole and absolute discretion, determines that redemptions will begin, or (b) at a mutually agreeable pre-AGI alternative. Specifically, the fundraising rounds will be as follows:

1. Initial raise: \$500M, 100x target redemption
2. Employee pool: \$100B capped target redemption
 - Actual redemption amount will be based on the number of employee LP Interests issued, which will be determined (subject only to the aggregate cap) by the General Partner in its sole and absolute discretion.
 - The General Partner may increase the target redemption of the employee pool in its sole and absolute discretion.
3. Second raise: \$10+B, 15x expected target redemption.
4. There may be additional fundraising rounds and/or an additional employee pool, though the interests of employee and/or investor Limited Partners cannot be diluted absent approval of a majority-in-interest of the affected class of Limited Partner (employee/investor).

Distributions will be made as follows:

- First, distributions will be made to initial raise investors in proportion to their respective capital contributions until each investor has received an amount equal to its aggregate capital contributions, with such distributions counting against the investors' target redemption amount;
- Next, 25% of all distributions will be made to initial raise investors and employees in proportion to their respective target redemption amounts, and 75% will go to second raise investors in proportion to their respective capital contributions until each investor has received an amount equal to its aggregate capital contributions. Subsequently, all future payments will be in proportion to each investor/employee's respective target redemption amount. Distributions will continue until all investors and employees have received their respective target redemption amounts; and
- Any amounts in excess of the target redemption amounts shall be distributed to or for the benefit of the Nonprofit in its capacity as a Limited Partner.

For the first 10 years after the initial raise, the inflation rate will be determined by reference to the U.S. Consumer Price Index or such alternative U.S. governmental index as determined by the General Partner with the consent of a majority-in-interest of the Limited Partners (determined by reference to target redemption amounts). Subsequently, starting in year 11, the target redemption amounts will increase annually at the greater of the inflation rate or a simple interest rate equal to the London Interbank Offered Rate plus 2%.

Any earlier redemption date must be approved by a majority-in-interest of the investor Limited Partners, as determined by reference to their target redemption amounts.

Prior to redemption, all items of profits and loss will be allocated among the Partners in proportion to their target redemption amounts.

Revenue

At some future date, OpenAI LP may generate revenue, primarily in the form of interest, dividends, capital gains, and royalties. To the extent it seeks to commercialize technology in a way that will generate significant revenue subject to the Unrelated Business Income Tax, OpenAI LP will create taxable "corporate blocker" subsidiaries to commercialize such technology.

Should we generate revenue (through OpenAI LP or its subsidiaries) that is not reserved for Research & Development activities and/or other related expenses, at the sole discretion of the General Partner, such surplus revenue will be distributed to our Limited Partners. These periodic distributions will be allocated based on the Limited Partner's targeted redemption amounts and will not count against the targeted redemption amounts.

Tax Distributions

OpenAI LP will make customary tax distributions to all Partners in respect of the net taxable income of OpenAI LP, if any. The amount and timing of any tax distribution will be determined in the discretion of the General Partner.

OpenAI LP exists to advance our mission using the principles expounded in OpenAI's Charter. We will employ the Partnership's assets to help us achieve these goals, such as through funding research and development activities, investing in companies and assets that may further the mission, and other related activities. The General Partner shall have the sole right to determine the amount of the Partnership's assets that shall be spent on and reserved for the expenditures in advancement of the mission.

Fiduciary Duties

Our duty to these principles and the advancement of our mission takes precedence over any obligation to generate a profit. We may never make a profit, and we are under no obligation to do so. We are free to re-invest any or all of our cash flow into research and development activities and/or related expenses without any obligation to the Limited Partners. The General Partner will have discretion to establish and maintain reserves for such expenditures, consistent with the mission, so long as such discretion is exercised reasonably.

The fiduciary duties of the Nonprofit Board of Directors flow exclusively to the Nonprofit, not to the Limited Partners.

Employee Comp.

In addition to their salary, employees will be granted profit interests in an employee holding vehicle (detailed below) in exchange for their services. Certain employees of the Nonprofit will also be granted profits interests in the target redemption amount, if any, held by Nonprofit. It is intended that the profits interests will not give rise to tax liability at the time of grant. Grants will vest over 6 years, with 0% in the first year, followed by

20% vesting each year for the next five years.

Employee interests will not be transferrable until the employee has been at OpenAI for one year. Employees are limited to selling a maximum of 25% of their vested interests per year.

Each employee recipient will make a timely Section 83(b) election with respect to their profits interests.

Employee Holding Company

We will create a separate holding company (OpenAI Holdings, L.P.) to issue profit interests to employees of OpenAI LP. OpenAI LP will issue mirror profits interests to OpenAI Holdings, and, as a result, OpenAI Holdings will be a partner in OpenAI LP (along with the investors). The profits interests held by employees in OpenAI Holdings thus will represent an indirect economic interest in OpenAI LP.

By receiving profits interests in OpenAI Holdings rather than OpenAI LP, it is intended that employees will be spared certain negative tax consequences of holding a partnership interest directly in a partnership in which they are also employees (such as quarterly estimated tax filings, self-employment taxes and no longer being eligible to participate in certain employee benefit plans). Employees will receive W-2s from OpenAI LP and otherwise will be treated as employees at that level for tax purposes. Employees will receive K-1s from OpenAI Holdings in connection with their profits interests and otherwise will be treated as partners at that level for tax purposes.

Limitations Re: Transfer of Interests

Notwithstanding any of the preceding, the transferability of LP Interests will be subject to limitations including:

1. transfers will be limited to a period occurring once per year;
2. transfers can only be between and among Limited Partners absent prior approval by the General Partner;
3. transfers will be limited so as to avoid material burdens upon the partnership or the General Partner under securities and other applicable laws; and
4. transfers will be limited to avoid partnership tax liability under US federal "publicly traded partnership" and other similar laws.

Information Rights

Each Limited Partner will be furnished with annual financial statements within 90 days following the end of each financial year.

All scheduled payments to Limited Partners will be considered confidential, and Limited Partners are not entitled to information regarding the interests of other Partners, provided that the Limited Partners will be entitled to such information as they need to self-organize for purposes of exercising their voting and other rights under the governing Limited Partnership Agreement. Confidentiality will be subject to customary exclusions for required governmental reporting; for example, the Nonprofit will be required to disclose certain information regarding OpenAI LP on its IRS Form 990,

which will be available to the public.

Strategic Investment Agreements

OpenAI LP is empowered to enter into strategic investment agreements with companies that become Limited Partners subject only to the caveat that those agreements must not clearly subvert the redemption waterfall described above nor clearly detract from the advancement of our mission.

About this email: Six months after Musk left the board, Altman sends him the summary term sheet for the planned capped-profit OpenAI LP (\$500M initial raise at a 100x redemption cap, nonprofit-controlled general partner) and says his current thought is to take no equity himself so as to be completely unconflicted.

Notes: Text taken from the PDF text layer of the court exhibit via CourtListener/RECAP. Header: Fri, 31 Aug 2018 11:30:15 +0200. The OCR renders the subject as 'OpenAI term sheet'; it is normalised here to 'OpenAI term sheet'. Paragraph breaks in the email body and line unwrapping in the attachment are reconstructed from the line layout; the term sheet's left-column section labels (Mission, Corporate Structure, Funding/Revenue, Fiduciary Duties, Employee Comp., etc.) are kept as separate lines, and its nested bullets are flattened. The bracketed line is an editorial label. No reply from Musk appears in the exhibit; the same email is Ex. 61 to the Wiener Declaration and Ex. 63 to the Hawes Declaration, which were not retrievable.

Sources: [Musk v. Altman, Dkt. 379-46 \(Ex. 19 to Cohen Declaration; Musk deposition Ex. 8; Bates SPX-001642 to SPX-001647; unsealed Jan 6, 2026\)](#); [Musk v. Altman, Dkt. 379-2 \(OpenAI statement of undisputed facts, citing the term sheet as Wiener Ex. 61\)](#)

Tags: capped-profit, openai-lp, term-sheet, equity, fundraising, musk-v-openai

Altman's Dec 2018 update: Dota, robot hands, move to Microsoft

From: Sam Altman

Date: December 17, 2018

To: Elon Musk

Subject: Re: OpenAI update

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

Hey Elon-

In Q1 of next year, we plan to do a final Dota tournament with any pro team that wants to play for a large cash prize, on an unrestricted game. After that, we'll call model-free RL complete, and a subset of the team will work on re-solving 1v1 Dota with model-based RL.

We also plan to release a number of robot hand demos in Q1-Rubiks cube, pen tricks, and Chinese ball rotation. The learning speed for new tasks should be very fast. Later in the year, we will try mounting two hands on two arms and see what happens...

We are also making fast progress on language. My hope is that next year we can generate short stories and a good dialogue bot.

[redacted] The hope is that we can use this unsupervised learning to build models that can do hard things, eg never make an image classification mistake a human wouldn't, which to me would imply some level of conceptual understanding.

We are also making good progress in the multi-agent environment, with multiple agents now collaborating to build simple structure, play laser tag, etc.

Finally, I am working on a deal to move our computing from Google to Microsoft (in addition to our own data centers).

Also happy to talk about fundraising (we should have enough for the next ~2 years even with aggressive growth) and evolving hardware thoughts if helpful, would prefer not to put either of those in email but maybe next time you're at Pioneer?

Sam

About this email: Altman briefs Musk on plans for a final Dota tournament, robot-hand demos, language progress and multi-agent work, says he is working on a deal to move compute from Google to Microsoft, and offers to discuss fundraising in person.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Dec 17, 2018 3:42 PM. One sentence is blacked out. Published in OpenAI's Dec 2024 blog post.

Sources: [OpenAI blog: Elon Musk wanted an OpenAI for-profit \(Dec 13, 2024\)](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: status-update, dota, robotics, microsoft, fundraising

Musk: 'Could probably meet Wed eve in SF'

From: Elon Musk

Date: December 17, 2018

To: Sam Altman

Subject: Re: OpenAI update

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

Sounds good

Could probably meet Wed eve in SF

About this email: Musk acknowledges Altman's update and offers to meet in San Francisco that Wednesday evening.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Dec 17, 2018 3:47 PM.

Sources: [OpenAI blog: Elon Musk wanted an OpenAI for-profit \(Dec 13, 2024\)](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: status-update, meeting, short-reply

Musk: OpenAI's chance of being relevant 'is 0%. Not 1%'

From: Elon Musk

Date: December 26, 2018

To: Ilya Sutskever, Greg Brockman; cc Sam Altman, Shivon Zilis

Subject: I feel I should reiterate

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

My probability assessment of OpenAI being relevant to DeepMind/Google without a dramatic change in execution and resources is 0%. Not 1%. I wish it were otherwise.

Even raising several hundred million won't be enough. This needs billions per year immediately or forget it.

Unfortunately, humanity's future is in the hands of Demis.

<https://nyti.ms/2Rjetd3?smid=nytcare-ios-share>

And they are doing a lot more than this.

OpenAI reminds me of Bezos and Blue Origin. They are hopelessly behind SpaceX and getting worse, but the ego of Bezos has him insanely thinking that they are not!

I really hope I'm wrong.

Elon

About this email: Ten months after leaving the board, Musk tells OpenAI's leaders that without a dramatic change in execution and resources its probability of being relevant to DeepMind/Google is zero, that it needs billions per year immediately, that humanity's future is in the hands of Demis Hassabis, and compares OpenAI to Bezos's Blue Origin.

Notes: Sent Dec 26, 2018 12:07 PM (quoted in the 2026 trial exhibit chain as 'Wed, 26 Dec 2018 at 20:07'). OpenAI's 2024 blog posts and the LessWrong archive redact the name and the sentence after the link; the unredacted text here is from the Musk v. Altman trial exhibit as transcribed by Internal Tech Emails (May 2026). That exhibit also contains a reply from Sam Altman dated Dec 30, 2018 2:04 PM, whose text is behind the newsletter's paywall and could not be retrieved.

Sources: [OpenAI blog: OpenAI and Elon Musk \(Mar 5, 2024\)](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [OpenAI blog: Elon Musk wanted an OpenAI for-profit](#); [Internal Tech Emails: Musk v. Altman begins \(May 2, 2026\)](#)

Tags: google, deepmind, funding, billions, blue-origin, demis-hassabis

Altman sends Musk the draft capped-profit announcement

From: Sam Altman

Date: March 6, 2019

To: Elon Musk; cc Sam Teller, Shiron Zilis

Subject: OpenAI

FULL TEXT · EXECUTIVE & INTERNAL · PUBLISHED BY RECIPIENT · CONFIRMED

Elon—

Here is a draft post we are planning for Monday. Anything to add/edit?

TL;DR:

- We've created the capped-profit company and raised the first round, led by Reid and Vinod.
- We did this is a way where all investors are clear that they should never expect a profit.
- We made Greg chairman and me CEO of the new entity.
- We have tested this structure with potential next-round investors and they seem to like it.

Speaking of the last point, we are now discussing a multi-billion dollar investment which I would like to get your advice on when you have time. Happy to come see you some time you are in the bay area.

Sam

Draft post

We've created OpenAI LP, a new "capped-profit" company that allows us to rapidly increase our investments in compute and talent while including checks and balances to actualize our mission.

Our mission is to ensure that artificial general intelligence (AGI) benefits all of humanity, primarily by attempting to build safe AGI and share the benefits with the world.

Due to the exponential growth of compute investments in the field, we've needed to scale much faster than we'd planned when starting OpenAI. We expect to need to raise many billions of dollars in upcoming years for large-scale cloud compute, attracting and retaining talented people, and building AI supercomputers.

We haven't been able to raise that much money as a nonprofit, and though we considered becoming a for-profit, we were afraid that doing so would mean giving up our mission.

Instead, we created a new company, OpenAI LP, as a hybrid for-profit and nonprofit — which we are calling a "capped-profit" company.

The fundamental idea of OpenAI LP is that investors and employees can get a fixed return if we succeed at our mission, which allows us to raise investment capital and attract employees with startup-like equity. But any returns beyond that amount — and if we are successful, we expect to generate orders of magnitude more value than we'd owe to people who invest in or work at OpenAI LP — are owned by the original OpenAI Nonprofit entity.

Going forward (in this post and elsewhere), "OpenAI" refers to OpenAI LP (which now employs most of our staff), and the original entity is referred to as "OpenAI Nonprofit".

The mission comes first

We've designed OpenAI LP to put our overall mission — ensuring the creation and adoption of safe and beneficial AGI — over generating returns for investors.

To minimize conflicts of interest with the mission, OpenAI LP's primary fiduciary obligation is to advance the aims of the OpenAI Charter, and the company is controlled by OpenAI Nonprofit's board. All investors and employees sign agreements that OpenAI LP's obligation to the Charter always comes first, even at the expense of some or all of their financial stake.

Our employee and investor paperwork starts like this. The general partner refers to OpenAI Nonprofit (whose official name is "OpenAI Inc"); limited partners refers to investors and employees.

Only a minority of board members can hold financial stakes in the partnership. Furthermore, only board members without such stakes are allowed to vote on decisions where the interests of limited partners and the nonprofit's mission may conflict — including any decisions about making payouts to investors and employees.

Corporate structure

Another provision from our paperwork specifies that the nonprofit retains control. (The paperwork uses OpenAI LP's official name "OpenAI, L.P.".)

As mentioned above, economic returns for investors and employees are capped (with the cap negotiated in advance on a per-limited partner basis). Any excess returns are owned by the nonprofit. Our goal is to ensure that most of the value we create if successful is returned to the world, so we think this is an important first step. Returns for our first round of investors are capped to 100x their investment, and we expect this multiple to be lower for future rounds.

What OpenAI does

Our day-to-day work remains the same. Today, we believe we can build the most value by focusing exclusively on developing new AI technologies, not commercial products. Our structure gives us flexibility for how to make money in the long term, but we hope to figure that out only once we've created safe AGI (though we're open to non-distracting revenue sources such as licensing in the interim).

OpenAI LP currently employs around 100 people organized into three main areas: capabilities (advancing what AI systems can do), safety (ensuring those systems are aligned with human values), and policy (ensuring appropriate governance for such systems). OpenAI Nonprofit governs OpenAI LP, runs educational programs such as Scholars and Fellows, and hosts policy initiatives. OpenAI LP is continuing (at increased pace and scale) the development roadmap started at OpenAI Nonprofit, which has yielded breakthroughs in reinforcement learning, robotics, and language.

Safety

We are concerned about AGI's potential to cause rapid change, whether through machines pursuing goals misspecified by their operator, malicious humans subverting deployed systems, or an out-of-control economy that grows without resulting in improvements to human lives. As described in our Charter, we are willing to merge with a value-aligned organization (even if it means reduced or zero payouts to investors) to avoid a competitive race which would make it hard to prioritize safety.

Who's involved

OpenAI Nonprofit's board consists of OpenAI LP employees Greg Brockman (Chairman & CTO), Ilya Sutskever (Chief Scientist), and Sam Altman (CEO), and non-employees Adam D'Angelo, Holden Karnofsky, Reid Hoffman, Sue Yoon, and Tasha McCauley.

Elon Musk left the board of OpenAI Nonprofit in February 2018 and is not involved with OpenAI LP.

Our investors include Reid Hoffman and Khosla Ventures.

We are traveling a hard and uncertain path, but we have designed our structure to help us positively affect the world should we succeed in creating AGI. If you'd like to help us make this mission a reality, we're hiring :)!

About this email: Altman sends Musk the draft blog post announcing OpenAI LP, summarizing that a capped-profit company has been created with a first round led by Reid Hoffman and Vinod Khosla, and says a multi-billion dollar investment is under discussion on which he would like Musk's advice.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Mar 6, 2019 3:13 PM. The LessWrong archive marks this email as sourced from OpenAI's blog; the draft post had embedded images of legal paperwork that are not reproduced. 'We did this is a way' is as transcribed. The court exhibit copy (Dkt. 379-88; header Wed, 6 Mar 2019 15:12:20 -0800) shows the second TL;DR bullet ending '(see purple box below)', which the blog/LessWrong transcription omits; no reply from Musk appears in that exhibit.

Sources: [OpenAI blog: Elon Musk wanted an OpenAI for-profit \(Dec 13, 2024\)](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#); [Musk v. Altman, Dkt. 379-88 \(Ex. 64 to Hawes Declaration, Bates 2024MUSK-0005585\)](#)

Tags: capped-profit, openai-lp, fundraising, announcement, microsoft

Musk: 'be explicit that I have no financial interest' in the for-profit

From: Elon Musk

Date: March 11, 2019

To: Sam Altman

Subject: Bloomberg: AI Research Group Co-Founded by Elon Musk Starts For-Profit Arm

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

Please be explicit that I have no financial interest in the for-profit arm of OpenAI

AI Research Group Co-Founded by Elon Musk Starts For-Profit Arm

Bloomberg

OpenAI, the San Francisco-based artificial intelligence research group co-founded by Elon Musk and several other prominent Silicon Valley entrepreneurs, is starting a for-profit arm that will allow it to raise more money. Read the full story

Shared from Apple News

About this email: On the day OpenAI LP was announced, Musk forwards a Bloomberg headline to Altman and asks that OpenAI state explicitly that he has no financial interest in the for-profit arm.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Mar 11, 2019 3:04 PM. Everything after the first line is the Apple News share card for the Bloomberg story.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits, CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: capped-profit, openai-lp, press, disassociation, musk-v-openai

Altman: 'on it'

From: Sam Altman

Date: March 11, 2019

To: Elon Musk

Subject: Re: Bloomberg: AI Research Group Co-Founded by Elon Musk Starts For-Profit Arm

FULL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

on it

About this email: Altman's two-word reply agreeing to make clear Musk has no financial interest in OpenAI LP.

Notes: Text taken from the LessWrong transcription (habryka, Nov 16 2024), which the compiler states was checked against the originals. Sent Mar 11, 2019 3:11 PM.

Sources: [Musk v. Altman \(N.D. Cal.\) court exhibits](#), [CourtListener docket](#); [LessWrong: OpenAI Email Archives \(from Musk v. Altman and OpenAI blog\)](#)

Tags: capped-profit, press, short-reply, musk-v-openai

Twitter/X

The offer letter, emails to staff after the takeover, and emails to reporters.

30 emails, April 13, 2022 to January 2025

Best-and-final offer letter to buy Twitter at \$54.20

From: Elon Musk

Date: April 13, 2022

To: Bret Taylor, Chairman of the Board, Twitter

FULL TEXT · BOARD, INVESTORS & ADVISERS · PUBLISHED BY MUSK · CONFIRMED

Bret Taylor

Chairman of the Board,

I invested in Twitter as I believe in its potential to be the platform for free speech around the globe, and I believe free speech is a societal imperative for a functioning democracy.

However, since making my investment I now realize the company will neither thrive nor serve this societal imperative in its current form. Twitter needs to be transformed as a private company.

As a result, I am offering to buy 100% of Twitter for \$54.20 per share in cash, a 54% premium over the day before I began investing in Twitter and a 38% premium over the day before my investment was publicly announced. My offer is my best and final offer and if it is not accepted, I would need to reconsider my position as a shareholder.

Twitter has extraordinary potential. I will unlock it.

/s/ Elon Musk

Elon Musk

About this email: Musk tells Twitter's chairman he invested because he believes in Twitter as the platform for free speech, says it must be transformed as a private company, and offers \$54.20 a share in cash as a best and final offer.

Notes: Letter delivered to Twitter on April 13, 2022 and filed by Musk with the SEC the next day; the filing does not say whether it was transmitted by email. The same exhibit also contains a script Musk sent by text/voice, which is excluded here as a text message. Paragraph breaks reconstructed from the retrieval tool's numbered paragraph listing.

Sources: [SEC EDGAR - Schedule 13D/A \(Amendment No. 2\)](#), [Exhibit B](#); [TechSpot](#)

Tags: acquisition, twitter-offer, free-speech, sec-filing

Musk to Kara Swisher: 'You're an asshole'

From: Elon Musk

Date: November 2022 (disclosed by Swisher on her podcast in mid-November 2022; send date not given)

To: Kara Swisher

Subject: You're an asshole

EXCERPT · PRESS & JOURNALISTS · PUBLISHED BY RECIPIENT · WIDELY REPORTED

You're an asshole

About this email: Swisher said Musk sent her an email with this subject line and a screenshot of one of her tweets, ending their working relationship; she said the tweet had actually been supportive of him.

Notes: Only the subject line was made public; the body reportedly contained a screenshot of Swisher's tweet. The email itself has not been published, so the wording rests on Swisher's account.

Sources: [Business Today](#) (reporting Swisher's 'On with Kara Swisher' podcast); [Fox News](#) (Spanish edition); [Out](#) (on Swisher's memoir 'Burn Book')

Tags: kara-swisher, twitter, press, insult

First all-staff email: remote work ends, dire economy

From: Elon Musk

Date: November 9, 2022 (Sent late on Nov. 9 Pacific time (about 2:39 a.m. ET Nov. 10 per Business Insider))

To: All Twitter employees

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Sorry that this is my first email to the whole company but there is no way to sugarcoat the message.

Frankly, the economic picture ahead is dire especially for a company like ours that is so dependent on advertising in a challenging economic climate. Moreover, 70% of our advertising is brand, rather than specific performance, which makes us doubly vulnerable!

That is why the priority over the past ten days has been to develop and launch Twitter Blue Verified subscriptions (huge props to the team!).

Without significant subscription revenue, there is a good chance Twitter will not survive the upcoming economic downturn. We need roughly half of our revenue to be subscription.

Of course, we will still then be significantly reliant on advertising, so I am spending time with our sales & partnerships teams to ensure that Twitter continues to be appealing to advertisers.

This is the Spaces discussion that Robin, Yoel and I hosted today: [Link to Elon Q&A: Advertising & the Future removed]

The road ahead is arduous and will require intense work to succeed. We are also changing Twitter policy such that remote work is no longer allowed unless you have a specific exception. Managers will send the exception lists to me for review and approval.

Starting tomorrow (Thursday), everyone is required to be in the office for a minimum of 40 hours per week. Obviously, if you are physically unable to travel to an office or have a critical personal obligation, then your absence is understandable.

I look forward to working with you to take Twitter to a whole new level. The potential is truly incredible!

Thanks,

Elon

About this email: In his first company-wide email, Musk warns the economic picture is dire, says Twitter needs half its revenue from subscriptions to survive, and ends remote work with a 40-hour in-office minimum starting the next day.

Notes: CNBC removed the Spaces link. Business Insider's transcription differs trivially (e.g. 'exceptions lists', 'review an approval').

Sources: [CNBC \(via NBC Washington\)](#); [Bloomberg](#); [Business Insider \(via Yahoo\)](#)

Tags: remote-work, subscriptions, advertising, twitter-blue, economy

Second email: top priority is suspending verified bots and spam

From: Elon Musk

Date: November 9, 2022 (Sent Wednesday night, Nov. 9, shortly after the first all-staff email (Bloomberg))

To: All Twitter employees

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Over the next few days, the absolute top priority is finding and suspending any verified bots/trolls/spam

About this email: In a follow-up to his first all-staff email, Musk tells employees the top priority for the coming days is finding and suspending verified bots, trolls and spam.

Notes: Only this sentence was published.

Sources: [Bloomberg \(via The Peninsula\)](#); [Bloomberg](#); [TechCrunch](#); [Bloomberg \(via Boston Globe\)](#)

Tags: spam, bots, verification, twitter-blue

Twitter will keep granting stock, SpaceX-style

From: Elon Musk

Date: November 14, 2022 (Date of CNBC report; exact send date not given)

To: Twitter employees

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Even though Twitter is now a private company we absolutely will continue to provide stock and options as part of our ongoing compensation plan.

The stock plan will be much like that of SpaceX, which has been very successful. As with SpaceX, exceptional amounts of stock will be awarded for exceptional performance.

Thanks,

Elon

About this email: Musk assures staff that although Twitter is now private it will keep granting stock and options under a plan modeled on SpaceX's, with exceptional awards for exceptional performance.

Sources: [CNBC](#); [NBC San Diego \(CNBC syndication\)](#)

Tags: compensation, stock-options, spacex

'A Fork in the Road': extremely hardcore ultimatum

From: Elon Musk

Date: November 16, 2022

To: All Twitter employees

Subject: A Fork in the Road

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Going forward, to build a breakthrough Twitter 2.0 and succeed in an increasingly competitive world, we will need to be extremely hardcore. This will mean working long hours at high intensity. Only exceptional performance will constitute a passing grade.

Twitter will also be much more engineering-driven. Design and product management will still be very important and report to me, but those writing great code will constitute the majority of our team and have the greatest sway.

At its heart, Twitter is a software and servers company, so I think this makes sense.

If you are sure that you want to be part of the new Twitter, please click yes on the link below:

[Link removed]

Anyone who has not done so by 5pm ET tomorrow (Thursday) will receive three months of severance.

Whatever decision you make, thank you for your efforts to make Twitter successful.

Elon

About this email: Musk tells staff that Twitter 2.0 will require being 'extremely hardcore', with long hours at high intensity, and gives them until 5pm ET the next day to click yes or leave with three months of severance.

Notes: Sent late on the night of Nov. 15-16, 2022; first reported by The Washington Post. Link to the opt-in form removed by CNBC. The email later featured in severance litigation and arbitration by former Twitter employees.

Sources: [CNBC](#); [CNN](#); [Mediaite](#); [Entrepreneur](#); [PBS NewsHour \(AP\)](#)

Tags: work-ethic, ultimatum, hardcore, severance, twitter-2.0

Remote work allowed if manager vouches for excellence

From: Elon Musk

Date: November 17, 2022

To: All Twitter employees

Subject: re: Fork in the Road

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Regarding remote work, all that is required for approval is that your manager takes responsibility for ensuring that you are making an excellent contribution. It is also expected that you have in-person meetings with your colleagues on a reasonable cadence, ideally weekly, but not less than once per month.

About this email: Hours before the ultimatum deadline, Musk softens his remote-work stance: remote work is approved if a manager takes responsibility for the employee's excellent contribution, with in-person meetings at least monthly.

Notes: Body as printed by CNBC; greeting/sign-off, if any, not shown.

Sources: [CNBC](#); [Fortune](#); [Bloomberg \(via Yahoo\)](#)

Tags: remote-work, managers, policy

Managers who falsely vouch for staff will be exited

From: Elon Musk

Date: November 17, 2022

To: All Twitter employees

Subject: re: Fork in the Road

FULL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

At risk of stating the obvious, any manager who falsely claims that someone reporting to them is doing excellent work or that a given role is essential, whether remote or not, will be exited from the company.

About this email: In a second follow-up minutes later, Musk warns that any manager who falsely claims a report is doing excellent work or that a role is essential will be fired.

Notes: Body as printed by CNBC; greeting/sign-off, if any, not shown.

Sources: [CNBC](#); [The Statesman](#)

Tags: remote-work, managers, warning

'Anyone who actually writes software' report to 10th floor

From: Elon Musk

Date: November 18, 2022 (Sent Friday morning Pacific time)

To: Twitter employees (software engineers)

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Anyone who actually writes software, please report to the 10th floor at 2 pm today. Before doing so, please email a bullet point summary of what your code commands have achieved in the past ~6 months, along with up to 10 screenshots of the most salient lines of code

About this email: The morning after the ultimatum deadline triggered mass resignations, Musk summons anyone who writes software to the 10th floor of Twitter HQ at 2pm and asks for a bullet-point summary of six months of code commits plus up to 10 screenshots of code.

Notes: Text as given in the Ars Technica feed excerpt ('email a bullet point summary of what your code commands'); Fortune and others print it as 'please email me a bullet point summary of what your code commits have achieved in the last ~6 months'. Ars says Business Insider published the emails in full, but Business Insider, The Verge and Ars Technica were blocked for retrieval in this session. Fortune reports this was sent as two emails about 20 minutes apart.

Sources: [Ars Technica \(via Harvard TagTeam feed\)](#); [Fortune](#); [Reuters \(via Al Jazeera\)](#); [CNBC](#); [Mediaite](#); [Business Insider \(published the emails in full; not retrievable in this session\)](#); [The Daily Beast](#)

Tags: engineering, code-review, in-person, layoffs-aftermath

Flying to SF 'appreciated, but is not essential'

From: Elon Musk

Date: November 18, 2022 (About an hour after the earlier follow-ups, per Fortune)

To: Twitter employees (software engineers)

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · SINGLE-SOURCE

flying out to SF would be appreciated, but is not essential

About this email: About an hour later Musk softens the travel request, telling engineers that flying out to San Francisco would be appreciated but is not essential.

Notes: Only this fragment was published (Fortune).

Sources: [Fortune](#)

Tags: engineering, remote-work, in-person

Remote engineers: short technical interviews by video

From: Elon Musk

Date: November 18, 2022 (Minutes after the 10th-floor summons)

To: Twitter employees (software engineers working remotely)

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Only those who cannot get to Twitter HQ or have a family emergency are excused

[...]

These will be short, technical interviews that allow me to better understand the Twitter tech stack

About this email: Musk tells engineers working remotely to email their code summaries anyway so he can try to speak with them by video, excusing only those who cannot get to HQ or have a family emergency, and describes the sessions as short technical interviews.

Notes: Fragments from Reuters ('These will be short...') and CNBC ('Only those who cannot get to Twitter HQ...'); their order within the email is not confirmed, and The Daily Beast renders the excusal line as 'only those who cannot physically get to Twitter HQ are excused'. Sources disagree on whether this was a separate email or part of the next one.

Sources: [Reuters \(via Devdiscourse\)](#); [CNBC](#); [Fortune](#); [The Daily Beast](#)

Tags: engineering, remote-work, code-review

Fly to SF if possible; 'at Twitter HQ until midnight'

From: Elon Musk

Date: November 18, 2022 (Shortly after the remote-work follow-up)

To: Twitter employees (software engineers)

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

If possible, I would appreciate it if you could fly to SF to be present in person

[...]

I will be at Twitter HQ until midnight and then back again tomorrow morning

About this email: Musk asks engineers outside San Francisco to fly in to be present in person if they can, adding that he will be at Twitter HQ until midnight and back the next morning.

Notes: First sentence as quoted by Reuters, second by Fortune, which reports both as part of the same short email.

Sources: [Fortune](#); [Reuters \(via Devdiscourse\)](#); [Reuters \(via Al Jazeera\)](#); [Reuters \(via The Peninsula\)](#)

Tags: engineering, in-person, work-ethic

Prepare for 'brief' code reviews as Musk walks the office

From: Elon Musk

Date: November 18, 2022 (About 5 p.m. PT, per Fortune)

To: Twitter employees

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · SINGLE-SOURCE

brief

[...]

walking around the office

About this email: In a late-afternoon email Musk tells staff to be ready for brief code reviews as he walks around the office.

Notes: Only these two quoted fragments were published; the rest of the email is paraphrased by Fortune.

Sources: [Fortune](#)

Tags: engineering, code-review

World Cup support is the week's top priority

From: Elon Musk

Date: November 18, 2022 (About 6 p.m. PT, per Fortune)

To: Twitter employees

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · SINGLE-SOURCE

whatever it takes

About this email: In an evening email Musk tells staff that making sure Twitter supports the World Cup with the best possible user experience is the top priority for the week.

Notes: Only this quoted fragment was published; the rest of the email is paraphrased by Fortune.

Sources: [Fortune](#)

Tags: engineering, world-cup, priorities, stability

Leak warning: violate your NDA and Twitter will seek damages

From: Elon Musk

Date: December 10, 2022 (Reported Dec. 10 by Zoe Schiffer; Platformer says it was sent Friday Dec. 9, with a pledge due by 5 p.m. Saturday)

To: All Twitter employees

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

As evidenced by the many detailed leaks of confidential Twitter information, a few people at our company continue to act in a manner contrary to the company's interests and in violation of their NDA

[...]

This will be said only once: If you clearly and deliberately violate the NDA that you signed when you joined, you accept liability to the full extent of the law & Twitter will immediately seek damages

[...]

but breaking your word by sending detailed info to the media [...] will receive the response it deserves

About this email: Musk warns staff that detailed leaks show a few people are violating their NDAs, threatens to seek damages from deliberate leakers, and requires employees to sign a pledge acknowledging the policy.

Notes: Schiffer deliberately did not publish the full email to protect sources, so only these passages are public. The third passage is quoted by sources around a paraphrase (sending detailed info to the media 'with the intent to harm Twitter').

Sources: [The News Minute \(IANS\)](#); [Platformer](#); [Rolling Stone](#); [Zoe Schiffer thread \(Thread Reader\)](#); [AFP \(via TechXplore\)](#); [Fortune](#); [TMZ](#)

Tags: leaks, nda, confidentiality, legal-threat

Pause new features for stability before Super Bowl

From: Elon Musk

Date: February 8, 2023 (Sent after the Feb. 8, 2023 outage, per The Information; exact time unknown)

To: Twitter employees

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · SINGLE-SOURCE

Please pause for now on new feature development in favor of maximizing system stability and robustness, especially with the Super Bowl coming up.

About this email: After a widespread Twitter outage, Musk tells staff to pause new feature development in favor of system stability and robustness ahead of the Super Bowl.

Notes: Email first reported by The Information (not retrievable in this session); only this sentence was quoted.

Sources: [Awful Announcing \(citing The Information\)](#); [Fortune](#); [Data Center Dynamics](#)

Tags: outage, stability, engineering, super-bowl

'Performance Awards' email after weekend layoffs

From: Elon Musk

Date: February 27, 2023 (Monday Feb. 27, 2023, 10:48 a.m. per The Verge (via APH Networks))

To: Remaining Twitter employees

Subject: Performance Awards

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

This past week, we completed a difficult organizational overhaul focused on improving future execution, using as much feedback as we could gather from the entire company. Those who remain are highly regarded by those around them.

About this email: After another round of weekend layoffs, Musk tells remaining staff the company completed a difficult organizational overhaul and that they will receive very significant stock and performance awards on March 24.

Notes: First reported by The Verge (blocked for retrieval in this session). The remainder, promising 'very significant' stock and performance awards on March 24, is only paraphrased or partially quoted in retrievable sources.

Sources: [WION](#); [APH Networks \(citing The Verge\)](#); [HR Katha](#)

Tags: layoffs, compensation, stock-awards, reorganization

2:30 a.m. email: 'office is not optional'

From: Elon Musk

Date: March 22, 2023 (Sent about 2:30 a.m. on a Wednesday; Business Insider/Yahoo give Wednesday March 22-23, 2023)

To: Twitter employees

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

office is not optional

[...]

was half empty yesterday

About this email: In an email sent around 2:30 a.m., Musk reminds Twitter staff that working from the office is not optional and notes the San Francisco office had been half empty the day before.

Notes: Known only from a tweet by Platformer's Zoe Schiffer; just these fragments were quoted. The retrieved Yahoo page gave the date as Wednesday March 23, 2023, but March 22 was the Wednesday of that week.

Sources: [Business Insider \(via Yahoo\)](#); [Fortune](#); [iHeart](#); [Fox/KTVU](#)

Tags: return-to-office, remote-work, work-ethic

Stock grants at \$20 billion valuation; 'inverse start-up'

From: Elon Musk

Date: March 24, 2023 (Sent Friday March 24, 2023 (some reports say March 25))

To: Twitter employees

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Twitter is being reshaped rapidly

[...]

can be thought of as an inverse startup

[...]

I see a clear, but difficult, path to a >\$250B valuation

[...]

liquid stock, but without the stock price chaos and lawsuit burdens of a public company

About this email: Announcing new equity grants, Musk tells staff the awards value Twitter at about \$20 billion, less than half what he paid, says the company came close to running out of money, and describes a path to a \$250 billion valuation with liquidity events every six months.

Notes: Email reviewed by The Information, Platformer, The Wall Street Journal and The New York Times; no outlet retrievable in this session printed the full text. Fragments are stitched from several reports and wording varies slightly between outlets ('inverse start-up' / 'inverse startup').

Sources: [WION](#) (citing [The Wall Street Journal](#)); [Rolling Stone](#) (citing [NYT](#)); [Nexstar/The Hill](#) (via [KOIN](#)); [Fortune](#); [Ars Technica](#) (reposted); [The Information](#) (first report; listed by [Techmeme](#), not opened); [The Wall Street Journal](#) (listed by [Techmeme](#), not opened)

Tags: compensation, stock-grants, valuation, finances, x-corp

To NPR: 'marinating in the Kool-Aid'

From: Elon Musk

Date: April 2023 (Sent between April 6 and April 12, 2023; quoted in NPR's April 12 story)

To: Bobby Allyn (NPR technology reporter)

PARTIAL TEXT · PRESS & JOURNALISTS · PUBLISHED BY RECIPIENT · CONFIRMED

If you really think that the government has no influence on the entity they're funding then you've been marinating in the Kool-Aid for too long

About this email: Pressed on how he justifies labeling NPR given its small share of government funding and editorial independence, Musk tells the reporter that anyone who thinks government funding brings no influence has been 'marinating in the Kool-Aid'.

Notes: NPR also reports that in these exchanges Musk said he was relying on a Wikipedia page of publicly funded broadcasters, and that he later tweeted a screenshot of Allyn's April 12 email asking for his reaction, followed by 'Defund @NPR' (a tweet, not archived here).

Sources: [NPR](#)

Tags: npr, media-labels, government-funded, press

To NPR: 'Well, then we should fix it'

From: Elon Musk

Date: April 5, 2023

To: Bobby Allyn (NPR technology reporter)

PARTIAL TEXT · PRESS & JOURNALISTS · PUBLISHED BY RECIPIENT · CONFIRMED

Well, then we should fix it

[...]

What's the breakdown of NPR annual funding?

About this email: Told by NPR's reporter that government support is about 1% of NPR's finances, Musk replies that Twitter should then fix the 'state-affiliated media' label and asks for NPR's funding breakdown.

Notes: Quoted by the recipient in his own reporting; NPR presents the second sentence as something Musk 'added', so it may be the same or a follow-up email.

Sources: [NPR \(via Texas Public Radio\)](#); [NPR \(via WAMC\)](#); [Fortune](#)

Tags: npr, media-labels, state-affiliated, press

To NPR: 'fair and equal treatment' for media labels

From: Elon Musk

Date: April 5, 2023 (Sent April 5-6, 2023; NPR does not give the exact day)

To: Bobby Allyn (NPR technology reporter)

PARTIAL TEXT · PRESS & JOURNALISTS · PUBLISHED BY RECIPIENT · CONFIRMED

The operating principle at new Twitter is simply fair and equal treatment, so if we label non-US accounts as govt, then we should do the same for US, but it sounds like that might not be accurate here

About this email: Musk explains that new Twitter's operating principle is fair and equal treatment, so US outlets should be labeled like non-US government media, while conceding the label might not be accurate for NPR.

Sources: [NPR \(via Texas Public Radio\)](#); [NPR \(via WAMC\)](#); [Fortune](#)

Tags: npr, media-labels, state-affiliated, press

To NPR: label still under review, 'We're digging into it'

From: Elon Musk

Date: April 6, 2023

To: Bobby Allyn (NPR technology reporter)

EXCERPT · PRESS & JOURNALISTS · PUBLISHED BY RECIPIENT · CONFIRMED

any given rule fairly to all

[...]

We're digging into it

About this email: In Thursday emails Musk restates that he wants any rule applied fairly to all, says NPR's label is still being evaluated, and when asked whether Twitter stands by the decision replies that they are digging into it.

Notes: NPR quotes a fragment from a Thursday email and a short reply; it is unclear whether these were one email or two.

Sources: [NPR \(via Texas Public Radio\)](#); [NPR \(via WAMC\)](#); [Fortune](#)

Tags: npr, media-labels, state-affiliated, press

To NPR: should we reassign @NPR to another company?

From: Elon Musk

Date: May 2, 2023

To: Bobby Allyn (NPR technology reporter)

FULL TEXT · PRESS & JOURNALISTS · PUBLISHED BY RECIPIENT · CONFIRMED

So is NPR going to start posting on Twitter again, or should we reassign @NPR to another company?

About this email: In an unprompted email after NPR stopped posting on Twitter, Musk asks whether NPR will start posting again or whether Twitter should reassign the @NPR handle to another company.

Notes: Appears to be the whole message as quoted by the recipient.

Sources: [NPR](#); [AP \(via Boston.com\)](#); [PBS NewsHour](#)

Tags: npr, account-handles, press, threat

To NPR: policy is to recycle dormant handles

From: Elon Musk

Date: May 2, 2023

To: Bobby Allyn (NPR technology reporter)

FULL TEXT · PRESS & JOURNALISTS · PUBLISHED BY RECIPIENT · CONFIRMED

Our policy is to recycle handles that are definitively dormant. Same policy applies to all accounts. No special treatment for NPR.

About this email: Musk tells NPR's reporter that Twitter recycles definitively dormant handles and that NPR will get no special treatment.

Notes: Appears to be the whole message as quoted by the recipient.

Sources: [NPR](#); [AP \(via Boston.com\)](#); [PBS NewsHour](#)

Tags: npr, account-handles, press, policy

To NPR: 'National Pumpkin Radio'

From: Elon Musk

Date: May 2, 2023

To: Bobby Allyn (NPR technology reporter)

PARTIAL TEXT · PRESS & JOURNALISTS · PUBLISHED BY RECIPIENT · CONFIRMED

National Pumpkin Radio [fire emoji] [laughing emoji]

[...]

NPR isn't tagged as government-funded anymore, so what's the beef?

About this email: Asked who might take over the @NPR handle, Musk jokes 'National Pumpkin Radio' with fire and laughing emojis and asks what NPR's complaint is now that the government-funded tag is gone.

Notes: NPR describes rather than reproduces the two emojis.

Sources: [NPR](#); [AP \(via Boston.com\)](#); [PBS NewsHour](#)

Tags: npr, account-handles, press, trolling

To NPR: 'You suck.' (subject line only)

From: Elon Musk

Date: May 3, 2023 (Sent 2:19 a.m. EDT after NPR's story was published)

To: Bobby Allyn (NPR technology reporter)

Subject: You suck.

FULL TEXT · PRESS & JOURNALISTS · PUBLISHED BY RECIPIENT · CONFIRMED

[Subject line] You suck.

[No text in the body]

About this email: After NPR published its story on his threat to reassign the @NPR handle, Musk sent the reporter an email with no body text and the subject line 'You suck.'

Sources: [NPR](#); [AP \(via Boston.com\)](#); [PBS NewsHour](#)

Tags: npr, press, insult

Twitter is becoming X; last email from a Twitter address

From: Elon Musk

Date: July 23, 2023 (Sent late on the night of Sunday July 23, 2023)

To: Twitter employees

NO TEXT PUBLISHED · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

No verbatim text of this email has been published.

About this email: On the night of the rebrand, Musk emails employees that the company is changing to X and that this is the last email he will ever send from a Twitter address, signing off with a salute emoji.

Notes: NO VERBATIM TEXT PUBLISHED: the email is known only from Zoe Schiffer's description ('the last email he'd ever send from a Twitter email address. He signed off with this emoji: [salute]'). full_text left empty rather than paraphrased.

Sources: [Engadget \(citing Zoe Schiffer of Platformer\)](#); [Techmeme archive \(Schiffer post\)](#); [Gigazine](#)

Tags: rebrand, x, branding

Stock options to be based on impact; submit one-page summary

From: Elon Musk

Date: August 23, 2024 (Sent overnight Aug. 22-23, 2024, per The Verge)

To: X employees

NO TEXT PUBLISHED · COMPANY-WIDE MEMO · LEAKED TO PRESS · SINGLE-SOURCE

No verbatim text of this email has been published.

About this email: Musk tells X staff that stock options will be awarded according to each person's impact and that employees must submit a one-page summary of their contributions to receive them.

Notes: NO VERBATIM TEXT RETRIEVED: The Verge (blocked in this session) saw the email; every retrievable report only paraphrases it. full_text left empty rather than paraphrased.

Sources: [Business Insider \(via AOL\)](#), citing [The Verge](#); [Fortune](#); [Benzinga](#); [TipRanks/The Fly](#); [IANS \(via DT Next\)](#), [Nov. 2024 follow-up](#); [The Verge \(original report; listed by Techmeme, blocked in this session\)](#); [Boing Boing](#)

Tags: compensation, stock-options, performance, x

X growth stagnant, 'we're barely breaking even'

From: Elon Musk

Date: January 2025 (Sent in January 2025; reported by The Wall Street Journal on Jan. 24, 2025)

To: X employees

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Over the last few months, we've witnessed the power of X in shaping national conversations and outcomes. We are also seeing other platforms begin to adopt our commitment to free speech and unbiased truth.

[...]

Our user growth is stagnant, revenue is unimpressive, and we're barely breaking even.

About this email: Musk tells X employees that the platform has shown its power in shaping national conversations but that user growth is stagnant, revenue is unimpressive and the company is barely breaking even.

Notes: Musk publicly denied sending the email after the WSJ report; The Verge then published direct quotes. WSJ and Verge URLs come from Techmeme's listing and could not be opened in this session. The order of the two passages within the email is not confirmed.

Sources: [TheWrap](#) (citing [The Wall Street Journal](#)); [The Wall Street Journal](#) (first report); [The Verge](#); [TechCrunch](#); [TechCentral.ie](#); [Pixel Envy](#)

Tags: finances, user-growth, revenue, x, debt

xAI

Emails to xAI staff.

1 emails, September 16, 2025 to September 16, 2025

xAI staff: send one-page summary of last four weeks

From: Elon Musk

Date: September 16, 2025 (Sent Tuesday afternoon, Sept. 16, 2025)

To: All xAI employees

PARTIAL TEXT · COMPANY-WIDE MEMO · LEAKED TO PRESS · WIDELY REPORTED

Send a one-page summary of what you've accomplished in the past four weeks and what you intend to accomplish in the next four weeks. This is due by noon on Thursday.

About this email: Days after xAI laid off hundreds of data annotators, Musk orders every xAI employee to send a one-page summary of what they accomplished in the past four weeks and plan for the next four, due within 48 hours.

Notes: CNN quotes the instruction in two fragments ('send a one page summary...' and 'This is due by noon on Thursday'); the joined wording follows Storyboard18's rendering. Any further text of the email was not published.

Sources: [CNN \(via KVIA\)](#); [Storyboard18](#)

Tags: performance, accountability, xai, layoffs

SolarCity

Emails from the SolarCity acquisition litigation.

3 emails, July 10, 2016 to September 18, 2016

Lyndon Rive warns Musk of SolarCity cash 'domino effect'

From: Lyndon Rive (SolarCity CEO)

Date: July 10, 2016

To: Elon Musk

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

cash forecast [he] gave the [SolarCity Board] in April

[...]

domino effect

[...]

delay[s] [in] funding

About this email: SolarCity's CEO sends Musk the cash forecast he gave SolarCity's board, warns of a 'domino effect' from funding delays and asks to discuss a \$200 million bridge loan; the court found Musk told him Tesla's board would not authorise a bridge loan.

Notes: Email TO Musk; only fragments are quoted in Vice Chancellor Slight's opinion (bracketed alterations are the court's). The opinion says Musk responded that Tesla's board would not authorise a bridge loan, citing trial testimony rather than a written reply.

Sources: [In re Tesla Motors, Inc. Stockholder Litigation, C.A. No. 12711-VCS \(Del. Ch. Apr. 27, 2022\)](#), [post-trial opinion \(citing JX 1455\)](#); [Delaware Supreme Court affirmance \(June 6, 2023\)](#)

Tags: solarcity, liquidity, bridge-loan, lyndon-rive

Musk: investors 'very negative on SolarCity' - show solar roof before vote

From: Elon Musk

Date: September 16, 2016

To: Peter Rive (SolarCity co-founder and CTO)

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · WIDELY REPORTED

Latest feedback from major investors is very negative on SolarCity. We need to show them what the integrated product looks like. They just don't get it. Needs to happen before the vote, so maybe aim for October 28th for a joint solar roof and Powerwall 2 unveiling.

About this email: Two months before the shareholder vote on the SolarCity deal, Musk tells Peter Rive that major investors are very negative and that a joint solar roof and Powerwall 2 unveiling must happen before the vote.

Notes: Unsealed in the Delaware SolarCity litigation in autumn 2019. The article quotes the email continuously; it is unclear whether this is the entire message. The PlainSite docket (primary documents) could not be reached in this session.

Sources: [Business Insider \(republished by Entrepreneur\)](#), reporting on unsealed filings in *In re Tesla Motors, Inc. Stockholder Litigation* (Del. Ch. C.A. No. 12711-VCS); PlainSite docket for *In re Tesla Motors, Inc. Stockholder Litigation*

Tags: solarcity, solar-roof, powerwall, investors, merger-vote

Musk: SolarCity must solve its 'liquidity crisis' before the vote

From: Elon Musk

Date: September 18, 2016 (date and recipient per Business Insider; CNBC says only 'September' 2016)

To: Brad Buss (former SolarCity CFO, Tesla director) and SolarCity executives

PARTIAL TEXT · BOARD, INVESTORS & ADVISERS · COURT RECORD · WIDELY REPORTED

Three things need to happen to change investor sentiment: SolarCity solving its liquidity crisis, an LOI with Panasonic to address solar cell production risk, and a joint product demo. Should be able to do all those before the shareholder vote.

About this email: Musk lists three things needed to change investor sentiment about the SolarCity acquisition: fixing SolarCity's liquidity crisis, a Panasonic letter of intent on solar cells, and a joint product demo.

Notes: CNBC quotes the text and says Musk 'wrote to SolarCity execs in September' 2016; Business Insider describes an email dated Sept. 18, 2016 to Brad Buss saying SolarCity needed to solve its liquidity crisis to persuade investors. The two descriptions are treated here as the same email; that identification is an inference.

Sources: [CNBC](#), citing unsealed court documents in *In re Tesla Motors, Inc. Stockholder Litigation*; [Business Insider via Entrepreneur](#) (dates the email Sept. 18, 2016 and names Brad Buss as recipient)

Tags: solarcity, liquidity, panasonic, investors, merger-vote

Neuralink

Messages to Neuralink staff.

3 emails, October 2017 to February 8, 2022

Musk tells Neuralink it can hire 'directly from OpenAI'

From: Elon Musk

Date: October 2017 (MIT Technology Review gives only the year 2017; a summary of the trial exhibit list (Ex. 719) dates it October 2017.)

To: A Neuralink cofounder (unnamed in coverage)

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · WIDELY REPORTED

hire independently or directly from OpenAI

About this email: Musk told a Neuralink cofounder that the company could hire independently or directly from OpenAI; OpenAI's lawyers used the email at trial to argue Musk was recruiting from the nonprofit for his own companies.

Notes: Trial exhibit 719 in Musk v. Altman (2026). Only this phrase was quoted in coverage I could retrieve; the full exhibit could not be accessed.

Sources: [MIT Technology Review: Musk v. Altman week 1 \(May 1, 2026\)](#); [Neura News summary of trial exhibits \(Ex. 719\)](#)

Tags: neuralink, hiring, openai, poaching, musk-v-openai

Neuralink: 'We could enable people to ... walk again'

From: Elon Musk

Date: February 8, 2022 (6:37 a.m. Pacific Time, per Reuters)

To: Neuralink staff

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · SINGLE-SOURCE

We could enable people to use their hands and walk again in daily life!

About this email: Musk forwarded Neuralink staff a news article about Swiss researchers whose electrical implant helped a paralyzed man walk again, adding that Neuralink could do the same.

Notes: Reuters says he 'wrote to staff' while sending a news article; reported from internal documents/messages reviewed by Reuters in its Dec 2022 investigation of Neuralink animal testing. Reuters does not use the word 'email' in the passage as syndicated, so the medium is not fully certain.

Sources: [Reuters \(Dec 5, 2022\)](#), as carried by [The Korea Times](#); [Forklog summary of the Reuters report](#)

Tags: neuralink, urgency, pace, competition

Neuralink: 'We are simply not moving fast enough'

From: Elon Musk

Date: February 8, 2022 (Sent ten minutes after the 6:37 a.m. PT message, per Reuters)

To: Neuralink staff

EXCERPT · COMPANY-WIDE MEMO · LEAKED TO PRESS · SINGLE-SOURCE

In general, we are simply not moving fast enough. It is driving me nuts!

About this email: Ten minutes after forwarding the Swiss implant article, Musk followed up to Neuralink staff complaining about the company's pace.

Notes: Follow-up to the 6:37 a.m. message; medium described by Reuters as written to staff (see note on companion entry).

Sources: [Reuters \(Dec 5, 2022\)](#), as carried by [The Korea Times](#); [Xataka \(Spanish summary\)](#)

Tags: neuralink, urgency, pace

Personal

Correspondence outside the companies: the Thai cave rescue and BuzzFeed emails from Unsworth v. Musk, and the 2012-2013 emails with Jeffrey Epstein released by the Department of Justice.

54 emails, September 24, 2012 to September 4, 2018

Musk to Epstein: 'very enjoyable conversation!'

From: Elon Musk

Date: September 24, 2012 (TIME dates the exchange to 24 Sep 2012; NBC says 'late September 2012')

To: Jeffrey Epstein

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

very enjoyable conversation!

About this email: The earliest Musk-Epstein email in the DOJ files: after a conversation, Musk tells Epstein he enjoyed it. Epstein replied 'Hopefully the first of many.'

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Epstein's reply, as quoted by TIME and NBC: "Hopefully the first of many."

Sources: [TIME](#); [NBC News](#); [DOJ Epstein files](#), EFTA00689860 (age-gated)

Tags: epstein-files, introductions

Epstein replies: 'Hopefully the first of many'

From: Jeffrey Epstein

Date: September 24, 2012 (TIME dates the exchange to 24 Sep 2012; NBC says 'late September 2012')

To: Elon Musk

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Hopefully the first of many.

About this email: Epstein's reply to Musk's note that they had a 'very enjoyable conversation'.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name." Email sent TO Musk in a thread he started.

Sources: [TIME](#); [NBC News](#)

Tags: epstein-files, introductions

Epstein invites Musk to his Caribbean island

From: Jeffrey Epstein

Date: September 25, 2012

To: Elon Musk

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

If you find some time, come visit me on my island in the caribbean. bring your friend or friends. i will be there most of nov and dec.

About this email: Epstein invites Musk to visit his private island and to bring friends, saying he will be there most of November and December.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Email sent TO Musk; included because Musk replied the same day.

Sources: [Fortune](#); [Business Insider \(via AOL\)](#)

Tags: epstein-files, island-invitation

Musk replies to island invitation: 'will try to make it'

From: Elon Musk

Date: September 25, 2012

To: Jeffrey Epstein

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Sounds good, will try to make it.

About this email: Musk's same-day reply to Epstein's invitation to visit his Caribbean island.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Appears to be the whole reply as quoted, but the DOJ original was not retrieved.

Sources: [Fortune](#); [Business Insider \(via AOL\)](#)

Tags: epstein-files, island-invitation

Musk forwards Epstein's solar query: 'Are we in New Mexico?'

From: Elon Musk

Date: October 2012 (forwarded after Epstein's 4 Oct 2012 email; exact date not given)

To: Peter Rive (SolarCity)

EXCERPT · EXECUTIVE & INTERNAL · GOVERNMENT RELEASE · WIDELY REPORTED

Are we in New Mexico?

About this email: Musk forwards Epstein's request for a SolarCity contact to his cousin Peter Rive, asking whether SolarCity operates in New Mexico.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Fortune identifies the recipient of the forward as Peter Rive; no further replies are recorded in the reporting.

Sources: [404 Media](#); [Fortune](#)

Tags: epstein-files, solarcity

Epstein offers to send a helicopter to St Barts

From: Jeffrey Epstein

Date: October 2012 (reply to Musk's 20 Oct 2012 email; exact date not given)

To: Elon Musk

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

an easy but correct assumption... we will catch up either in ny, or i will send a heli to st bart for you, 45 minutes away.

About this email: Epstein confirms he will be on his island and offers to meet in New York or send a helicopter to St Barts for Musk.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Email sent TO Musk.

Sources: [Fortune](#); [Lead Stories](#)

Tags: epstein-files, st-barts

Epstein asks Musk for a SolarCity contact

From: Jeffrey Epstein

Date: October 4, 2012

To: Elon Musk

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Is there any one at Solar City that my guys can talk to about electrifying t=e [sic] caribbean island? or the new mexico ranch.

About this email: Epstein asks whether someone at SolarCity could talk to his staff about electrifying his Caribbean island or New Mexico ranch.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Email sent TO Musk. Per Fortune, Musk forwarded it to his cousin Peter Rive of SolarCity (see next entry). 404 Media prints a slightly normalized spelling of the same sentence.

Sources: [Fortune](#); [404 Media](#); [TIME](#); [DOJ Epstein files, EFTA01894511 \(age-gated\)](#)

Tags: epstein-files, solarcity

Musk tells Epstein he and Talulah Riley are headed to St Barts

From: Elon Musk

Date: October 20, 2012

To: Jeffrey Epstein

PARTIAL TEXT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

The world needs more romance

[...]

Talulah and I are headed to St Barth's at the end of the year. I assume yo= will most likely be on your island?

About this email: Musk tells Epstein that he and Talulah Riley will be in St Barts at year-end and asks whether Epstein will be on his island.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." DOJ document EFTA02702478 per Lead Stories. The opening phrase 'The world needs more romance' is quoted by 404 Media with an ellipsis before the rest; 'yo=' is an encoding artifact for 'you'.

Sources: [Lead Stories](#); [404 Media](#); [Fortune](#)

Tags: epstein-files, st-barts

Musk asks when to come over between Dec 23 and 30

From: Elon Musk

Date: November 2, 2012

To: Jeffrey Epstein

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

What's a good time to come over between Dec 23 and 30?

About this email: Musk asks Epstein for a good time to visit during Christmas week 2012.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Epstein's reply as quoted by Fortune: "Probably after Xmas day. Any day, lets choose a sunny one" and "Glad you are coming."

Sources: [Fortune](#); [Lead Stories](#)

Tags: epstein-files, island-invitation

Musk: 'Let's aim for Dec 28, 29 or 30'

From: Elon Musk

Date: November 2, 2012 (follow-up in the 2 Nov 2012 exchange; Fortune does not give a separate date)

To: Jeffrey Epstein

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Let's aim for Dec 28, 29 or 30. Looking forward to it.

About this email: After Epstein suggests a day after Christmas, Musk proposes December 28, 29 or 30 for the visit.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes."

Sources: [Fortune](#); [Business Insider \(via AOL\)](#)

Tags: epstein-files, island-invitation

Epstein: 'Probably after Xmas day ... Glad you are coming'

From: Jeffrey Epstein

Date: November 2, 2012 (reply in the 2 Nov 2012 exchange; Fortune gives no separate date)

To: Elon Musk

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Probably after Xmas day. Any day, lets choose a sunny one

[...]

Glad you are coming.

About this email: Epstein answers Musk's question about when to visit during Christmas week 2012.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name." Email sent TO Musk. Fortune quotes the two fragments separately.

Sources: [Fortune](#); [Business Insider](#) (via AOL)

Tags: epstein-files, island-invitation

Musk asks about the 'wildest party' on Epstein's island

From: Elon Musk

Date: November 24, 2012

To: Jeffrey Epstein

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Probably just Talulah and me. What day/night will be the wildest party on =our island?

About this email: Replying to Epstein's question about how many people would need the helicopter, Musk says probably just he and Talulah Riley and asks which night will have the wildest party.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Lead Stories cites DOJ document EFTA02521947. The released text reads '=our island' (an encoding artifact); most outlets render it as 'your island'. Epstein's preceding question: "how many people will you be for the heli to island".

Sources: [TIME](#); [Lead Stories](#); [404 Media](#); [NBC News](#); [Gizmodo](#); [DOJ Epstein files, EFTA01762434](#) (age-gated)

Tags: epstein-files, island-invitation, talulah-riley

Epstein asks how many people for the helicopter

From: Jeffrey Epstein

Date: November 24, 2012

To: Elon Musk

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

how many people will you be for the heli to island

About this email: Epstein asks Musk how many people will need the helicopter to his island; Musk answered 'Probably just Talulah and me' and asked about the 'wildest party'.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name." Email sent TO Musk. Lead Stories cites DOJ document EFTA02521947.

Sources: [Lead Stories](#); [404 Media](#); [Business Insider \(via AOL\)](#); [TIME](#); [DOJ Epstein files, EFTA01762434 \(age-gated\)](#)

Tags: epstein-files, island-invitation

Musk on Christmas Day: 'Do you have any parties planned?'

From: Elon Musk

Date: December 25, 2012

To: Jeffrey Epstein

PARTIAL TEXT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Do you have any parties planned? I've been working to the edge of sanity this year and so, once my kids head home after Christmas, I really want to hit the party scene in St Barts or elsewhere and let loose. The invitation is much appreciated, but a peaceful island experience is the opposite of what I'm looking for.

About this email: Musk tells Epstein he has worked 'to the edge of sanity' and wants to party in St Barts after his kids go home, and that a peaceful island visit is not what he is looking for.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Lead Stories cites EFTA01912578; TIME says it was sent at 6am. The quoted passage may be the whole message body, but that could not be checked against the DOJ file. Gizmodo quotes an earlier Epstein line in the chain: "you are welcome to stay or just come for the day, plenty of room i will=send heli to get you".

Sources: [Lead Stories](#); [Fortune](#); [TIME](#); [Gizmodo](#); [NBC News](#); [DOJ Epstein files, EFTA02706510 \(age-gated\)](#)

Tags: epstein-files, st-barts, parties

Epstein: 'the ratio on my island' might make Riley uncomfortable

From: Jeffrey Epstein

Date: December 25, 2012 (reply in the 25 Dec 2012 chain; exact timestamp not given)

To: Elon Musk

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Understood , I will see you on st Barth, the ratio on my island might make Talilah uncomfortable

About this email: Epstein replies that he will see Musk in St Barts instead, saying the 'ratio' on his island might make Talulah Riley uncomfortable.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Email sent TO Musk; spelling as in the released document.

Sources: [Lead Stories](#); [TIME](#); [Fortune](#); [Gizmodo](#)

Tags: epstein-files, st-barts, talulah-riley

Musk: 'Ratio is not a problem for Talulah'

From: Elon Musk

Date: December 25, 2012 (reply in the 25 Dec 2012 chain; exact timestamp not given)

To: Jeffrey Epstein

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Ratio is not a problem for Talulah

About this email: Musk's reply to Epstein's remark about the 'ratio' on his island.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes."

Sources: [Lead Stories](#); [TIME](#); [Fortune](#); [Gizmodo](#)

Tags: epstein-files, talulah-riley

Musk invites Epstein for drinks in St Barts on the 30th

From: Elon Musk

Date: December 28, 2012

To: Jeffrey Epstein

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

join us (and some others) for drinks on the 30th around 10 o= 11

[...]

Back afternoon of the 30th

About this email: In a late-December scheduling exchange after Epstein asked 'how do I find you?', Musk invites him to drinks on December 30 and says he will be back that afternoon.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." DOJ document EFTA02706068 per Lead Stories. The two fragments come from two separate Musk messages in the chain; between them Epstein said he would only be on St Barts on the 29th before returning to 'my island in the evening'.

Sources: [Lead Stories](#); [TIME](#)

Tags: epstein-files, st-barts

Musk to Epstein on requests to bring Tesla to Israel

From: Elon Musk

Date: 2013 (Fortune gives only the year)

To: Jeffrey Epstein

PARTIAL TEXT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Not a week passes without a request from someone important in Israel (and other countries) wanting to bring Tesla there. What they don't appreciate is that Tesla is production constrained, not demand constrained. When production has increased to the point of addressing the major markets, then we will start shipping to tiny markets.

About this email: After Epstein said former Israeli prime minister Ehud Barak wanted to discuss bringing Tesla to Israel, Musk answered that Tesla was production constrained and would serve small markets later.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Epstein had written that Ehud Barak "is very clever about many things that you have in the past shown interest in, when you are next on my coast let me know."

Sources: [Fortune](#); [Business Insider \(via AOL\)](#)

Tags: epstein-files, israel, tesla-production

Epstein tells Musk that Ehud Barak wants to talk

From: Jeffrey Epstein

Date: 2013 (Fortune gives only the year)

To: Elon Musk

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · SINGLE-SOURCE

is very clever about many things that you have in the past shown interest in, when you are next on my coast let me know.

About this email: Epstein tells Musk that former Israeli prime minister Ehud Barak, who wanted to discuss bringing Tesla to Israel, is worth meeting.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name." Email sent TO Musk; the quoted fragment begins mid-sentence (its subject is Ehud Barak). Musk's reply about Tesla being production constrained is a separate entry.

Sources: [Fortune](#)

Tags: epstein-files, israel, ehud-barak

Epstein: 'sorry we didn't connect' — invites Musk to lunch via St Thomas

From: Jeffrey Epstein

Date: January 1, 2013 (Year uncertain: the New York Post gives 'Jan. 1' in a passage about the December 2013 exchange (implying 2014), while TIME dates Musk's next-day reply to 2 Jan 2013)

To: Elon Musk

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · SINGLE-SOURCE

sorry we didn't connect, why don't you consider clearing customs into the states in st thomas, come for lunch with the wife[?]

About this email: Epstein suggests Musk clear customs in St Thomas and come for lunch with his wife; Musk replied the next day that logistics would not work.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name." Email sent TO Musk. Bracketed question mark is the New York Post's. Filed under 2013 to sit with Musk's reply as dated by TIME; see the date conflict noted on that entry.

Sources: [New York Post \(via AOL\)](#); [TIME](#)

Tags: epstein-files, island-invitation

Musk: 'Logistics won't work this time around'

From: Elon Musk

Date: January 2, 2013 (TIME dates this to 2 Jan 2013; the New York Post places the same line on 2 Jan 2014)

To: Jeffrey Epstein

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Logistics won't work this time around.

About this email: Musk tells Epstein the planned meet-up will not happen on this trip.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Date conflict between outlets: TIME (linking EFTA01912905) says 2 January 2013; the New York Post attributes it to 2 January 2014, following an Epstein message of 1 January ("sorry we didn't connect, why don't you consider clearing customs into the states in st thomas, come for lunch with the wife[?]"). Not resolved against the DOJ original.

Sources: [TIME](#); [New York Post \(via AOL\)](#); [DOJ Epstein files, EFTA01912905 \(age-gated\)](#)

Tags: epstein-files, island-invitation

Musk replies 'I see' after Epstein's SpaceX tour

From: Elon Musk

Date: February 26, 2013

To: Jeffrey Epstein

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

I see

About this email: The day after Epstein and three assistants toured SpaceX, Epstein thanked Musk and said he 'would have had fun at xmas'; Musk replied 'I see' with a smiley face.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Musk's reply is reported as 'I see' followed by a smiley-face emoticon. Epstein's email of 26 Feb 2013, as quoted: "thanks for the tour" and "You would have had fun at xmas." Fortune also reports a 22 Feb 2013 email from Epstein's assistant to Musk sending passport copies for the visit, and a 25 Feb note that 'Jeffrey Epstein and 3 of his assistants' would visit SpaceX at 1:00.

Sources: [Fortune](#); [TIME](#); [NBC News](#); [DOJ Epstein files, EFTA00955694 \(age-gated\)](#)

Tags: epstein-files, spacex-tour

Epstein thanks Musk for the SpaceX tour

From: Jeffrey Epstein

Date: February 26, 2013

To: Elon Musk

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

thanks for the tour

[...]

You would have had fun at xmas.

About this email: The day after visiting SpaceX with three assistants, Epstein thanks Musk for the tour and says Musk would have had fun at Christmas.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name." Email sent TO Musk; he replied 'I see' with a smiley face. The two fragments are quoted separately by outlets.

Sources: [Fortune](#); [TIME](#); [NBC News](#); [DOJ Epstein files](#), [EFTA00955694 \(age-gated\)](#)

Tags: epstein-files, spacex-tour

Musk to Epstein on sleep and productivity

From: Elon Musk

Date: March 2013 (Business Insider says 'early March' 2013)

To: Jeffrey Epstein

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · SINGLE-SOURCE

Normally I would agree, as I have found that my total daily productivity is optimal at around 6 to 6.5 hours of sleep.

About this email: After Epstein wrote 'now its time for fun' and suggested Tesla would do better if Musk slept more, Musk replied that his productivity is optimal at about 6 to 6.5 hours of sleep.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name." Epstein's lines in the same exchange, per Business Insider: "now its time for fun" and "benefit analysis would probably show , tesla doing better with you getting more sleep". Business Insider also quotes a February 2013 email from Musk's assistant: "Shall we organize a lunch for Elon and Jeffrey to get together at SpaceX in the coming weeks?" and "Elon is generally available at SpaceX on Mon, Thurs and Fri each week."

Sources: [Business Insider \(via AOL\)](#)

Tags: epstein-files, sleep, work-ethic

Epstein to Musk: 'now its time for fun' and get more sleep

From: Jeffrey Epstein

Date: March 2013 (Business Insider says 'early March' 2013)

To: Elon Musk

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · SINGLE-SOURCE

now its time for fun

[...]

benefit analysis would probably show , tesla doing better with you getting more sleep

About this email: Epstein asks about Musk's availability and suggests Tesla would do better if Musk got more sleep.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name." Email(s) sent TO Musk; the two fragments may come from separate messages in the same early-March 2013 exchange.

Sources: [Business Insider](#) (via AOL)

Tags: epstein-files, sleep

Musk declines Epstein's UN General Assembly invitation

From: Elon Musk

Date: September 11, 2013 (The Independent dates the exchange to 11 Sep 2013; Fortune says only September 2013)

To: Jeffrey Epstein

PARTIAL TEXT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

I run and lead product design/engineering for two complicated companies. Moreover, SpaceX is about to launch what is arguably the most advanced rocket in history. Flying to NY to see UN diplomats do nothing would be an unwise use of time.

About this email: Invited by Epstein to New York during the UN General Assembly, Musk declines, saying he runs two complicated companies and the trip would be an unwise use of time.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Epstein's invitation, as quoted: "Any plans for ny" and "the opening of the general assembly has many interesting people coming to the house." Epstein's later reply: "just kidding, there is no one over 25 and all very cute."

Sources: [Fortune](#); [Business Insider](#) (via AOL); [The Independent](#) (via AOL)

Tags: epstein-files, new-york, time-management

Epstein invites Musk to New York for the UN General Assembly

From: Jeffrey Epstein

Date: September 11, 2013 (The Independent dates the exchange to 11 Sep 2013; Fortune says September 2013)

To: Elon Musk

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Any plans for ny

[...]

the opening of the general assembly has many interesting people coming to the house.

About this email: Epstein asks whether Musk has plans for New York, saying the opening of the UN General Assembly brings interesting people to his house. Musk declined.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name." Email sent TO Musk. After Musk declined, Epstein wrote: "just kidding, there is no one over 25 and all very cute."

Sources: [Fortune](#); [The Independent \(via AOL\)](#)

Tags: epstein-files, new-york

Epstein's reply: 'no one over 25 and all very cute'

From: Jeffrey Epstein

Date: September 11, 2013 (The Independent dates the exchange to 11 Sep 2013; the reply may have been sent later)

To: Elon Musk

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Do you think i am retarded

[...]

Just kidding , there is no one over 25 and all very cute.

About this email: After Musk dismissed a trip to New York to see UN diplomats, Epstein replied that he had been joking about the guests.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name." Email sent TO Musk; text as quoted by The Independent (Fortune prints the second sentence in lower case).

Sources: [The Independent \(via AOL\)](#); [Fortune](#)

Tags: epstein-files, new-york

Musk answers 'Yes' to coming to the Caribbean at Christmas

From: Elon Musk

Date: November 2013 (Fortune gives only the month)

To: Jeffrey Epstein

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Yes.

About this email: Asked by Epstein whether he would come to the Caribbean for Christmas 2013, Musk replied with one word.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Epstein's question, as quoted by Fortune: "will you come to caribbean this xmas?" (he also mentioned that Woody Allen would be there). Epstein answered Musk's 'Yes.' with "great."

Sources: [Fortune](#)

Tags: epstein-files, island-invitation

Epstein: 'will you come to caribbean this xmas?'

From: Jeffrey Epstein

Date: November 2013 (Fortune gives only the month)

To: Elon Musk

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

will you come to caribbean this xmas?

About this email: Epstein asks Musk whether he will come to the Caribbean for Christmas 2013, mentioning that Woody Allen would be there; Musk answered 'Yes.'

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name." Email sent TO Musk; the rest of the message (the Woody Allen reference) is paraphrased, not quoted in full, by Fortune.

Sources: [Fortune](#)

Tags: epstein-files, island-invitation

Musk: 'When should we head to your island on the 2nd?'

From: Elon Musk

Date: December 2013 (between 15 and 30 Dec 2013; exact date not given)

To: Jeffrey Epstein

Subject: Christmas and New Years

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Actually, I could fly back early on the 3rd. We will be in St Bart's. When should we head to your island on the 2nd?

About this email: Musk adjusts his holiday plans and asks Epstein when he should head to the island on January 2.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." The New York Post also quotes an earlier short Musk reply in this chain: "probably the 1st then".

Sources: [404 Media](#); [Fortune](#); [Business Insider \(via AOL\)](#); [New York Post \(via AOL\)](#)

Tags: epstein-files, island-invitation, st-barts

Epstein: 'I will send heli for you'

From: Jeffrey Epstein

Date: December 2013 (between 15 and 30 Dec 2013; exact date not given)

To: Elon Musk

Subject: Christmas and New Years

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

[T]he 2 or 3 would be perfect. I will come and get you.

[...]

I will send heli for you

About this email: Epstein tells Musk January 2 or 3 would be perfect and that he will send a helicopter; Musk replied 'Thanks.'

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name." Email(s) sent TO Musk. The first sentence is as printed by the New York Post (bracketed capital is the Post's); 'I will send heli for you' is quoted by TIME and 404 Media and may be a separate message in the chain. Musk's reply per TIME: "Thanks."

Sources: [New York Post \(via AOL\)](#); [TIME](#); [404 Media](#); [Fortune](#); [DOJ Epstein files, EFTA02518374 \(age-gated\)](#)

Tags: epstein-files, island-invitation

Musk: 'probably the 1st then'

From: Elon Musk

Date: December 2013 (between 15 and 30 Dec 2013; exact date not given)

To: Jeffrey Epstein

Subject: Christmas and New Years

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · SINGLE-SOURCE

probably the 1st then

About this email: Musk's short reply after Epstein said any day from January 1 to 8 would work for an island visit.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name." Lead Stories says the two then settled on a 2 January arrival.

Sources: [New York Post \(via AOL\)](#); [Lead Stories](#)

Tags: epstein-files, island-invitation

Musk: 'Thanks.' after Epstein offers helicopter

From: Elon Musk

Date: December 2013 (between 15 and 30 Dec 2013; exact date not given)

To: Jeffrey Epstein

Subject: Christmas and New Years

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · SINGLE-SOURCE

Thanks.

About this email: Musk's one-word reply after Epstein wrote that he would send a helicopter for him.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name."

Sources: [TIME](#); [DOJ Epstein files](#), EFTA02518374 (age-gated)

Tags: epstein-files, island-invitation

Musk asks Epstein: 'Is there a good time to visit?'

From: Elon Musk

Date: December 13, 2013 (Lead Stories and the New York Post give 13 Dec 2013; TIME gives 14 Dec)

To: Jeffrey Epstein

Subject: Christmas and New Years

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Will be in the BVI/St Bart's area over the holidays. Is there a good time to visit?

About this email: Musk tells Epstein he will be in the British Virgin Islands and St Barts over the 2013 holidays and asks when to visit.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Subject line per Lead Stories (404 Media renders it 'Christmas and New Year's').

Sources: [Lead Stories](#); [TIME](#); [NBC News](#); [404 Media](#); [New York Post \(via AOL\)](#); [DOJ Epstein files, EFTA00980008 \(age-gated\)](#)

Tags: epstein-files, island-invitation, st-barts

Epstein: 'always space for you'

From: Jeffrey Epstein

Date: December 15, 2013

To: Elon Musk

Subject: Christmas and New Years

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

any day 1st - 8th . play it by ear if you want. always space for you

About this email: Epstein tells Musk any day from January 1 to 8 would work for an island visit.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Email sent TO Musk. Later Epstein lines in the chain, per the New York Post and TIME: "[T]he 2 or 3 would be perfect. I will come and get you." and "I will send heli for you" (to which Musk replied "Thanks.").

Sources: [Lead Stories](#); [Fortune](#); [NBC News](#); [New York Post \(via AOL\)](#)

Tags: epstein-files, island-invitation

Epstein cancels the January 2014 island visit

From: Jeffrey Epstein

Date: December 30, 2013 (TIME says Epstein cancelled on 30 Dec 2013; Lead Stories says three days before the planned 2 Jan arrival)

To: Elon Musk

Subject: Christmas and New Years

PARTIAL TEXT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · WIDELY REPORTED

Bad news- Unfortunately , my schedule will keep me in New York . I was really looking forward to finally spending some time together with just fun as the agenda. so i am very disappointed. Hopefully we can schedule another time in the near future.

About this email: Epstein tells Musk he must stay in New York and cancels the planned island visit; Musk replied 'No problem.'

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files (about 3 million pages). The text here is only what news outlets quoted from the DOJ PDF; the DOJ file itself sits behind an age-verification page and could not be retrieved, and '=' characters are encoding artifacts in the released files. Being named in the files is not evidence of wrongdoing, and no record of Musk actually visiting the island appears in the reporting. Musk's public response on X (31 Jan 2026, as quoted by TIME): "No one pushed harder than me to have the Epstein files released and I'm glad that has finally happened" and "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express'"; NBC also quotes him: "I have never been to any Epstein parties ever and have many times call for the prosecution of those who committed crimes." Email sent TO Musk. Musk's reply, per TIME: "No problem."

Sources: [Lead Stories](#); [TIME](#); [Fortune](#); [NBC News](#)

Tags: epstein-files, island-invitation

Musk: 'No problem.' after Epstein cancels

From: Elon Musk

Date: December 30, 2013 (reply to Epstein's cancellation, which TIME dates to 30 Dec 2013)

To: Jeffrey Epstein

Subject: Christmas and New Years

EXCERPT · OUTSIDE & PERSONAL · GOVERNMENT RELEASE · SINGLE-SOURCE

No problem.

About this email: Musk's reply after Epstein cancelled the planned early-January 2014 island visit.

Notes: Released by the U.S. Department of Justice on 2026-01-30 as part of the Epstein files. The text here is only what news outlets quoted from the DOJ PDFs; the DOJ files sit behind an age-verification page and could not be retrieved. Being named in the files is not evidence of wrongdoing. Musk's public response on X (31 Jan 2026): "I had very little correspondence with Epstein and declined repeated invitations to go to his island or fly on his 'Lolita Express', but was well aware that some email correspondence with him could be misinterpreted and used by detractors to smear my name."

Sources: [TIME](#); [DOJ Epstein files](#), [EFTA00980008](#) (age-gated)

Tags: epstein-files, island-invitation

Chief of staff drafts 'pedo guy' apology for Musk

From: Sam Teller

Date: July 17, 2018

To: Elon Musk

EXCERPT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

I understand that my recent behavior has probably embarrassed you and I'm sorry. I apologize to Vernon Unsworth and am reaching out to him to personally convey this, as well as my deep appreciation for his work in saving the kids. ...

About this email: Two days after the 'pedo guy' tweet, Sam Teller sends Musk a recommended apology to employees and to Vernon Unsworth, based on feedback from trusted team members.

Notes: Only the beginning of Teller's proposed apology text is quoted in Plaintiff's Additional Material Fact 4, citing Wilson Decl. Ex. 16 (MUSK_1706-07). The quoted words are the draft apology Teller recommended Musk send.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Plaintiff's Statement of Genuine Disputes, Dkt. 76](#) (via [CourtListener/RECAP](#))

Tags: unsworth-case, pedo-guy, apology, sam-teller

Musk rejects apology advice after 'pedo guy' tweet

From: Elon Musk

Date: July 17, 2018

To: Sam Teller

PARTIAL TEXT · EXECUTIVE & INTERNAL · COURT RECORD · CONFIRMED

After sleeping on this, I'm not happy about the suggested approach. It would have been particularly foolish and craven of my to do it last night, right after a share drop, as my apology would simply have been dismissed as a disingenuous and cowardly attempt to restore the stock price.

About this email: Musk tells his chief of staff he is unhappy with the proposed apology, saying that apologising right after a share-price drop would have looked like a cowardly attempt to restore the stock.

Notes: Quoted in Plaintiff's Additional Material Fact 5, citing Wilson Decl. Ex. 16; the filing may not reproduce the whole email. 'craven of my' is as printed. Musk tweeted an apology the next day, July 18, 2018.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Plaintiff's Statement of Genuine Disputes, Dkt. 76 \(via CourtListener/RECAP\)](#)

Tags: unsworth-case, pedo-guy, apology, sam-teller, stock-price

BuzzFeed's Ryan Mac asks Musk about Unsworth's legal letter

From: Ryan Mac (BuzzFeed News)

Date: August 29, 2018

To: Elon Musk

Subject: BuzzFeed News: Unsworth legal letter

FULL TEXT · PRESS & JOURNALISTS · COURT RECORD · CONFIRMED

Hi Elon,

Ryan from BuzzFeed News here. We're reporting a story out about you receiving a letter from a lawyer representing British diver Vernon Unsworth. The letter, dated August 6, was sent to your Los Angeles home and discusses potential legal proceedings against you for libel. Given the Twitter conversation yesterday, I was hoping you could talk about the letter and whether you had seen it yet. I'm happy to chat on the phone if that works better for you.

Best,

Ryan Mac

About this email: BuzzFeed reporter Ryan Mac asks Musk for comment on a letter from Vernon Unsworth's lawyer threatening a libel claim.

Notes: Musk Decl. Ex. L (also Wilson Decl. Ex. 19). The exhibit prints the chain newest-first with nested 'On ... wrote:' lines; times given here are a best reading of those lines. Sent Wed Aug 29, 2018, about 7:40 AM. Signature block omitted.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [BuzzFeed News](#); [Unsworth opposition to summary judgment, Dkt. 75](#)

Tags: unsworth-case, buzzfeed, ryan-mac, defamation, off-the-record

Musk to BuzzFeed: 'Have you actually done any research at all?'

From: Elon Musk

Date: August 29, 2018

To: Ryan Mac (BuzzFeed News)

Subject: Re: BuzzFeed News: Unsworth legal letter

FULL TEXT · PRESS & JOURNALISTS · COURT RECORD · CONFIRMED

Have you actually done any research at all? For example, you incorrectly state that he is a diver, which shows that you know essentially nothing and have not even bothered to research basic facts.

About this email: Musk's first reply to Ryan Mac disputes that Unsworth is a diver and accuses the reporter of not researching basic facts.

Notes: Musk Decl. Ex. L (also Wilson Decl. Ex. 19). The exhibit prints the chain newest-first with nested 'On ... wrote:' lines; times given here are a best reading of those lines. Sent Wed Aug 29, 2018 (about 9 AM).

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [BuzzFeed News](#); [Unsworth opposition to summary judgment, Dkt. 75](#)

Tags: unsworth-case, buzzfeed, ryan-mac, defamation, off-the-record

Ryan Mac replies: Unsworth does cave dive

From: Ryan Mac (BuzzFeed News)

Date: August 29, 2018

To: Elon Musk

Subject: Re: BuzzFeed News: Unsworth legal letter

FULL TEXT · PRESS & JOURNALISTS · COURT RECORD · CONFIRMED

Hey Elon, thanks for getting back. Actually he prefers to be called a 'spelunker' and we've confirmed that he actually does do cave diving. But do you have any comment on the letter your received?

About this email: Mac replies that Unsworth prefers 'spelunker' and does cave dive, and again asks for comment on the legal letter.

Notes: Musk Decl. Ex. L (also Wilson Decl. Ex. 19). The exhibit prints the chain newest-first with nested 'On ... wrote:' lines; times given here are a best reading of those lines. 'your received' is as written. Signature block omitted.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [BuzzFeed News](#); [Unsworth opposition to summary judgment, Dkt. 75](#)

Tags: unsworth-case, buzzfeed, ryan-mac, defamation, off-the-record

'Stop defending child rapists' email to BuzzFeed's Ryan Mac

From: Elon Musk

Date: August 30, 2018

To: Ryan Mac (BuzzFeed News)

Subject: RE: BuzzFeed News: Unsworth legal letter

FULL TEXT · PRESS & JOURNALISTS · PUBLISHED BY RECIPIENT · CONFIRMED

Off the record

I suggest that you call people you know in Thailand, find out what's actually going on and stop defending child rapists, you fucking asshole. He's an old, single white guy from England who's been traveling to or living in Thailand for 30 to 40 years, mostly Pattaya Beach, until moving to Chiang Rai for a child bride who was about 12 years old at the time. There's only one reason people go to Pattaya Beach. It isn't where you'd go for caves, but it is where you'd go for something else. Chiang Rai is renowned for child sex-trafficking.

He may claim to know how to cave dive, but he wasn't on the cave dive rescue team and most of the actual dive team refused to hang out with him. I wonder why ...

As for this alleged threat of a lawsuit, which magically appeared when I raised the issue (nothing was sent or raised beforehand), I fucking hope he sues me.

About this email: Prefacing the email 'Off the record' (a condition BuzzFeed had not agreed to), Musk tells reporter Ryan Mac to stop 'defending child rapists', makes unsupported accusations about British caver Vernon Unsworth, and says he hopes Unsworth sues.

Notes: Sent Thu, Aug 30, 2018 at 6:43 PM. Published by BuzzFeed News on Sept 4, 2018 and later central to Unsworth v. Musk (C.D. Cal. 2:18-cv-08048), where it was a trial exhibit; Musk testified he believed the email was off the record. The accusations in the email were unsubstantiated; Unsworth denied them. The court exhibit (Musk Decl. Ex. L) shows the email also contained a Google search link for 'chiang rai child trafficking' between the second and third paragraphs, which BuzzFeed's published version omits.

Sources: [BuzzFeed News](#); [Vice](#); [The Drive](#); [Unsworth v. Musk](#), [Musk Declaration Ex. L/M](#), Dkt. 60 (CourtListener)

Tags: unsworth-case, thai-cave-rescue, defamation, buzzfeed, pedo-guy

'On background' follow-up to BuzzFeed on the mini-sub

From: Elon Musk

Date: August 30, 2018

To: Ryan Mac (BuzzFeed News)

Subject: Fwd: Letters

FULL TEXT · PRESS & JOURNALISTS · PUBLISHED BY RECIPIENT · CONFIRMED

On background

Unsworth also said I was asked to leave by the Thai govt, which is utterly false. Thai Prime Minister thanked me personally per attached docs. I went all the way to area 3 with the Thai SEAL team, who were awesome. Never saw Unsworth at any point. Was told he was banned from the site.

It is also total bs that the mini-sub wouldn't fit through the caves. It was designed and built to specifications provided to me directly by Stanton and the actual dive team. The only reason it wasn't used was that they were able to drain almost all the water out of the caves, so the underwater portion was very short, and the monsoon arrived later than expected.

Those pumps were critical. Some of the Tesla team helped with electrical power, but major credit to whoever provided those pumps. They were amazing. I'm told they were from some company in India.

About this email: In a second email marked 'On background', Musk disputes Unsworth's claims that he was asked to leave the cave site and that the mini-submarine would not have fit, and credits the pumps that drained the cave.

Notes: Sent Thu, Aug 30, 2018 at 7:16 PM, forwarding letters from Thai officials. BuzzFeed said it had not agreed to 'on background' terms either.

Sources: [BuzzFeed News](#); [Unsworth v. Musk](#), [Musk Declaration Ex. L/M](#), [Dkt. 60 \(CourtListener\)](#)

Tags: unsworth-case, thai-cave-rescue, mini-sub, buzzfeed

Ryan Mac follows up after doing 'due diligence'

From: Ryan Mac (BuzzFeed News)

Date: August 30, 2018

To: Elon Musk

Subject: Re: BuzzFeed News: Unsworth legal letter

FULL TEXT · PRESS & JOURNALISTS · COURT RECORD · CONFIRMED

Hey Elon, just wanted to make sure I did my due diligence to research basic facts and follow up here.

Thanks,
Ryan Mac

About this email: Mac follows up the next day; Musk's reply to this message was the 'child rapists' email.

Notes: Musk Decl. Ex. L (also Wilson Decl. Ex. 19). The exhibit prints the chain newest-first with nested 'On ... wrote:' lines; times given here are a best reading of those lines. Sent Thu Aug 30, 2018, about 6:07 PM. Signature block omitted.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [BuzzFeed News](#); [Unsworth opposition to summary judgment, Dkt. 75](#)

Tags: unsworth-case, buzzfeed, ryan-mac, defamation, off-the-record

Musk to PR adviser: 'Still, I'm a fucking idiot'

From: Elon Musk

Date: September 4, 2018

To: Juleanna Glover (outside PR adviser)

PARTIAL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

Yes, this is extremely bad. I didn't expect BuzzFeed to publish an off the record email. My intent was to have them investigate and come to their own conclusions, not publish my email directly. Still, I'm a fucking idiot.

About this email: After BuzzFeed published his 'child rapist' email, Musk tells his outside PR adviser the situation is 'extremely bad' and that he had meant for BuzzFeed to investigate, not publish.

Notes: Quoted 'in material part' in Plaintiff's Additional Material Fact 65, citing Wilson Decl. Ex. 34 (email correspondence dated Sept 4-5, 2018 between Musk, Juleanna Glover and others). The filing identifies the recipient only as Musk's outside public relations contact; press reports name Juleanna Glover. The full exhibit could not be retrieved.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Plaintiff's Statement of Genuine Disputes, Dkt. 76 \(via CourtListener/RECAP\)](#); [Unsworth opposition to summary judgment, Dkt. 75](#); [BuzzFeed News](#); [Gizmodo](#); [Gizmodo](#)

Tags: unsworth-case, buzzfeed, pr, juleanna-glover, regret

Musk: 'one of the dumbest things I've ever done'

From: Elon Musk

Date: September 4, 2018

To: Juleanna Glover (outside PR adviser)

PARTIAL TEXT · OUTSIDE & PERSONAL · COURT RECORD · CONFIRMED

Completely agree. I was told a lot of bad things this guy did and was upset that no one cared to investigate, so responded in what I felt was off the record. In the past, BuzzFeed has respected emails prefaced with 'off the record', but this time they did not. It was still one of the dumbest things I've ever done and this distraction couldn't come at a worse time.

About this email: In a second email to his PR adviser, Musk says he had been told bad things about Unsworth, was upset nobody investigated, and calls his email to BuzzFeed one of the dumbest things he has done.

Notes: Quoted 'in material part' in Plaintiff's Additional Material Fact 65, citing Wilson Decl. Ex. 34. The full exhibit could not be retrieved.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Plaintiff's Statement of Genuine Disputes, Dkt. 76 \(via CourtListener/RECAP\)](#); [Unsworth opposition to summary judgment, Dkt. 75](#); [BuzzFeed News](#); [Gizmodo](#)

Tags: unsworth-case, buzzfeed, pr, juleanna-glover, regret

Ryan Mac: 'I didn't agree for the conversation to be off the record'

From: Ryan Mac (BuzzFeed News)

Date: September 4, 2018

To: Elon Musk

Subject: Re: BuzzFeed News: Unsworth legal letter

FULL TEXT · PRESS & JOURNALISTS · COURT RECORD · CONFIRMED

Hi Elon,

I didn't agree for the conversation to be off the record, but appreciate the response. To follow up, I've tried to report out some of these accusations on my own but have not found anything to corroborate the claims. Are you able to share anything that you've found about Vernon Unsworth? Do you have any evidence or documentation showing he took a 12-year-old child bride, that he is a child rapist, or that he was kicked off the rescue site, as you stated in your other email? Also are you able to share your correspondence with Rick Stanton showing your discussion of the submarine specs?

With regards to your statement about the legal threat not coming up until you raised the issue on Twitter, the legal letter was dated on Aug. 6 and sent to your Los Angeles home and to one of your SpaceX emails. Did you not see the letter prior to your tweets?

Thank you,
Ryan

About this email: Mac tells Musk he never agreed to off-the-record terms, says he could not corroborate the accusations and asks for evidence.

Notes: Musk Decl. Ex. L (also Wilson Decl. Ex. 19). The exhibit prints the chain newest-first with nested 'On ... wrote:' lines; times given here are a best reading of those lines. Sent Tue Sep 4, 2018, about 8:49 AM. 'that be

was kicked off' is as printed.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [BuzzFeed News](#); [Unsworth opposition to summary judgment, Dkt. 75](#)

Tags: unsworth-case, buzzfeed, ryan-mac, defamation, off-the-record

Musk: publishing would 'destroy your journalistic credibility'

From: Elon Musk

Date: September 4, 2018

To: Ryan Mac (BuzzFeed News)

Subject: Re: BuzzFeed News: Unsworth legal letter

FULL TEXT · PRESS & JOURNALISTS · COURT RECORD · CONFIRMED

Off the record

We haven't had a conversation at all. I sent you an off the record email, which very clearly and unambiguously said 'off the record'. If you want to publish off the record comments and destroy your journalistic credibility, that's up to you.

As for answering more questions, I would be happy to do so, but not with someone who just told me that they will not honor accepted rules of journalism.

About this email: Musk insists his earlier email was unambiguously off the record and warns Mac that publishing it would destroy his credibility.

Notes: Musk Decl. Ex. L (also Wilson Decl. Ex. 19). The exhibit prints the chain newest-first with nested 'On ... wrote:' lines; times given here are a best reading of those lines. Sent Tue Sep 4, 2018, about 10:53 AM.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [BuzzFeed News](#); [Unsworth opposition to summary judgment, Dkt. 75](#)

Tags: unsworth-case, buzzfeed, ryan-mac, defamation, off-the-record

Ryan Mac offers to go off the record going forward

From: Ryan Mac (BuzzFeed News)

Date: September 4, 2018

To: Elon Musk

Subject: Re: BuzzFeed News: Unsworth legal letter

FULL TEXT · PRESS & JOURNALISTS · COURT RECORD · CONFIRMED

Hi Elon,

While I'd rather chat on the record, I'm happy to go off record with you moving forward so you can answer specific questions regarding the allegations you've made.

Thanks,

Ryan

About this email: Mac offers to speak off the record from that point so Musk can answer questions about his allegations.

Notes: Musk Decl. Ex. L (also Wilson Decl. Ex. 19). The exhibit prints the chain newest-first with nested 'On ... wrote:' lines; times given here are a best reading of those lines. Sent Tue Sep 4, 2018, about 12:22 PM.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [BuzzFeed News](#); [Unsworth opposition to summary judgment, Dkt. 75](#)

Tags: unsworth-case, buzzfeed, ryan-mac, defamation, off-the-record

Musk: ask Unsworth about his history in Thailand

From: Elon Musk

Date: September 4, 2018

To: Ryan Mac (BuzzFeed News)

Subject: Re: BuzzFeed News: Unsworth legal letter

FULL TEXT · PRESS & JOURNALISTS · COURT RECORD · CONFIRMED

Off the record

I suggest you ask Unsworth to describe his whole - 30 year history of visiting Thailand. What was he doing in Pattaya Beach for the better part of a decade when there are no caves of note in the area?

About this email: In a further 'off the record' email on the day BuzzFeed published, Musk tells Mac to question Unsworth about his decades of visits to Thailand.

Notes: Musk Decl. Ex. L (also Wilson Decl. Ex. 19). The exhibit prints the chain newest-first with nested 'On ... wrote:' lines; times given here are a best reading of those lines. Sent Tue Sep 4, 2018, about 1:18 PM. '- 30 year' is as transcribed (possibly '~30 year' in the original).

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [BuzzFeed News](#); [Unsworth opposition to summary judgment, Dkt. 75](#)

Tags: unsworth-case, buzzfeed, ryan-mac, defamation, off-the-record

Ryan Mac after publication: still happy to talk

From: Ryan Mac (BuzzFeed News)

Date: September 4, 2018

To: Elon Musk

Subject: Re: BuzzFeed News: Unsworth legal letter

FULL TEXT · PRESS & JOURNALISTS · COURT RECORD · CONFIRMED

Hey Elon, I'm sure you've seen the story at this point. Still happy to talk with you on whatever terms you want as long as we set them beforehand. Let me know if you want to do a phone call.

Best,

R

About this email: After BuzzFeed published Musk's emails, Mac says he is still willing to talk on agreed terms.

Notes: Musk Decl. Ex. L (also Wilson Decl. Ex. 19). The exhibit prints the chain newest-first with nested 'On ... wrote:' lines; times given here are a best reading of those lines. Sent Tue Sep 4, 2018, about 5:30 PM.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [BuzzFeed News](#); [Unsworth opposition to summary judgment, Dkt. 75](#)

Tags: unsworth-case, buzzfeed, ryan-mac, defamation, off-the-record

Musk to BuzzFeed reporter: 'Get lost, you creep'

From: Elon Musk

Date: September 4, 2018

To: Ryan Mac (BuzzFeed News)

Subject: Re: BuzzFeed News: Unsworth legal letter

FULL TEXT · PRESS & JOURNALISTS · COURT RECORD · CONFIRMED

Get lost, you creep

About this email: Musk's final reply to Ryan Mac after BuzzFeed published his emails.

Notes: Musk Decl. Ex. L (also Wilson Decl. Ex. 19). The exhibit prints the chain newest-first with nested 'On ... wrote:' lines; times given here are a best reading of those lines. Exhibit header: Tuesday, September 04, 2018 5:31:45 PM.

Sources: [Unsworth v. Musk \(C.D. Cal. 2:18-cv-08048\)](#), [Declaration of Elon Musk ISO summary judgment, Dkt. 60 \(via CourtListener/RECAP\)](#); [BuzzFeed News](#); [Unsworth opposition to summary judgment, Dkt. 75](#)

Tags: unsworth-case, buzzfeed, ryan-mac, defamation, off-the-record